



USERS GUIDE AND TECHNICAL MANUAL

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For Windows 10/8/7

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For

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FOREWORD

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We look forward to continued user support to allow us to improve AFGROW as a commercial product.

1.0 INTRODUCTION

1.1 Historical Information

AFGROW's history traces back to a crack growth life prediction program (ASDGRO), which was written in BASIC for IBM-PCs by Mr. Ed Davidson at ASD/ENSF in the early 1980's. In 1985, ASDGRO was used as the basis for crack growth analysis for the Sikorsky H-53 Helicopter under contract to Warner-Robins ALC. The program was modified to utilize very large load spectra, approximate stress intensity solutions for cracks in arbitrary stress fields and use a tabular crack growth rate relationship based on the Walker equation on a point-by-point basis (Harter T-Method). The point loaded crack solution from the Tada, Paris, and Irwin Stress Intensity Factor Handbook was originally used to determine K (for arbitrary stress fields) by integration over the crack length using the unflawed stress distribution independently for each crack dimension. After discussions with Dr. Jack Lincoln (ASD/ENSF), a new method was developed by Mr. Frank Grimsley (AFWAL/FIBEC) to determine stress intensity, which used a 2-D Gaussian integration scheme with Richardson Extrapolation, which was optimized by Dr. George Sendeckyj (AFWAL/FIBEC). The resulting program was named MODGRO since it was a modified version of ASDGRO.

In 1987, James Harter came to work for the Air Force Wright Aeronautical Laboratories (AFWAL/FIBEC) and rewrote MODGRO, Version 1.X (still in BASIC for PC DOS). Over the next 2 years, a tabular crack growth rate database was added. Decreasing-increasing crack growth rate tests were performed to obtain data below $1.0\text{E-}08$ inches/cycle for 7075-T651 Aluminum and 4340 Steel. During that period, MODGRO, Version 1.X [1] included part-through flaw solutions from Newman and Raju, and standard closed-form solutions for symmetrical through-cracks (center, single edge, and double edge cracks). These solutions could also be modified for arbitrary stress fields using a Gaussian integration method with a stress distribution defined by the ratio of the unflawed stress field of interest divided by the unflawed stress field for the baseline geometry. The error in this method, of course, increases with crack length, but error in life is minor since the majority of life is consumed while the crack lengths are relatively short.

In 1989, MODGRO, Version 2.0 was rewritten in Turbo Pascal for PC-DOS as a move to a more structured computer language. At that time, Dr. George Sendeckyj provided MUCH assistance in debugging and optimizing the arithmetic operations. George was also learning the C language and was practicing by translating the BASIC code to Structured BASIC and then C at the same time I was coding it in Turbo Pascal. Runtime comparisons were made in the spirit of friendly competition. Actually, George's C version of MODGRO, Version 1.0 was faster. George was the first to have written a version of MODGRO in the C language. Additions to version 2.0 of the code included a plasticity based closure model, which was based on work by Erdogan, Irwin, Elber, M. Creager, and Sunder [2, 3, and 4]. The model is a variable amplitude closure model and more detail is contained in this report. There is also credit due to Mitch Kaplan [5] because of his good suggestion to only recalculate the beta (or alpha) values at user defined crack growth increments. It was decided to simply use the user-input value for the Vroman integration percentage, which is

normally used when analyzing blocked spectra. A real-time crack length plotting capability was also added to the program. The code was totally changed in the process, but the name MODGRO remained.

From 1990-1993, the code changed very little (still released in Turbo Pascal). Small changes/repairs were made based on errors that were discovered. The code was used to help manage the flight test program for the X-29. During high angle-of-attack maneuvers, the vertical tail experienced severe buffeting. MODGRO, Version 2.0 was used by NASA/Dryden to estimate the vertical tail life from actual flight test data collected for each flight. The use of the code allowed the Program Managers to assess the effect of various flight maneuvers on the vertical tail, and in some cases, flights were re-arranged to maximize the amount of flight data and minimize tail damage accumulation.

In 1993, the Navy was interested in using MODGRO to assist in a program to assess the effect of certain (classified) environments on the damage tolerance of aircraft. The Navy wanted to build a user-friendly code to be used in the program and initiated an agreement with WL/FIBEC to develop a state-of-the-art user interface with the added capability to perform life analysis under adverse environments. This effort required additional manpower for software development and baseline crack growth testing. On-site contract support was used to meet this requirement. Work began at that time to convert the MODGRO, Version 3.0 to the C language for UNIX to provide performance and portability to several UNIX Workstations [6]. The workstation platform was chosen to provide additional computational power for MODGRO.

In 1994, a research contract with Analytical Services and Materials was established to provide support for the Navy effort and assist in future research and development requirements of WL/FIBEC. This was when the current UNIX interface was born. In July 1994, a presentation of the results for the Navy project was given to the Navy sponsor and WL/FIBE management. After the presentation, the WL/FIBE Branch Chief (Mr. Jerome Pearson) requested that the code be renamed AFGROW, Version 3.0. Work on the Windows 95 version of AFGROW was started in October of 1996.

A composite bonded repair crack growth analysis capability was added during 1996-97. The bonded repair capability was based entirely on work by Dr. Mohan Ratwani [7]. In addition, a strain-life based crack initiation analysis capability was added. The strain-life initiation analysis capability was taken from APES, Inc. [8]. During reorganizations at Wright-Patterson AFB in 1997, it was decided that AFGROW would not receive further research and development funds. As a result, the on-site software development support provided by Analytical Services and Materials was reduced significantly. Since the Windows95 version of AFGROW had become most widely used, it was decided to discontinue the UNIX support. Recent advances in windows hardware capability has made it possible for AFGROW to equal and even surpass the performance capabilities of many UNIX systems. The Air Force organization responsible for AFGROW development was changed from WL/FIBEC to AFRL/VASE during a reorganization in 1998.

In late 1997 and early 1998, the U.S. Navy provided AFGROW funding to support a fleet tracking database development effort (FLEETLIFE) for the AV-8 Harrier. It was decided to add the Microsoft Component Object Model (COM) server technology [9] to AFGROW. This capability allows AFGROW to be used by any Windows software. Since the FLEETLIFE code was being written for the Windows platform, this provided an efficient means for the fleet tracking database to use AFGROW for structural life analyses.

An experimental Power Macintosh version of AFGROW was released in late 1998 for evaluation purposes. This version was discontinued shortly thereafter due to maintenance costs and the lack of demand.

The AFGROW user base continued to grow dramatically in 1998. Air Force Air Logistic Center (ALC) use and strong support for the code was greatly responsible for additional funding, provided in late 1998, for multiple crack and time dependent analysis capabilities. The Air Force Aging Aircraft Office (ASC/SMS) provided these funds. As a result of this funding, these new features were added to the code. Since AFGROW treats each crack tip as a separate object, it was fairly straight forward to accommodate the analysis of a large number of cracks.

One of the biggest challenges to the multiple crack analysis capability was the design of the user interface. An intuitive drag and drop design interface was developed for the multiple crack capability.

The multiple crack capability allowed AFGROW to analyze two independent cracks in a plate (including hole effects), non-symmetric corner cracked holes under axial, bending, and bearing loading (corner cracks only for now). Finite element based solutions were available for two through or corner cracks at holes, and through cracks in plates¹ under axial loading. These solutions and more information were available in the open literature [10, 11].

The COM capabilities in AFGROW have allowed it to be used with an external K-solver program to communicate with AFGROW to perform real time crack growth analysis for multiple cracks (more than two) and cracks growing in complex and/or unique structure. Additional stress intensity solutions and spectrum load interaction models were added to AFGROW, and user-defined plug-in capabilities have been added to allow users to include proprietary or unique stress intensity solutions.

¹ including the effect of any adjacent hole(s)

1.2 Commercial Development

AFGROW became a commercial product of LexTech, Inc. in 2008, when the U.S. Air Force Research Laboratory ended all development, training, and support effort. The graphical user interface has been changed to include several new enhancements, features, and remain compliant with the Microsoft Windows operating environment. The new GUI included “dock-able” frames that may be moved to different locations as desired. Extensive copy/paste improvements and additions have been made, and user-input tabular data may be used with no limitation on the number of entries. The Advanced Model Interface included a new pull-down menu to allow users to select/delete objects. A new notification window was also added to provide additional information/warnings to users. This included notification when geometric limitations have been violated for a given K-solution.

LexTech also developed a database search engine for an extensive fracture mechanics database (AFMAT). This capability is available for all AFGROW users with a current license².

The tabular solution for corner crack(s) at countersunk fastener holes have been one of the most requested newer solutions for incorporation in AFGROW. The solution matrix is quite large and required a lot more time than anticipated for implementation. The current release includes solutions for the axial and bending load cases. Additional work is required before releasing the solutions for the bearing load case.

1.3 Current Development

AFGROW, release 5.03.01.18 and later are compliant with 64 bit Windows operating systems. AFGROW now includes the ability to use different tabular crack growth rate data as a function of temperature and/or crack tip location (a or c-direction). A new bearing solution has been added to the Advanced Model Interface for non-symmetric through-the-thickness cracks at holes. The solution for a corner crack at the “knuckle” of a countersunk hole has been added to the Advanced Model Interface. The current version also includes a new weight function solution for corner cracks with a non-uniform stress distribution in the c-direction.

A new spectrum management tool has been developed to simplify spectrum development, expand AFGROW’s capabilities, and provide several additional features for spectrum analysis. A new XML based spectrum format has been developed to allow tagging of individual spectrum levels to identify temperature or track damage. The ability to develop spectra for three channels of loading (axial, bending, and bearing) is also included. Damage tracking and multiple load channels will be implemented in a future interim release of AFGROW.

Finally, there have been numerous bug fixes and capability enhancements. Many of the enhancements have be related to the COM capability of AFGROW.

² One year from initial license installation or under a yearly maintenance agreement

1.4 Future Plans

Updates to the Advanced Solution for multiple through cracks will be incorporated, and a solution for multiple through cracks in a row of holes (MSD scenario) is under development.

The ability to analyze cracks that grow out-of-plane is an area that will require a substantial effort. It is hoped that work can begin as soon as possible in this area as time and funding permit. For now, this capability may be accomplished using the existing AFGROW Plug-In Interface.

As always, the developers of AFGROW will continue to listen to user comments and suggestions to improve the code.

1.4 Installing AFGROW for Windows

AFGROW, for Windows 8 (Windows 7, VISTA & XP, Service Pack 2), is available for download³ as a single self-extracting executable file. This file is approximately 670 MB in size. When installed, the code with data is approximately 2 GB⁴, so be patient because the installation process may be slow (especially in VISTA – allow some time for the installation to start).

Always remember to remove any previous version of AFGROW before installing the latest release.

³ www.afgrow.net

⁴ Most of this space is required by the Advanced Solution database

1.4.1 The Installation Process

AFGROW uses the Install Shield© program to generate the installation program required to copy and register the required program files to an individual PC. The installation program must be run by a user with Administrator privileges⁵. If the single file method is used, the dialog shown in Figure 1 appears:

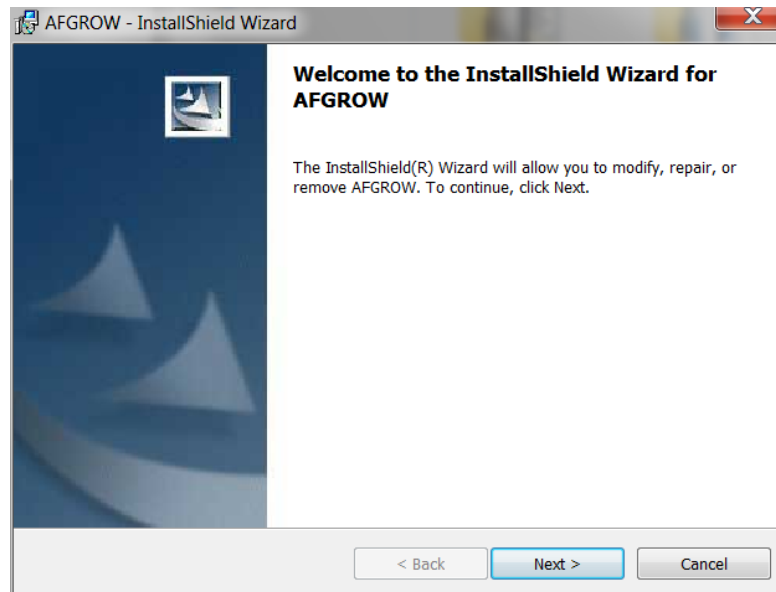


Figure 1: AFGROW Self-Extracting Setup Dialog

The installation procedure is started when the user selects the setup button in the above dialog box (Figure 1).

⁵ VISTA users should not right-click the install file to run as administrator

Once the installation has been started (using the single or multiple file methods), the following dialog (Figure 2) is displayed:



Figure 2: AFGROW Splash Screen

A blue background also appears with logo for AFRL/RBSM. This splash screen is also used each time AFGROW is opened. The installation process proceeds as the user selects next (or back) on each dialog.

One of the installation dialogs (Figure 3) provides users with the option to select the directory path for the new AFGROW installation.



Figure 3: AFGROW Installation Directory

The default directory path is C:\Program Files\AFGROW (or Program Files(x86) on 64 bit operating systems).

The installation software will display the user-defined settings in the following dialog (Figure 4):

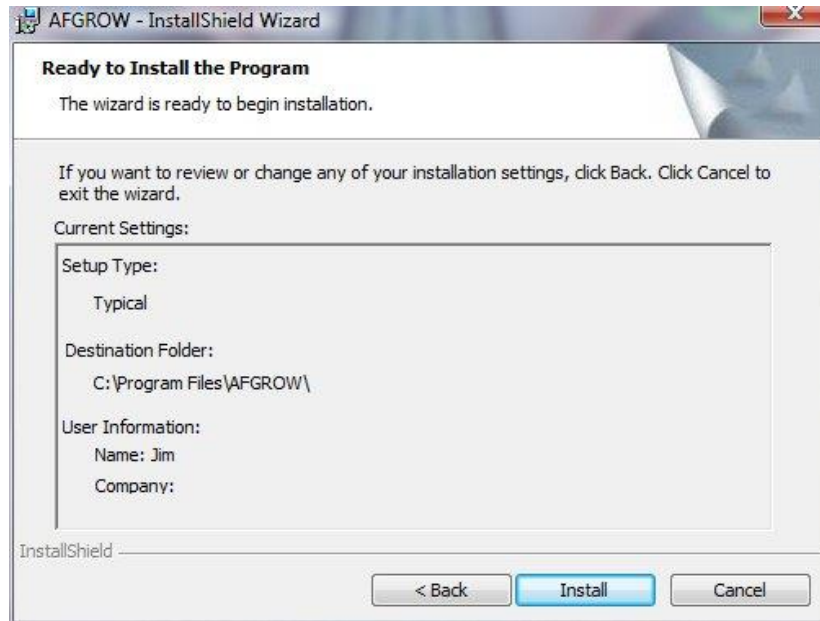


Figure 4: AFGROW Program Folder Name

At this point, there is an option to go back to change certain settings or to proceed with the installation.

The final dialog box notifies the user that the required files have been copied to the computer as shown in Figure 5.

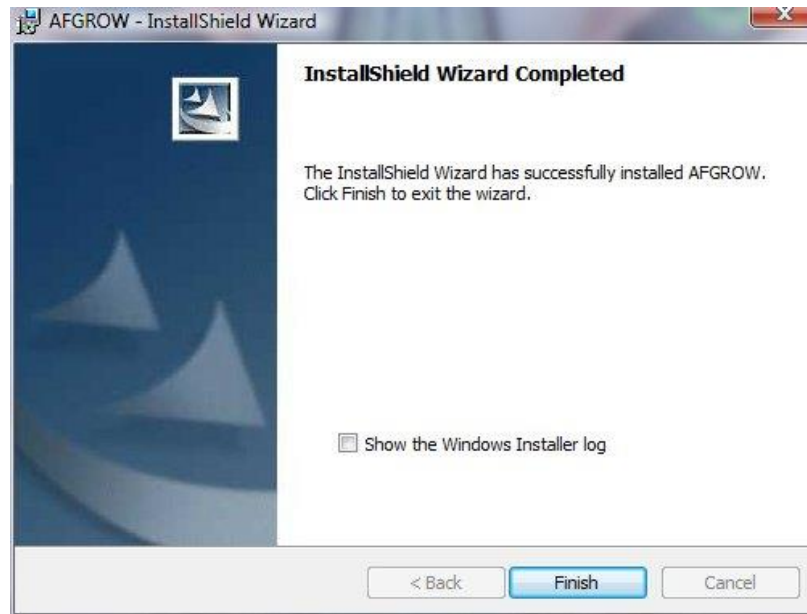


Figure 5: Final Installation Dialog

When the finish button is selected, the last thing that the installation program does is to register the required Microsoft® Dynamic Link Libraries (DLLs) and other control files. The installation program may prompt the user to reboot so that certain system related libraries can be reloaded.

AFGROW will not operate properly unless this registration is successful. This may occur if the user does not have full control of the Windows© system directories.

The installation software will attempt to complete all registration tasks. If there are any problems with this process, AFGROW functionality may be affected. There may be problems saving input files or the COM server may not function properly. If any of these problems occur, notify the developers of AFGROW for help in resolving the problem.

In order to simplify the process of obtaining the host ID for the installation PC (to obtain a permanent user license), a program is included to perform this task. It is accessible in the AFGROW startup menu.

1.5 Uninstalling AFGROW for Windows

AFGROW is a fairly complex code that includes several files, libraries, and registrations that need to be properly removed before installing a new version (or simply to clear AFGROW out of a computer). The proper way to remove AFGROW is to use the Add/Remove Programs⁶ dialog in the Windows® control panel. The Add/Remove Programs dialog is shown in Figure 6.



Figure 6: Add/Remove Programs Dialog

Simply select AFGROW, as shown in Figure 6, and click on the Add/Remove button and follow the subsequent instructions to complete the process.

⁶ Programs and Features in MS Windows 7

2.0 INTERFACE FEATURES

There are three types of graphical user interface (GUI) options available to AFGROW users. The most common interface is the interactive user interface for classic specimen geometries. A second interface is used for more complicated geometries for which more advanced stress intensity solutions are used. These advanced solutions are generally developed using finite element models (curve fit or table look-up). Users can drag and drop objects (cracks, and holes) on a cross-section of a plate. The third interface is called the component object model (COM), or dispatch interface. This interface allows the use of AFGROW for life prediction calculations from most windows compatible software.

The interactive GUI for AFGROW now includes the ability for each tabbed frame to be resized and moved (docked) to any location within AFGROW dialog window as desired. Frames are moved by moving the mouse to the title bar of the frame, holding down the left button, and moving the frame as desired. As the frame is moved, possible docking locations are displayed as shown in Figure 7. Simply move the mouse cursor to the indicated location and release the left button.

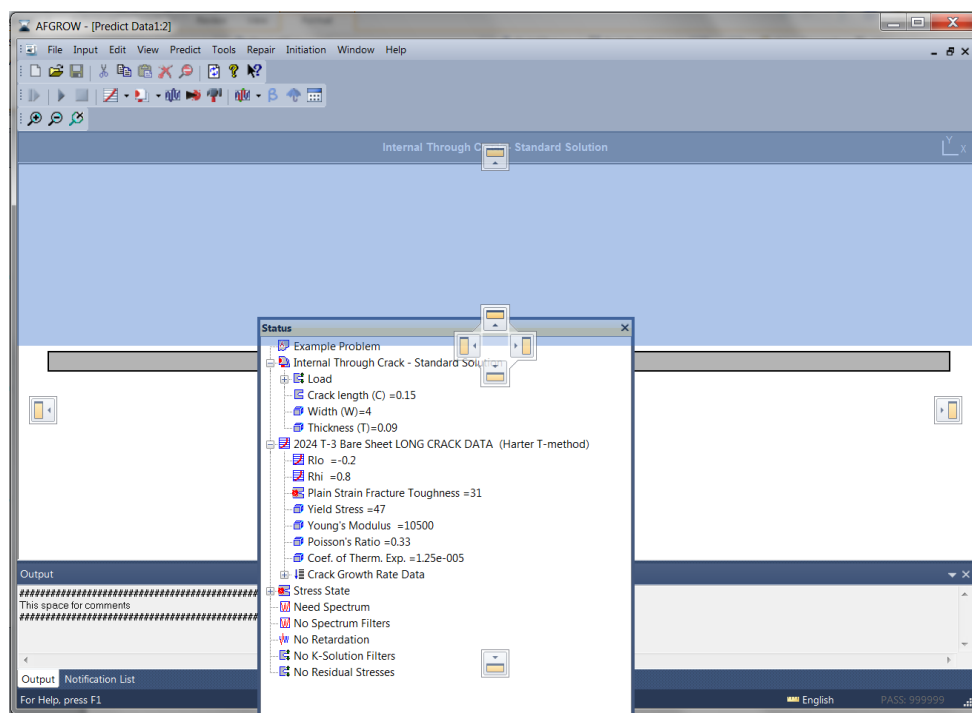


Figure 7: Moving or Docking AFGROW Frames

The frames can also be separated from the main AFGROW dialog box and displayed anywhere on the monitor (or a second monitor, if available).

In addition to the ability to dock/move a particular frame, tabbed windows may also be removed and docked to other locations as desired. To move a tabbed frame, move the

mouse cursor over the desired tab, press & hold the left button, and drag the view to the desired location. A tabbed frame may also be extracted as a separate window by double-clicking in the desired tab. To replace any tabbed frame in an existing tabbed window, simply drag the title bar of the frame to the desired tab bar. If there is only one frame in the window, dragging the frame to the title bar of the desired location will place the frame in this window and a new tab bar will be created for this window.

2.1 Classic Model Interface

The classic AFGROW user interface is divided in three frames, Figure 8:

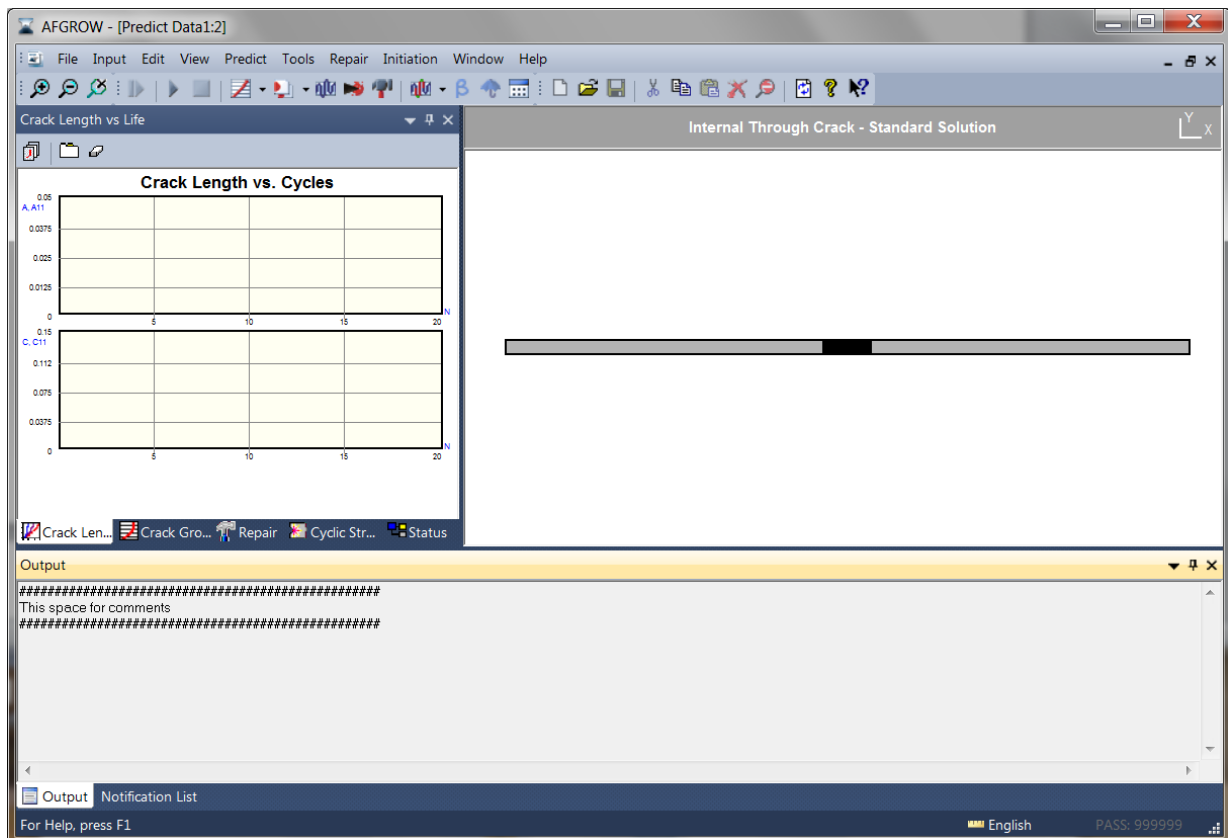


Figure 8: AFGROW Windows Graphical User Interface

Note: The frames are resized by clicking on a frame boundary and dragging it to the desired position.

2.1.1 Main Frame

We will refer to the upper left-hand frame as the main frame since it is used as the workhorse frame of AFGROW. The main frame has several functions, as indicated in the tab bar shown in Figure 9.

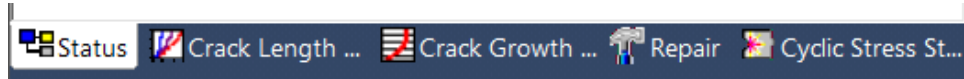


Figure 9: Mainframe Functions

The desired view may be selected using the tabbed list as shown above.

2.1.1.1 Status View

The status view shows users the values of the input variables to be used in any life prediction, Figure 10.

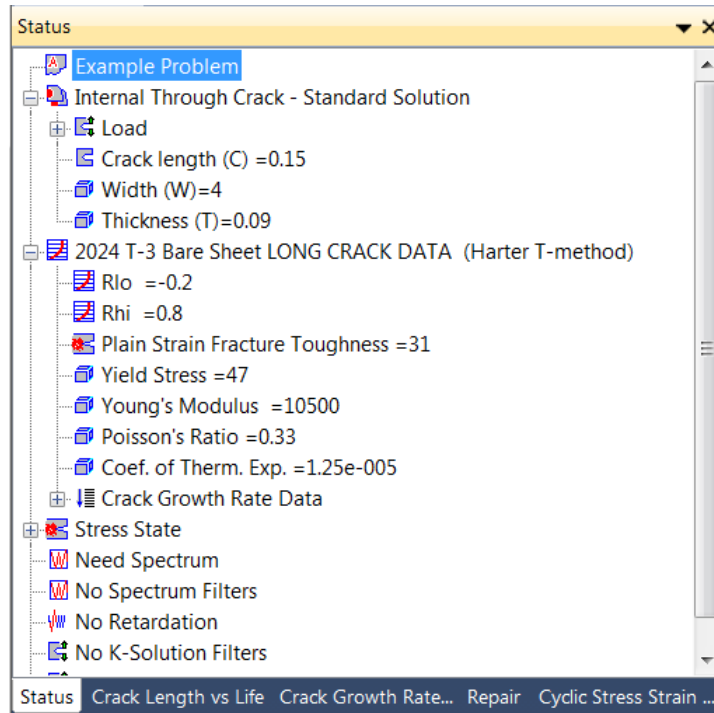


Figure 10: Status View

This view shows the major input variables with the option to expand certain variables to show more detail. A tree structure is used to expand or contract the view. Clicking on a plus (+) symbol will expand a variable list (showing more details) and clicking on a minus (-) symbol will contract the list (hiding the details).

2.1.1.2 Crack Growth Plot View

The crack growth vs life plot view, Figure 11, is provided to give users a real-time view of the crack length vs. cycles in the two possible directions of crack growth. In the case of a part through-the-thickness cracks, crack length in the thickness direction (a-direction) is displayed on the upper plot. The crack length in the width direction (c-direction) is shown in the lower plot.

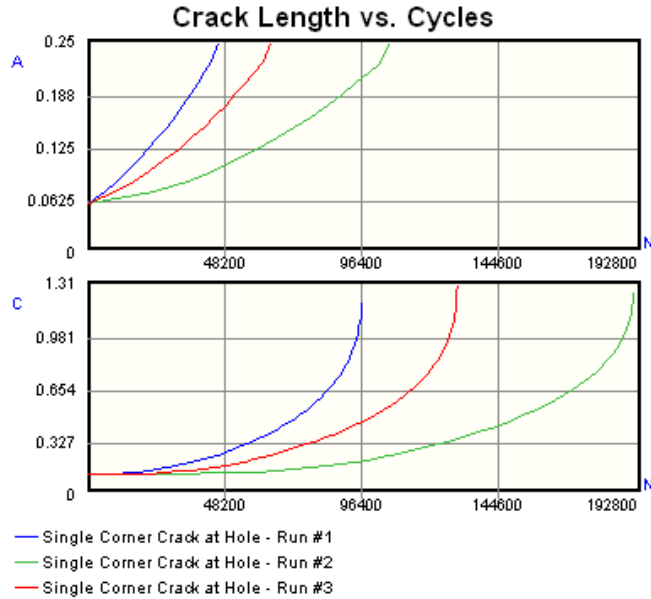


Figure 11: Crack Growth Plot View

There are several features incorporated in this view. These features are accessible through the icons at the top of the frame, Figure 12.

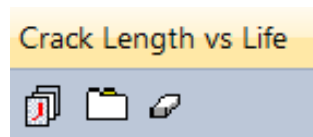


Figure 12: Crack Growth Plot Tools

User's may also access these features by right-clicking in the plot window and selecting the desired option. The tabular data for any plot may also be copied by selecting the legend item, right-clicking, and selecting "Copy Data" as indicated in Figure 13. These copied data may be copied into any Windows applications (e.g. Excel). The selected curve may also be deleted.

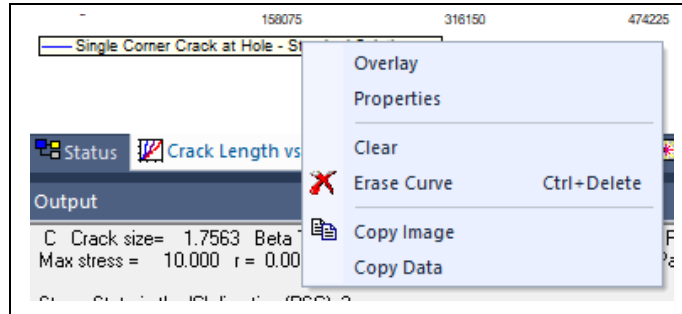


Figure 13: Copying Plot Data

2.1.1.2.1 Overlay Tool



The first tool is the overlay tool. Clicking on this toggle-type button causes the crack length plots for each prediction (up to the last eight runs) to appear on the same plot for comparison purposes. If this button is not activated, the plot only displays data for the latest analysis.

2.1.1.2.2 Chart Property Tool



The second tool is the property tool. It allows the user to select various plot properties such as the plot legend, black and white plots, reverse plotting⁷, and the width of the graph lines.

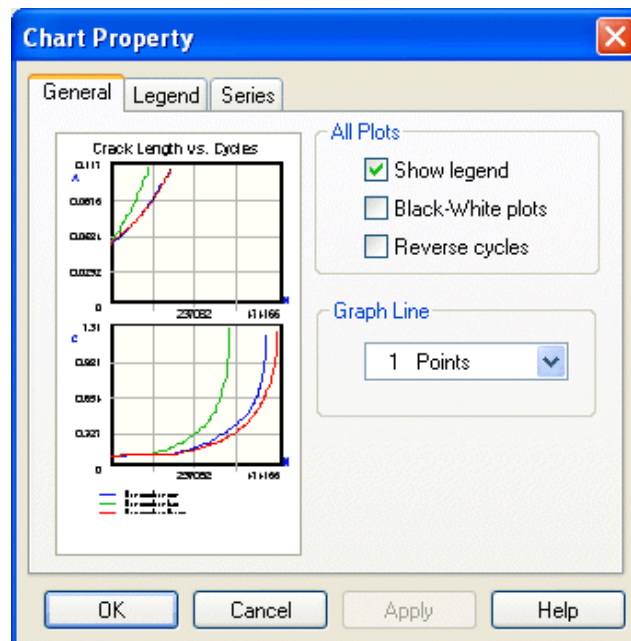


Figure 14: General Plot Properties

⁷ Reverse plotting shows the number of cycles remaining until failure on the x-axis.

The default legend for each plot is the name of the model being analyzed. Users may also edit the default legend in the chart property dialog.

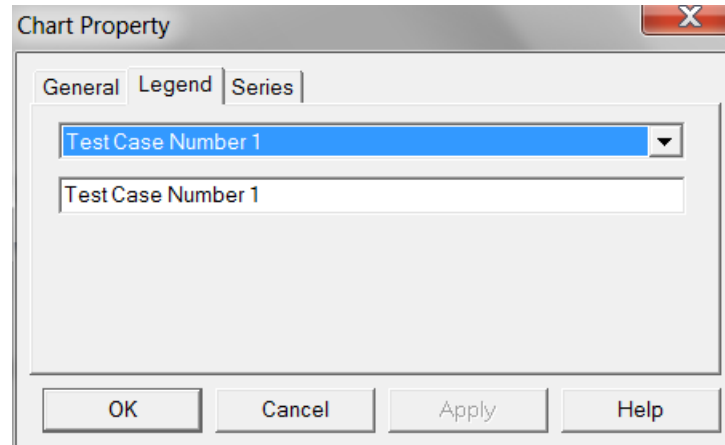


Figure 15: Plot Legend Editor

Finally, the chart property dialog may be used to control which crack length data are plotted in either graph.

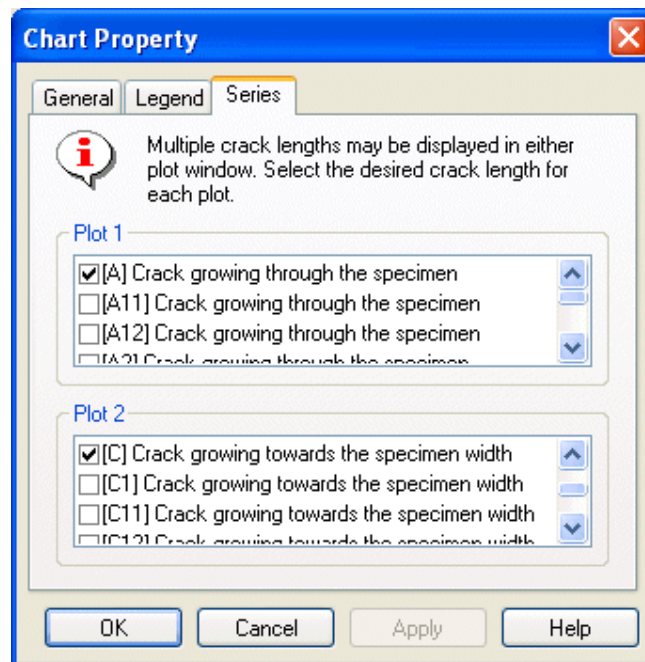


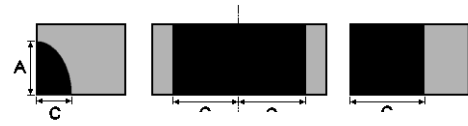
Figure 16: Plot Series Selection

Since AFGROW can analyze multiple cracks, the various crack lengths are identified using the following labeling system described below

Classic Models

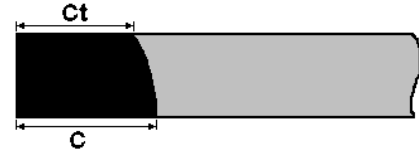
Single or Symmetric Cracks⁸

C - Length in the width direction
A - Length in the thickness direction



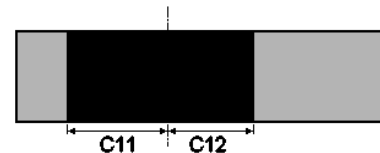
Oblique Through-the-Thickness Crack

C - Length in the width direction (longest)
Ct - Length in the width direction (shortest)



Offset Through Crack

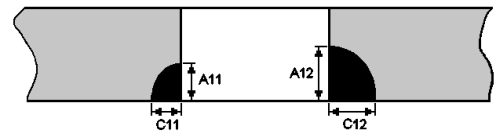
C11 - Length in the width direction (left tip)
C12 - Length in the width direction (right tip)



Advanced Models

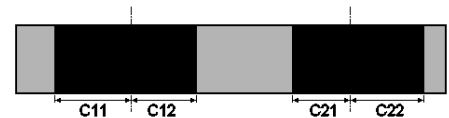
Single⁹ or Double¹⁰ Cracks at a Hole

A11 - Length in the thickness direction (left tip)
A12 - Length in the thickness direction (right tip)
C11 - Length in the width direction (left tip)
C12 - Length in the width direction (right tip)



Single¹¹ or Double Through Cracks (not attached to a hole)

C11 - Length in the thickness direction (left crack, left tip)
C12 - Length in the thickness direction (left crack, right tip)
C21 - Length in the thickness direction (right crack, left tip)
C22 - Length in the thickness direction (right crack, right tip)



Single⁵ or Double Edge Cracks

C11 - Length in the thickness direction (left crack)
C21 - Length in the thickness direction (right crack)



⁸ This is the most general case and includes corner cracks, surface cracks, embedded cracks, edge cracks, and centered through cracks.

⁹ Single cracks at a hole are enumerated as a left (1) crack tip.

¹⁰ Double cracks at a hole are attached on opposite sides of a given hole and are enumerated by the left (1) and right (2) crack tips.

¹¹ Single cracks not attached to a hole are enumerated as a left (1) crack.

2.1.1.2.3 Erase Tool



The third tool is the erase tool. This tool simply erases all plots from the plot view.

2.1.1.2.4 Copy Image

The plot view may be copied into the Windows clip board and pasted in other applications (Word, Excel, etc.) by right-clicking in the view and selecting the option “Copy Image.”

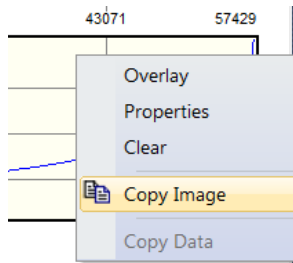


Figure 17: Crack Growth Plot View Tool Menu

2.1.1.3 da/dN vs. Delta K Plot View

This view displays the crack growth rate versus ΔK relationship for the given material and crack growth rate method being used (Forman, Walker, Tabular, etc.). The relationship for negative R-values may be handled differently for each crack growth rate model. This information is displayed at the bottom of the plot, Figure 18.

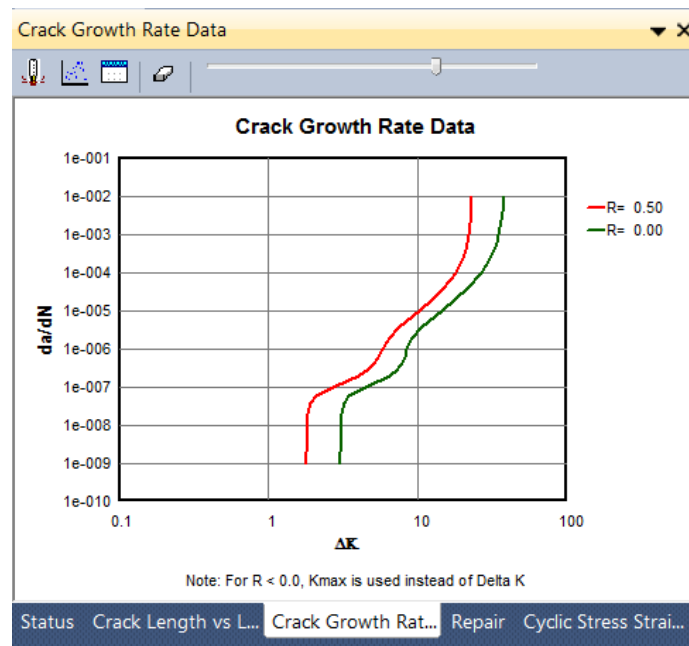


Figure 18: da/dN vs. Delta K Plot View

This view shows EXACTLY what crack growth rate data are being used for a given analysis. A “slider bar” is located at the top of the view which may be used to change the stress ratio (R) that is displayed. Users can also “left-click” on the numeric value of R at the top of the legend and enter the desired R-value to be displayed.

When multiple data sets are being used with tabular lookup data for different environmental conditions (e.g. temperature), users may select which data set is being plotted by using the pull-down menu as indicated in Figure 19.

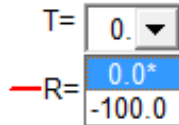


Figure 19: Data Set Selection for Tabular Lookup Growth Rate Data

There are a few more tools available for this view as described in the following sections.

2.1.1.3.1 Freeze Tool



The left most icon is a thermometer that is used to “freeze” a given curve so that data at several (up to 8) R-values may be displayed on the same plot. The R-value for each curve is displayed on the right side of the plot. Users may double-click on the numeric value on the top element of the R legend and enter an exact value (instead of using the slider tool).

2.1.1.3.2 Data Overlay Tool



The next icon (second from the left) allows tabular crack growth rate data from a text file to be overlaid on the plot for comparison. The format for this file is given in the on-line manual, or may be determined from the example file included with the AFGROW installation. The next icon allows the material data to be changed by opening the AFGROW material dialog window.

2.1.1.3.3 Material Data Tool



The material data icon is used to open the dialog box for the currently selected crack growth rate model.

2.1.1.3.4 Erase All Tool



The fourth icon is the erase all tool. This tool is used to erase any data or growth rate curves in the crack growth rate plot view except for the last crack growth rate curve that was displayed.

2.1.1.3.5 Additional Tools

There are a few additional tools that are not located at the top of the crack growth rate plot view. All of the tools are available by right-clicking inside the crack growth rate view as indicated in Figure 20.

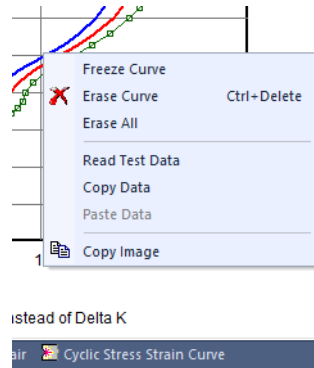


Figure 20: Crack Growth Rate View Tool Menu

In addition to the freeze, erase curve, and erase all tools, other useful tools are available. The option to read test data allows tabular crack growth rate data from a text file to be overlaid on the plot for comparison. When this option is selected, the Windows open file dialog is displayed, and the appropriate file may be selected by navigating to the directory where the file is located. The format for this file is given in the on-line manual or may be determined from the example file included with the AFGROW installation. Prior to displaying the data, AFGROW opens a dialog box showing the minimum and maximum values of crack growth rate and ΔK and provides a means to switch the values if they are in the wrong order.

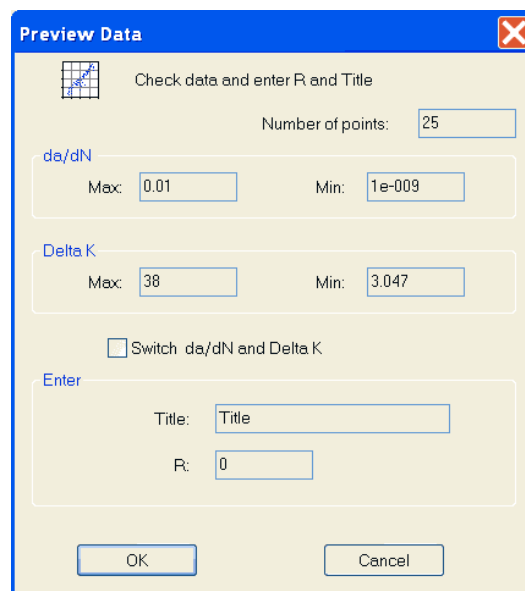


Figure 21: Rate Data Preview Dialog

The copy and paste data tools allow data to be transferred between other Windows applications (e.g. Excel). If there is only one curve (for a single R-ratio) shown on the plot, the copy data tool will copy that data into the Windows clipboard. If more than one curve is displayed on the plot, the data to be copied is selected by clicking on the curve of choice. Once data is in the clipboard, the paste data capability will be activated, and the data may be pasted in other Windows applications. Data copied from other Windows applications may be pasted in the crack growth rate plot view by selecting the paste data tool. These data merely need to exist in two columns (crack growth rate and ΔK). Finally, the copy image tool is used to copy the plot image itself so that it can be pasted in another windows application.

2.1.1.4 Repair Plot View

The repair plot view, Figure 22, shows the stress intensity correction as a function of crack length for a crack under a bonded repair.

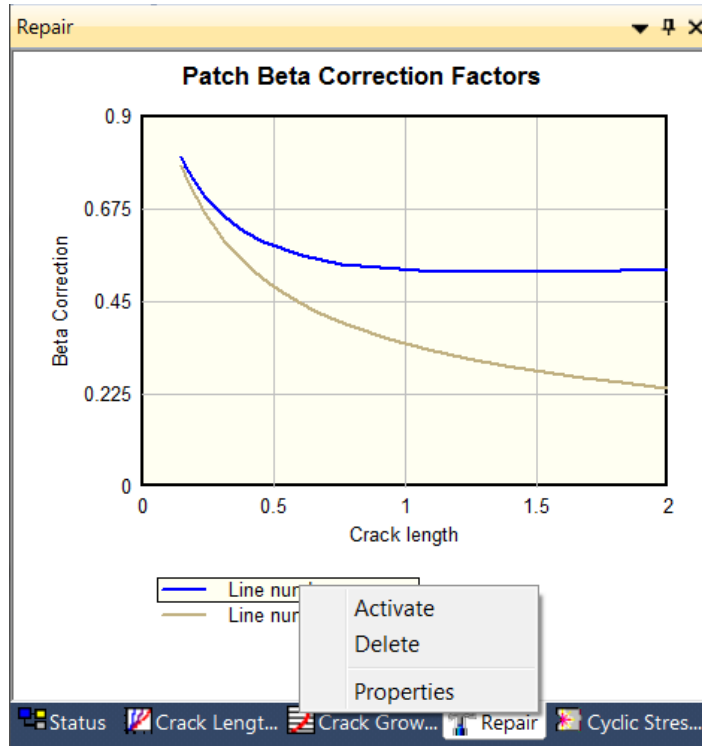


Figure 22: Repair Plot View

The correction at a given crack length (at this time, AFGROW only allows the repair option to be applied to through-the-thickness cracks) is multiplied by the applied beta factor (see section 3.2.3 on beta factors). There are no tools for this view. However, up to eight repair design curves are displayed on this plot. The user may select the curve of choice by either left clicking on the desired curve on the plot, or by right clicking on the legend for the desired curve. Three options are available: Activate, Delete, or Properties. Choosing properties will open a series of windows showing the details of the repair design for the selected curve.

As is the case with the other main frame views, the repair image may be copied into the Windows clipboard by left clicking in the view and selecting the option to copy the image.

2.1.1.5 Initiation Plot View

AFGROW uses a strain-life based crack initiation analysis method to predict crack initiation life. The initiation plot view, Figure 23, displays the cyclic stress strain or the strain-life data to be used for a given analysis.

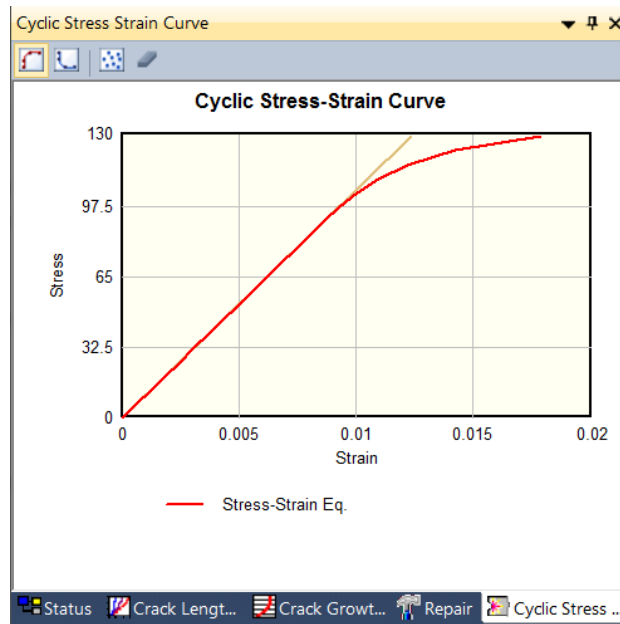


Figure 23: Initiation Plot View

The cyclic stress-strain plot includes a line representing the current Young's modulus to allow the user to verify that the appropriate modulus value is being used for the input cyclic data. If it is not correct, this must be changed in the appropriate material data dialog box.

There are several tools available for the initiation plot view. The first (left most) activates the cyclic stress-strain plot. The cyclic stress-strain curve is the locus of the endpoints of stable hysteresis loops for the given material. The next tool (second from the left) activates the strain-life plot for the given material. The strain-life data is usually obtained for small round bar specimens but is only applicable for the given lives to a specified initial crack size. The next tool allows tabular cyclic stress-strain or strain-life data from a text file to be overlaid on the plot for comparison. The format for these files is given in the on-line manual or may be determined from the example files included with the AFGROW installation. The last tool on the top of the dialog erases any overlaid data from the plot.

As in the other main frame views, other less frequently used tools may be accessed by right-clicking inside the view and selecting the desired option. These options include the ability to copy the image and pasting data on the plot from another Windows application. These data merely need to exist in two columns (stress and strain or strain and life). Prior to displaying the data, AFGROW opens a dialog box showing the minimum and maximum values and provides a means to switch the values if they are in the wrong order.

2.1.2 Animation Frame

The upper right-hand frame will be referred to as the animation frame since this frame shows a view of the crack plane (*the standard AFGROW GUI interface assumes planar crack growth*), and the crack growth is animated during the prediction process. This allows users to visualize the crack growth prediction process. The specimen view may be magnified by dragging an area with the mouse (or using the view, zoom menu option) enlarged or diminished by simply resizing the animation frame, Figure 24.

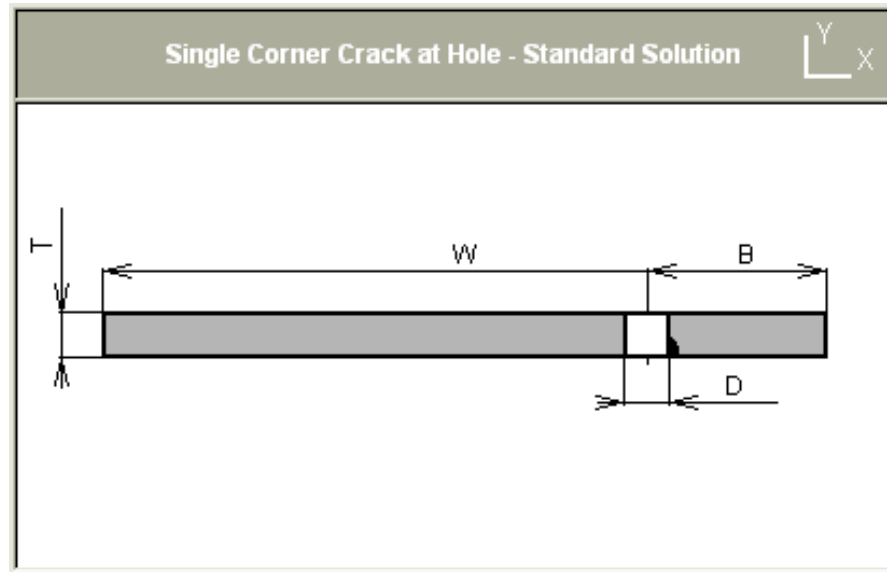


Figure 24: Animation Frame

2.1.2.1 Showing Specimen Dimensions

Specimen dimension definitions are displayed in the animation frame by selecting Dimensions in the View menu. The actual dimensions will not be shown since they are given in the status view, but the definitions of width (W), thickness (T), Offset (B), ..., etc., will be indicated in the frame. Selecting Dimensions again in the View menu will turn off this option.

2.1.2.2 Refreshing the Specimen View



After an analysis, the crack will remain at the failure length in the animation frame. The specimen view may be reset to the initial crack length, by selecting Refresh in the View menu. Users may also select the refresh icon in the toolbar (see Section 3.3.11).

2.1.3 Output Frame

The lower frame will be referred to as the output frame, Figure 25, since it is the default location for the results of life analyses and any notifications related to a given analysis.

2.1.3.1 Output View

By default, output data for a given life prediction is displayed in the output view. This view may be selected in the tab bar for the output frame as shown in Figure 25.

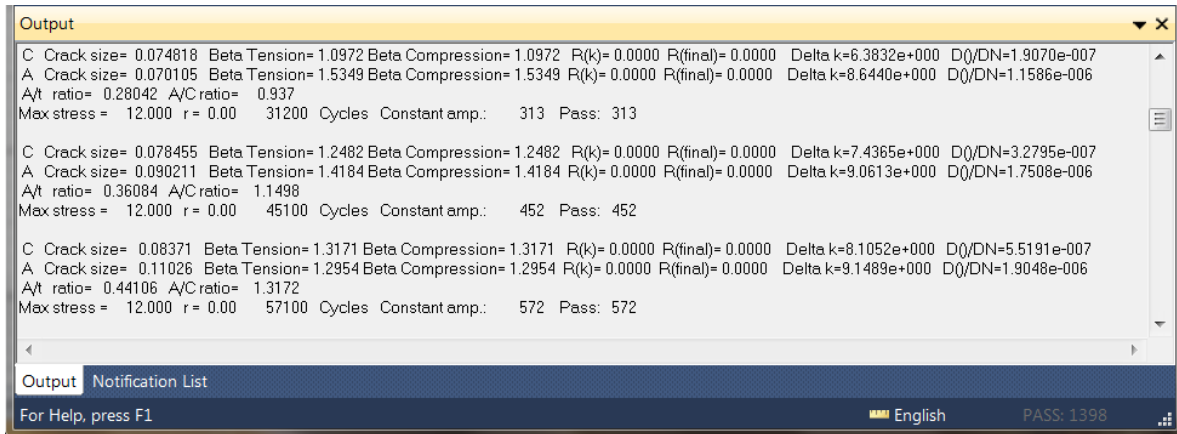


Figure 25: Output Frame

Output data consists of crack length, beta values, stress ratio, stress intensity, crack growth rate, and spectrum data.

2.1.3.2 Notification List

The notification list view has been added to the output frame to provide notifications relevant to the life prediction process. Users have requested that warning messages be displayed whenever a crack length limit of a given K-solution has been exceeded. One example is shown below in Figure 26.

Figure 26: Notification List

Additional notifications are expected in future releases.

2.1.4 Menu Bar

The menu bar, Figure 27, provides access to all of the features of AFGROW.

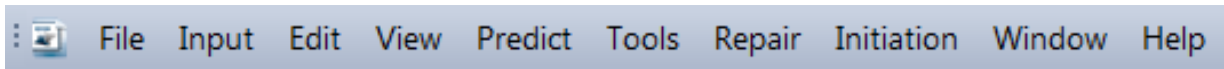


Figure 27: Menu Bar

Complete descriptions of all of the items in the menu bar are provided in Section 3.0.

2.1.5 Tool Bars

The tool bar used in the classic model interface is divided in two parts; the predict tool bar, and the standard Windows tool bar (see Figure 28).

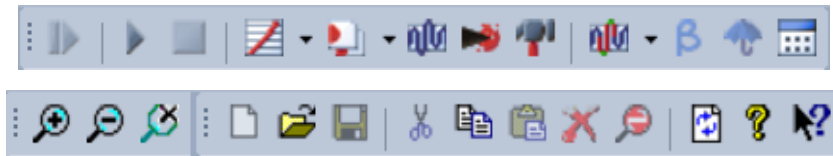


Figure 28: Tool Bars

The tool bars are dockable – meaning that they can be moved and placed in other areas in the AFGROW window. To move a tool bar: left-click/hold on the 3 vertical dots for a given tool bar, drag it to the desired location, and release the mouse button.

The predict tool bar is designed to provide shortcuts to many of AFGROW's most commonly used features, and the standard tool bar consists of icons that are used by most Windows programs. The icons in the tool bar are designed to give a visual depiction of their purpose¹². Tool bar icons are associated with the appropriate menu item (see Section 3).

2.1.6 Status Bar

The status bar, Figure 29, is located at the bottom of the AFGROW window.



Figure 29: Status Bar

¹² A short description of tool bar icons is displayed when the mouse cursor is held over an icon.

The status bar is used as the location for messages related to the status of AFGROW. A message is printed telling users that the prediction is executing or has finished. The current system of units is displayed and may be changed by clicking (right or left) on the units icon and selecting the units of choice. Finally, the status bar prints the number of times the input spectrum has been repeated (spectrum passes) while the prediction is being executed. This may be useful for cases that require long run times since this will let users know that the code is still running.

2.2 Advanced Model Interface

The advanced model interface uses all of the features of the classic interface with the addition of the specimen properties and tool box menu bars as shown in Figure 30.

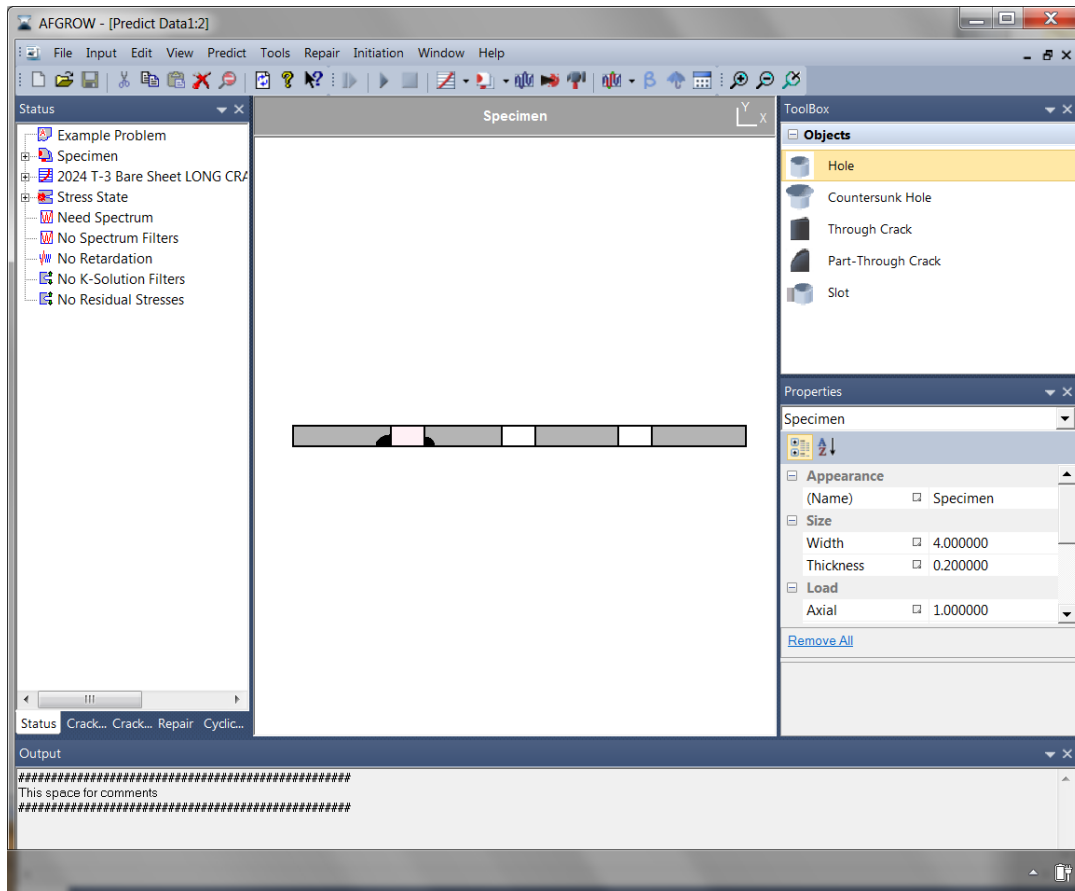


Figure 30: Advanced Model Interface

2.2.1 Specimen Properties

The specimen design bar provides a means to change the attributes of any object (plate, crack, or hole) in the animation frame. The default object is the plate itself which appears when the advanced model option is selected (see Sections 3.2.3.2 and 3.3.1.3 for more details). The plate is the only object that can't be deleted since it represents the crack plane and is essentially the “canvas” upon which the model is constructed. The plate width and thickness are always displayed when the plate object is selected.

2.2.1.1 Load Cases

This property is assigned to the specimen and is accessed through the property view for the plate itself. Advanced solutions for one or two corner cracks at a straight shank hole are available for axial, bending, and bearing load cases. The beta solution for combined loading cases is determined using the axial, bending, and bearing stress fraction properties. The

stress fractions are determined as documented in Section 3.2.3.1.5. The reference stresses for each load case are as indicated below:

Axial Load Case: Remote Gross Stress

Bending Load Case: Maximum Gross Bending Stress

Bearing Load Case: Bearing Stress at the Hole

The bearing stress fraction calculator provided in the Classic Model Interface (see Figure 90) is not available in the Advanced Model Interface, so care should be taken when setting the bearing fraction. The concept of an effective width should be considered, and the bearing fraction is easily determined as shown below when the gross remote stress is used as the reference for the combined load case.

$$\text{Bearing Fraction} = (1 - \text{Axial Fraction} - \text{Bending Fraction}) * W_{\text{effective}} / \text{Hole Diameter}$$

For wide plates, it would not make sense to assume that the local bearing load at a given hole would be applied over the entire width (for the purpose of converting the bearing beta values to a gross reference stress). It may be more practical to use the hole spacing as the effective width, or another value based on Engineering judgment¹³. However this is determined, it is most important to be sure that the actual bearing stress at the hole is accounted for in the conversion calculation. This can be done by dividing the fastener load for the reference load case by the effective width and plate thickness, then dividing that result by the total reacted gross reference stress.

The stress fraction properties are currently associated with the specimen since corner cracks may be placed at any hole, but may not be placed on different holes. Also, these properties will not be visible to the user until corner crack(s) objects have been placed on a hole. However, since bending and bearing solutions are currently available for these limited cases, the load fraction values are changed to 100 percent axial when the corner crack(s) transition to a specimen boundary. The load fraction values in the Advanced Model interface will be assumed to be 100 percent axial for all geometries that do not include K-solutions for bending and bearing. Currently, the advanced, continuing damage solution (crack at a u-shaped notch) does not include a bearing solution. It will be added in a future release. For all advanced cracks at a hole (corner and through-the-thickness), bending and bearing solutions are available.

2.2.1.2 Multiple Point K-Solution

The default solution for the Advanced Model for two cracks at a hole is a two-point solution for each crack object. However, the option to use a multiple point solution (11 points along the crack front) is available under the Property View for the specimen under the Load Case option. This option is currently only available for two corner cracks at a hole.

¹³ It is recommended that the edge distance modeled on either side of a bearing loaded hole be $\leq 6D$

2.2.1.3 Additional Properties Assigned to Crack Objects

Some options are available through the Property View for a crack object. For example, Beta Correction, Residual Stress, and AoC Constant are capabilities that may be used for corner cracks at a hole. These additional capabilities will be indicated in the Property View if they are available for a given geometric case.

2.2.2 Modifying Properties for Objects

As crack and/or hole objects are added to the model, they may be selected by simply left-clicking on them in the animation frame. Crack and hole attributes may be changed by left-clicking in the value column next to the attribute to be changed and typing the desired value. These attributes may also be changed by using the mouse to drag a selected object using “handles” as shown in Figure 31.

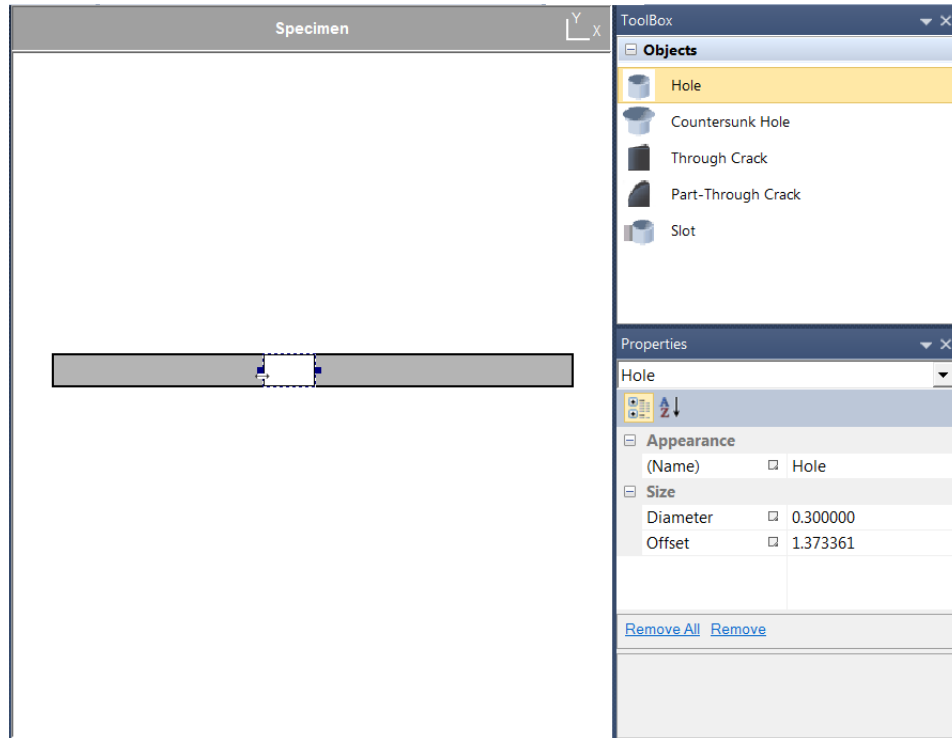


Figure 31: Moving and/or Resizing Objects with the Mouse

The position (offset) of an object is measured from the left edge of the plate to the center of the object. This is consistent with the definition of hole and crack offset that is used in “classic” models that include an offset crack or hole option. Part-through (corner) cracks must be attached to a hole, so there is no offset option for these objects. Once objects are added to a plate, the position does not change if the plate dimensions are changed. Therefore, the plate dimensions may not be reduced to a size that does not extend beyond any crack or hole object.

2.2.3 Tool Box Menu Bar

The tool box is the “palette” used to create the desired model in the advanced crack interface. Objects in the tool box may be placed on the plate by dragging them to the plate with the mouse. The current general limitations¹⁴ for the advanced model interface are listed below.

Maximum number of cracks = 2

Maximum number of holes = 4

Part-Through cracks are attached to the lower edge on either side of one hole

Through cracks may be placed anywhere on a plate (internal, edge, or at any hole)

When dragging a crack to a hole, it is important to release the mouse button when the cursor is very close to the edge of the hole (within 2-3 pixels). This placement is easier if the view, zoom option is used to magnify the appearance of the hole (see Section 3.3.12).

¹⁴ Note: These limitation do not apply to User-Defined Plug-In Models

2.3 Plug-In Model Interface

The Plug-In interface allows users to add new models to AFGROW as required. The Classic and Advanced models included with AFGROW may not include a specific geometry of interest. These models may be shared between users, or maintained as proprietary models. A sample plug-in model was developed to demonstrate this capability.

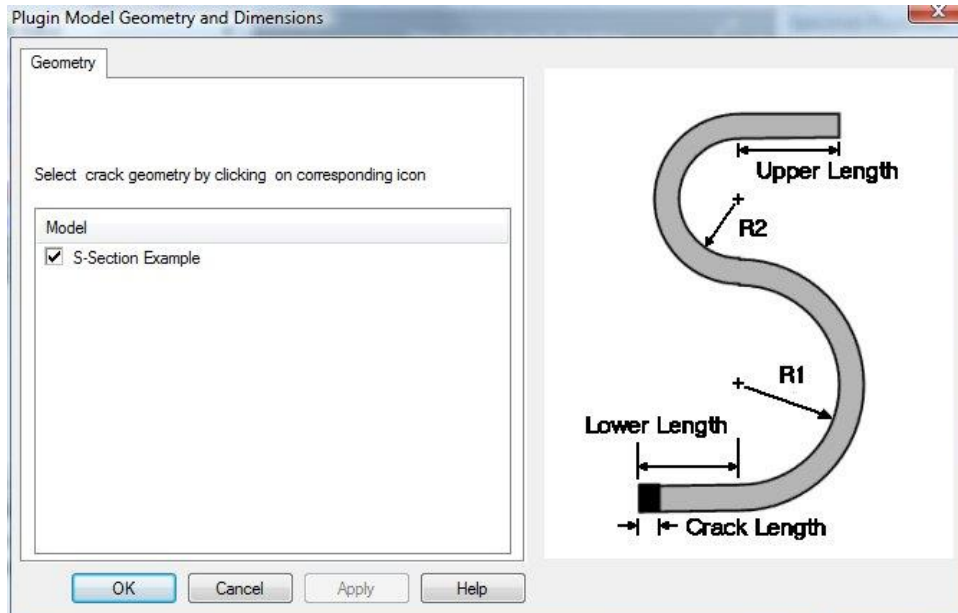


Figure 32: Example Plug-In Model

This relatively simple plug-in model was intended to demonstrate that arbitrary geometries can be modeled and displayed using the drawing tools available in AFGROW. In this case, the s-section example is the only model available for the plug-in interface. If more models were available, the user would select the desired model from the list of all available plug-in models as indicated in Figure 32.

Any number of cracks (part through, through-the-thickness, and/or combinations of the two) may be modeled. Each crack tip is a separate object, and the developer is responsible for the individual K-solutions, error checking, drawing, etc.

Plug-in models are compiled Dynamic Link Library (DLL) files that pass information to and from AFGROW. AFGROW includes drawing tools to allow the plug-in model to be displayed and animated in the animation frame. Step-by-step instructions for this example are downloadable on the AFGROW web page as a Power Point presentation.

The Plug-In interface uses the Advanced Model interface features as shown in Figure 33.

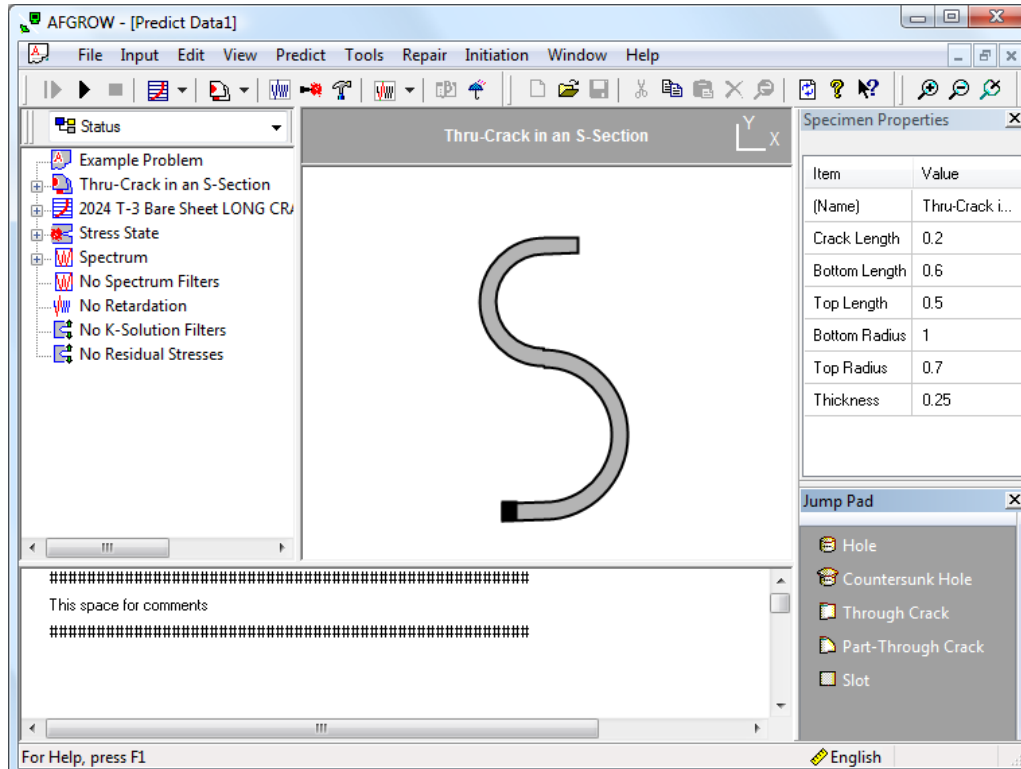


Figure 33: Plug-In Model Interface

2.3.1 Specimen Properties

The parameters for each plug-in model are listed as specimen properties. Their values may be changed by selecting the appropriate value with a mouse click and entering the desired value.

2.3.2 Tool Box Menu Bar

The tool box menu bar is not currently used for the Plug-In interface.

3.0 AFGROW MENU SELECTIONS

All of the analytical features in AFGROW are accessible through the main menu. The following sections will provide the details of all the available menu selections.

3.1 File Menu

The AFGROW file menu, Figure 34, contains several options as shown below.

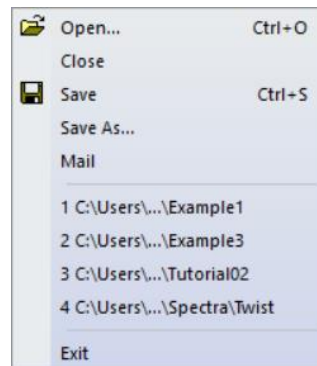


Figure 34: File Menu

As is the case with most Windows software, AFGROW stores the last few opened files that may be recalled by clicking on any one of the numbered items in the file menu. The standard selections in the file menu are described below.

3.1.1 File Open

Toolbar Icon: 

This action allows you to choose a previously saved file to be opened in AFGROW.

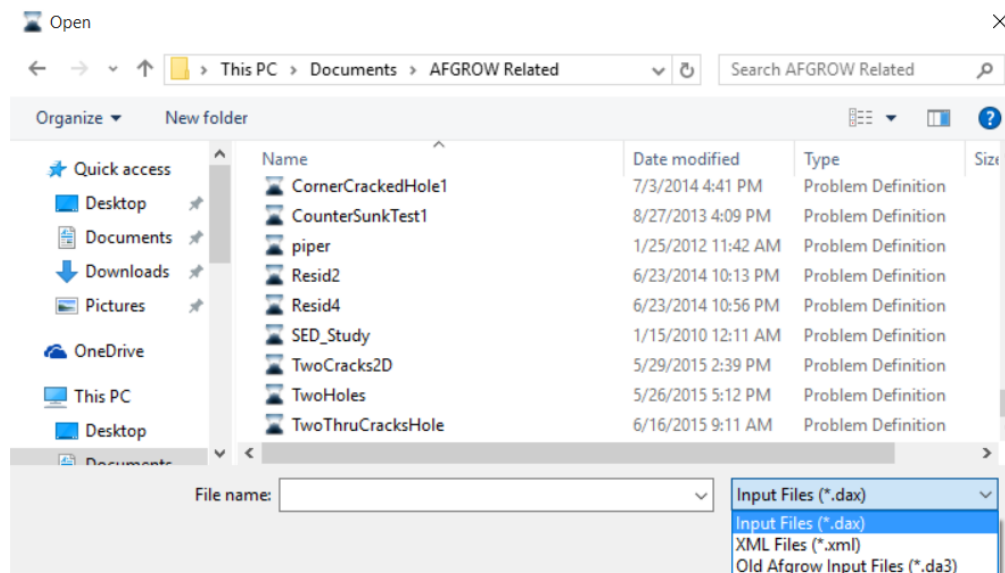


Figure 35: File Open Dialog

AFGROW now supports the Microsoft Extensible Markup Language (XML) format. This format uses data tags for all input parameters and will allow new features and capabilities to be added to AFGROW without requiring a new input file format. The default input file format (*.dax) is in XML format and is the file type created when input data are saved by the latest version of AFGROW. Users also have the option to open AFGROW output files that have been saved in XML format. Old AFGROW input files (*.da3) files may also be opened to allow backward compatibility. Users can double-click on the desired file, single-click on the desired file and click the Open button, drag the file icon to the AFGROW window, or simply type in the file name in the file name box.

Users should not attempt to manually edit input files for use in AFGROW. While it is possible, it is also very easy to make mistakes.

3.1.2 File Close

This action closes the active window. There are several possible windows in AFGROW. There are the classic (three-frame) and advanced (five-frame) views that have been discussed in previous sections. There are also spectrum and exceedance plot views. The spectrum and exceedance plot views will be discussed in a later section. If there is only one active window, closing it will leave a gray background until another file is opened or a new file is selected.

Files may also be closed by clicking on the standard windows “X” icon in the upper right hand corner of the application window. This icon should not be confused with the large “X” icon inside a red background which will close AFGROW entirely.

3.1.3 File Save

Toolbar Icon: 

This action allows you to save a current input file. This option can only be used AFTER a user has either opened a file or has saved the current input data with the save as option. There must be a file name and location associated with a given file before the file save option can be used.

If the saved input data file includes a reference to a spectrum file, the spectrum file must be available in the same location to open the same spectrum file when the input file is re-opened. An error will occur if the spectrum files have been deleted or relocated since the last save.

3.1.4 File Save As

This action allows users to save their current input data to a file as shown in Figure 36 below¹⁵.

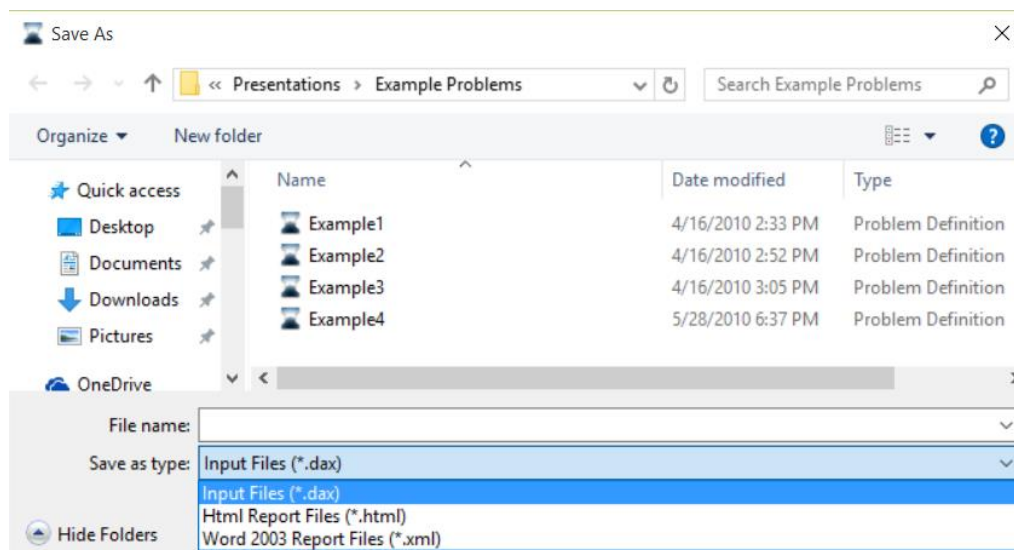


Figure 36: File Save As Dialog

Simply choose an existing file (it will be overwritten) or use the dialog tools to go to another location on your computer to save the input information. You can double-click on the desired file you want to overwrite, single-click on the desired file and click the Save button, or you can type in the file name you would like to save to in the File name box and single-click the Save button. All input files are now saved in XML format using the default file extension (*.dax). Files may also be saved for reporting purposes using a web browser (*.html) or MS Word¹⁶ (*.xml). Users may customize the appearance of these reports by editing the style sheets (*.xsl) provided in the AFGROW installation directory.

If the saved input data file includes a reference to a spectrum file, the spectrum file must be available in the same location to open the same spectrum file when the input file is re-opened. An error will occur if the spectrum files have been deleted or relocated since the last save.

3.1.5 File Mail

This action will activate the users default e-mail client and open a new message addressed to support@afgrow.net. This is provided as a convenience for any comments or inquiries related to AFGROW. Current AFGROW license holders receive 1 hour of support per year.

¹⁵ Files must be saved to a directory with User read/write privileges

¹⁶ MS Word 2003 or later is required to use this option.

3.1.6 File Exit

This action will terminate AFGROW and completely close all open AFGROW related files.

3.2 Input Menu

The input menu, Figure 37, is the gateway for all of the information required for a standard crack growth life prediction.

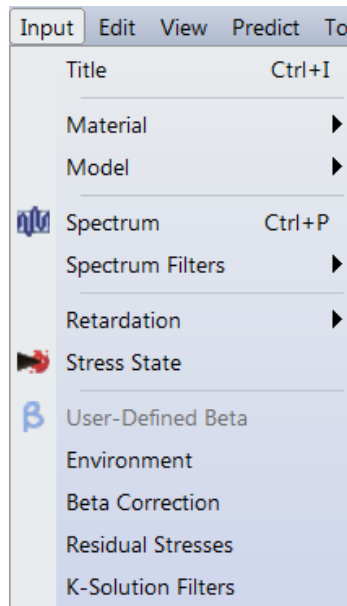


Figure 37: Input Menu

If a tool bar short cut is available for a given option, the appropriate icon is displayed next to the selection. More details of the input menu are given in the following sections.

3.2.1 Input Title

The title option, Figure 38, is provided as a documentation tool. The title line is used to describe the problem being modeled. An additional 1000 characters may be stored in the comments area. The title dialog is shown below:

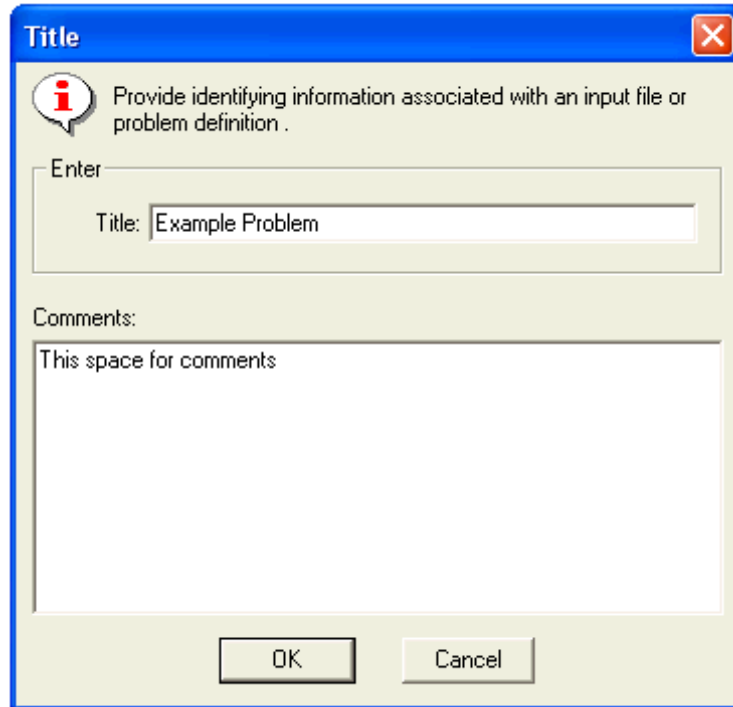


Figure 38: Input Title Dialog

3.2.2 Input Material

Toolbar Icon: 

The material selection pull down menu provides a means of specifying the crack growth material properties to be used by AFGROW. The material model pull down is available in the input, material menu, or through the toolbar icon as indicated in Figure 39.

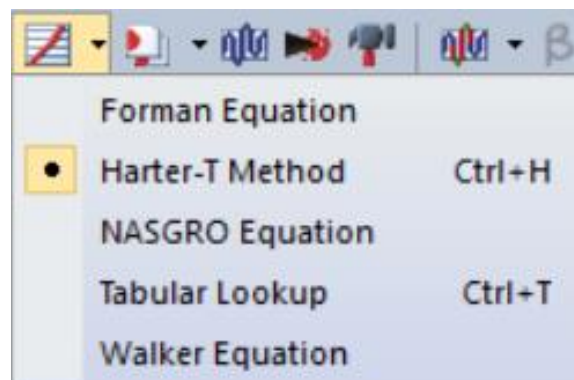


Figure 39: Input Material Model Selection

The toolbar icon may be used in two ways. If a user left-clicks on the icon itself, the currently selected material model dialog will be displayed. The pull down menu is displayed when a user left-clicks on the pull down symbol (). The following sections contain detailed descriptions of each of the methods used to determine crack growth material properties.

Crack growth rate data are represented in terms of Delta K ($K_{max} - K_{min}$) at a given stress ratio ($R = K_{min}/K_{max}$) for $R \geq 0.0$. When $R < 0.0$, K_{max} is used instead of Delta K for all models except the Forman equation.

Please note:

AFGROW depends on the applicability of similitude¹⁷ and operates under the assumption that crack growth rate data monotonically increases with Delta K (or K_{max} when $R < 0$). This makes a great deal of physical sense. If crack growth rate data monotonically increases, then the resulting stress vs. life curve will monotonically decrease (short of any strange behavior that may result from load interaction model irregularities). Most of the crack growth rate models used in AFGROW will not allow this assumption to be violated. When it is permitted, users should exercise caution when attempting to model data that does not monotonically increase with stress intensity.

Also, when modeling small cracks (< 0.03 in.), the microstructure of a given material can have a significant effect on the actual crack growth behavior. LEFM methods may not be appropriate for use with small cracks – especially when the applied K-values are in the threshold region (Region I). In this region significant differences can result due to rate sensitivity, and the user is cautioned to be cognizant of the potential for high variations of results.

¹⁷ Crack growth rate is a unique function of ΔK (or K_{max}) and R .

3.2.2.1 Forman Equation

Material- Forman Equation

Material Properties | Forman Constants

Enter

Delta K threshold value @R=0:

Upper limit on R shift, RHI (Max: 1.0):

Lower limit on R shift, RLO (0... -4.0):

☒ Do not Use RCUT

RCUT :

Select

☐ Map R | if <= R <= , R =

Forman Curve for RLO <= R <= RCUT

Number of Sets:

Set	C	n	K cut
1	2.6e-007	3.2	=KC
2	1	1	1
3	1	1	1

Forman Curve for RCUT < R <= RHI

Number of Sets:

Set	C	n	K cut
1	1	1	=KC
2	1	1	1
3	1	1	1

OK Cancel Apply Save Read

Figure 40: Forman Equation Dialog

The Forman equation [12], named for Dr Royce Forman, included a means to account for the upper portion of the da/dN vs. Delta K curve where the data become asymptotic to the value of Delta K at fracture (see Figure 41).

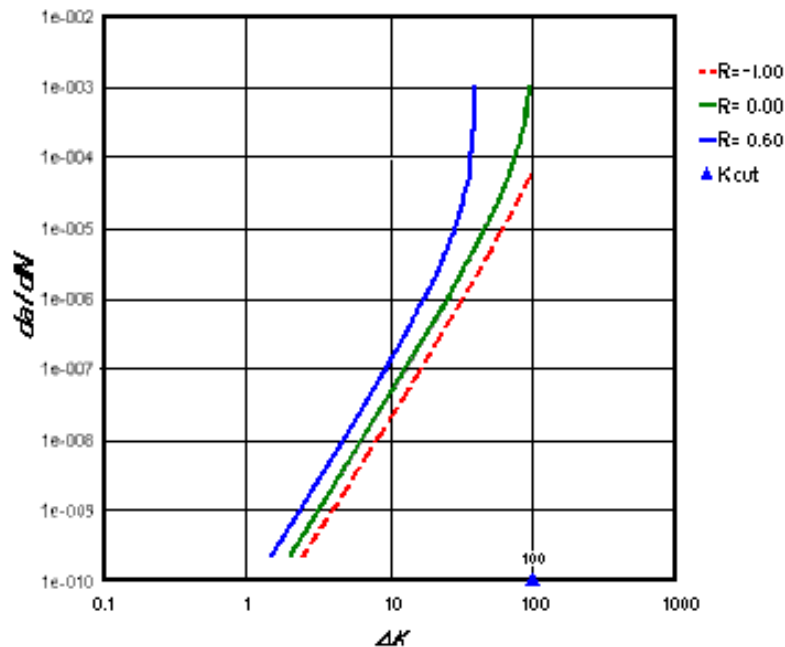


Figure 41: Forman Equation

The form of the Forman equation used in AFGROW is shown here.

$$\frac{da}{dN} = \frac{C \Delta K^n}{((1-R)K_C - \Delta K)}$$

A weakness of the Forman equation lies in a lack of flexibility in modeling data shifting as a function of stress ratio (R). There is no parameter to adjust the R shift directly. The amount of shifting is controlled by the plane stress fracture toughness of a given material.

The material properties, used with the Forman equation, are accessible in a separate tab of the Forman dialog box as shown in Figure 42 (simply click on the material properties tab):

The dialog box is titled "Material- Forman Equation" and has two tabs: "Material Properties" (selected) and "Forman Constants".

Material Properties Tab:

- Information:** The Forman equation has been used by several investigators to characterize da/dN vs. Delta K behavior. The equation results in linear behavior in log-log space in the low and mid-range, and allows the upper end of the data to curve upward, modeling the near-failure behavior of metals.
- Enter:**
 - Material Name: User defined data
 - Coefficient of Thermal Expansion: 1.25e-005
 - Young's Modulus: 10500
 - Poisson's Ratio: 0.33
 - Yield Strength, YLD : 56
 - Plane Strain Fracture Toughness, KIC: 35
 - Plane Stress Fracture Toughness, KC: 100

Buttons: OK, Cancel, Apply, Save, Read

Figure 42: Forman Equation Material Property Dialog

AFGROW allows up to 3 Forman segments (or sets – see Figure 40) to provide the best possible fit to actual crack growth rate data. Users are permitted to define up to 2 fits (3 segments each) as a function of stress ratio (R). If a second fit is desired for R greater than a given value (Rcut), simply uncheck the [Do not Use RCUT] box and enter the desired Rcut value in the appropriate field. AFGROW also allows users to map the Forman fit for a given R to a range of R-values. This option may be useful if users would like to limit the R shift to a certain value. It should be noted that although the Forman equation uses the Paris equation in its numerator, it IS NOT equivalent to the Paris equation because of the terms in the denominator. It is important to note here that when using the Forman equation, AFGROW allows the use of Delta K to include negative K when $R < 0.0$. **This is the**

ONLY exception to the normal standard in AFGROW. This exception results in a shift in crack growth rate data to the right of the $R = 0.0$ data when $R < 0.0$.

The current Forman dialog provides a GREAT deal of flexibility in handling crack growth rate data with a closed-form equation.

The following parameters are ONLY used in the analysis of bonded composite repairs:

Coefficient of Thermal Expansion: (Temperature)-1 Used in the calculation of the thermal effect of patch cure temperature on the stress intensity factor of the patched metal.

Young's Modulus: (Stress) Used in the calculation of the stress intensity factor correction due to the presence of the bonded patch (*also used in the initiation module*).

Poisson's Ratio: (Non-Dimensional) Used in the calculation of the stress intensity factor correction due to the presence of the bonded patch.

The following parameters are used in the standard crack growth analysis:

C: (**Stress⁽¹⁻ⁿ⁾, Length^{((3-n)/2)}**) Value of $da/dN * (Kc-1)$ when $R=0$ and $\Delta K=1$.

n: (**Non-Dimensional**) Paris Exponent (in this case, limit in da/dN slope as ΔK approaches 0.0).

Kcut: (**Non-Dimensional**) Value of Stress Ratio (R) defining the highest R allowed for the first Forman curve fit (leftmost curve fit in Forman Constants dialog box).

Kcut: (**Stress, Length^{0.5}**) Value of ΔK (at $R=0$) defining the highest ΔK allowed for the given segment (upper segment boundary) - Note, the K_{cut} for the last defined segment is assumed to be equal to the plane stress fracture toughness of the metal being analyzed.

Plane Strain Fracture Toughness (KIC): (**Stress, Length^{0.5}**) Value of Fracture Toughness to be used under pure plane strain conditions.

Plane Stress Fracture Toughness (KC): (**Stress, Length^{0.5}**) Value of Fracture Toughness to be used under pure plane stress conditions.

Yield Strength, YLD: (**Stress**) Yield stress (0.2% offset strain) for the metal being analyzed.

The following parameters may be used in the retardation models in AFGROW:

Delta K Threshold Value @ R=0, THOLD: (Stress, Length^{0.5}) Threshold stress intensity value at R=0 (this parameter is required by the Willenborg retardation model). It is NOT currently used in crack growth rate calculations. At this time, there is no lower bound on da/dN in the Forman equation in AFGROW. The only limit occurs when the total crack growth after one spectrum pass is less than the user-specified limit set in the Propagation Limits tab of the predict, preferences dialog. The default value is 1.0E-13 (in whatever length units are being used).

Lower limit on R shift, Rlo: (Non-Dimensional) R-value below which no further R shifting is calculated.

Upper limit on R shift, Rhi: (Non-Dimensional) R-value above which no further R shifting is calculated.

Buttons:

OK: Accept the current parameters and close the dialog box.

CANCEL: Cancel the dialog box.

APPLY: Apply the current parameters.

SAVE: Save the current parameters to a file.

READ: Read a file containing Forman parameters.

3.2.2.2 Harter T-Method

Dr. Joseph Gallagher (ASC/ENF) first coined the name (Harter T-Method [13]) in 1994 and it has since replaced the original name – “Point-by-Point Walker Shift Method.” In 1983, James A. Harter first developed the method as a means to interpolate and/or extrapolate crack growth rate data using a limited amount of tabular crack growth rate test data. The Harter T-Method dialog is shown in Figure 43.

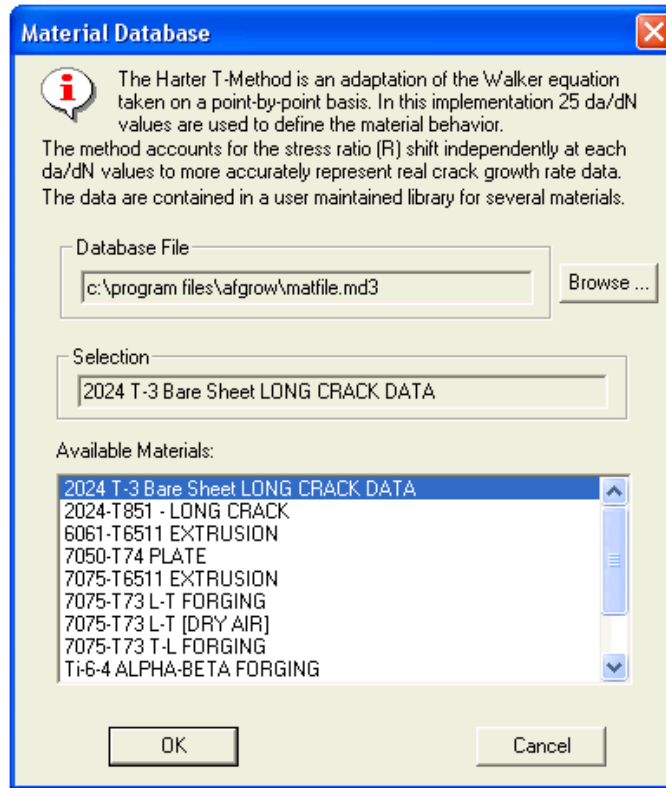


Figure 43: Harter T-Method Dialog

Tabular crack growth rate data are provided in a database file (of sorts). The tabular data are provided as an option to users and users are encouraged to create their own file or files using data of your own choice. AFGROW provides the ability to browse your system to look for tabular material data files. The file extension (.md3) is used since the file has a set format and is the tabular material data format used by AFGROW, Version 3.X. The tabular data utilizes the Walker equation on a point-by-point basis (Harter T-Method) to extrapolate/interpolate data for any R value.

AFGROW uses the Walker equation on a point-by-point basis (Harter T-Method) to determine crack growth rate shifting as a function of stress ratio. Using standard AFGROW practices, K_{max} is used in place of ΔK when $R < 0$. The data shifting is handled as follows:

$$\frac{da}{dN} = C[\Delta K(1 - R)^{(m-1)}]^n ; \text{ Walker Equation}$$

At a given da/dN , the relationship reduces to:

$$\Delta K = \Delta K_{R=0}(1-R)^{(1-m)} ; \text{ for } R \geq 0.0$$

$$K_{\max} = \Delta K_{R=0}(1-R)^{(m-1)} ; \text{ for } R < 0.0$$

Note that $K_{\max}$ is used in place of ΔK when $R < 0$. Although not algebraically correct, it is important that the proper trend in R shift be maintained. This trend is that as m increases, the R shift decreases. This method is simply a way to interpolate/extrapolate data in log-log scale by using the exponential form. This method has given very good results over the years.

It is usually very difficult to obtain crack growth rate data over a sufficient range of crack growth rate and R values to allow the use of simple interpolation methods to accurately model material behavior. A matrix large enough to allow that would consist of actual test data for at least 7 decades of crack growth rate, several R values (positive and negative), and cover the entire range of rate and R values required for the spectrum being analyzed. The Harter T-Method, Figure 44, allows the use of as much data as is available (of course, more data is better) and experience is very useful when data are limited.

Here's how it works:

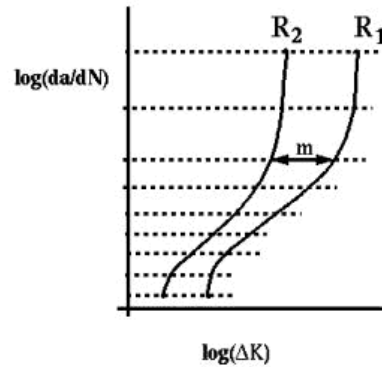


Figure 44: Harter T-Method Crack Growth Rate Shifting as a Function of R

Using the Walker equation (see above) at a single crack growth rate for two positive R values, the following relationship is seen:

$$\Delta K_1(1-R_1)^{(m-1)} = \Delta K_2(1-R_2)^{(m-1)}$$

Solving for m yields:

$$m = 1 + \left[\log_{10} \left(\frac{\Delta K_1}{\Delta K_2} \right) / \log_{10} \left(\frac{(1-R_2)}{(1-R_1)} \right) \right] ; \text{ for } R_1 \text{ and } R_2 \geq 0$$

For the reasons stated above, the method to handle negative stress ratios simply involves using $K_{\max}$ in place of ΔK and switching the exponent for the negative R as follows:

$$K_{\max 1}(1 - R_1)^{(1-m)} = \Delta K_2(1 - R_2)^{(m-1)}$$

Solving for m yields:

$$m = 1 + \left[\log_{10} \left(\frac{K_{\max 1}}{\Delta K_2} \right) / \log_{10} ((1 - R_1)(1 - R_2)) \right] ; \text{ Where } R_1 < 0.0 \text{ and } R_2 \geq 0.0$$

For two negative R-values, the relationship becomes:

$$K_{\max 1}(1 - R_1)^{(1-m)} = K_{\max 2}(1 - R_2)^{(1-m)}$$

Solving for m yields:

$$m = 1 - \left[\log_{10} \left(\frac{K_{\max 1}}{K_{\max 2}} \right) / \log_{10} \left(\frac{(1 - R_2)}{(1 - R_1)} \right) \right] ; \text{ Where } R_1 < 0.0 \text{ and } R_2 < 0.0$$

It is important to know the significance of the value of m. The m-value is non-dimensional and has no real physical significance. The value of m is merely a mathematical means of controlling the shift of the crack growth rate data as a function of stress ratio (R).

The n (slope) value in the Walker Equation gets cancelled when the equations for 2 R-values are set equal at a given da/dN. All m does is provide a means to determine the R shift on a point-by-point basis. All that is required is to take ΔK (or $K_{\max}$ if $R < 0$) for two R-values at the same crack growth rate, apply the appropriate equation, and an appropriate m may be calculated for the given crack growth rate. This method may be repeated at several rate values to describe the tabular data for any R-value. AFGROW uses da/dN and Delta K (for $R=0$) and m at 25 crack growth rate values (da/dN) to recreate the da/dN, Delta K (or $K_{\max}$) curve for any R desired using the method described above. However, the recreated data are determined for the same rate values in the input table. AFGROW calculates the curve (really just the ΔK s or $K_{\max}$) for each rate until the K value exceeds the current K value of interest. Then it just does a logarithmic interpolation between the last two points in the curve (points on each side of the current stress intensity) to give the current rate. This can save a great deal of CPU time.

There are a few RULES that should be adhered to:

- $K_{\max}$ is used in place of Delta K when $R < 0.0$ - All curves shift left of $R=0.0$
- Normally, the R shift for negative R values will stop for $R < [-0.2 \text{ to } -0.5]$ (Rlo)
- It is NOT advisable to use data for $R < Rlo$ to determine m values

- Normal range for m is: ($0 < m \leq 1$)
- Rlo may be determined using m values determined for R values > -0.2 by finding which negative R returns the curve for $R \leq -0.5$
- Shift for positive R is $>$ negative R values for the same absolute R value
- For adjacent points, m should not change abruptly

The format that is required for the material data file [filename.md3] is as follows (space delimited):

[*Title*] (up to 35 characters - should include units being used)

[*da/dN*] [*Delta K @ R=0.0*] [*m*] (25 lines of these data – EXACTLY 25 lines)

[*Rlo*] [*Rhi*] [*KIC*] [*Yield*]

[*Modulus*] [*Poisson's ratio*] [*Coefficient of Thermal Expansion*]

The above is repeated for each material in the file. The LAST line requires the word, *END*, to denote the end of material data. See additional notes¹⁸ on the use of this method in AFGROW

The following parameters are ONLY used in the analysis of bonded composite repairs:

Coefficient of Thermal Expansion: (Temperature)-1 Used in the calculation of the thermal effect of patch cure temperature on the stress intensity factor of the patched metal.

Young's Modulus: (Stress) Used in the calculation of the stress intensity factor correction due to the presence of the bonded patch (*also used in the initiation module*).

Poisson's Ratio: (Non-Dimensional) Used in the calculation of the stress intensity factor correction due to the presence of the bonded patch.

¹⁸ When using the Harter T-Method in AFGROW, the threshold value of Delta K is taken to be the Delta K value (for R=0) corresponding to the lowest rate value of the table. AFGROW handles the shifting for the current R-value internally. The maximum Delta K value for R=0.0 in the tabular data is assumed to be the plane stress fracture toughness (K_c) which is used to determine fracture under pure Plane Stress conditions. AFGROW expects 25 values of crack growth rate, Delta K (at R=0.0), and m. Please be sure to use 25 points, no more or less! For now, the units for this method MUST be English (Ksi, inches, degrees F) when editing data in matfile.dat. The conversion to metric units will be done by AFGROW internally if required.

The following parameters are used in the standard crack growth analysis:

Walker Exponent, m: (Non-Dimensional) Normal Range ($0 < m \leq 1$), Controls shift in crack growth rate data - curve shift decreases as m increases.

Plane Strain Fracture Toughness (KIC): (Stress, Length^{0.5}) Value of Fracture Toughness to be used under pure plane strain conditions.

Delta K Threshold Value @ R=0, THOLD: (Stress, Length^{0.5}) Threshold stress intensity value at R=0 - no crack growth will be calculated when Delta K is below threshold for a given R value.

Yield Strength, YLD: (Stress) Yield stress (0.2% offset strain) for the metal being analyzed.

Rlo: (Non-Dimensional) R value below which no further R shifting is calculated.

Rhi: (Non-Dimensional) R value above which no further R shifting is calculated.

Buttons:

BROWSE: Browse system to find *.md3 files.

CANCEL: Cancel the dialog box.

OK: Accept the current choice and close the dialog box.

3.2.2.3 NASGRO Equation

The NASGRO equation [14], used in NASA's crack growth life prediction program, NASGRO, Version 3.0 is now available in AFGROW. Those who are familiar with the NASGRO equation may notice a few additional parameters in the NASGRO equation dialog (see Figure 45). The additional values are required by AFGROW (explained later in this section).

Figure 45: NASGRO Equation Dialog

Forman and Newman at NASA, De Koning at NLR, and Henriksen at ESA developed the elements of the NASGRO (Version 3.0) crack growth rate equation. It has been implemented in AFGROW as follows:

$$\frac{da}{dN} = C \left[\left(\frac{1-f}{1-R} \right) \Delta K \right]^n \frac{\left(1 - \frac{\Delta K_{th}}{\Delta K} \right)^p}{\left(1 - \frac{K_{max}}{K_{crit}} \right)^q}$$

Where C, n, p, and q are empirically derived, and

$$f = \frac{K_{op}}{K_{max}} = \begin{cases} \max(R, A_0 + A_1 R + A_2 R^2 + A_3 R^3) & R \geq 0 \\ A_0 + A_1 R & -2 \leq R < 0 \\ A_0 - 2A_1 & R < -2 \end{cases}$$

The coefficients are:

$$A_0 = \left(0.825 - 0.34\alpha + 0.05\alpha^2\right) \left[\cos\left(\frac{\pi}{2} S_{\max} / \sigma_0\right) \right]^{\frac{1}{\alpha}}$$

$$A_1 = (0.415 - 0.071\alpha) S_{\max} / \sigma_0$$

$$A_2 = 1 - A_0 - A_1 - A_3$$

$$A_3 = 2A_0 + A_1 - 1$$

Here, α is the plane stress/strain constraint factor, and $S_{\max}/\sigma_0$ is the ratio of the maximum applied stress to the flow stress. These values are provided by the NASGRO material database for each material.

$$\Delta K_{th} = \Delta K_0 \left(\frac{a}{a + a_0} \right)^{\frac{1}{2}} / \left(\frac{1 - f}{(1 - A_0)(1 - R)} \right)^{(1 + C_{th}R)}$$

Where:

- ΔK_0 - threshold stress intensity range at $R = 0$
- a - crack length (a or c in AFGROW)
- a_0 - intrinsic crack length (0.0015 inches or 0.0000381 meters)
- C_{th} - threshold coefficient

The values for ΔK_0 and C_{th} are provided by the NASGRO material database for each material.

The NASGRO equation accounts for thickness effects by the use of the critical stress intensity factor, K_{crit} .

$$\frac{K_{crit}}{K_{Ic}} = 1 + B_k e^{-\left(A_k \frac{t}{t_0}\right)^2}$$

Where:

- K_{Ic} - plane strain fracture toughness (Mode I)
- A_k - Fit Parameter
- B_k - Fit Parameter
- t - thickness
- t_0 - reference thickness (plane strain condition)

The plane strain condition is:

$$t_0 = 2.5(K_{Ic} / \sigma_{ys})^2$$

The values for K_{Ic} , A_k , and B_k are provided by the NASGRO material database for each material. Although the plane strain thickness, t_0 , is defined by the equation shown above, K_{crit} will asymptotically approach K_{Ic} as the actual thickness gets larger than t_0 .

For part-through cracks, the NASGRO equation uses a variable, K_{Ie} (in the database), in place of K_{crit} . The value, K_{Ie} , is a material constant since the developers of the NASGRO equation felt that the K_{crit} value of a part-through crack is not highly dependent on thickness. The value, K_{crit} , is calculated internally and is ONLY used by AFGROW to determine da/dN . It is NOT used as a failure criterion. The variable, K_c , printed in the dialog box is NOT the K_{crit} shown above (see note¹⁹ below).

The NASGRO equation constants are accessible in the equation constant tab, Figure 46, of the dialog box.

Material- NASGRO Equation

Material Properties | **NASGRO Equation Constants**

The parameters required for the NASGRO equation are given below. AFGROW requires additional parameters (Rhi and Rlo) to provide limits for the curve shifting. It has been demonstrated that these limits may be necessary for the NASGRO equation.

Enter

Paris crack growth rate constant, C:	1.49e-008
Paris exponent in NASGRO Equation, n:	3.321
Exponent in NASGRO Equation, p:	0.5
Exponent in NASGRO Equation, q:	1
Threshold stress intensity factor range at R = 0, DK0:	3
Threshold coefficient, Cth:	2
Plane stress/strain constraint factor, Alpha:	1.9
Ratio of the maximum applied stress to the flow stress, Smax/S0:	0.3
Upper limit on R shift, RHI (Max: 1.0):	0.7
Lower limit on R shift, RLO (0... -2.0):	-0.3

OK Cancel Apply Save Read... ▾

Figure 46: NASGRO Equation Constants

¹⁹ Please note that AFGROW uses the plane strain (K_{Ic}) and plane stress (K_c) fracture toughness values to interpolate a value for the critical stress intensity factor failure criterion. There is a difference between NASGRO and AFGROW in this regard. Therefore, the value (K_c) shown in the NASGRO dialog is really the value of K_{crit} determined by setting $t=0$ in the above equation for K_{crit} / K_{Ic} . This is done to provide a means of estimating the plane stress fracture toughness for a given material for use by AFGROW.

These values are set when a material is selected from the NASGRO material database. AFGROW requires a few parameters that are not directly required for the NASGRO equation. AFGROW uses the variables R_{lo} and R_{hi} to set stress ratio limits. It was discovered that the parameters for many of the materials in the NASGRO database would cause the crack growth rate curves to behave erratically above or below certain stress ratios. The crack growth rate curves can become vertical ($K_{th} = K_{crit}$). To avoid this, AFGROW will check for this problem and automatically set R_{hi} and R_{lo} when a material is selected. If parameters are edited manually, care should be taken to verify that this problem will not occur (use the da/dN vs. Delta K plot view in the main frame – see section 2.1.1.3).

The material database for the NASGRO equation is extensive (361 Materials). Selecting the READ button, Figure 47, at the bottom of the main dialog allows access the database:

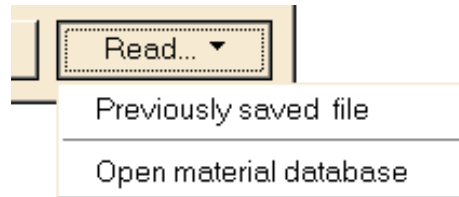


Figure 47: Opening the NASGRO Material Database

AFGROW allows you to open a previously saved file for a material which may not be available in the database (if a user has their own data or has modified data in the NASGRO database), or to open a special browser, Figure 48, to navigate through the large database.

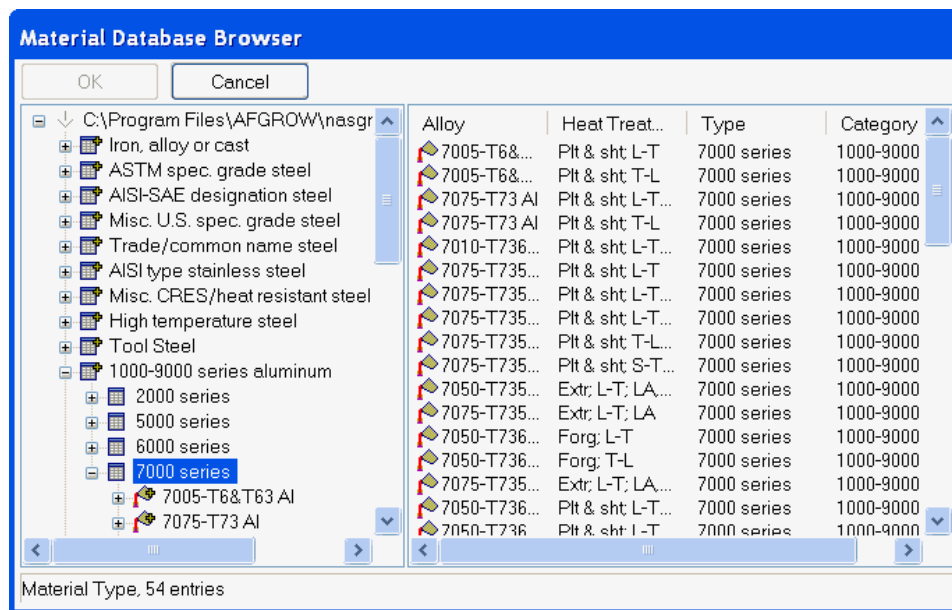


Figure 48: NASGRO Material Database Browser

The browser was designed using a tree structure to aid in locating a desired material. First, select the material by category (or alloy type) by double clicking on the name or clicking on the plus sign to expand the list of materials for the given category, sub-category, heat

treatment and material form. Once a material has been selected, the parameters are displayed as shown in Figure 49:

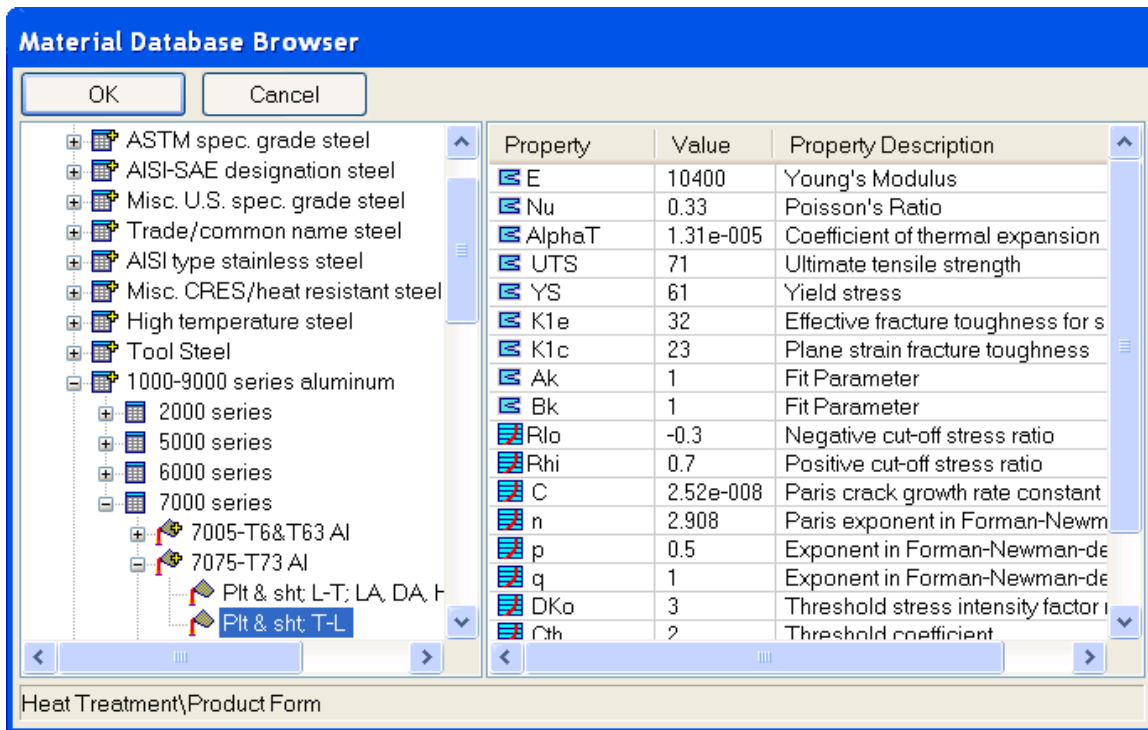


Figure 49: NASGRO Database Material Selection

At this point, pressing the OK button will complete the material selection process. It should be noted that this window may be inside the previous (parent) window and the OK button for the parent window could be visible. Remember that the OK button for the material database browser is at the upper left-hand corner of its window.

The following parameters are ONLY used in the analysis of bonded composite repairs:

Coefficient of Thermal Expansion: (Temperature)-1 Used in the calculation of the thermal effect of patch cure temperature on the stress intensity factor of the patched metal.

Young's Modulus: (Stress) Used in the calculation of the stress intensity factor correction due to the presence of the bonded patch (*also used in the initiation module*).

Poisson's Ratio: (Non-Dimensional) Used in the calculation of the stress intensity factor correction due to the presence of the bonded patch.

The following parameters are used in the standard crack growth analysis:

C: (Stress⁽⁻ⁿ⁾, Length^(1-n/2)) Paris Coefficient.

n: (Non-Dimensional) Paris Exponent.

p: (Non-Dimensional) NASGRO Equation Exponent.

q: (Non-Dimensional) NASGRO Equation Exponent.

C_{th}: (Non-Dimensional) Threshold Coefficient.

Alpha: (Non-Dimensional) Plane stress/strain constraint factor.

S_{max}/σ₀: (Non-Dimensional) Maximum applied stress to flow stress ratio.

Yield Strength, YLD: (Stress) Yield stress (0.2% offset strain) for the metal being analyzed.

Plane Strain Fracture Toughness (KIC): (Stress, Length^{0.5}) Value of Fracture Toughness to be used under pure plane strain conditions.

Plane Stress Fracture Toughness (KC): (Stress, Length^{0.5}) Value of Fracture Toughness to be used under pure plane stress conditions.

K_{Ic}: (Stress, Length^{0.5}) Effective fracture toughness for part through-the-thickness cracks - ONLY used in place of K_{crit} in the NASGRO equation for crack growth rate calculations for part through-the-thickness cracks (not a failure criterion).

A_k: (Non-Dimensional) Fit parameter in K_{crit}/K_{Ic} vs. thickness equation.

B_k: (Non-Dimensional) Fit parameter in K_{crit}/K_{Ic} vs. thickness equation.

The following parameters may be used in the retardation models in AFGROW:

ΔK₀: (Stress, Length^{0.5}) Threshold stress intensity factor range at R=0.

R_{lo}: (Non-Dimensional) Lower limit on R shift.

R_{hi}: (Non-Dimensional) Upper limit on R shift.

Buttons:

OK: Accept the current parameters and close the dialog box.

CANCEL: Cancel the dialog box.

APPLY: Apply the current parameters.

SAVE: Save the current parameters to a file.

READ: Read the NASGRO material database OR a file containing NASGRO equation parameters.

3.2.2.4 Tabular Look-Up

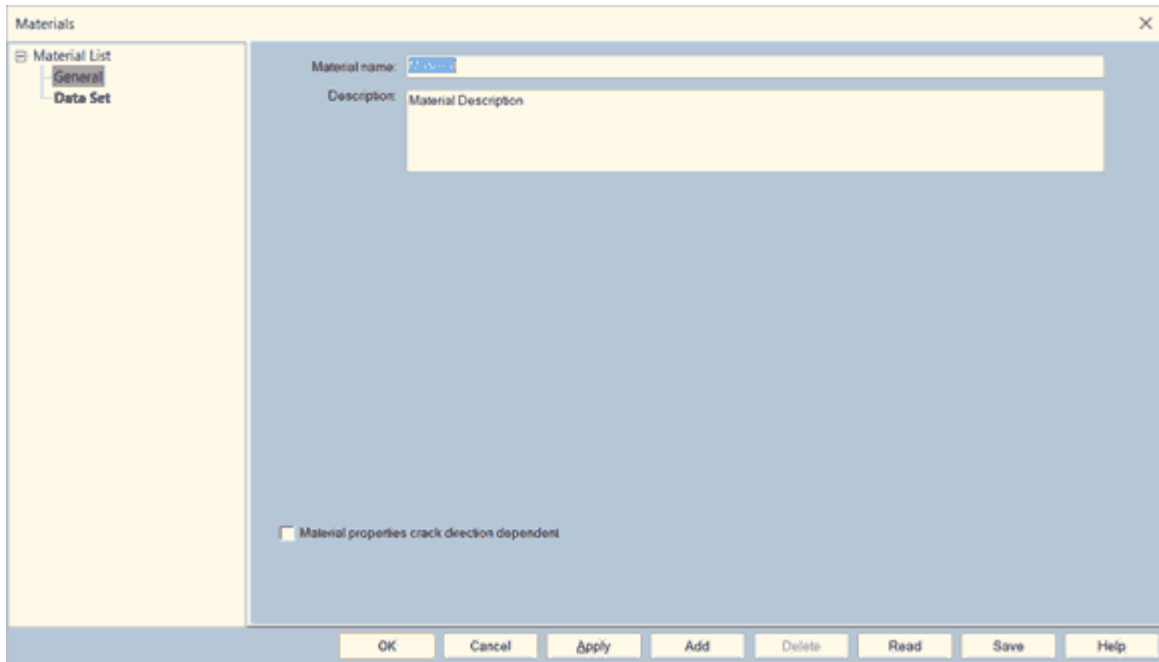


Figure 50: Tabular Look-Up Dialog

A tabular look-up crack growth rate capability is provided in AFGROW to allow users to input their own tabular crack growth rate curves. This model has now been expanded²⁰ to allow users to input tabular data for different environmental conditions (e.g., temperature) and/or crack growth directions (a or c-direction). When the dialog box opens, users may enter general information (Material name, and Description) as documentation information for the material. In this general view, users may also select whether material properties are direction dependent so that data for each crack growth direction will be entered in the Data Set(s).

The tabular look-up model utilizes the Walker equation on a point-by-point basis (Harter T-Method) to extrapolate/interpolate data for any two adjacent R-values (see Section 3.2.2.2). The difference in the tabular lookup method is that the user doesn't have to calculate all m values (AFGROW does it internally between each possible pair of input R curves). Data is interpolated/extrapolated using the m values determined from data for the nearest two R curves. It is very important to note that data for each R curve must be entered from left to right for increasing R-values.

AFGROW will also allow users to enter data for a single R-value. In this case, the user-defined data will be used regardless of the stress ratio for a given analysis. This may be useful in cases where rate data are scarce, and the user is only interested in predicting

²⁰ It is important to note that these new capabilities are optional, and tabular growth rate data used in previous versions of AFGROW are upward compatible.

results for constant amplitude loading (constant R). It may also be used in conjunction with the FASTRAN retardation model to define the da/dN vs. ΔK_{eff} curve. It should also be noted that this dialog includes ultimate strength as a parameter. This is because it is needed when the FASTRAN model is selected, and this is the only crack growth rate model that may be used with the FASTRAN model.

3.2.2.4.1 Entering Data for Multiple Crack Directions

If the option for direction dependent material properties is selected, the general dialog box will include an option to control how crack growth rate data are interpolated between the two orthogonal directions as indicated below.

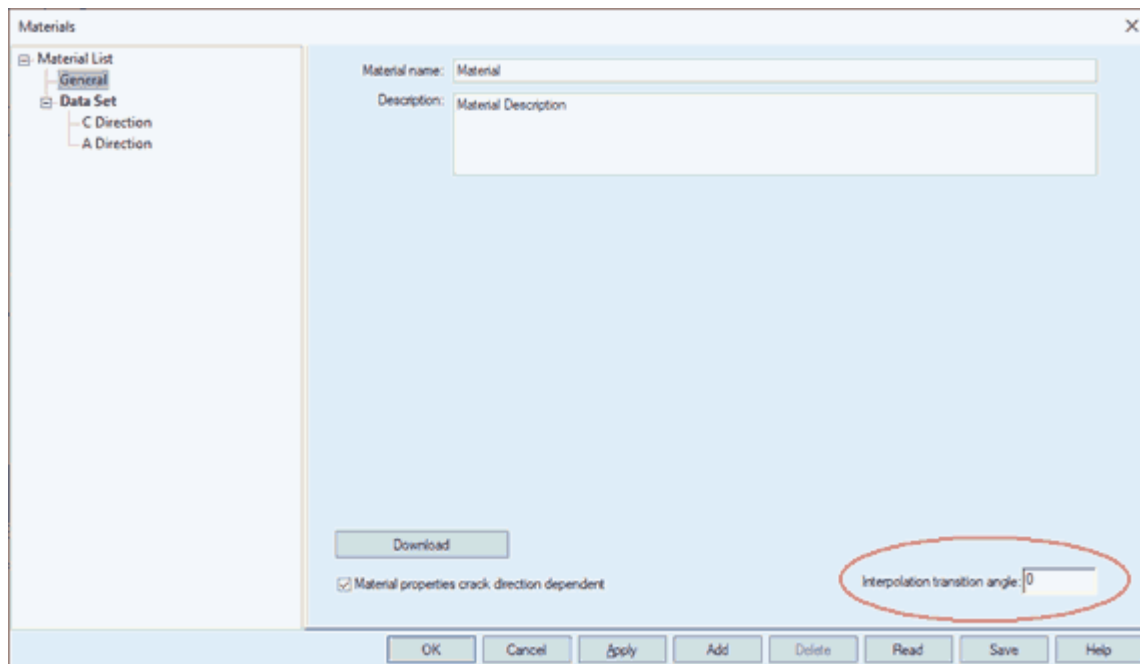


Figure 51: Entering Data for Multiple Crack Directions

If this option is used, crack growth rates in each growth direction will be determined from the tabular crack growth rate data provided in the corresponding material data set.

For crack models with multiple points along the crack front, linear interpolation is performed from the c-direction (0 degrees) toward the a-direction (90 degrees). The transition to the a-direction data will be complete at the user-defined interpolation transition angle (0 – 90 degrees). The linear interpolation between the crack growth rate data for the c-direction and a-direction will be applied as a function of the material orientation angle (γ) between zero degrees and the interpolation transition angle. The orientation angle should not be confused with the parametric angle (ϕ) that is used in the Newman/Raju Classic Models to identify the crack tip location around the crack front (see Figure 70). The material orientation angle provides a reference for possible differences in

crack growth rate data as the position around the crack front moves between grain orientations (e.g.: L-T to L-S). The material orientation angle is shown in Figure 52.

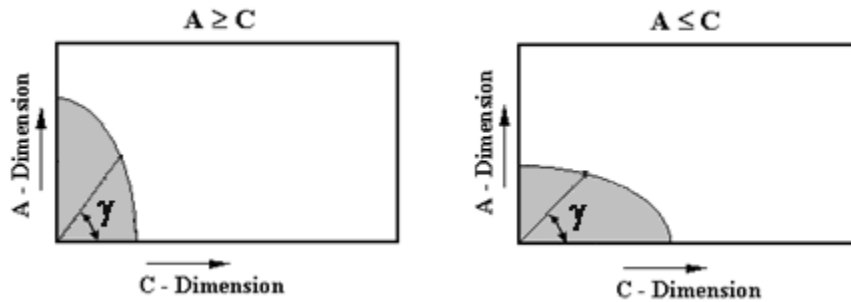


Figure 52: Material Orientation Angle

Therefore, crack growth rate data entered for the a-direction will be used for all points beyond the user-specified interpolation transition angle (γ). The capability to interpolate crack growth rate data along a crack front is currently only available for use with plug-in models. We plan to add this functionality to our Advanced Model interface in a future AFGROW release.

The user will enter data for each direction Data Set by clicking on either direction for that Data Set as indicated in Figure 53.

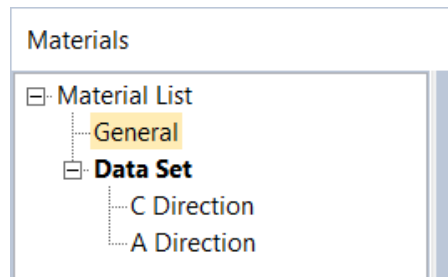


Figure 53: Data Set for Direction Dependent Information

Data are entered for each growth direction in the Data Set view as shown in Figure 54.

Materials

Material List

- General
- Data Set
 - C Direction
 - A Direction

Input values of Delta_K for da/dN values and up to 10 different R(stress ratio) values. Matrix must have at least two R values and two da/dN values. Input Delta_K for R >= 0, input Kmax for R < 0.0

Name: 7075-T7651 L-T LHA Plate

Number of da/dN Sets: 16 Number of R Sets: 1

		R[1]
da/dN[1]	1.00e-006	0.1
da/dN[2]	2.00e-006	5.525
da/dN[3]	4.00e-006	6.7
da/dN[4]	6.00e-006	8.2
da/dN[5]	1.00e-005	9.35
da/dN[6]	2.00e-005	11
da/dN[7]	4.00e-005	14.8
da/dN[8]	6.00e-005	21
da/dN[9]	1.00e-004	25
da/dN[10]	2.00e-004	29
da/dN[11]	4.00e-004	34
da/dN[12]	1.00e-003	38
da/dN[13]	2.00e-003	41
da/dN[14]	1.00e-002	44
da/dN[15]	2.00e-002	50
da/dN[16]	1.00e-001	52
		55

Ultimate Strength: 75

Young's Modulus: 10400

Coefficient of Thermal Expansion: 1.31e-005

Poisson's Ratio: 0.33

Upper limit on da/dN, DADNHI: 0.1

Lower limit on da/dN, DADNLO: 4e-007

Plane Stress Fracture Toughness, KC: 62

Plane Strain Fracture Toughness, KIC: 30

Delta K threshold value @R=0: 4.7

Yield Strength, YLD: 64

Lower limit on R shift (Max: 0): -0.33

Upper limit on R shift (0, 1): 0.72

OK Cancel Apply Add Delete Read Save Help

Figure 54: Data Set View

For each data set, the user may enter a name to identify the material (optional), the number of crack growth rate (da/dN) sets, number of stress ratio (R) sets, and all of the material properties shown on the right hand side of Figure 54. These material properties are defined in Section 3.2.2.4.7.

3.2.2.4.2 Entering Data for Different Environments

If data for different environments are desired for a given analysis, the additional data are entered by clicking on the “Add” button at the bottom of the dialog box.

Materials

Material List

- General
- Data Set
- Data Set, Temperature: 0

Input values of Delta_K for da/dN values and up to 10 different R(stress ratio) values. Matrix must have at least two R values and two da/dN values. Input Delta_K for R >= 0, input Kmax for R < 0.0

Name: Data for another environment

Number of da/dN Sets: 27 Number of R Sets: 3

		R[1]	R[2]	R[3]
da/dN[1]	1.00e-009	2.29	1.5	1
da/dN[2]	2.00e-009	2.3	1.51	1.01
da/dN[3]	1.00e-008	2.35	1.56	1.07
da/dN[4]	2.00e-008	2.39	1.6	1.14
da/dN[5]	4.00e-008	2.55	1.72	1.3
da/dN[6]	6.00e-008	2.8	1.9	1.55
da/dN[7]	1.00e-007	3.5	2.4	2.01
da/dN[8]	2.00e-007	4.65	3.25	2.7
da/dN[9]	3.00e-007	5.5	3.9	3.05
da/dN[10]	4.00e-007	6	4.3	3.22
da/dN[11]	6.00e-007	6.5	4.73	3.5
da/dN[12]	1.00e-006	7	5.19	3.85
da/dN[13]	2.00e-006	7.9	6.05	4.6
da/dN[14]	4.00e-006	9.4	7.4	5.8
da/dN[15]	1.00e-005	12	9.7	8
da/dN[16]	2.00e-005	14.45	11.78	9.7
da/dN[17]	4.00e-005	18.65	15.236	12
da/dN[18]	1.00e-004	27.5	22.3	15
da/dN[19]	2.00e-004	36	28.6	17.5
da/dN[20]	4.00e-004	45.2	35	20
da/dN[21]	6.00e-004	52	39.8	21.2

Ultimate Strength: 78

Young's Modulus: 10400

Coefficient of Thermal Expansion: 1.31e-005

Poisson's Ratio: 0.33

Upper limit on da/dN, DADNHI: 0.1

Lower limit on da/dN, DADNLO: 1e-009

Plane Stress Fracture Toughness, KIC: 80

Plane Strain Fracture Toughness, KIC: 40

Delta K threshold value @R=0: 2.29

Yield Strength, YLD: 68

Lower limit on R shift (Max 0): -0.33

Upper limit on R shift (0, 1): 0.7

OK Cancel Apply Add Delete Read Save Help

Figure 55: Tabular Environmental Data

The environmental tag used for these data is found in the Material List/General view after the data have been added. Currently, the environmental tag is based on temperature.

Materials

Material List

- General
- Data Set
- Data Set, Temperature: -40

Material name: Material

Description: Material Description

D..	Name	Temperature
1	☐ 7075-T6 L-T Lab air	
2	☒ Data for another environment	-40

☐ Material properties crack direction dependent

OK Cancel Apply Add Delete Read Save Help

Figure 56: Editing the Environmental Tag

In the example shown in Figure 56, the environmental (Temperature) tag is -40, and it is also set to be the default growth rate data by the check box. The default crack growth rate data will always be used when there are no tags in the spectrum or when a spectrum tag does not match any other tag for the data sets that have been added. The tags in the spectrum and material list must match exactly for AFGROW to use the additional environmental data.

The following sections contain information related to the implementation of the tabular lookup model that is the same for each data set. Remember to set the desired default data set by checking the appropriate box.

3.2.2.4.3 Use of a Common Set of Rate Values for All R Curves

The first thing to notice in this implementation is that there is a single column for crack growth rates (within each data set), which apply for the stress intensity factor data at each stress ratio. There have been many questions from users over the years about the use of a single column of rates for all stress ratios. The following paragraphs will provide more information and hopefully, explain why things are done this way.

It is usually very difficult to obtain crack growth rate data over a sufficient range of crack growth rates and R-values for simple interpolation methods to accurately model material behavior. In most cases, growth rate data for one stress ratio will cover a different range of growth rates than data obtained for a different stress ratio. This can cause problems when attempting to interpolate/extrapolate data using rate curves for different stress ratios. These data need to be adjusted to cover a common range of crack growth rate that is appropriate for a given analysis. While it is possible to extend the rate curve for a given R value, it is also important that no two positive or negative R curves cross each other. It is also much easier to check for crossed growth rate curves if a common set of rate values are used for all user-defined data.

Another issue is related to the way that the ΔK threshold (ΔK for which the growth rate is assumed to be zero) values are calculated for various stress ratios. Since it is not practical to ask users to enter these values for all possible stress ratios, AFGROW calculates the threshold values for any given stress ratio based on the user-defined threshold at $R = 0.0$. This is accomplished by determining the growth rate at $R = 0$ for the user-defined threshold – call it the Threshold Growth Rate (TGR). The assumption is that the threshold at any stress ratio will be the corresponding ΔK (or K_{max} for most models when $R < 0$) value at the TGR. This is equivalent to shifting the growth rate curve horizontally as a function of R since growth rate values are given on the y-axis of the da/dN vs. ΔK plot. The use of a common set of rate values for all stress ratios facilitates this process.

It should also be noted that crack growth rate data tend to exhibit a fair amount of scatter. Data for the same material can differ by as much as a factor of 2 in terms of rate. Normally, users should attempt to capture the mean of the data (when there are enough data to make this determination). Since data are typically scarce, it is necessary to use engineering judgment for data that are often obtained from multiple sources. The user should examine

growth rate data for different R-values carefully. There are a few “rules” that AFGROW uses to determine whether user-defined data are valid. These rules are given in Section 3.2.2.4.5. The best way to examine data (prior to use in an analysis) is to plot the data and look for anomalies. Once the data are plotted, it is usually fairly easy to select a common set of rate values to use for all of the data. A spreadsheet is well suited for this since because of the ability to plot the data. The resulting tabular data may be copied from the spreadsheet to AFGROW as indicated in Section 3.2.2.4.4.

3.2.2.4.4 General Data Entry

The tabular look-up option provided in AFGROW allows a user to enter crack growth rate vs. stress intensity data for up to ten R-values. If data for a single R-value is entered, AFGROW will use that data for any R-value since there is not enough information to allow for interpolation or extrapolation based on R-value. When data for two or more R-values are provided, the Harter T-method is used to interpolate/extrapolate data for any R-value (see Section 3.2.2.2). The Harter T-method allows the use of a minimal amount of data (of course, more data will allow a better model) and can be used to interpolate and extrapolate data within user specified limits. Once data for the appropriate R-value has been determined, the resulting points define a rate vs. ΔK curve for that R-value. The growth rate for a given ΔK is calculated using log-linear interpolation or extrapolation based on the nearest two stress intensity values for the given curve. Please note: when $R < 0$, K_{max} is used instead of ΔK . The following paragraphs describe how to use the table look-up option in AFGROW.

First, obtain da/dN vs. stress intensity data for up to ten R-values that provide a satisfactory fit to test data for the material of interest. It is necessary to input stress intensity data for each R-value at the same crack growth rate values (see Section 3.2.2.4.1). Using the same rate values for all stress ratios ensures that the data covers the same range of growth rate. The data for each stress ratio must be entered from left to right according to increasing R-value. In other words, the data for the lowest numerical R-value is entered in the column labeled R[1], and data for all other R-values are entered for each higher stress ratio in the columns to the right. If K_{max} data for $R < 0$ are entered in the table, data for $R = 0$ should also be included. If not readily available, these data can be extrapolated using the method described in Section 3.2.2.2 using existing data for the nearest pair of R-values. Including data for $R = 0$ provides a “bridge” between data for negative and positive R-values and allows for a proper transition between the use of Delta K and K_{max} for these values.

Set the values for the number of da/dN and R sets at the top of the dialog (see Figure 54). These values are changed by clicking on the up or down arrows next to the value. Then, enter the appropriate crack growth rate data in the matrix.

Enter the appropriate material property data for all fields on the right side of the data set view. A material name may be entered to identify the material and provide any additional information for documentation purposes. AFGROW provides optional default material property values if the user needs help with these parameters. This help is available by clicking on the default icon as shown in Figure 57. The default values are merely typical

properties for the materials listed. There is also a choice to add zeros to all these data fields. Setting the material properties to zero resets all the properties and helps to prevent the use of data from a previous analysis. The zeros are easy to see at a glance, so the user will be alerted to the need for data in this dialog box. A description of the material properties is given in Section 3.2.2.4.7.

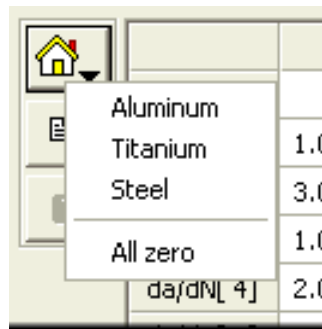


Figure 57: Tabular Look-Up Default Material Data

AFGROW includes the option to copy crack growth rate data to the Windows clipboard and allow it to be pasted into another Windows application (i.e., Microsoft Excel®, see Figure 58). When copying a column or row, the user must first click on the row or column to be copied²¹.

		R[1]	R[2]
		0.1	0.6
da/dN[1]	1.00e-009	2.606	1.38
da/dN[2]	3.00e-009	2.636	1.409

Figure 58: Tabular Look-Up Copy Option

Data that have been placed in the Windows clipboard from another Windows application may be pasted into AFGROW (see Figure 59). The number of data sets for da/dN and R must match the data to be pasted. If the application is not in Excel format, be sure that the data are tab delimited in each row. When pasting a column or row, you must first click on the row or column where the data is to be pasted.

²¹ To select a row, click on a cell in the rate column. A column is selected by a mouse click in the top row (containing R(1), R(2), ...). The entire table is selected by clicking the top, left cell in the table.

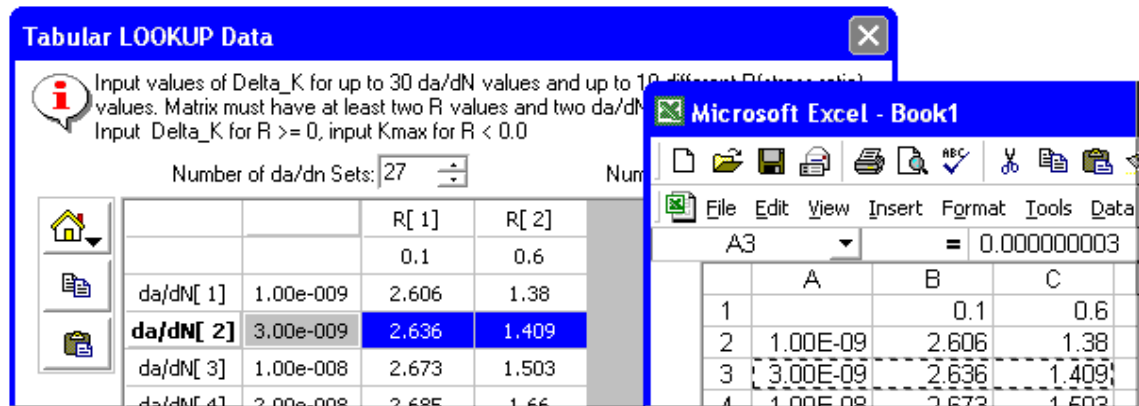


Figure 59: Tabular Look-Up Paste Choices

If Excel is used to create the crack growth rate data table, the required format is shown in Figure 60:

dadN/R	0.1	0.6
1.00E-09	2.606	1.38
3.00E-09	2.636	1.409
1.00E-08	2.673	1.503
2.00E-08	2.685	1.66
4.00E-08	2.729	1.897
6.00E-08	2.792	2.089
1.00E-07	2.954	2.355

Figure 60: Excel Spreadsheet Example for Crack Growth Rate Data

When copying the entire table, it is important to include something in the upper, left cell. In the example shown in Figure 60, 'dadN/R' is used as a convenient way to label the data to the right as R and below as da/dN.

3.2.2.4.5 Error and Warning Checking

AFGROW performs several tests on tabular input data to help ensure that growth rate values may be interpolated or extrapolated correctly. Errors are conditions that would result in serious difficulties in growth rate determination and life prediction. Warnings are less serious and are based on trends observed for crack growth rate data. AFGROW will not accept tabular input data that contains errors and users are encouraged to carefully examine any warning conditions.

The following error checks are performed:

- Positive R curves may NOT cross each other in the domain of the crack growth rate and R limits input by the user
- Negative R curves may NOT cross each other in the domain of the rate and R limits input by the user

- ΔK (or K_{max}) values for a given R MUST increase with increasing rate
- ΔK values at a given growth rate for increasing positive R must decrease for increasing R
- K_{max} values at a given growth rate for decreasing negative R must decrease for decreasing R
- K_{max} values at a given growth rate for negative R values must be less than ΔK for $R = 0.0$
- Threshold ΔK value at $R=0$ must be in the range of possible ΔK values for $R=0$ (within the crack growth rate limits input by the user – DADNLO and DADNHI)
- K_{IC} must be less than K_C
- RLO must be less than or equal to 0.0
- RHI must be greater than 0.0 AND less than 1.0

The following warning checks are performed:

- Data for negative R (K_{max}) should be greater than the data (ΔK) at the same positive R
- K_{max} values for negative R should be greater than data at $R=0$ when converted to ΔK ($\Delta K = K_{max} * (1-R)$ - AFGROW will do this conversion internally)

User defined input data is used to interpolate and extrapolate growth rate data for any stress ratio (R) and stress intensity value that falls within the boundaries defined as follows:

Lower Limit on R Shift (RLO)
 Upper Limit on R Shift (RHI)
 Lower Limit on da/dN ($DADNLO$)
 Upper Limit on da/dN ($DADNHI$)

All error and warning checking is performed within these limits based on the user-defined tabular input data. It is important to remember that all data will be shifted to the “left” of the curve for $R=0$ since AFGROW used K_{max} when $R < 0$. Users are not required to input data for $R=0$, since AFGROW can calculate it using data at the two closest R -values. If data for a single R is entered, it makes no difference, since that data will be used for all R -values and many of the tests are not required or performed. An example is shown below in Figure 61.

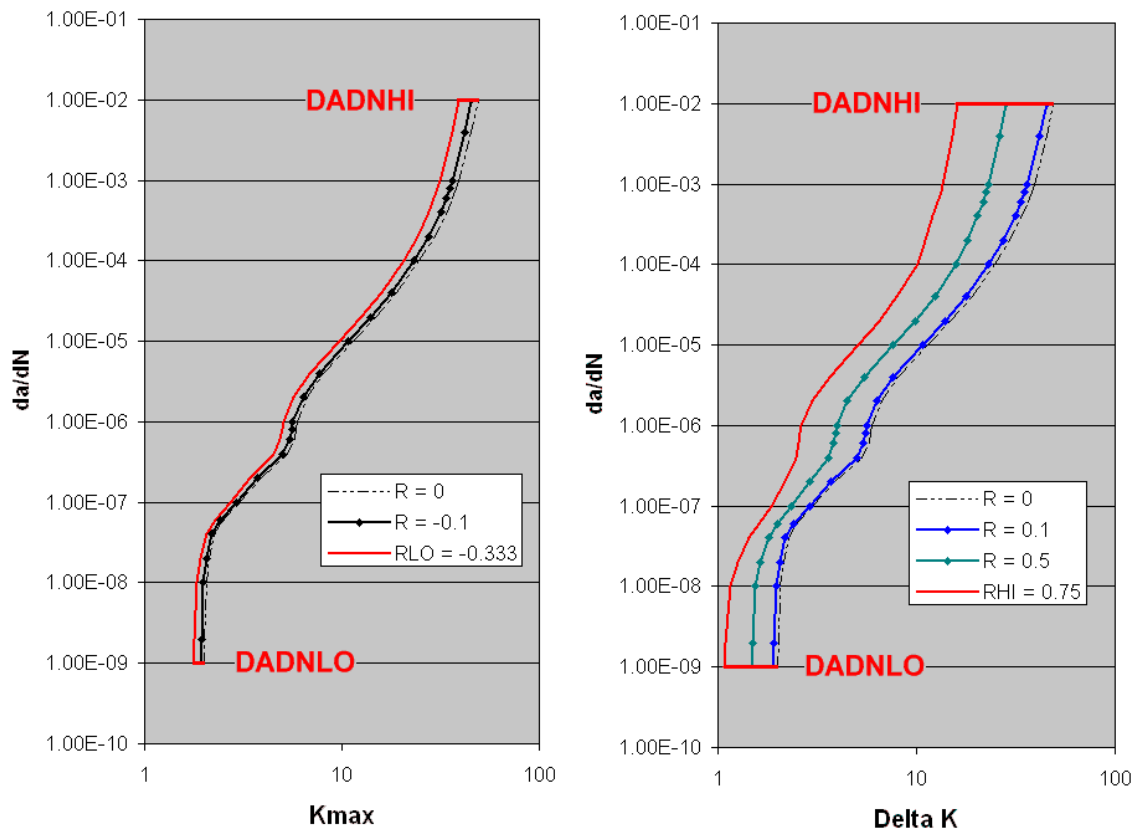


Figure 61: Example Rate Plot Showing Boundaries

In the example above, data were entered for three R-values (-0.1, 0.1, and 0.5). Data for R=0 were calculated from the data entered for R=-0.1 and 0.1 using the Harter T-method (see Section 3.2.2.2 for details). The following boundaries were set in the example:

RLO = -0.333
RHI = 0.75
DADNLO = 1e-09
DADNHI = 1e-02

Data for RLO were calculated from the data entered for R=-0.1 and 0.1, and data for RHI were calculated from data for R=0.1 and 0.5 (closest two input curves). The information presented in Figure 61 is divided in two separate plots to make it easier to visualize since the data shifts to the left of R=0.0 since Kmax is used when R<0.0. The tabular data used in this example are shown below:

da/dN	-0.1	0	0.1	0.5
1.00E-09	1.924	2.008	1.915	1.47
2.00E-09	1.933	2.016	1.925	1.486
1.00E-08	1.983	2.064	1.975	1.543
2.00E-08	2.058	2.134	2.05	1.64
4.00E-08	2.196	2.266	2.189	1.803
6.00E-08	2.417	2.492	2.409	1.996
1.00E-07	2.902	2.999	2.891	2.354
2.00E-07	3.734	3.887	3.719	2.905
4.00E-07	5.01	5.280	4.983	3.606
6.00E-07	5.439	5.754	5.407	3.823
8.00E-07	5.563	5.885	5.53	3.91
1.00E-06	5.636	5.960	5.603	3.968
2.00E-06	6.352	6.713	6.315	4.491
4.00E-06	7.652	8.081	7.608	5.436
1.00E-05	10.798	11.413	10.736	7.634
2.00E-05	13.995	14.804	13.912	9.835
4.00E-05	17.903	19.000	17.791	12.329
1.00E-04	23.227	24.700	23.077	15.795
2.00E-04	27.572	29.501	27.377	18.046
4.00E-04	31.654	34.000	31.417	20.217
6.00E-04	33.966	36.500	33.709	21.628
8.00E-04	35.348	38.001	35.08	22.454
1.00E-03	36.268	39.000	35.991	22.997
4.00E-03	41.95	45.200	41.621	26.268
1.00E-02	45.403	49.000	45.039	28.143

Figure 62: Example Tabular Input Data

When errors or warnings are detected in the user-input data, AFGROW displays messages as indicated below:

Error Tests:

Properties	
Message	Explanation
RLO cannot be greater than 0.0... cannot apply	RLO is defined to be less than or equal to zero
Youngs Modulus cannot be <= 0.0... cannot apply	Young's modulus must be a positive value
Poisson Ratio cannot be = 0.0... cannot apply	Poisson's ratio must not be zero
RHI must be in the interval (0.0, 1.0)... cannot apply	RHI is defined to be greater than zero and less than one.

DADNLO must be greater than 0.0... cannot apply	Zero is not defined in log scale
DADNHI must be greater than DADNLO... cannot apply	There must be a non-zero range in possible growth rates.
KC must be greater than KIC... cannot apply	Plane stress toughness must be greater than plane strain in order to interpolate the local apparent toughness.
YIELD must be positive... cannot apply	Yield stress is positive by definition.
THRESHOLD must be positive... can not apply	Threshold is positive by definition

Crack Growth Rate Data	
Message	Explanation
dadN must be positive... cannot apply values.	All rate values must be positive - zero is not defined in log scale
dadN values must be in ascending order... can not apply values.	Rate values must be in ascending order to ensure a one-to-one relationship to ΔK or K_{max} for any stress ratio
delta K must be positive... cannot apply values.	All ΔK values must be positive - zero is not defined in log scale
Kmax must be positive... cannot apply values.	All K_{max} values must be positive - zero is not defined in log scale
delta K values in each column must be in ascending order...	ΔK values must be in ascending order to ensure a one-to-one relationship to rate at any stress ratio
Kmax values in each column must be in ascending order...	K_{max} values must be in ascending order to ensure a one-to-one relationship to rate at any stress ratio
R[X] not in bounds [RLO, RHI]... cannot apply	Currently, AFGROW does not allow any of the input R-values to be outside user-defined boundaries. [X] - tells you which value is out of bounds (1 - 10)
R values must be discrete... $R[X] = R[X]$	No two input R-values can be equal - [X] tells the user which R-values are equal (1 - 10).

Crack Growth Rate Data	
Message	Explanation
Error! $K_{max} @ R = [X]$ must be less than $K_{max} @ R = [X]$ for $DADN = [X]$	For a given growth rate, user-input ΔK (or K_{max} , when $R < 0$) values must decrease as the absolute value of R increases (curves shift to the left). If this error is detected, AFGROW will give the rate and user-input R-values where this occurs. This also ensures that no two input curves cross inside the domain of the user-defined growth rate values.
Error! $\Delta K @ R = [X]$ must be less than $\Delta K @ R = [X]$ for $DADN = [X]$	
Threshold is less than DeltaK at DADNLO... cannot apply	Since no data can be extrapolated outside of the user-defined boundaries, the user-input threshold

Threshold is greater than DeltaK at DADNHI... cannot apply	(at R=0) must be between DADNLO and DADNHI. In these messages, Delta K means Delta K at R=0, and AFGROW calculates it if R=0 is not one of the user-input curves.
Delta K values not in ascending order at R = RLO... Minimum possible value for RLO = [X]	When AFGROW extrapolates data for RLO and RHI, it is possible that the extrapolated Delta K (or Kmax) values may not increase with growth rate. If this occurs, AFGROW calculates the value of maximum RHI and/or minimum RLO that prevents this problem.
Kmax values not in ascending order at R = RLO... Minimum possible value for RLO = [X]	
Delta K values not in ascending order at R = RHI... Maximum possible value for RHI = [X]	AFGROW checks to be sure that all input Delta K or Kmax data (including data that may be extrapolated to the boundaries, RLO and RHI) is less than DK at R = 0 for the same rate value. This is a similar check to the one that makes sure that each input curve is shifted to the left as the absolute value of R increases. These tests simply make sure that data for each R-value is also to the "left" of the data for R=0. Again, AFGROW will calculate the data for R=0 internally if it is not one of the R-values entered by the user.
Delta K value @ RLO is > Delta K @ R = 0 at DADNLO... cannot apply	
Kmax value @ RLO is > Delta K @ R = 0 at DADNLO... cannot apply	
Delta K value @ R = [X] is > Delta K @ R = 0 at DADNLO... cannot apply	
Kmax value @ R = [X] is > Delta K @ R = 0 at DADNLO... cannot apply	
Delta K value @ RHI is > Delta K @ R = 0 at DADNLO... cannot apply	
Delta K at RLO is > Delta K at R = 0 at DADN[X]	
Kmax at RLO is > Delta K at R = 0 at DADN[X]	
Delta K at R = [X] is > Delta K at R = 0 at DADN[X]	
Kmax at R = [X] is > Delta K at R = 0 at DADN[X]	
Delta K at RHI is > Delta K at R = 0 at DADN[X]	

Crack Growth Rate Data	
Message	Explanation
Delta K value @ RLO is > Delta K @ R = 0 at DADNHI... cannot apply	AFGROW Checks to be sure that all input Delta K or Kmax data (including data that may be extrapolated to the boundaries, RLO and RHI) is less than DK at R = 0 for the same rate value. This is a similar check to the one that makes sure that each input curve is shifted to the left as the absolute value of R increases. These tests simply make sure that data for each R-value is also to the "left" of the data for R=0. Again, AFGROW will calculate the data for R=0 internally if it is not one of the R-values entered by the user.
Kmax value @ RLO is > Delta K @ R = 0 at DADNHI... cannot apply	
Delta K value @ R = [X] is > Delta K @ R = 0 at DADNHI... cannot apply	
Kmax value @ R = [X] is > Delta K @ R = 0 at DADNHI... cannot apply	
Delta K value @ RHI is > Delta K @ R = 0 at DADNHI... cannot apply	
Error! Curves crossing between DADNLO and DADN1.	Although all of the user-input curves have been checked so that they don't cross each other inside the domain of the user-defined growth rate values. AFGROW checks to be sure that no curves cross when data are extrapolated to either DADNLO or DADNLO.
Error! Curves crossing between DADN[last input value] and DADNHI	

Warning Tests:

The following possible discrepancies exist in the tabular lookup data. AFGROW has accepted these data, but they may warrant further examination	
Message	Explanation
Kmax*(1-R) at RLO is not greater than Delta K at R = 0 at DADNLO	AFGROW converts any user-input Kmax data for negative R-values (including data that may have been internally extrapolated for RLO) to Delta K ($\Delta K = K_{max}(1-R)$), and checks to see whether the converted ΔK value is greater than ΔK at R = 0. Delta K values for negative R, should be greater than ΔK for R=0 at the same growth rate. These messages should alert a user to a problem with the input data.
Kmax*(1-R) is not greater than Delta K at R = 0 at DADNLO at R = [X]	
Kmax*(1-R) at RLO is not greater than Delta K at R = 0 at DADN[X]	
Kmax*(1-R) at R = [X] is not greater than Delta K at R = 0 at DADN[X]	
Kmax*(1-R) at RLO is not greater than Delta K at R = 0 at DADNHI	
Kmax*(1-R) at R = [X] is not greater than Delta K at R = 0 at DADNHI	
Delta K @ R = [X] is greater than Kmax @ R = [X] at DADNLO	Normally, crack growth rate data for a given positive R-value will be shifted to the left of rate data for a negative R-value of the same magnitude since Kmax is used for negative Rs. This means that DK should be less than Kmax at a given growth rate for the same absolute R-value. This is not universally accepted, so it is given as a warning.
Delta K @ R = [X] is greater than Kmax @ R = [X] at DADN[X]	
Delta K @ R = [X] is greater than Kmax @ R = [X] at DADNHI	

Finally, there may be cases where multiple error or warning messages will be issued. If there are more than 25 messages, AFGROW will only print the first 25 messages (because of limitations in the size of the message dialog). As mentioned earlier, it is always a good practice to plot the growth rate data and look for errors prior to entering it in table look-up dialog box.

3.2.2.4.6 Saving Tabular Lookup Data to a File

Once tabular data have been entered and applied (error checked), these data may be saved in a file by clicking on the save button in the tabular look-up dialog. The test-based format used for tabular lookup data (single data set) [filename.lkp] is as follows (space delimited):

[*No. of da/dN values*] (2 min., 30 max.) [*No. of R values*] (1 min., 10 max.)

[*R₁*] [*R₂*] [*R_{max}*]

[*da/dN₁*] [*DK @ R₁*] [*DK @ R₂*] ... [*DK @ R_{max}*]

[*da/dN₂*] [*DK @ R₁*] [*DK @ R₂*] ... [*DK @ R_{max}*]

.....

[*da/dN_{max}*] [*DK @ R₁*] [*DK @ R₂*] ... [*DK @ R_{max}*]

[*R_{lo}*] [*K_{IC}*] [*DADNLO*] [*Yield*]

[*R_{hi}*] [*K_C*] [*DADNHI*] [*THOLD*]

[*Poisson's ratio*] [*Coefficient of Thermal Expansion*] [*Modulus*]

Remember that K_{max} is required in place of ΔK for $R < 0.0$.

If multiple data sets are used, files are saved in XML format [filename.lkpx].

3.2.2.4.7 Material Parameters

The following parameter is ONLY used for the FASTRAN retardation model:

Ultimate Strength: (Stress) Used to determine the material flow stress as needed for the FASTRAN retardation model.

The following three parameters are ONLY used in the analysis of bonded composite repairs:

Coefficient of Thermal Expansion: (Temperature)-1 Used in the calculation of the thermal effect of patch cure temperature on the stress intensity factor of the patched metal.

Young's Modulus: (Stress) Used in the calculation of the stress intensity factor correction due to the presence of the bonded patch (*also used in the initiation module*).

Poisson's Ratio: (Non-Dimensional) Used in the calculation of the stress intensity factor correction due to the presence of the bonded patch.

The following parameters are used in standard crack growth analyses:

DADNLO: (Length) Lower limit for da/dN extrapolation (uses log-log linear extrapolation based on the first two user input points for the appropriate R).

DADNHI: (Length) Upper limit for da/dN extrapolation (uses log-log linear extrapolation based on the last two user input points for the appropriate R).

Plane Stress Fracture Toughness (KC): (Stress, Length^{0.5}) Value of Fracture Toughness to be used under pure plane stress conditions.

Plane Strain Fracture Toughness (KIC): (Stress, Length^{0.5}) Value of Fracture Toughness to be used under pure plane strain conditions.

Delta K Threshold Value @ R=0: (Stress, Length^{0.5}) Threshold stress intensity value at $R=0$ - no crack growth will be calculated when Delta K is below threshold for a given R value.

Yield Strength, (YLD): (Stress) Yield stress (0.2% offset strain) for the metal being analyzed.

Lower Limit on R Shift: (Non-Dimensional) R value below which no further R shifting is calculated.

Upper Limit on R Shift: (Non-Dimensional) R value above which no further R shifting is calculated.

Buttons:

OK: Load the data in AFGROW (if errors are detected) and close the dialog box.

CANCEL: Cancel the dialog box.

APPLY: Apply the current input values to check for any errors.

ADD: Add a data set.

DELETE: Delete the currently displayed data set.

READ: Read a previously saved file (*.lkp or *.lkpx).

SAVE: Save the current data to a user specified file.

HELP: Open user help for the tabular lookup model.

3.2.2.5 Walker Equation

Walker Equation Data

The Walker equation extended the early Paris equation by allowing the shift in da/dN vs. ΔK as a function of stress ratio (R). The equation may be used in several segments to attempt to model the sigmoidal shape of the data.

Use up to 5 sets of values of 'C', 'n', and 'm'

Number of Sets:

Set	C	n	m
1	2.6e-010	7	0.3
2	1e-008	3	0.5
3	2e-019	9	0.7
4	1e-008	3	0.5
5	1e-008	3	0.5

Material name:

Coefficient of Thermal Expansion: Young's Modulus:

Yield Strength, YLD: Poisson's Ratio:

Plane Stress Fracture Toughness, KIC:

Plane Strain Fracture Toughness, KIC: Lower limit on R shift (0.. -1):

Delta K threshold value @R=0: Upper limit on R shift (< 1):

OK Cancel Save Read Apply

Figure 63: Walker Equation Dialog

The Walker equation [15] was essentially an enhancement of the Paris Equation that included a means to account for the effect of Stress Ratio (Minimum Stress/Maximum Stress) on crack growth rate (see Figure 64).

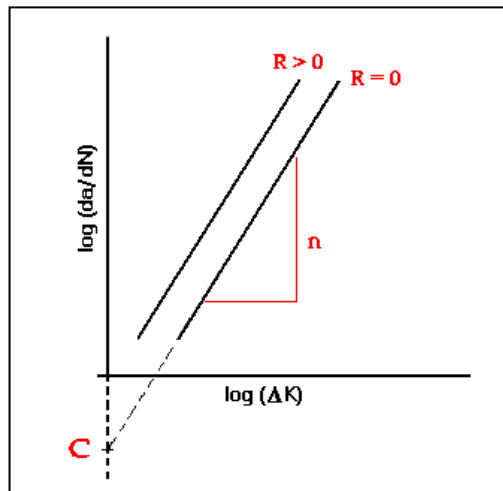


Figure 64: Walker Equation

$$\frac{da}{dN} = C[\Delta K(1 - R)^{(m-1)}]^n ; \text{ for } R \geq 0$$

$$\frac{da}{dN} = C[K_{\max} (1 - R)^{(1-m)}]^n ; \text{ for } R < 0$$

There are three reasons for using a different form of the Walker equation when R is less than 0.

First, it is more convenient to use $K_{\max}$ in place of ΔK for negative R's. If ΔK were used for negative R values, the crack growth rate curves would continue to shift to the right as R decreases and eventually converge to a factor $(1 - R)$ of ΔK at $R=0$.

Second, the shift in crack growth rate is controlled by the term $(1 - R)^{(m-1)}$ when $R \geq 0$. In this case, $(1 - R)$ is less than 1 so that as m increases, the shift decreases. Conversely, as m decreases, the shift increases. Note: m is in the range $(0 < m \leq 1)$. It is important that the trend in the data shifting be consistent with respect to m. Therefore, AFGROW uses the modified form of the standard Walker equation shown above for R less than 0. There seems to be a practical limit to the R shifting as R decreases below 0.0 (based on actual test data plotting da/dN vs. $K_{\max}$). This is why AFGROW provides the capability to set limits for R shifting (R_{lo} , R_{hi}).

Third, since AFGROW uses $K_{\max}$ in place of ΔK for $R < 0$, the relative shifting should follow the trend that the magnitude of the shifting for a given negative R will be less than the shift for the corresponding positive R (ΔK is used for the positive R). An explanation for this may be seen in the ratio of the crack opening stress to maximum stress ratio (C_f) as a function of R (stress ratio). The change in the opening stress ratio, Figure 65, tends to decrease as R decreases causing the change in effective stress intensity (and growth rate) to decrease. This trend forces the shifting of growth rate to be less for negative R values than for the corresponding positive values. The use of the exponent $(1-m)$ applied to $(1-R)$ ensures that the appropriate trend in rate shifting will be maintained.

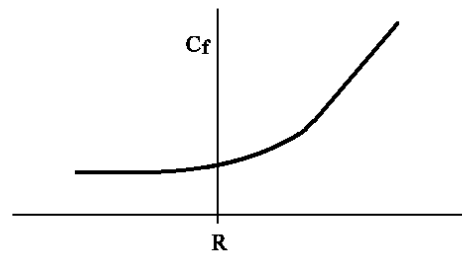


Figure 65: Closure Factor vs. Stress Ratio

AFGROW allows up to 5 Walker line segments to provide the best possible fit to actual crack growth rate data. The current implementation of the Walker equation allows users to assign different m-values for each segment. (Figure 66).

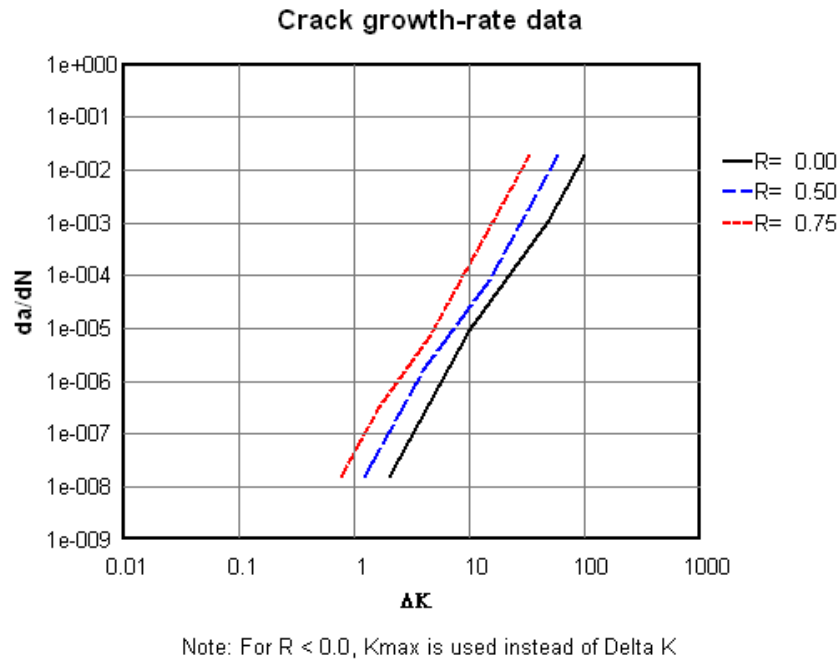


Figure 66: Using the Walker Equation with Multiple Segments

AFGROW automatically calculates the intersection points for each segment at any R value. Users are responsible for the integrity of the input data, but AFGROW will check to ensure that the following conditions are true:

- Intersection Points (for $R = 0$) are Monotonically Increasing (in terms of da/dN and ΔK)
- Segment Slopes are Always Positive
- Adjacent Slope Values (n) Must NOT Match
- Adjacent Intercept Values (C) Must NOT Match
- Threshold ΔK Values Must be Less Than ΔK_c for all R Values

The use of unequal m values may result in discontinuous crack growth rate curves. Although AFGROW checks to be sure that the intersection points for the Walker segments are monotonically increasing for $R=0$, it is possible that the intersection points will NOT be increasing in terms of da/dN and ΔK for other R values. This is an important issue since it has a large impact on the crack growth rates that will result in these cases. AFGROW will NOT allow any crack growth rate curve to result in multiple crack growth rates for a given ΔK . If AFGROW detects this condition for any R value, users will have the option to limit the range of possible R values or allow portions of the curve that fall below the ΔK value for the intersection of previous line segment to be ignored (as shown in Figure 67). The crack growth rate will jump to the value for the appropriate line segment that corresponds to the ΔK value for the intersection point prior to the error condition. If the resulting crack growth rate exceeds the maximum rate allowed for a given analysis,

AFGROW will only plot (and use) the data to the last intersection and assign the maximum rate to any ΔK values that exceed the value at the last intersection.

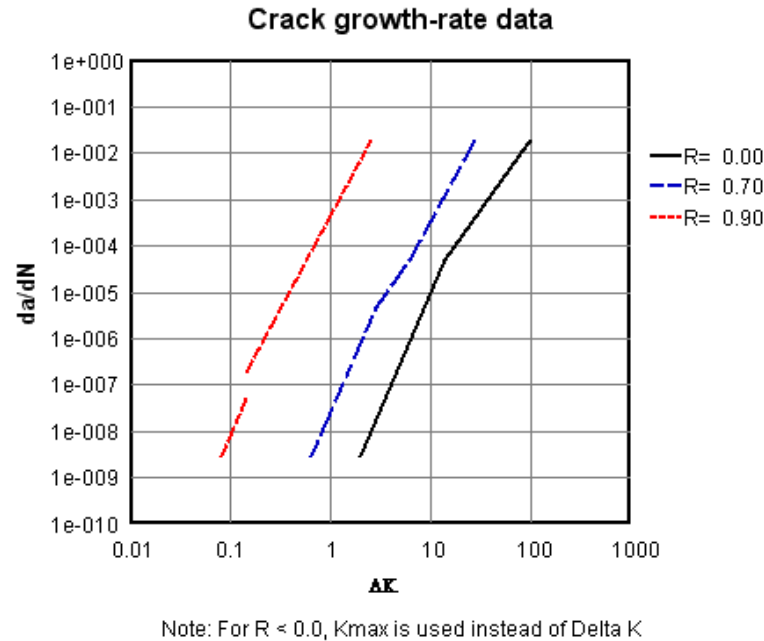


Figure 67: Discontinuous Crack Growth Rate Curves

Remember that the range of possible crack growth rate values is controlled by ΔK threshold and the plane stress fracture toughness (K_{IC}) – both at $R=0$. AFGROW calculates ΔK threshold and ΔK_{IC} for each R value using the crack growth rate for each term at $R = 0$. These crack growth rates determine the lower and upper bounds on crack growth rate values. Points below the lower limit ($< \Delta K$ threshold) will be assumed to result in no crack growth rate, and those above the upper limit will be assigned a crack growth rate value equal to the upper limit. Regardless of the number of segments used, **only data in the current range of possible crack growth rates will be used or shown in the crack growth rate plots.**

The following is a description of the terms used in the Walker dialog box.

The following parameters are ONLY used in the analysis of bonded composite repairs:

Coefficient of Thermal Expansion: (Temperature)⁻¹ Used in the calculation of the thermal effect of patch cure temperature on the stress intensity factor of the patched metal.

Young's Modulus: (Stress) Used in the calculation of the stress intensity factor correction due to the presence of the bonded patch (*also used in the initiation module*).

Poisson's Ratio: (Non-Dimensional) Used in the calculation of the stress intensity factor correction due to the presence of the bonded patch.

The following parameters are used in the standard crack growth analysis:

C: (Stress⁽⁻ⁿ⁾, Length^(1-n/2)) Value of da/dN when R=0 and Delta K=1 (da/dN intercept).

n: (Non-Dimensional) Paris Exponent (da/dN slope).

Walker Exponent, m: (Non-Dimensional) Normal Range ($0 < m \leq 1$) Controls shift in crack growth rate data, curve shift decreases as m increases.

Plane Stress Fracture Toughness (KC): (Stress, Length^{0.5}) Value of Fracture Toughness to be used under pure plane stress conditions.

Plane Strain Fracture Toughness (KIC): (Stress, Length^{0.5}) Value of Fracture Toughness to be used under pure plane strain conditions.

Delta K Threshold Value @ R=0, THOLD: (Stress, Length^{0.5}) Threshold stress intensity value at R=0 - no crack growth will be calculated when Delta K is below threshold for a given R value.

Yield Strength, YLD: (Stress) Yield stress (0.2% offset strain) for the metal being analyzed.

Lower limit on R shift, Rlo: (Non-Dimensional) R value below which no further R shifting is calculated.

Upper limit on R shift, Rhi: (Non-Dimensional) R value above which no further R shifting is calculated.

Buttons:

APPLY: Apply the current parameters.

READ: Read a file containing Walker parameters.

SAVE: Save the current parameters to a file.

CANCEL: Cancel the dialog box.

OK: Accept the current parameters and close the dialog box.

3.2.3 Input Model

Toolbar Icon: 

Nearly every crack growth life prediction program available today is capable of predicting the life of a number of structural geometries with single (or symmetric) cracks using closed-form stress intensity (K) solutions. AFGROW has taken a step forward to allow users to predict the lives of more complex (single and double, un-symmetric) crack cases. The new models are curve-fit or table look-up solutions based on finite element models (FEMs). For the purpose of differentiating these capabilities, relatively simple single (or symmetric) crack models are being called “Classic” models in AFGROW. The majority of these solutions are available in the open literature. The more complex solutions are called “Advanced” models and require a unique user interface (see Section 2.0). AFGROW also allows users to develop plug-in solutions. As the name implies, plug-in models are separate modules that are not part of the standard AFGROW installation. The model interface is selected using a pull down menu available in the menu bar (input, model) or the toolbar icon as shown in Figure 68.

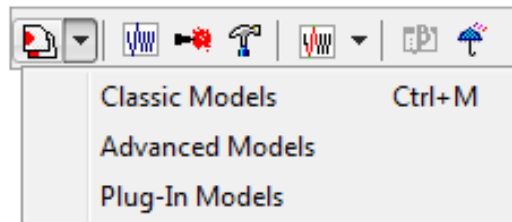


Figure 68: Model Interface Selection

If a user left-clicks on the icon itself, the classic model interface dialog will be displayed. The model interface pull down menu is displayed when a user left-clicks on the pull down symbol (▼). The following sections contain detailed descriptions of the models available in each interface.

3.2.3.1 Classic Models

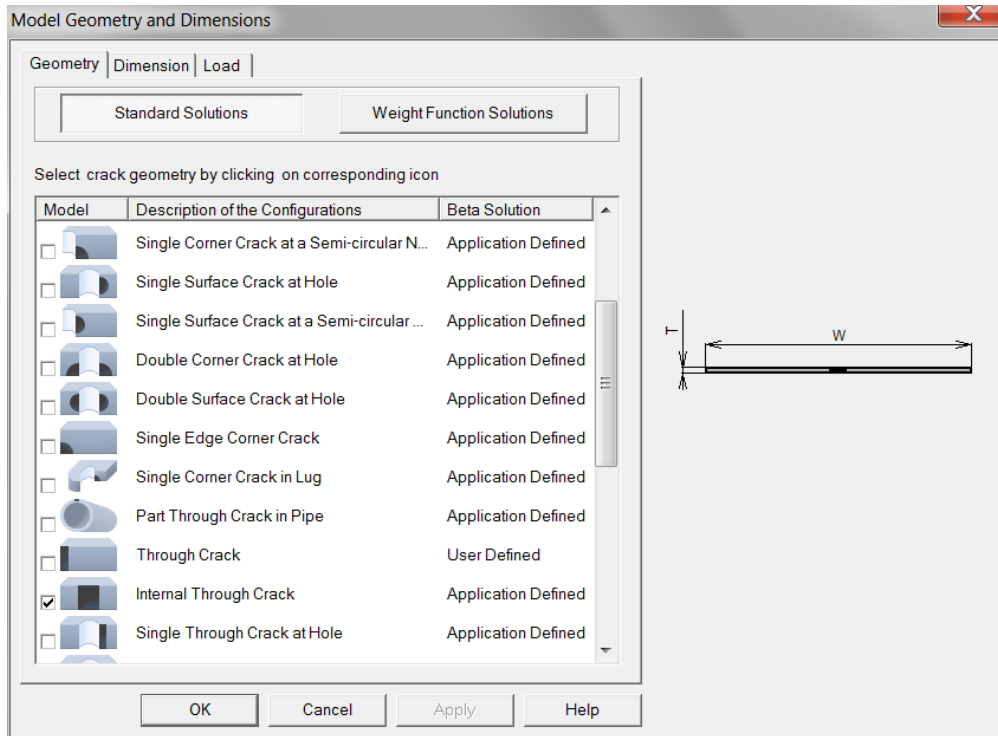


Figure 69: Classic Input Model Dialog

There are two types of “classic” stress intensity factor solutions available in AFGROW:

- Standard Stress Intensity Solutions
- Weight Function Stress Intensity Solutions

In addition to these solutions, users can input their own solutions through the user input beta option. However, to use this option, the user must first select either the 1-D or 2-D user defined geometry from the Standard Solutions dialog. The user can also choose to use one of the Standard Solutions and apply a beta correction based on the ratio of the actual stress distribution to the standard stress distribution.

3.2.3.1.1 Standard Stress Intensity Solutions

The standard crack geometries in AFGROW consist of several models for which closed form or tabular stress intensity factor solutions are available. Solutions for several geometries are built into the code and are referred to as application defined solutions. AFGROW also allows user defined stress intensity solutions to be input in the form of beta factors at various crack lengths. Beta factors are defined as follows:

$$\beta = \frac{K}{\sigma \sqrt{\pi x}} ; \text{ Where } x \text{ is the appropriate crack length}$$

The stress reference for each classic model is shown in the load tab for each available loading case. The crack length in the thickness direction is the a-dimension, and the crack length in the width direction is the c-dimension. Many of the standard, classic stress intensity solutions in AFGROW use the popular Newman and Raju curve fit solutions to finite element results [16]. An angle, ϕ , is used in these equations to determine the stress intensity value for the crack growth dimensions (a, and c). This angle is defined as shown in Figure 70:

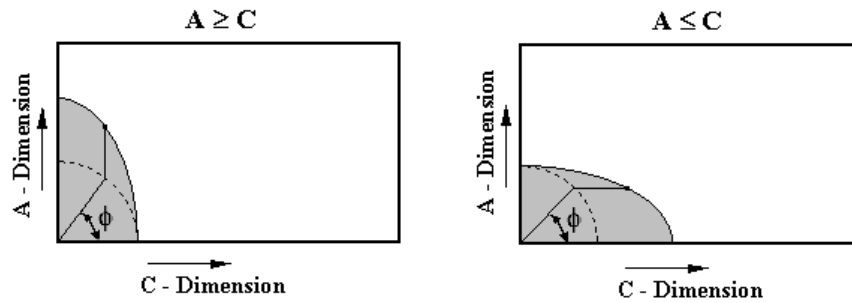


Figure 70: Parametric Angle Used in Newman and Raju Solutions

The angle is measured from a line in the c-direction beginning at the crack origin. The closed-form Newman and Raju solutions do not necessarily match the finite element results at the free edges. Care was taken in AFGROW to use the angle for each crack dimension that tends to match the published finite element results near the free edges. The default angles used in the Newman and Raju solutions for each crack dimension are documented in the following sections for models that use these solutions.

Many advanced users of AFGROW have requested the ability to change these default angles to provide greater flexibility for life predictions. In order to provide a means to change these angles and to discourage less advanced users from making changes, we decided to place this information in the Windows Registry (run, regedit) in the following location:

My Computer\HKEY_CURRENT_USER\Software\LexTech\AFGROW\AngleInit

The default angles are shown in Figure 71 for each classic part-thru crack model in AFGROW. The codes used by AFGROW for each part-thru crack are given below:

- 1010 Center Semi-elliptic Surface Flaw
- 1015 Center Semi-elliptic Edge Surface Flaw
- 1020 Center Full-elliptic Embedded Flaw
- 1030 Single Corner Cracked at a Hole
- 1035 Single Corner Crack at a Notch
- 1040 Single Surface Crack at a Hole
- 1045 Single Surface Crack at a Notch
- 1050 Double Corner Crack at a Hole
- 1060 Double Surface Crack at a Hole

- 1070 Single Edge Corner Crack
- 1080 Single Corner Crack in a Lug
- 1090 Part Through Crack in a Pipe
- 1091 Internal Axial Crack in Thick Pipe
- 1092 External Axial Crack in Thick Pipe

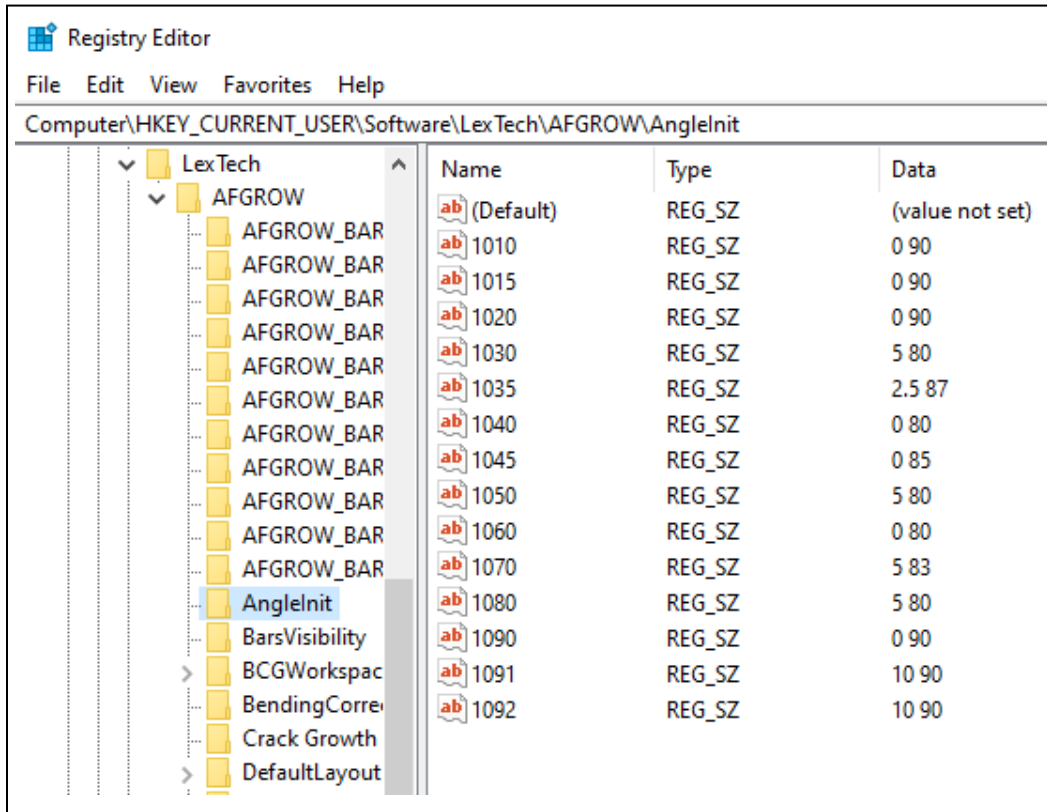


Figure 71: Using the Registry Editor to Change Default Parametric Angles

To change a value in the registry, simply select the item to change (in the right hand window in regedit) and select edit from the menu to change the values.

To return to the default angles, just delete the AngleInit folder from the registry, save and exit the registry, and run AFGROW as a stand-alone code. A new AngleInit folder will then be created by AFGROW with the default values.

Whenever one of these models is used in AFGROW, the appropriate angles will be shown in the output file. This is useful when comparing analyses for the same input file that have been run on different computers. Different angles will produce different results.

Application and user defined solutions are identified under the beta solution column in the geometry tab of the model dialog (see Figure 69). There are only two user-defined models

among the standard solutions since AFGROW currently models only 1-D or 2-D cracks. The currently available standard solutions are described in the following sections.

3.2.3.1.1.1 Part Through-the-Thickness Crack (User Defined)



This model is used when a user has an existing stress intensity factor solution (in the form of a beta table) for any 2-D crack, which may be described with two length dimensions (2-D) to input in AFGROW.

The geometric beta values are NOT calculated by AFGROW, but are merely interpolated from a two-dimensional user-defined table of beta values. Users must supply beta values at various crack lengths so that the appropriate value at a given crack length may be interpolated. This model is shown as a corner cracked plate in the animation frame. The representation of the model is merely meant to indicate the two dimensional nature of the crack. It was not possible to create representations of all possible geometries that may be modeled using user defined beta factors

For the [a] crack length dimension: $K = \sigma \sqrt{\pi a} \beta(a)$

For the [c] crack length dimension: $K = \sigma \sqrt{\pi c} \beta(c)$

Once this model is selected, AFGROW will add a user input beta icon, , in the AFGROW toolbar (if active). A notice will also be provided in the status view of the main frame window indicating that user-defined beta information is required. Users may choose any external source to calculate stress intensity factors and convert them to beta values. The details of the 2-D user-defined beta option is given in section 3.2.7.2.

Once the 2-D beta information has been entered, the user will be prompted to enter beta values for the 1-D case (see section 3.2.7.1). The 1-D user-defined beta table is used after the 2-D crack transitions to become a 1-D (through-the-thickness) crack.

3.2.3.1.1.2 Center Semi-Elliptical Surface Crack (Application Defined)



Axial Loading:

Reference [16]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 90°

This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

$$2c/W < 0.5$$

Bending Loading:

Reference [16]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 90°

This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

$$2c/W < 0.5$$

3.2.3.1.1.3 Center Semi-Elliptical Edge Surface Crack (Application Defined)



Axial Loading:

Reference [16]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 90°

This solution is valid for the following dimensions:

$$0 < a/t \leq 0.5$$

$$0.2 \leq a/c \leq 2.0$$

$$c/W < 0.5$$

3.2.3.1.1.4 Center Full-Elliptical Embedded Crack (Application Defined)



Axial Loading:

Reference [16]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 90°

This solution is valid for the following dimensions:

$$0 < a/t \leq 0.5$$

$$0.2 \leq a/c \leq 2.0$$

$$2c/W < 0.5$$

3.2.3.1.1.5 Single Corner Crack at a Hole (Application Defined)



Axial Loading:

Reference [16]

Default angle (ϕ) used for the C Dimension: 5°

Default angle (ϕ) used for the A Dimension: 80°

This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

$$(D+2c)/W < 0.5$$

Note: An additional width correction factor (F_{ww}) is applied based in the C dimension as explained in Section 3.2.3.1.1.19

Bending Loading:

References [16, 17, 18]

Default angle (ϕ) used for the C Dimension: 5°

Default angle (ϕ) used for the A Dimension: 80°

This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

$$(D+2c)/W < 0.5$$

Note: An additional width correction factor (F_{ww}) is applied based in the C dimension as explained in Section 3.2.3.1.1.19

Bearing Loading:

References [16, 19]

Default angle (ϕ) used for the C Dimension: 5°

Default angle (ϕ) used for the A Dimension: 80°

Until recently (May, 2007), it had been assumed that the finite width correction proposed by Dr. Newman [16] for open holes could also be applied for the bearing load case. Dr. Newman's paper never made such a claim, but the equation has been used by others [19] in developing closed form solutions for corner cracked holes under bearing loading. Some significant discrepancies were discovered while we were verifying our latest advanced multiple corner cracked hole solutions for this load case (Section 3.2.3.2.2). These advanced solutions are actually tabular FEM solutions for a very wide plate ($W/D=100$). It was thought that the wide plate solutions could simply be multiplied by the finite width correction to obtain solutions for any plate width. Upon checking, it was discovered that this was not a valid assumption.

The bearing solution used for this classic solution in previous releases of AFGROW was found to be relatively good ($< 3\text{-}5\%$ difference) for very narrow ($W/D < 2$) and wide ($W/D > 50$) plates when compared to detailed 3-D FEM models using StressCheck [31]. However, the previous solution is consistently higher (as much as 50%) than the FEM cases for intermediate plate widths that make up the majority of practical cases.

A good deal of effort has gone into developing a better bearing solution. While we think that the proposed solution can be improved in time, the following method is currently being used to determine the bearing stress intensity solution for part-through cracks at holes under bearing load:

First, the bearing solution can be determined using the principle of superposition.

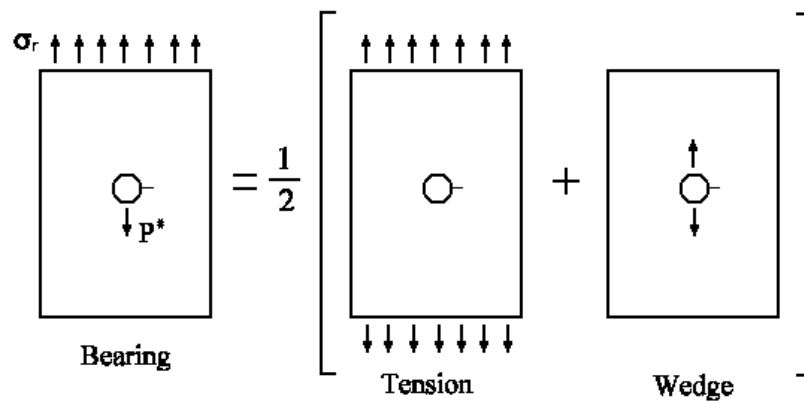


Figure 72: Determining the Bearing Solution Using Superposition

The pin load (P^*) may be modeled using any boundary condition (i.e. nodal springs, $\cos(\theta)$, $\cos^2(\theta)$, etc.) if a FEM is used to determine the stress intensity value. The solution will ultimately depend on the boundary condition, but the principal of superposition applies in any case.

Secondly, we have the classic solution for the axial case, and the bearing solution used in the earlier releases of AFGROW [19] has been verified for the wide plates ($W/D > 50$). The bearing solution for an infinitely wide plate is exactly half of the solution for the corresponding wedge solution. This is true because the gross axial stress (σ_r) reacting the pin load would go to zero as the plate width goes to infinity²². Therefore, if we assume that the bearing solution for very wide plate (i.e. $W/D=100$), is equivalent to the infinite plate case, we can use the existing wide plate bearing solution in the following way:

$$\beta_{\text{Bearing}} = \left(\frac{D}{2W} \right) \beta_{\text{Axial}} + \beta_{\text{Bearing}(W/D=100)} * F_{\text{wp}} ; \text{ (for any given plate width)}$$

Where, F_{wp} = Finite Plate Correction for the infinite plate bearing solution

Finally, to determine F_{wp} , a large number of FE analyses had been performed as part of the effort to develop the advanced bearing solution for two, non-symmetric corner cracks (Section 3.2.3.2.2). As a result of this effort, the following relationship was determined for F_{wp} :

$$F_{\text{wp}} = \left[0.96 + 0.1 \left(\frac{a}{b} \right) + 0.75 \left(\frac{a}{b} \right)^2 + 2.25 \left(\frac{a}{b} \right)^3 + 1.8 \left(\frac{a}{b} \right)^4 \right] \left[1 + \Phi \left(\frac{a}{b} \right)^{2.1} \right]$$

Φ is the parametric crack angle (Section 3.2.3.1.1), in radians.

$\frac{a}{b}$ is the cracked fraction of the plate as measured from the center of the hole in the c-direction (width direction). This value changes along the crack front.

For example:

$$\text{If } \Phi = 0 ; \frac{a}{b} = \frac{2(R+C)}{W} \text{ (for a crack at a centered hole in a plate)}$$

$$\text{If } \Phi = \frac{\pi}{2} ; \frac{a}{b} = \frac{2R}{W} = \frac{D}{W} \text{ (for a crack at a centered hole in a plate)}$$

²² Note, β_{Axial} is based on the bearing stress in the bearing equation. The standard axial beta is converted to a bearing stress reference when multiplied by (D/W) .

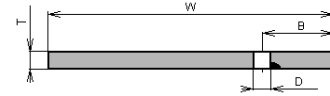
This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

$$\frac{a}{b} < 0.6 \text{ (refer to Section 3.2.3.2.2 for clarification)}$$

Offset Correction:



For the Axial and Bending Load Cases:

The solution for an offset (non-centered hole) uses the centered hole solution in AFGROW with the width adjusted to be equal to twice the distance from the center of the hole to the right edge (2B). AFGROW now includes an offset correction for a crack growing to the near edge ($B < W/2$) and an offset correction for a crack growing to the far edge ($B > W/2$). The offset corrections are given below:

For $B < W/2$:

$$F_{offset} = \frac{\sin \left(\sqrt{\frac{a}{t}} \left(\frac{D+c}{B-c/2} \right) \left(\frac{W-2B}{W} \right) \right)}{\sqrt{\frac{a}{t}} \left(\frac{D+c}{B-c/2} \right) \left(\frac{W-2B}{W} \right)} F_C$$

Reference [20]

Where:

$$F_C = \left(1 - (0.45F_G - 0.021) \left(\frac{c}{B-D/2} \right)^{16} \right)$$

$$F_G = \left(\frac{2B}{W} + \frac{D}{2B} \right)$$

Note: The following limitations apply to F_C and F_G

$$\text{If } F_G < 0.0468, F_C = 1.0$$

$$\text{If } F_G > 0.7, F_G = 0.7$$

This solution is valid for the following dimensions:

$$\left(\frac{D+c}{2B-c} \right) \leq 0.7$$

The solution tends to be slightly conservative (1 to 3%) when the limit is exceeded.

For $B > W/2$:

This correction is more complex than the previous case since the stress intensity factor may be affected by the proximity of the hole to the edge of the plate as well as the fact that the crack is growing to the far edge of the plate. The offset correction is given below:

$$F_{offset} = F_{AHFB} F_{B/W}$$

$$F_{AHFB} = 1 + \frac{\sqrt{\sec\left(\frac{\pi D}{14}\left(\frac{2}{W-B} + \frac{1.5}{B}\right)\right)} - 1}{\left[1 + 0.21 \sin\left(8 \tan^{-1}\left(\left(\frac{2B-W}{W}\right)^{0.9}\right)\right)\right]} \quad \text{Reference [21]}$$

Note: The above equation has been modified to reflect the definition of the parameter, B, used by AFGROW for this geometry.

The factor (F_{AHFB}) accounts for the effect of the proximity of the hole to the edge of the plate.

$$F_{B/W} = 1 + \left(F_{\max} \sin\left(\pi \tanh\left(2\delta^{1.1} + (1.18\delta)^7\right)\right) \right) \quad \text{Reference [22]}$$

Where:

$$\delta = \sqrt{\frac{a}{t}} \left(\frac{D+c}{2B-c} \right)$$

$$F_{\max} = 0.5 e^{-(10\gamma + 4.2\gamma^2 + (3\gamma)^{14})}$$

$$\gamma = 1 - \frac{B}{W}$$

The factor ($F_{B/W}$) adjusts the offset correction as a function of the ratio of the offset to the plate width. This empirical curve fit was made using finite element results for a single

through cracked hole. It is assumed that this correction is also valid for part through flaws. A sample beta solution is shown below in Figure 73.

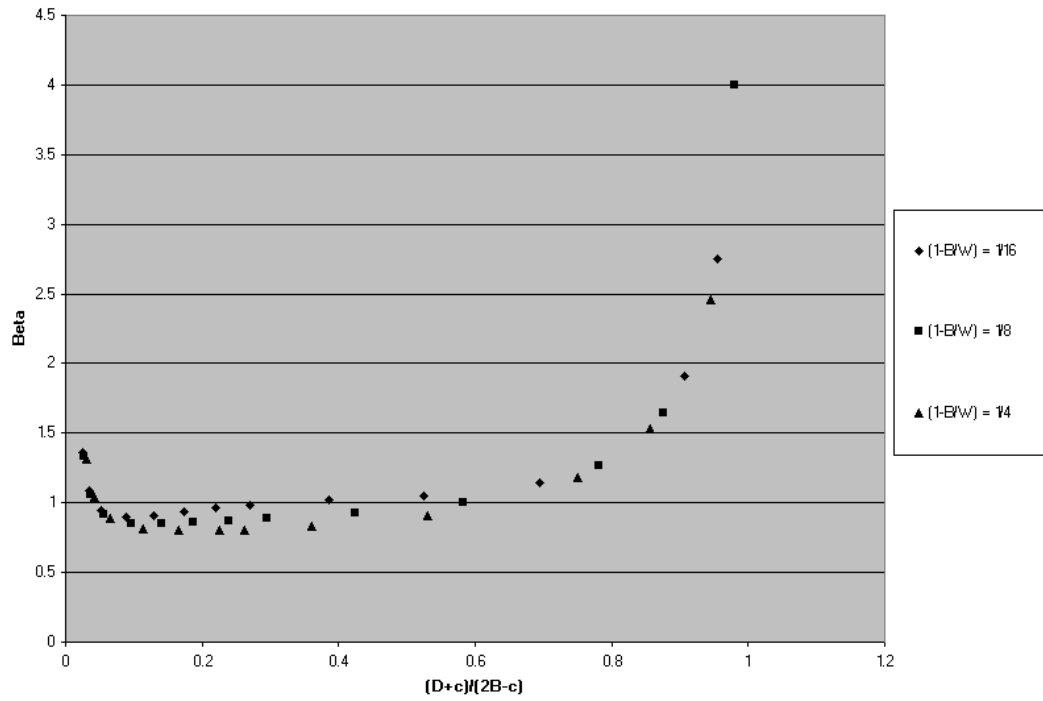


Figure 73: Sample Beta Solutions for an Offset Hole, $B > W/2$

For the Bearing Load Case:

The offset correction was developed using FEMs for the through crack case as described in Section 3.2.3.2.2.16.

3.2.3.1.1.6 Single Corner Crack at a Semi-Circular Notch (Application Defined)



Axial Loading:

Reference [23]

Default angle (ϕ) used for the C Dimension: 2.5°

Default angle (ϕ) used for the A Dimension: 87°

This solution is valid for the following dimensions:

$$0 < a/t < 1.0$$

$$0.2 < a/c < 2$$

$$1 < r/t < 2.0$$

$$(r+c)/w < 0.5$$

$$r/w = 1/16 \rightarrow K_t = 3.17$$

Where, r is the notch radius

This solution was developed using fixed grip boundary conditions.

3.2.3.1.1.7 Single Surface Crack at a Hole (Application Defined)



Axial Loading:

Reference [16]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 80°

This solution is valid for the following dimensions:

$$0 < a/t \leq 0.5$$

$$0.2 \leq a/c \leq 2.0$$

$$(D+2c)/W < 0.5$$

Note: An additional width correction factor (F_{ww}) is applied based in the C dimension as explained in Section 3.2.3.1.1.19

Bearing Loading:

References [16, 19]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 80°

The method used to determine the bearing solution for all classic part-through cracked holes in plates is explained in Section 3.2.3.1.1.5.

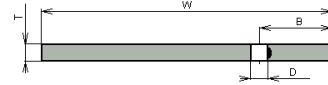
This solution is valid for the following dimensions:

$$0 < a/t \leq 0.5$$

$$0.2 \leq a/c \leq 2.0$$

$$\frac{a}{b} < 0.6 \text{ (refer to Section 3.2.3.2.2.1 for clarification)}$$

Offset Correction:



The solution for an offset (non-centered hole) uses the centered hole solution in AFGROW with the width adjusted to be equal to twice the distance from the center of the hole to the right edge ($2B$). AFGROW now includes an offset correction for a crack growing to the near edge ($B < W/2$) and an offset correction for a crack growing to the far edge ($B > W/2$). The offset corrections are given below:

For $B < W/2$:

$$F_{offset} = \frac{\sin \left(\sqrt{\frac{2a}{t}} \left(\frac{D+c}{B-c/2} \right) \left(\frac{W-2B}{W} \right) \right)}{\sqrt{\frac{2a}{t}} \left(\frac{D+c}{B-c/2} \right) \left(\frac{W-2B}{W} \right)} F_C$$

Reference

[20]

Where:

$$F_C = \left(1 - (0.45F_G - 0.021) \left(\frac{c}{B-D/2} \right)^{16} \right)$$

$$F_G = \left(\frac{2B}{W} + \frac{D}{2B} \right)$$

Note: The following limitations apply to F_C and F_G

If $F_G < 0.0468$, $F_C = 1.0$

If $F_G > 0.7$, $F_G = 0.7$

This solution is valid for the following dimensions:

$$\left(\frac{D+c}{2B-c} \right) \leq 0.7$$

The solution tends to be conservative (~1-3% too high) when the above limit is exceeded.

For $B > W/2$:

This correction is more complex than the previous case since the stress intensity factor may be affected by the proximity of the hole to the edge of the plate as well as the fact that the crack is growing to the far edge of the plate. The offset correction is given below:

$$F_{offset} = F_{AHFB} F_{B/W}$$

$$F_{AHFB} = 1 + \frac{\sqrt{\sec\left(\frac{\pi D}{14} \left(\frac{2}{W-B} + \frac{1.5}{B}\right)\right)} - 1}{\left[1 + 0.21 \sin\left(8 \tan^{-1}\left(\left(\frac{2B-W}{W}\right)^{0.9}\right)\right)\right]} \quad \text{Reference [21]}$$

Note: The above equation has been modified to reflect the definition of the parameter, B , used by AFGROW for this geometry.

The factor (F_{AHFB}) accounts for the effect of the proximity of the hole to the edge of the plate.

$$F_{B/W} = 1 + \left(F_{\max} \sin\left(\pi \tanh\left(2\delta^{1.1} + (1.18\delta)^7\right)\right)\right) \quad \text{Reference [22]}$$

Where:

$$\delta = \sqrt{\frac{2a}{t}} \left(\frac{D+c}{2B-c}\right)$$

$$F_{\max} = 0.5 e^{-(10\gamma + 4.2\gamma^2 + (3\gamma)^{14})}$$

$$\gamma = 1 - \frac{B}{W}$$

The factor ($F_{B/W}$) adjusts the offset correction as a function of the ratio of the offset to the plate width. This empirical curve fit was made using finite element results for a single through cracked hole. It is assumed that this correction is also valid for part through flaws. A sample beta solution is shown in Figure 73.

3.2.3.1.1.8 Single Surface Crack at a Semi-Circular Notch (Application Defined)



Axial Loading:

Reference [23]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 85°

This solution is valid for the following dimensions:

$$0 < a/t < 0.5$$

$$0.2 < a/c < 2$$

$$1 < r/t < 3.5$$

$$(r+c)/w < 0.5$$

$$r/w = 1/16 \rightarrow K_t = 3.17$$

Where, r is the notch radius

This solution was developed using fixed grip boundary conditions.

3.2.3.1.1.9 Double Corner Crack at a Hole (Application Defined)



Axial Loading:

Reference [16]

Default angle (ϕ) used for the C Dimension: 5°

Default angle (ϕ) used for the A Dimension: 80°

This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

$$(D+2c)/W < 0.5$$

Note: An additional width correction factor (F_{ww}) is applied based in the C dimension as explained in Section 3.2.3.1.1.20

Bending Loading:

References [16, 17, and 18]

Default angle (ϕ) used for the C Dimension: 5°

Default angle (ϕ) used for the A Dimension: 80°

This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

$$(D+2c)/W < 0.5$$

Note: An additional width correction factor (F_{ww}) is applied based in the C dimension as explained in Section 3.2.3.1.1.20

Bearing Loading:

References [16, 19]

Default angle (ϕ) used for the C Dimension: 5°

Default angle (ϕ) used for the A Dimension: 80°

The method used to determine the bearing solution for all classic part-through cracked holes in plates is explained in Section 3.2.3.1.1.5.

This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

$$\frac{a}{b} < 0.6 \text{ (refer to Section 3.2.3.2.2 for clarification)}$$

3.2.3.1.1.10 Double Surface Crack at Hole (Application Defined)



Axial Loading:

Reference [16]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 80°

This solution is valid for the following dimensions:

$$0 < a/t \leq 0.5$$

$$0.2 \leq a/c \leq 2.0$$

$$(D+2c)/W < 0.5$$

Note: An additional width correction factor (F_{ww}) is applied based in the C dimension as explained in Section 3.2.3.1.1.20

Bearing Loading:

References [16, 19]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 80°

The method used to determine the bearing solution for all classic part-through cracked holes in plates is explained in Section 3.2.3.1.1.5.

This solution is valid for the following dimensions:

$$0 < a/t \leq 0.5$$

$$0.2 \leq a/c \leq 2.0$$

$$\frac{a}{b} < 0.6 \text{ (refer to Section 3.2.3.2.2 for clarification)}$$

3.2.3.1.1.11 Single Edge Corner Crack (Application Defined)



Axial Loading:

Reference [16]

Default angle (ϕ) used for the C Dimension: 5°

Default angle (ϕ) used for the A Dimension: 83°

This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

$$c/W < 0.5$$

3.2.3.1.1.12 Single Corner Crack in Lug (Application Defined)



Bearing Loading:

Reference [24]

This solution was developed directly from detailed StressCheck finite element models (FEMs). The tabular solutions are stored in AFGROW, and B-spline interpolation is used to obtain the solution for any given geometric case.

The pin bearing load was applied to the FEMs using either a spring or bearing (cosine² loading) type of pin load distribution. The spring loading case is intended to model a neat fit pin, and the bearing case more closely represents a clearance fit pin. The spring condition utilized a spring constant to plate modulus ratio of 3 to represent a steel pin in an aluminum plate²³. The major difference in the two boundary conditions is the fact that the bearing condition allows the hole to deform, and the spring condition constrains the hole. The default boundary condition in AFGROW is the bearing condition for crack lengths for which $a/t \leq 0.7$. The spring condition is used for $a/t \geq 0.8$, and linear superposition is used for intermediate crack lengths. This provided the best correlation to lug test data currently available. Users may change the default condition using the Predict, Preference (Lug Boundary Condition) option in the main menu (Section 3.4.1.6). More information about the boundary conditions is also located in that section.

Default angle (ϕ) used for the C Dimension: Variable²⁴

Default angle (ϕ) used for the A Dimension: Variable

The tabular solution was developed for the following dimensions:

$$a/t = 0.005, 0.01, 0.08, 0.3, 0.6, \text{ and } 0.9$$

$$a/c = 0.5, 1.0, 2.0, \text{ and } 4.0$$

$$D/t = 0.25, 0.5, 1.0, 2.0, \text{ and } 4.0$$

$$W/D = 1.3, 1.5, 1.75, 2.0, 2.1, 2.25, 2.5, 3.0, 3.75, \text{ and } 5.0$$

²³ Spring to plate modulus ratios as low as 1.0 resulted in nearly the same K-solutions.

²⁴ The extraction point for the K-solution varied with the specimen/crack geometry. The exact point for each dimension was taken as the local maximum K-value within 10 degrees of each free surface.

There is no extrapolation outside of these boundaries. The nearest value is used for any case outside of a given limit.

3.2.3.1.1.13 Part Through Crack in Pipe (Application Defined)



Axial Loading:

Reference [25]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 90°

This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

Bending Loading:

Reference [25]

Default angle (ϕ) used for the C Dimension: 0°

Default angle (ϕ) used for the A Dimension: 90°

This solution is valid for the following dimensions:

$$0 < a/t \leq 1.0$$

$$0.2 \leq a/c \leq 2.0$$

The solutions for both loading cases are empirical curve fits to FE solutions provided in the above reference.

3.2.3.1.1.14 Internal Axial Crack in Pipe (Application Defined)



This solution assumes an “infinite” pipe length, but AFGROW includes a width parameter that serves as a crack length limit in the c-direction.

Axial Loading:

Reference [84]

Default angle (ϕ) used for the C Dimension: 10°

Default angle (ϕ) used for the A Dimension: 90°

This solution is valid for the following dimensions:

$$0 < a/t \leq 0.8$$

$$0.2 < a/c \leq 1.0$$

$$(D_o - D_i)/D_i \leq 0.25$$

3.2.3.1.1.15 External Axial Crack in Pipe (Application Defined)



This solution assumes an “infinite” pipe length, but AFGROW includes a width parameter that serves as a crack length limit in the c-direction.

Axial Loading:

Reference [84]

Default angle (ϕ) used for the C Dimension: 10°

Default angle (ϕ) used for the A Dimension: 90°

This solution is valid for the following dimensions:

$$0 < a/t \leq 0.8$$

$$0.2 < a/c \leq 1.0$$

$$(D_o - D_i)/D_i \leq 0.25$$

3.2.3.1.16 Through Crack (User Defined)



This model is used when a user has an existing stress intensity factor solution (in the form of a beta table) for any 1-D crack, which may be described with one length dimension (1-D) to input in AFGROW.

The geometric beta values are NOT calculated by AFGROW but are merely interpolated from a one-dimensional user defined table of beta values. Users must supply beta values at various crack lengths so that the appropriate value at a given crack length may be interpolated. This model is shown as an edge cracked plate in the animation frame. The representation of the model is merely meant to indicate the one-dimensional nature of the crack. It was not possible to create representations of all possible geometries that may be modeled using user defined beta factors

For the [c] crack length dimension: $K = \sigma \sqrt{\pi c} \beta(c)$

Once this model is selected, AFGROW will add a user input beta icon, , in the AFGROW toolbar (if active). A notice will also be provided in the status view of the main frame window indicating that user-defined beta information is required. Users may choose any external source to calculate stress intensity factors and convert them to beta values. Details of the through crack (1-D) user-defined beta option are given in Section 3.2.7.1.



3.2.3.1.17 Interdependent Through Cracks (User Defined)

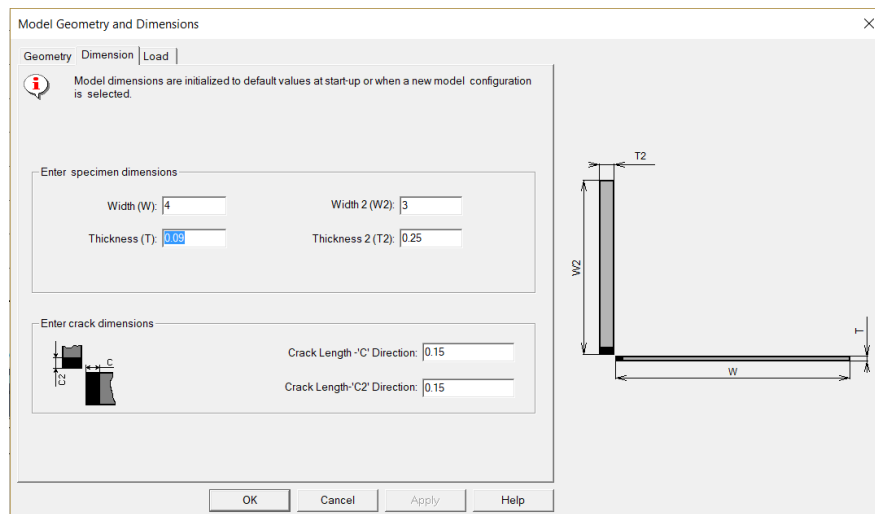


Figure 74: Interdependent Through Cracks

This model is used for cases with two crack dimensions that are interdependent (the K-solution for each dimension is dependent on the length of the both crack dimensions). This

model could be used to model a part-through crack with two growth dimensions or two through-the-thickness cracks growing in adjacent webs/flanges.

This model also allows each growth dimension to be assigned a thickness value (if required) to more accurately predict failure of either dimension (based on the local stress state), retardation effects (also may be dependent on stress state), and crack growth rate calculations (when the NASGRO Equation is used).

When this model is selected, user-defined beta information must be input using the Input, User-Defined Beta menu selection. When the dialog box opens for this model, it will appear as shown in Figure 75.

Dialog

Enter 'C1' beta sets

Number of sets: 4

		C1[1]	C1[2]	C1[3]	C1[4]
		0	0	0	0
C2[1]	0	0	0	0	0
C2[2]	0	0	0	0	0

Enter 'C2' beta sets

Number of sets: 2

		C1[1]	C1[2]	C1[3]	C1[4]
		0	0	0	0
C2[1]	0	0	0	0	0
C2[2]	0	0	0	0	0

OK Apply Read Save Cancel

Figure 75: User-Defined Beta Tables for Two, Interdependent Cracks

For the [C1] crack length dimension: $K_{(C1)} = \sigma \sqrt{\pi C_1} \beta_{(C1)}$

For the [C2] crack length dimension: $K_{(C2)} = \sigma \sqrt{\pi C_2} \beta_{(C2)}$

The notable difference between this set of user-defined beta tables and the tables used for a user-defined part through-the-thickness crack is that the matrices do not have to be square. Users may choose the number of crack lengths in each direction to be used in the tables for the solution in each direction. The solution ultimately used in AFGROW will be linearly interpolated from these tables. However, please remember that AFGROW will not extrapolated outside of the crack lengths included in the tables. If a crack length falls outside of the table, the nearest value in the table will be used.

3.2.3.1.1.18 Internal Through Crack (Application Defined)



Axial Loading:

$$\text{Beta} = \left[1.0 - 0.025 \left(\frac{2C}{W} \right)^2 + 0.06 \left(\frac{2C}{W} \right)^4 \right] \sqrt{\sec(\pi C / W)}$$

Reference [26]

This solution is valid for the following dimensions:

$$0 < C/W \leq 0.5$$

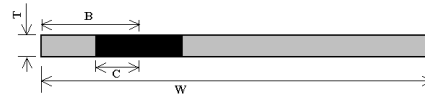
This solution is within 0.1% for all crack lengths

Bending Loading:

$$\text{Beta} = \text{Beta}_{(\text{axial})} * \text{Bending Correction}$$

It is important to be very clear that there is no way to provide a true solution for the out-of-plane (transverse) bending case since the actual stress intensity value will vary through the thickness. The bending correction value is simply being used to provide a solution for the straight through crack that provides reasonable continuity with the bending solution for the c-dimension of the part-through (surface) crack. The default value for the bending correction is 2/3, but this may be changed to any desired value between 1/3 and 2/3 in the Predict, Preferences, Bending dialog box (see Section 3.4.1.8 for more details). This is required to allow users to model bending for the surface crack case since an initial surface crack may transition to become a through crack. The most accurate way to model this case is to use an oblique through crack solution, which accounts for the changes in the stress intensity solution through the thickness. Unfortunately, no oblique internal crack solutions were found that are valid for the full range of crack shapes required.

Offset Correction:



The stress intensity solution for the offset internal through crack must be calculated at each crack tip. The offset case is non-symmetric, and the stress intensity values of each crack tip will be different. The offset parameter, B, is defined as the distance between the nearest plate edge and the center of the through crack. AFGROW measures this distance from the left edge of the plate and B must be less than one half of the plate width. Any offset case may be modeled in this manner. An offset crack on the right side of the plate will be on the left side if the plate is rotated 180 degrees.

The solution for the crack tip closest to the edge of the plate is:

Reference [28, 29]

$$\text{Beta} = \left(1 - 0.025\lambda^2 + 0.6\lambda^4 - \gamma\lambda^{11}\right) \sqrt{\sec\left(\frac{\pi\lambda}{2}\right) \frac{\sin\left(2\lambda - \frac{4C}{W}\right)}{2\lambda - \frac{4C}{W}}}$$

Where: $\lambda = \frac{C}{B}$

C = current half crack length

B = current distance from the near plate edge to the crack center

W= plate width

γ = function of B/W

B/W	Gamma (γ)
0.1	0.382
0.25	0.136
0.4	0.0
0.5	0.0

Values of γ for any B/W (note: by definition, B/W $\leq$ 0.5) are obtained by linear interpolation (extrapolation for cases where B/W < 0.1). The γ term in the polynomial was added to the solutions from [28, 29] to allow for a better fit at high λ values ($\lambda > 0.6$). This fit is shown in Figure 76 and was determined using the finite element code, StressCheck [31].

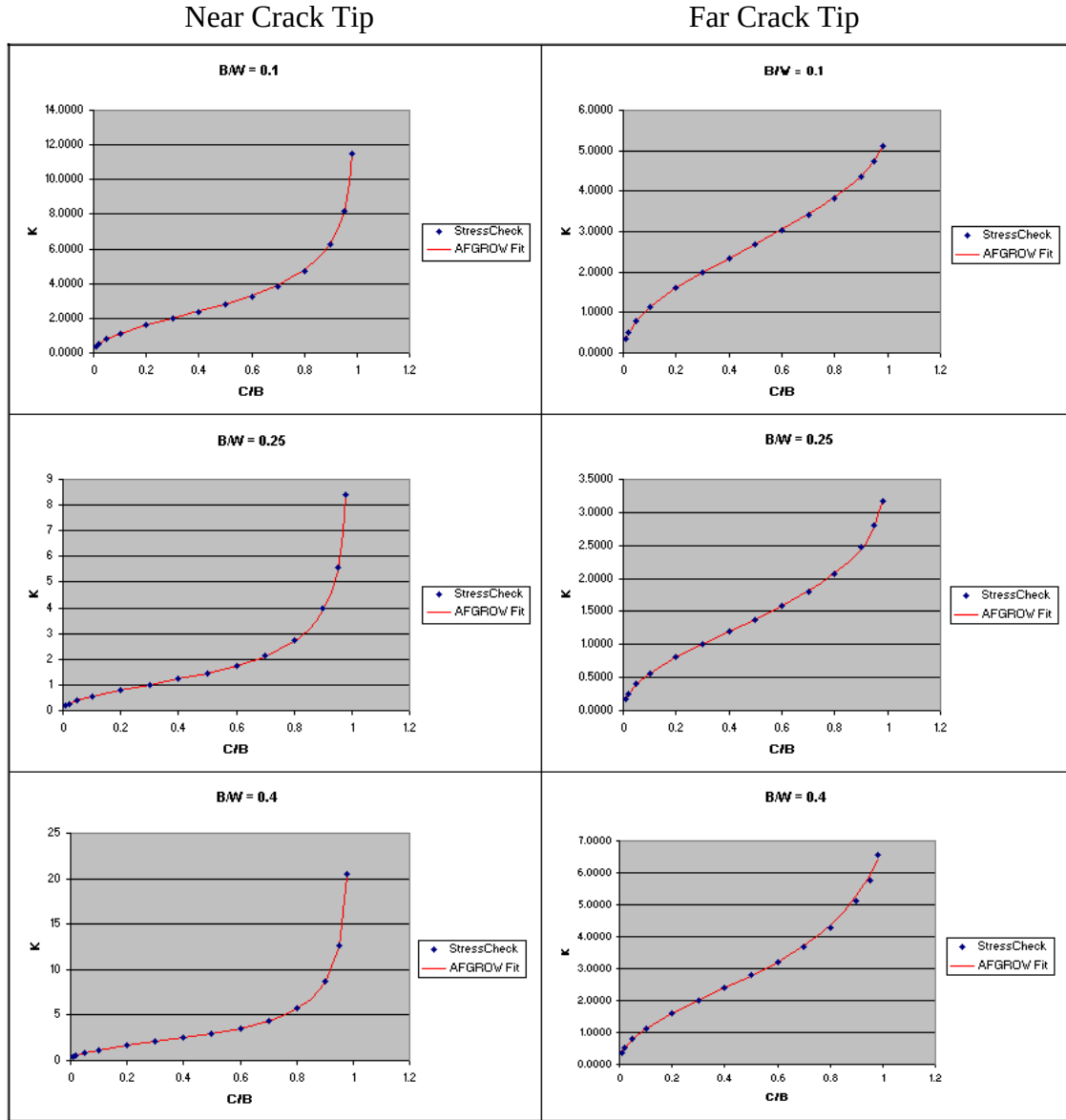


Figure 76: Offset Crack Solutions

The solution for the crack tip furthest from the plate edge is:

References [28, 29]

$$\text{Beta} = \left(1 - 0.025\delta^2 + 0.06\delta^4 + \gamma\lambda^{30}\right) \left(1 + \frac{\sqrt{\sec\left(\frac{1}{7}(2\pi\lambda + 1.5\pi\delta)\right)} - 1}{\left[1 + 0.21\sin\left(8\tan^{-1}\left(\left(\frac{\lambda - \delta}{\lambda + \delta}\right)^{0.9}\right)\right)\right]}\right)$$

Where: $\lambda = \frac{C}{B}$; $\delta = \frac{C}{W - B}$

C = current half crack length

B = current distance from the near plate edge to the crack center

W= plate width

γ = function of B/W

B/W	Gamma (γ)
0.1	0.114
0.25	0.286
0.4	0.0
0.5	0.0

Values of γ for any B/W (note: by definition, B/W $\leq$ 0.5) are obtained by linear interpolation (extrapolation for cases where B/W < 0.1). The γ term in the polynomial was added to the solutions from [28, 29] to allow for a better fit at high λ values ($\lambda > 0.8$). This fit is shown in Figure 76 and was determined using the p-version finite element code, StressCheck [31].

3.2.3.1.19 Single Through Crack at Hole (Application Defined)



AFGROW now allows for either straight or oblique through cracks to be analyzed for this geometry. As the name implies, straight through-the-thickness cracks are assumed to be one-dimensional cracks of constant length (C) through the thickness of a component (see Figure 77).

Enter crack dimensions

Crack Length -C Direction: 0.15

☐ Oblique through crack

Figure 77: Straight Through-the-Thickness Cracks

Oblique cracks are assumed to be elliptic in shape and are NOT of constant length through the thickness (see Figure 78).

Enter crack dimensions

Crack Length -C Direction: 0.15

Crack Length -Ct Direction: 0.05

☒ Oblique through crack

Figure 78: Oblique Through-the-Thickness Cracks

Axial Loading:

Infinite Plate Solution:

$$\text{Beta} = 0.7071 + 0.7548 \left(\frac{R}{R+C} \right) + 0.3415 \left(\frac{R}{R+C} \right)^2 + 0.642 \left(\frac{R}{R+C} \right)^3 + 0.9196 \left(\frac{R}{R+C} \right)^4$$

Reference [30]

Finite Width Correction:

$$F_w = \sqrt{\sec\left(\frac{\pi R}{W}\right) \sec\left(\frac{\pi (R + C/2)}{W - C}\right)}$$

The finite width correction was taken from reference [16], Equation 62 with a/t set to 1.0. Please refer to reference [16] for the finite width equation used for part-through cracks.

The above finite width correction has been shown to be from 0 to ~30% high for relatively narrow plates ($W/D < 6$) using STRESSCHECK [31] (P-Version FEM program). An additional correction has been added to AFGROW for all cases of single cracks at holes (part-through as well as through cracks).

$$F_{ww} = 1 - \left(\left(2.65 - 0.24 \left(2.75 - \frac{W}{D} \right)^2 \right)^{-\left(\frac{W}{D}\right)} \right) \left(\frac{2C}{W - D} \right)^{\frac{W}{D} + 0.5}$$

Note: $\left(2.65 - 0.24 \left(2.75 - \frac{W}{D} \right)^2 \right)$ must not be < 2.275 - if so, set it equal to 2.275

This correction, shown in Figure 79, below compares the ratio of the STRESSCHECK to the non-corrected AFGROW results to the additional AFGROW width correction (F_{ww}).

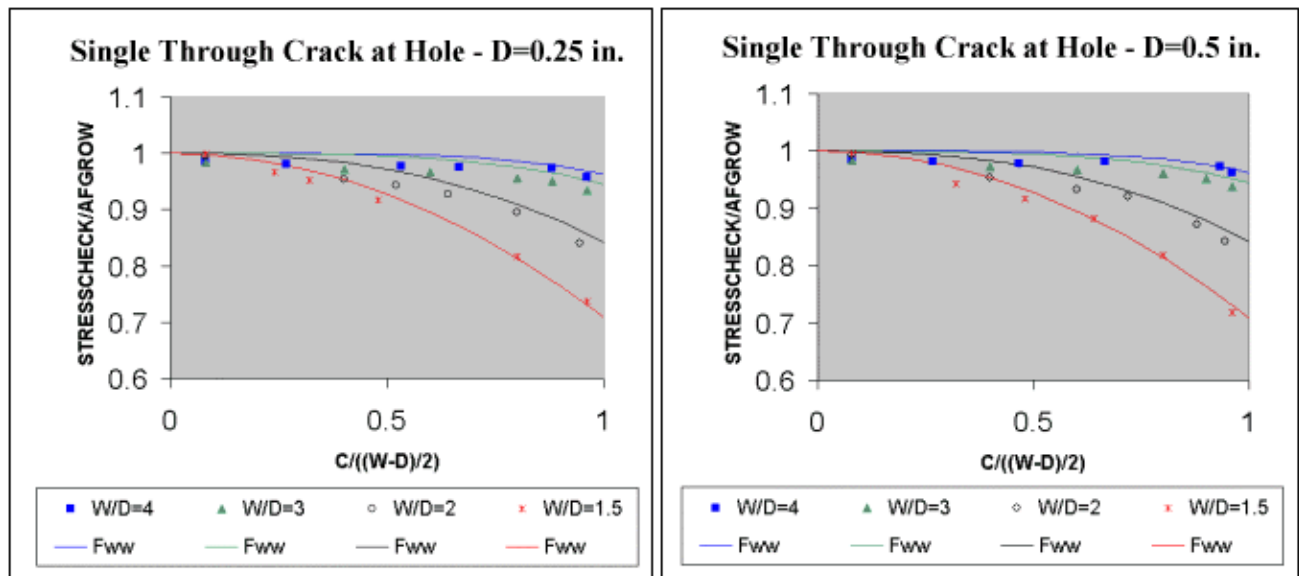


Figure 79: Finite Width Adjustments for a Single Cracked Hole

This additional width correction is simply an additional multiplication factor and greatly improves the accuracy of AFGROW when the hole is relatively close to a free edge ($W/D < 6$).

This solution is valid for the following dimensions:

$$0 < C/R \leq \text{infinite}$$

$$\frac{R + C/2}{W - C} < 0.5$$

Bending Loading:

Infinite Plate Solution:

Reference [32]

Note: Equation 9 (single crack solution) was used from this reference - The actual beta value was obtained by dividing by $\sqrt{\pi}$ since the reference left that value out of the calculation of stress intensity.

$$\text{Beta} = \sqrt{\frac{1}{2\pi}} \left(\frac{C/R + 2}{C/R + 1} \right)^{1.5} F_c F_w F_{ww} \quad (\text{Refer to Axial Loading Section Above for } F_{ww})$$

The factor, F_c , was added to correct equation 9 (which is a shear stress solution) to match the bending data provided in the above reference for a Poisson's ratio of 1/3. The error for any C/R was determined to be less than 1 percent for any C/R (for most values the error was MUCH less than 1 percent) according to the above reference. The difference between the data at Poisson's ratios of 1/3 and 1/4 is very small - other solutions use a correction for Poisson's ratio that is in great disagreement with this reference.

$$F_c = 0.9 + 0.083 \left(1 - 10^{-0.046(C/R)} \right) + 0.017 \left(1 - 10^{-3.0(C/R)} \right)$$

Finite Width Correction:

$$F_w = \sqrt{\sec\left(\frac{\pi R}{W}\right)\sec\left(\frac{\pi (R + C/2)}{W - C}\right)}$$

The finite width correction was taken from reference 16, Equation 46 with a/t set to 1.0.

This solution is valid for the following dimensions:

$$0 < C/R \leq \text{infinite}$$

$$\frac{R + C/2}{W - C} < 0.5$$

Bearing Loading:

Beta = (Table Lookup w/Linear Interpolation)

The stress intensity solution for the bearing load case was determined using StressCheck [31] FE models for a range of plate width to hole diameter (W/D) ratios. Crack lengths for each W/D value were normalized²⁵ with the maximum possible crack length for each case as indicated below. When normalized in this way, the resulting stress intensity factors retain a high resolution for all possible crack lengths for a given plate width and hole diameter

Norm C*	w/d												
	1.3	1.5	2	2.5	3	4	5	6	8	16	40	100	1000
0.000	4.5250	3.2850	2.3750	1.9450	1.7400	1.5260	1.3920	1.3150	1.2360	1.0980	1.0252	0.9968	0.9787
0.025	4.4275	3.1865	2.1870	1.8033	1.5959	1.3743	1.2539	1.1755	1.0847	0.9600	0.8814	0.8545	0.8391
0.050	4.3645	3.1134	2.0858	1.6927	1.4824	1.2616	1.1476	1.0779	0.9960	0.8844	0.8170	0.7916	0.7773
0.100	4.2725	3.0077	1.9350	1.5323	1.3201	1.1023	0.9914	0.9236	0.8451	0.7388	0.6802	0.6574	0.6455
0.150	4.2150	2.9200	1.8100	1.4000	1.1900	0.9925	0.8772	0.8050	0.7385	0.6375	0.5795	0.5571	0.5471
0.200	4.2003	2.8450	1.7215	1.3050	1.0950	0.8908	0.7854	0.7150	0.6490	0.5513	0.4987	0.4784	0.4698
0.250	4.2037	2.7850	1.6356	1.2150	1.0125	0.8125	0.7079	0.6375	0.5780	0.4825	0.4350	0.4132	0.4058
0.300	4.2198	2.7460	1.5643	1.1505	0.9447	0.7414	0.6412	0.5750	0.5136	0.4239	0.3760	0.3578	0.3514
0.350	4.2455	2.7228	1.4990	1.0854	0.8795	0.6850	0.5824	0.5253	0.4600	0.3725	0.3280	0.3095	0.3040
0.400	4.2876	2.7054	1.4435	1.0333	0.8235	0.6258	0.5301	0.4744	0.4101	0.3271	0.2835	0.2670	0.2622
0.450	4.3526	2.6950	1.3959	0.9860	0.7720	0.5729	0.4839	0.4287	0.3675	0.2850	0.2460	0.2292	0.2251
0.500	4.4414	2.6935	1.3562	0.9430	0.7271	0.5332	0.4414	0.3875	0.3273	0.2503	0.2102	0.1954	0.1919
0.550	4.5460	2.7046	1.3248	0.9037	0.6867	0.4950	0.4033	0.3504	0.2928	0.2175	0.1795	0.1649	0.1620
0.600	4.6575	2.7330	1.3025	0.8691	0.6523	0.4587	0.3687	0.3168	0.2597	0.1876	0.1507	0.1372	0.1347
0.650	4.7670	2.7829	1.2912	0.8417	0.6227	0.4275	0.3379	0.2865	0.2300	0.1585	0.1265	0.1130	0.1109
0.700	4.8895	2.8500	1.2936	0.8257	0.6001	0.4007	0.3104	0.2593	0.2042	0.1350	0.1020	0.0895	0.0879
0.750	5.1337	2.9620	1.3151	0.8200	0.5844	0.3790	0.2868	0.2354	0.1800	0.1120	0.0824	0.0692	0.0679
0.800	5.4893	3.1043	1.3652	0.8259	0.5803	0.3632	0.2676	0.2151	0.1596	0.0940	0.0625	0.0511	0.0502
0.825	5.6639	3.2000	1.4069	0.8399	0.5837	0.3582	0.2603	0.2066	0.1500	0.0840	0.0535	0.0429	0.0421
0.850	6.0283	3.3351	1.4654	0.8555	0.5933	0.3573	0.2550	0.1995	0.1421	0.0755	0.0460	0.0353	0.0346
0.875	6.8300	3.5963	1.5459	0.8800	0.6113	0.3600	0.2524	0.1943	0.1350	0.0677	0.0385	0.0280	0.0275
0.900	8.1821	4.0000	1.6728	0.9250	0.6430	0.3701	0.2540	0.1924	0.1299	0.0613	0.0318	0.0218	0.0210
0.925	9.9877	4.7500	1.8506	1.0500	0.6979	0.3922	0.2633	0.1954	0.1276	0.0555	0.0254	0.0175	0.0150
0.950	12.2500	5.9209	2.2500	1.2500	0.8026	0.4395	0.2881	0.2092	0.1314	0.0513	0.0205	0.0135	0.0097
0.975	15.0000	7.5000	3.1250	1.6500	1.0919	0.5659	0.3616	0.2400	0.1538	0.0521	0.0167	0.0115	0.0050
1.000	20.0000	10.0000	4.5000	2.5000	1.6000	0.8500	0.5000	0.3500	0.2500	0.1000	0.0400	0.0200	0.0000

²⁵ Values were normalized as: $\frac{C/(C+R)}{(1-D/W)}$

This normalization also allows for extrapolation beyond $W/D=1000$ since the solution converges as the width increases.

It was discovered that the standard, open hole finite width correction [16] did not provide correct stress intensity solutions for finite width plates when applied to the “infinite” plate solution for the bearing load case. The table lookup approach is an efficient method for calculating the solution, since the numbers of variables for the through crack geometry are manageable.

The current solution is based on FEM results for the pin loaded case using a spring boundary condition (BC) to model the pin loading. This does a couple of things. First, it attempts to model the actual pin load distribution around the bottom of the hole (180 degrees). Second, it provides constraint to the hole which prevents the hole from deforming as it would if any clearance exists between the pin and the hole. The resulting betas are lower than cases modeled with a typical pin pressure distribution (cosine or cosine²) without hole constraint. In the future we plan to include the option to model either BC, but are waiting until we have the corresponding solutions for the corner cracked bearing case (since the corner crack transitions to become a through crack).

The spring BC is well suited for cases in which the pin/hole fit is neat (tight). This is probably true in most practical applications (i.e. rivets).

This solution has been verified for the following dimensions:

$$1.3 \leq W/D \leq \text{infinity} \text{ (solution converges as } W/D \rightarrow \text{infinity)}$$

$$0 < C/(C + R) < 1 \text{ (tabular values for } C=0, \text{ and the maximum crack length were extrapolated)}$$

Oblique Through-the-Thickness Cracks

Dr. Scott Fawaz developed the finite element based oblique crack solutions for axial, bending, and bearing loading conditions [33, 34]. See Figure 78 for a description of the input requirements for the oblique crack.

The crack geometry is defined in Figure 80:

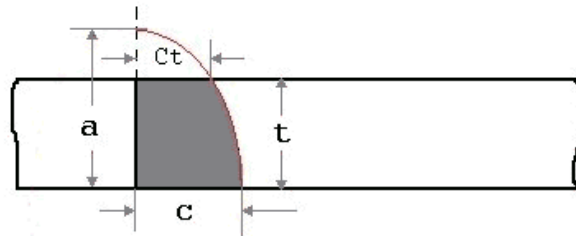


Figure 80: Oblique Through-the-Thickness Crack Geometry

The virtual corner crack is a quarter ellipse with the center at what would be the crack origin of a corner crack that has transitioned to an oblique through-the-thickness flaw at a hole in an infinite plate. The elliptical axes are defined by the a and c dimensions. While the a dimension is not input by the user, it is calculated from the [c, c_t, and t] dimensions which are input by the user. Dr. Fawaz's finite element solutions were calculated for the following range of dimensions:

a/c = 0.2, 0.3, 0.4, 0.6, 1.0, 2.0, 5.0, and 10.0

a/t = 1.05, 1.09, 1.13, 1.17, 1.21, 2.0, 5.0, and 10.0

R/t = 0.1, 0.25, 0.5, 1.0, and 2.0 - where R is the hole radius

Beta factors for each case were then calculated for the [c] and [c_t] dimensions as follows:

$$\text{Beta} = \frac{K}{\sigma \sqrt{\pi x}} ; \text{Where } x \text{ is the appropriate crack length}$$

In this case, crack length is the c or c_t dimension. AFGROW uses a cubic spline interpolation technique to determine the appropriate beta value during crack growth life prediction. The following rules are used in AFGROW when the oblique through crack option is selected:

- No extrapolation is made beyond the bounds of the finite element cases
- If a/c, or a/t goes below the limit of the finite element cases, the value will be held at that limit
- If R/t is beyond the limits, it will be maintained at the nearest limit value
- If a/c, or a/t goes above the limits, the crack will be transitioned to a straight through-the-thickness crack of length [C]

Finite Width Correction:

$$F_w = \sqrt{\sec\left(\frac{\pi R}{W}\right)\sec\left(\frac{\pi (R + C/2)}{W - C}\right)}$$

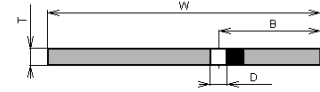
The finite width correction was taken from Reference 16, Equation 46 with a/t set to 1.0.

This solution is valid for the following dimensions:

$$\frac{R + C/2}{W - C} < 0.5$$

The Fawaz solutions were calculated for the double cracked hole case and were corrected for the single crack case by the Shah correction as follows:

$$\text{Beta (single crack)} = \text{Beta (double crack)} * \sqrt{\frac{8R + \pi C}{8R + 2\pi C}}$$



Offset Correction:

For Axial & Bending Load Cases:

The solution for an offset (non-centered hole) uses the centered hole solution in AFGROW with the width adjusted to be equal to twice the distance from the center of the hole to the right edge (2B). AFGROW now includes an offset correction for a crack growing to the near edge ($B < W/2$) and an offset correction for a crack growing to the far edge ($B > W/2$). The offset corrections are given below:

For $B < W/2$:

$$F_{offset} = \frac{\sin\left(\left(\frac{D + c}{B - c/2}\right)\left(\frac{W - 2B}{W}\right)\right)}{\left(\frac{D + c}{B - c/2}\right)\left(\frac{W - 2B}{W}\right)} F_C$$

Reference [20]

Where:

$$F_C = \left(1 - (0.45F_G - 0.021)\left(\frac{c}{B - D/2}\right)^{16}\right)$$

$$F_G = \left(\frac{2B}{W} + \frac{D}{2B}\right)$$

Note: The following limitations apply to F_C and F_G

If $F_G < 0.0468$, $F_C = 1.0$

If $F_G > 0.7$, $F_G = 0.7$

This solution is valid for the following dimensions:

$$\left(\frac{D+c}{2B-c} \right) \leq 0.7$$

The solution tends to be conservative (1 to 3%) when the limit is exceeded.

For $B > W/2$:

This correction is more complex than the previous case since the stress intensity factor may be affected by the proximity of the hole to the edge of the plate as well as the fact that the crack is growing to the far edge of the plate. The offset correction is given below:

$$F_{offset} = F_{AHFB} F_{B/W}$$

$$F_{AHFB} = 1 + \frac{\sqrt{\sec\left(\frac{\pi D}{14} \left(\frac{2}{W-B} + \frac{1.5}{B}\right)\right)} - 1}{\left[1 + 0.21 \sin\left(8 \tan^{-1}\left(\left(\frac{2B-W}{W}\right)^{0.9}\right)\right)\right]}$$

Reference [21]

Note: The above equation has been modified to reflect the definition of the parameter, B, used by AFGROW for this geometry.

The factor (F_{AHFB}) accounts for the effect of the proximity of the hole to the edge of the plate.

$$F_{B/W} = 1 + \left(F_{\max} \sin \left(\pi \tanh \left(2\delta^{1.1} + (1.18\delta)^7 \right) \right) \right) \quad \text{Reference [22]}$$

Where:

$$\delta = \frac{D + c}{2B - c}$$

$$F_{\max} = 0.5e^{-(10\gamma + 4.2\gamma^2 + (3\gamma)^{14})}$$

$$\gamma = 1 - \frac{B}{W}$$

The factor ($F_{B/W}$) adjusts the offset correction as a function of the ratio of the offset to the plate width. This empirical curve fit was made using finite element results for a single through cracked hole. A sample beta solution is shown in Figure 73.

For the Bearing Load Case:

The offset correction shown above for the axial and bending load cases has been shown to be incorrect for the bearing load case. A revised offset correction for the bearing load case was developed based on StressCheck FEMs for centered and offset holes.

The correction for the bearing load case was determined using FEMs for the offset and centered (baseline) hole models. A curve fit solution was developed from the results for offset holes in plates that were constrained to prevent in-plane bending due to loading asymmetry. The results for un-constrained, offset holes were not well suited for a curve fit solution, and interpolation would have required a very large number look-up table. Since most practical structural applications tend to be constrained by adjacent structure, the un-constrained solution was not included.

In the previous version of AFGROW, the centered hole solution used as the baseline for the offset correction was not constrained to prevent in-plane bending caused by asymmetry resulting from the single crack geometry. After discussions with users and comparisons with more FEMs, the centered hole baseline solution was changed to the constrained boundary condition to be more consistent with the offset correction. This lowers the K-solution, but since the offset correction itself was developed using the un-constrained centered hole solutions, it will still remain somewhat conservative for cases with a crack growing to the near edge of a plate.

3.2.3.1.1.20 Double Through Crack at a Hole (Application Defined)



AFGROW now allows for either straight or oblique through cracks to be analyzed for this geometry. As the name implies, straight through-the-thickness cracks are assumed to be one-dimensional cracks of constant length (C) through the thickness of a component (see Figure 77, Section 3.2.3.1.16). Oblique cracks are assumed to be elliptical in shape and are NOT of constant length through the thickness (see Figure 78, Section 3.2.3.1.1.16).

Axial Loading:

Infinite Plate Solution:

$$\text{Beta} = \text{Beta}_{\text{InfinitePlate}} F_w F_{ww}$$

The infinite plate solution in AFGROW has been obtained from detailed finite element analysis using STRESSCHECK [31] for a wide plate (40 inches wide - see Figure 81) with the standard finite with correction extracted from these values.

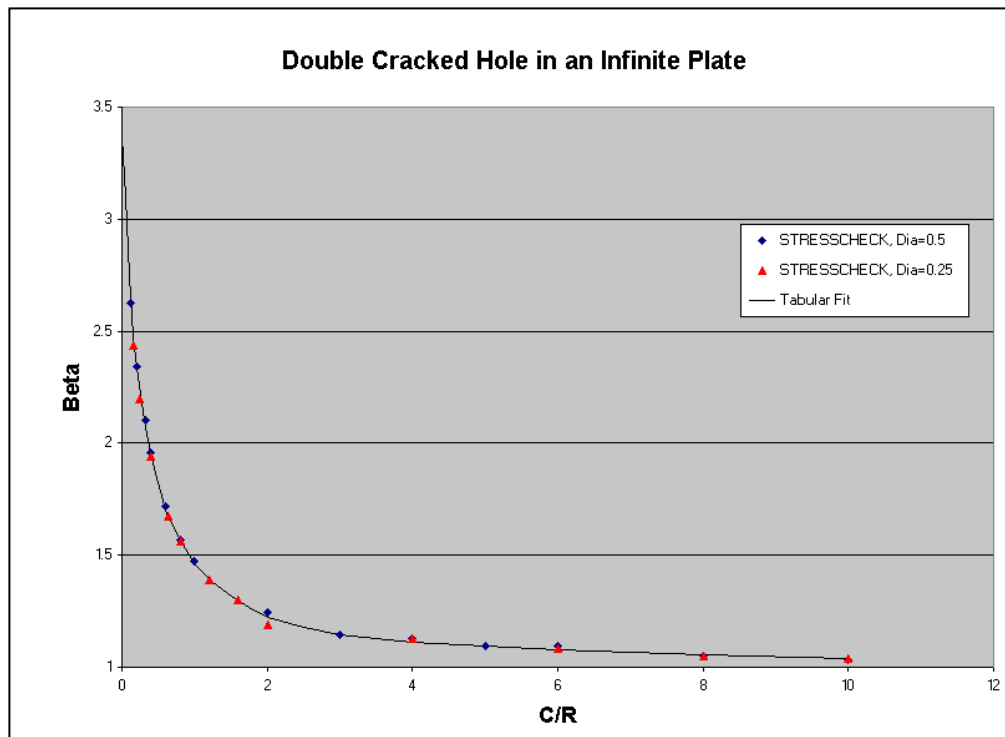


Figure 81: Beta Values for a Double Through Crack at Hole (Infinite Plate)

The boundary conditions were known for $C/R=0$ and $C/R>10$.

For $C/R = 0.0$, $\text{Beta} = 3.365$

This is the result of a combination of the beta value ($C=0$) for an edge-cracked plate (~ 1.122) and the stress concentration at a hole in an infinite plate (3.0). This value also appears to result in a smooth curve as shown in Figure 81.

For $C/R > 10$, $Beta = \sqrt{1 + R/C}$

When the crack is far from the hole, the hole has no influence on the crack. The solution then converges to the solution for an internal through crack in an infinite plate (1.0). The only difference is the definition of the crack length. The crack at a hole is measured from the edge of the hole, and the center crack length is measured from the center of the crack. This is determined as follows:

$$\sqrt{\pi C} \text{ Beta} = \sqrt{\pi (C + R)} \quad \leftarrow \text{Since, the internal crack length is } (C+R)$$

$$Beta = \frac{\sqrt{\pi (C + R)}}{\sqrt{\pi C}} = \sqrt{\frac{C + R}{C}} = \sqrt{1 + R/C}$$

The actual values of the tabular beta solution being used in AFGROW are as follows:

C/R	Beta
0	3.365
0.05	3.056
0.1	2.807
0.15	2.595
0.2	2.425
0.3	2.158
0.4	1.967
0.5	1.824
0.625	1.686
0.75	1.590
1	1.450
1.25	1.360
1.5	1.300
1.75	1.250
2	1.225
2.5	1.180
3	1.150
3.5	1.131
4	1.115
5	1.095
6	1.080
8	1.060
10	1.049
100	1.005
1000	1.0005

Reference [35]

The beta values used in AFGROW for this geometry are determined from the table above using a spline interpolation method.

Finite Width Correction:

$$F_w = \sqrt{\sec\left(\frac{\pi R}{W}\right)\sec\left(\frac{\pi (R + C)}{W}\right)}$$

The finite width correction was taken from reference [16], Equation 62 with a/t set to 1.0. Please refer to reference [16] for the finite width equation used for part-through cracks.

The above finite width correction has been shown to be from ~2% low to ~30% high using the STRESSCHECK [31] (P-Version FEM program). An additional correction has been added to AFGROW for all cases of double cracks at holes (part-through as well as through cracks).

$$F_{ww} = 1 - \left(\left(1.32 \frac{W}{D} - 0.14 \right)^{- \left(0.98 + \left(0.1 \frac{W}{D} \right)^{0.1} \right)} - 0.02 \right) \left(\frac{2C}{W-D} \right)^{\frac{W}{D} + 2.5}$$

Note: If the final exponent value $\left(\frac{W}{D} + 2.5 \right)$ is greater than 4.5, then use 4.5 instead

This correction, shown in Figure 82 below, compares the ratio of the STRESSCHECK to the non-corrected AFGROW results to the additional AFGROW width correction (F_{ww}).

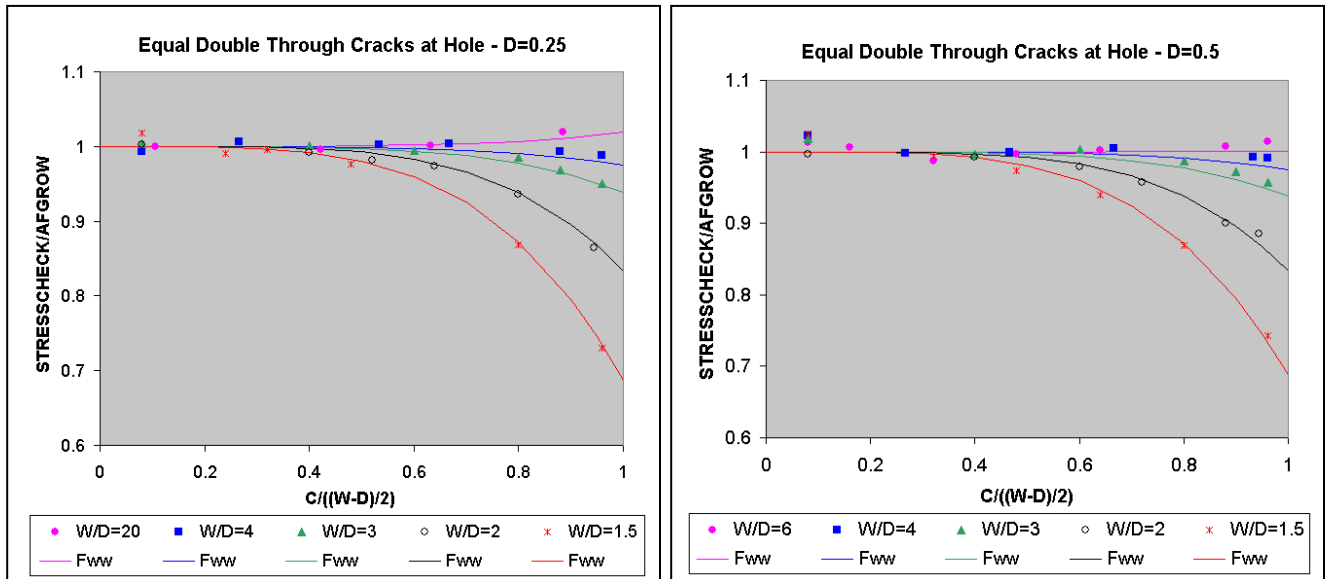


Figure 82: Finite Width Adjustment for a Double Cracked Hole

This additional width correction is simply an additional multiplication factor and greatly improves the accuracy of AFGROW for all plate widths.

This solution is valid for the following dimensions:

$$0 < C/R \leq \text{infinite}$$

$$\frac{R+C}{W} < 0.5$$

Bending Loading:

The double cracked hole solutions are corrected from the single-crack solutions using the Shah Correction (with a/t set = 1.0)

$$\text{Beta (double crack)} = \text{Beta (single crack)} * \sqrt{\frac{8R + 2\pi C}{8R + \pi C}}$$

Infinite Plate Solution:

Reference [32]

Note: Equation 9 (single crack solution) was used from this reference - The actual beta value was obtained by dividing by $\sqrt{\pi}$ since the reference left that value out of the calculation of stress intensity.

$$\text{Beta} = \sqrt{\frac{1}{2\pi} \left(\frac{C/R + 2}{C/R + 1} \right)^{1.5}} \sqrt{\frac{8R + 2\pi C}{8R + \pi C}} F_c F_w F_{ww} \quad (\text{Refer to Axial Loading Section Above for } F_{ww})$$

The factor, F_c , was added to correct equation 9 (which is a shear stress solution) to match the bending data provided in the above reference for a Poisson's ratio of 1/3. The error for any C/R was determined to be less than 1 percent for any C/R (for most values the error was MUCH less than 1 percent) according to the above reference. The difference between the data at Poisson's ratios of 1/3 and 1/4 are very small - other solutions use a correction for Poisson's ratio that is in great disagreement with this reference.

$$F_c = 0.9 + 0.083 \left(1 - 10^{-0.046(C/R)} \right) + 0.017 \left(1 - 10^{-3.0(C/R)} \right)$$

Finite Width Correction:

$$F_w = \sqrt{\sec\left(\frac{\pi R}{W}\right) \sec\left(\frac{\pi(R+C)}{W}\right)}$$

The finite width correction was taken from Reference 16, Equation 46 with a/t set to 1.0. This solution is valid for the following dimensions:

$$0 < C/R \leq \text{infinite}$$

$$\frac{R+C}{W} < 0.5$$

Bearing Loading:

Beta = (Table Lookup w/Linear Interpolation)

The stress intensity solution for the bearing load case was determined using StressCheck [31] FE models for a range of plate width to hole diameter (W/D) ratios. Crack lengths for each W/D value were normalized²⁶ with the maximum possible crack length for each case as indicated below. When normalized in this way, the resulting stress intensity factors retain a high resolution for all possible crack lengths for a given plate width and hole diameter. This normalization also allows for extrapolation beyond W/D=1000 since the solution converges as the width increases.

C/(C+R) ^a	W/D										
	1.3	1.5	2	2.5	3	4	8	16	40	100	1000
0.000	4.5150	3.2000	2.3000	1.9050	1.7000	1.5260	1.1970	1.0363	0.8898	0.8787	0.8700
0.025	4.4437	3.1195	2.1530	1.7367	1.5720	1.3912	1.1208	0.9789	0.8450	0.8304	0.8217
0.050	4.3769	3.0453	2.0281	1.6065	1.4558	1.2845	1.0535	0.9223	0.7930	0.7769	0.7682
0.100	4.2756	2.9400	1.8852	1.4682	1.2853	1.1307	0.9368	0.8145	0.6868	0.6691	0.6605
0.150	4.2551	2.8975	1.8186	1.3772	1.1791	1.0135	0.8371	0.7190	0.5944	0.5778	0.5698
0.200	4.2763	2.8671	1.7553	1.3017	1.1042	0.9204	0.7447	0.6349	0.5214	0.5048	0.4972
0.250	4.3115	2.8456	1.6775	1.2369	1.0418	0.8451	0.6589	0.5525	0.4571	0.4402	0.4331
0.300	4.3728	2.8456	1.6179	1.1844	0.9808	0.7805	0.5840	0.4861	0.4049	0.3884	0.3817
0.350	4.4456	2.8510	1.5697	1.1366	0.9209	0.7184	0.5194	0.4326	0.3576	0.3388	0.3323
0.400	4.5371	2.8628	1.5275	1.0934	0.8701	0.6614	0.4635	0.3803	0.3130	0.2954	0.2892
0.450	4.6802	2.8978	1.5010	1.0602	0.8301	0.6160	0.4132	0.3334	0.2740	0.2549	0.2488
0.500	4.8434	2.9372	1.4790	1.0284	0.7924	0.5745	0.3662	0.2915	0.2376	0.2163	0.2108
0.550	5.0511	3.0051	1.4720	1.0041	0.7630	0.5426	0.3253	0.2516	0.2054	0.1865	0.1815
0.600	5.2746	3.0951	1.4751	0.9842	0.7383	0.5164	0.2891	0.2135	0.1743	0.1559	0.1510
0.650	5.5110	3.2172	1.4927	0.9731	0.7199	0.4886	0.2569	0.1818	0.1473	0.1315	0.1275
0.700	5.8002	3.3904	1.5346	0.9794	0.7114	0.4723	0.2317	0.1566	0.1268	0.1122	0.1083
0.750	6.2606	3.6122	1.6037	1.0000	0.7127	0.4632	0.2135	0.1362	0.1065	0.0920	0.0881
0.800	6.9134	3.9097	1.7193	1.0402	0.7305	0.4572	0.1940	0.1159	0.0875	0.0734	0.0696
0.825	7.2800	4.1131	1.8083	1.0795	0.7503	0.4604	0.1857	0.1070	0.0792	0.0643	0.0604
0.850	7.9319	4.3883	1.9282	1.1369	0.7807	0.4719	0.1783	0.0998	0.0710	0.0552	0.0516
0.875	9.2298	4.8599	2.0891	1.1892	0.8261	0.4954	0.1715	0.0946	0.0621	0.0470	0.0441
0.900	11.3640	5.6764	2.3233	1.2536	0.8926	0.5291	0.1673	0.0888	0.0526	0.0395	0.0364
0.925	14.2696	6.9703	2.6438	1.6130	0.9970	0.5672	0.1701	0.0793	0.0452	0.0325	0.0274
0.950	17.8475	8.7072	3.2451	2.4575	1.1718	0.6471	0.1684	0.0692	0.0375	0.0261	0.0180
0.975	21.9865	10.8013	4.5117	3.6894	1.6544	0.9141	0.2231	0.1112	0.0356	0.0225	0.0094
1.000	26.5620	13.1563	6.0938	5.1719	2.5000	1.3297	0.4563	0.2156	0.1047	0.0484	0.0000

It was discovered that the standard, open hole finite width correction [16] did not provide correct stress intensity solutions for finite width plates when applied to the “infinite” plate solution for the bearing load case. The table lookup approach is an efficient method for calculating the solution, since the numbers of variables for the through crack geometry are manageable.

The current solution is based on FEM results for the pin loaded case using a spring boundary condition (BC) to model the pin loading. This does a couple of things. First, it attempts to model the actual pin load distribution around the bottom of the hole (180 degrees). Second, it provides constraint to the hole which prevents the hole from deforming as it would if any clearance exists between the pin and the hole. The resulting betas are lower than cases modeled with a typical pin pressure distribution (cosine or cosine²) without hole constraint. In the future we plan to include the option to model either BC, but

²⁶ Values were normalized as: $\frac{C/(C+R)}{(1-D/W)}$

are waiting until we have the corresponding solutions for the corner cracked bearing case (since the corner crack transitions to become a through crack).

The spring BC is well suited for cases in which the pin/hole fit is neat (tight). This is probably true in most practical applications (i.e. rivets).

This solution has been verified for the following dimensions:

$1.3 \leq W/D \leq \text{infinity}$ (solution converges as $W/D \rightarrow \text{infinity}$)

$0 < C/(C + R) < 1$ (tabular values for $C=0$, and the maximum crack length were extrapolated)

Oblique Through-the-Thickness Cracks:

Dr. Scott Fawaz developed the finite element based oblique crack solutions for axial, bending, and bearing loading conditions [33, 34].

The virtual corner crack is a quarter ellipse with the center at what would be the crack origin of a corner crack that has transitioned to an oblique through-the-thickness flaw at a hole in an infinite plate (see Figure 80, Section 3.2.3.1.1.16). The elliptical axes are defined by the A and C dimensions. While the A-dimension is not input by the user, it is calculated from the [C, Ct, and t] dimensions, which are input by the user. Dr. Fawaz's finite element solutions were calculated for the following range of dimensions:

$A/C = 0.2, 0.3, 0.4, 0.6, 1.0, 2.0, 5.0, \text{ and } 10.0$

$A/t = 1.05, 1.07, 1.09, 1.13, 1.17, 1.21, 2.0, 5.0, \text{ and } 10.0$

$R/t = 0.5, 1.0, \text{ and } 2.0$ - where R is the hole radius

Beta factors for each case were then calculated for the [C] and [Ct] dimensions as follows:

$$\text{Beta} = \frac{K}{\sigma \sqrt{\pi x}} ; \text{ Where } x \text{ is the appropriate crack length}$$

In this case, crack length is the C or Ct dimension. AFGROW uses a cubic spline interpolation technique to determine the appropriate beta value during crack growth life prediction. The following rules are used in AFGROW when the oblique through crack option is selected:

- No extrapolation is made beyond the bounds of the finite element cases
- If a/c, or a/t goes below the limit of the finite element cases, the value will be held at that limit
- If R/t is beyond the limits, it will be maintained at the nearest limit value
- If a/c, or a/t goes above the limits, the crack will be transitioned to a straight through-the-thickness crack of length [C]

Finite Width Correction:

$$F_w = \sqrt{\sec\left(\frac{\pi R}{W}\right) \sec\left(\frac{\pi(R+C)}{W}\right)}$$

The finite width correction was taken from Reference 16, Equation 46 with a/t set to 1.0.

This solution is valid for the following dimensions:

$$\frac{R+C}{W} < 0.5$$

3.2.3.1.1.21 Through Crack at a Semi-Circular Notch (Application Defined)



Axial Loading:

Reference [23]

This solution is valid for the following dimensions:

$$(R+C)/W < 0.6$$

$$r/w = 1/16 \rightarrow K_t = 3.17$$

Where, r is the notch radius

This solution was developed using fixed grip boundary conditions.

3.2.3.1.1.22 Single Edge Through Crack (Application Defined)



Axial Loading:

The standard solution for the edge cracked case accounts for in-plane bending caused by the specimen geometry as the crack grows. The specimen is assumed to be remotely pin loaded so there is no constraint to the in-plane bending as the crack grows.

$$\text{Beta} = \frac{\left(0.752 + 2.02(C/W) + 0.37 \left(1 - \sin \left(\frac{\pi C}{2W} \right) \right)^3 \right)}{\cos \left(\frac{\pi C}{2W} \right)} \sqrt{\frac{2W}{\pi C} \tan \left(\frac{\pi C}{2W} \right)}$$

Reference [36]

This solution is valid for the following dimensions:

$$0 < C/W < 1.0$$

This solution is within 0.5% for all crack lengths

AFGROW also includes an option to perform life predictions for edge crack cases where the in-plane bending is constrained (see Figure 83).

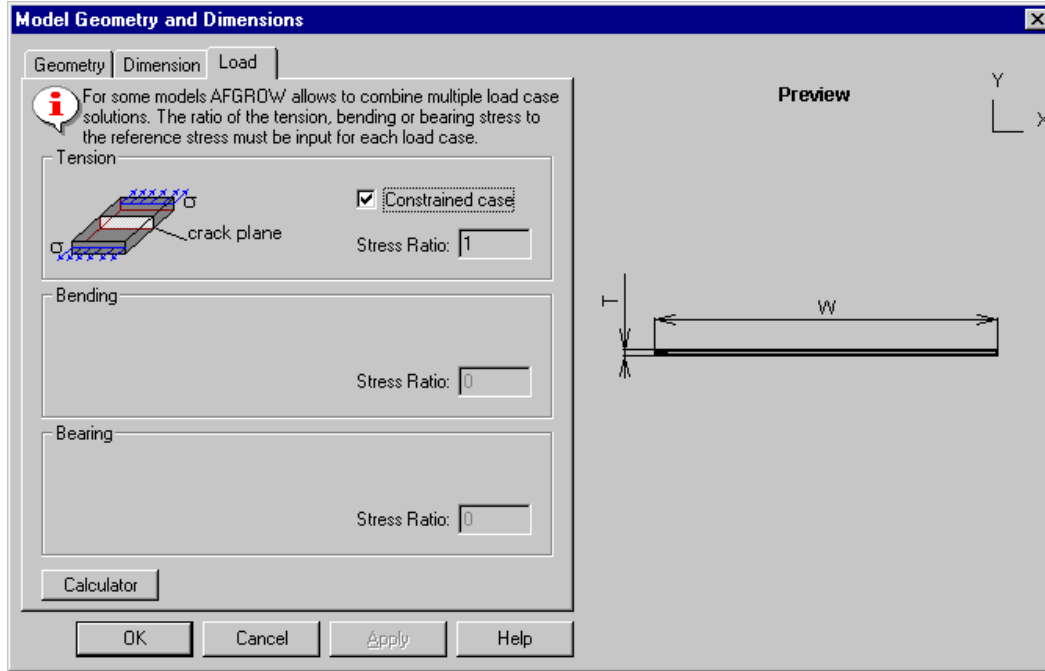


Figure 83: In-Plane Bending Constraint Option for the Edge Cracked Plate

To constrain (eliminate) the in-plane bending contribution from the single edge cracked plate, select the option for the constrained case in the load tab for the edge crack model as shown in Figure 83. The constrained solution was determined from numerous finite element models using FRANC2D/L [37].

The easiest way to eliminate in-plane bending from the edge crack case is to apply a uniform displacement to the finite element model. This method can be used to determine the stress intensity factor for a specific case. To be applicable to all edge crack cases, this solution should be in the form of a beta factor table.

$$Beta = \frac{K}{\sigma_{Ref} \sqrt{\pi x}} ; \text{ where } x \text{ is the crack length of interest}$$

If the stress intensity factor is known for a given edge crack case, the beta factor may be determined if the remote applied reference stress, σ_{Ref} , is known. The remote stress for the uniform displacement model can be extracted from the finite element model for relatively short cracks. When longer crack lengths are modeled, it becomes more difficult to determine the equivalent remote stress since the longer cracks cause large changes in the local stress distribution. Applying a uniformly distributed unit stress to the plate and constraining the displacement (normal to the applied stress) of the mid-plane nodes in the upper and lower portions of the plate model solved this problem. It was important to constrain only the mid-plane nodes to maintain a uniform stress field through the plate width. The nodes in the area of the crack plane were NOT constrained. The beta values

obtained using this approach were accurate within 0.1 percent of the uniform displacement method for the shorter crack lengths (where they could be compared). In addition, the stress distributions were in very good agreement for the long crack cases.

The beta solution for an edge crack in a semi-infinite plate is known to be equal to 1.122. This is true for both the constrained and unconstrained cases. The solution for the finite width cases is:

$Beta = 1.122 * F_w$; Where, F_w is the finite width correction

The finite width correction is simply a function of the ratio of the crack length to the plate width (C/W). This was verified by modeling various plate widths and comparing the betas at given C/W values. The resulting beta table is used in AFGROW to determine beta values (spline interpolation) when the in-plane bending constraint option is selected.

C/W	Beta
0	1.122
0.01	1.124
0.025	1.127
0.05	1.132
0.1	1.165
0.15	1.185
0.2	1.23
0.3	1.32
0.4	1.46
0.5	1.606
0.625	1.835
0.75	2.156
0.8	2.327
0.8333	2.499
0.875	2.789
0.9	3.005
0.916667	3.244
0.95	3.933
1	5.36

3.2.3.1.1.23 Edge Crack in a Finite Height Plate (Application Defined)



Axial Loading:

A tabular solution for this geometry was provided by Dr. Scott Fawaz for plate height to width (H/W) ratios between 1.0 and 10²⁷. Finite element models for the following H/W values were used for this solution with fixed end boundary conditions.

H/W = 1.0, 1.5, 2.0, 4.0, 6.0, 8.0, and 10.0.

The solution is assumed to be converged for any H/W > 10.

Crack length values in the table are normalized as C/W. The following C/W values are used in the tabular solution.

C/W = 0.01, 0.02, 0.03, 0.04, 0.05, 0.075, 0.1, 0.125, 0.15, 0.175, 0.2, 0.225, 0.25, 0.275, 0.3, 0.325, 0.35, 0.375, 0.4, 0.425, 0.45, 0.475, 0.5, 0.525, 0.55, 0.575, 0.6, 0.625, 0.65, 0.675, 0.7, 0.725, 0.75, 0.775, 0.8, 0.825, 0.85, 0.875, 0.9, 0.925, 0.95, and 0.975.

The solution is not extrapolated outside of the above range. For any crack length less than C/W = 0.01, the beta value for C/W = 0.01 is used. Likewise, for any crack length greater than C/W = 0.975, the value for C/W = 0.975 is used.

Reference [83]

This solution is valid for the following dimensions:

$$0.01 \leq C/W \leq 0.975$$

3.2.3.1.1.24 Double Edge Through Crack (Application Defined)



Axial Loading:

$$\text{Beta} = \left(1.0 + 0.122 \cos^4 \left(\frac{\pi C}{W} \right) \right) \sqrt{\frac{W}{\pi C} \tan \left(\frac{\pi C}{W} \right)}$$

Reference [38]

This solution is valid within 0.5% for the following dimensions:

$$0 < C/W < 0.5$$

²⁷ H is the total plate height

3.2.3.1.1.25 WOL/CT Specimen (Application Defined)



Note: The loading for this geometry is applied as pin loads through bolt holes in the specimen. Therefore, the input axial (stress) value is not really a stress value but is LOAD.

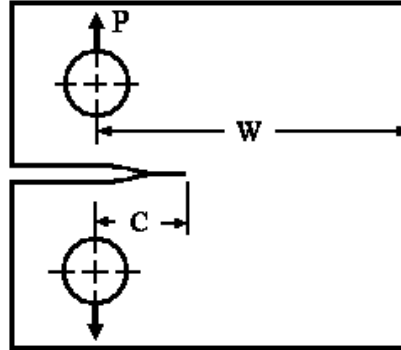


Figure 84: WOL/CT Specimen

When using this geometry, the user must input the applied LOAD instead of stress. The diagram in the loads tab indicates a remote stress input, but this is because all other geometries use remote stress as the input for the axial case. This geometry is an exception to that rule.

Axial Loading:

Reference [39]

This solution is valid for the following dimensions:

$$0.2 < C/W \leq 0.975$$

This solution is within 0.5% for all crack lengths in the range above. Since this geometry uses load instead of stress in the input spectrum, the beta values printed in the output include area units. This is required because of the definition of K used in AFGROW.

$$K = \sigma \sqrt{\pi x} \beta, \text{ where } x \text{ is the crack length of interest}$$

3.2.3.1.1.26 Constant K Specimen (Application Defined)



This model holds (K / applied reference stress) equal to 1.0. This means that Kmax and Kmin will be equal to the numeric value of the SMF multiplied by the maximum and minimum values in the input spectrum, respectively. This may be useful for determining the severity of a given input spectrum. When using this model, remember to set the crack growth increment to “Cycle by Cycle Beta and Spectrum calculation” in AFGROW Preferences so that the beta value is recalculated for each load cycle.

3.2.3.1.1.27 Single Edge Crack in Lug (Application Defined)



Bearing Loading:

Reference [40]

This solution was developed from detailed StressCheck finite element models (FEMs). The solution was the result of a curve fit to the FEMs performed by Mr. Dave Child at Purdue University.

The pin bearing load was applied to the FEMs using either a spring or bearing (cosine² loading) type of pin load distribution. The spring loading case is intended to model a neat fit pin, and the bearing case more closely represents a clearance fit pin. The spring condition utilized a spring constant to plate modulus ratio of 3 to represent a steel pin in an aluminum plate²⁸. The major difference in the two boundary conditions is the fact that the bearing condition allows the hole to deform, and the spring condition constrains the hole. The default boundary condition in AFGROW is the bearing condition for crack lengths for which $a/t \leq 0.7$. The spring condition is used for $a/t \geq 0.8$, and linear superposition is used for intermediate crack lengths. This provided the best correlation to lug test data currently available. Users may change the default condition using the Predict, Preference (Lug Boundary Condition) option in the main menu (Section 3.4.1.6). More information about the boundary conditions is also located in that section.

This solution is valid for the following dimensions:

$$1.5 < W/D \leq 40$$

$$0 < C/C_{\max} \leq 0.98$$

²⁸ Spring to plate modulus ratios as low as 1.0 resulted in nearly the same K-solutions.

3.2.3.1.1.28 Rod (Application Defined)



Axial Loading:

Reference [41]

Bending Loading:

Reference [41]

3.2.3.1.1.29 Circumferential Crack in a Bolt Thread (Application Defined)



Axial Loading:

Reference [77, 78]

Bending Loading:

Reference [77, 78]

3.2.3.1.1.30 Through Crack in Pipe (Application Defined)



Axial Loading:

Reference [42]

Bending Loading:

Reference [42]

3.2.3.1.1.31 External Circumferential Crack in Pipe (Application Defined)



Axial Loading:

Reference [79]

3.2.3.1.1.32 Internal Circumferential Crack in Pipe (Application Defined)



Axial Loading:

Reference [79]

3.2.3.1.1.33 Axial Through Crack in Pipe (Application Defined)



This solution assumes an “infinite” pipe length, but AFGROW includes a width parameter that serves as a crack length limit in the c-direction.

Axial Loading:

Reference [84]

This solution is valid for the following dimensions:

$$(D_o - D_i)/D_i \leq 0.25$$

3.2.3.1.2 Weight Function Stress Intensity Solutions

The weight function solutions [7] in AFGROW were provided under sub-contract to AS&M by Prof. G. Glinka, University of Waterloo, CA. Professor Glinka's solutions were translated to the C/C++ language and adapted for use in AFGROW.

3.2.3.1.2.1 Center Semi-Elliptical Surface Crack (Glinka's Weight Function)



This solution is valid for the following dimensions:

$$0 < A/t \leq 0.8$$

$$0 < A/C \leq 2.0$$

3.2.3.1.2.2 Single Corner Crack (Glinka's Weight Function)



This solution is valid for the following dimensions:

$$0 < A/t \leq 0.8$$

$$0.2 \leq A/C \leq 1.0$$

3.2.3.1.2.3 Internal Axial Crack in Thick Pipe (Glinka's Weight Function)



This solution is valid for the following dimensions:

$$1.1 \leq R_o/R_i \leq 2.0 ; \text{ Where } R_o : \text{ Outside Pipe Radius, } R_i : \text{ Inside Pipe Radius}$$

$$0 < A/t \leq 0.8$$

$$0.2 \leq A/C \leq 1.0$$

3.2.3.1.2.4 External Axial Crack in Thick Pipe (Glinka's Weight Function)



This solution is valid for the following dimensions:

$$1.1 \leq R_o/R \leq 2.0 ; \text{ Where } R_o : \text{ Outside Pipe Radius, } R_i : \text{ Inside Pipe Radius}$$

$$0 < A/t \leq 1.0$$

$$0.2 \leq A/C \leq 1.0$$

3.2.3.1.2.5 Center Through Crack (Glinka's Weight Function)



This solution is valid for the following dimensions:

$$0 < C/W < 0.45$$

3.2.3.1.2.6 Single Edge Through Crack (Glinka's Weight Function)



This solution is valid for the following dimensions:

$$0 < C/W < 0.9$$

3.2.3.1.2.7 Double Edge Through Crack (Glinka's Weight Function)



This solution is valid for the following dimensions:

$$0 < C/W < 0.45$$

3.2.3.1.2.8 Radial Edge Crack in Disc (Glinka's Weight Function)



This solution is valid for the following dimensions:

$$0 < C/\text{Diameter} < 0.9$$

3.2.3.1.2.9 Axial Through Crack in Thick Pipe (Glinka's Weight Function)



This solution is valid for the following dimensions:

$$1.1 \leq R_o/R_i \leq 2.0$$

$$0 < C/W < 0.45$$

3.2.3.1.3 Using the Weight Function Solutions

The 2-D solutions (part-through crack) currently allow the input stress field to vary in one direction only (currently the distribution in the thickness (y) direction). The origin of the x-y coordinate system is always at the crack origin, and the x and y values are always positive. The details of the stress field input are given in the stress distribution dialog. When the part through the thickness cracks transition to become through-the-thickness cracks, the model is automatically changed to the appropriate 1-D case and the applicable stress distribution is used to continue the life prediction. The stress distribution in the width (x) direction is always used for 1-D cases. Certain Tips and Tricks are available to provide additional guidance in the use of the weight function solutions.

3.2.3.1.3.1 Weight Function Stress Distribution

The decision whether to normalize the input stress distribution is really a personal preference as long as you have a clear understanding of the relationship between the input spectrum, stress multiplication factor, and the stress distribution you plan to use. The main thing to remember is that all three values are multiplied together by AFGROW to determine the stress values at each point where you input stress (or load).

For example, to simulate a double-crack at an open hole with a remote gross stress ($P/(W*t)$) of 14 ksi, $R = 0$, initial crack size = 0.07 in, (from the hole), the following was done:

- A normalized spectrum was used - Max Stress = 1.0
- Stress Multiplication Factor = 14
- Finite element results (FRANC2D) provided un-flawed K_t vs. crack length in the crack plane.
- Center cracked weight function model was chosen based on geometric similarity
- Open hole (0.5 in. diameter) modeled by a stress free zone
- Initial half crack length = 0.25 (radius) + $0.07 = 0.32$ in.

The stress distribution dialog is shown in Figure 85:

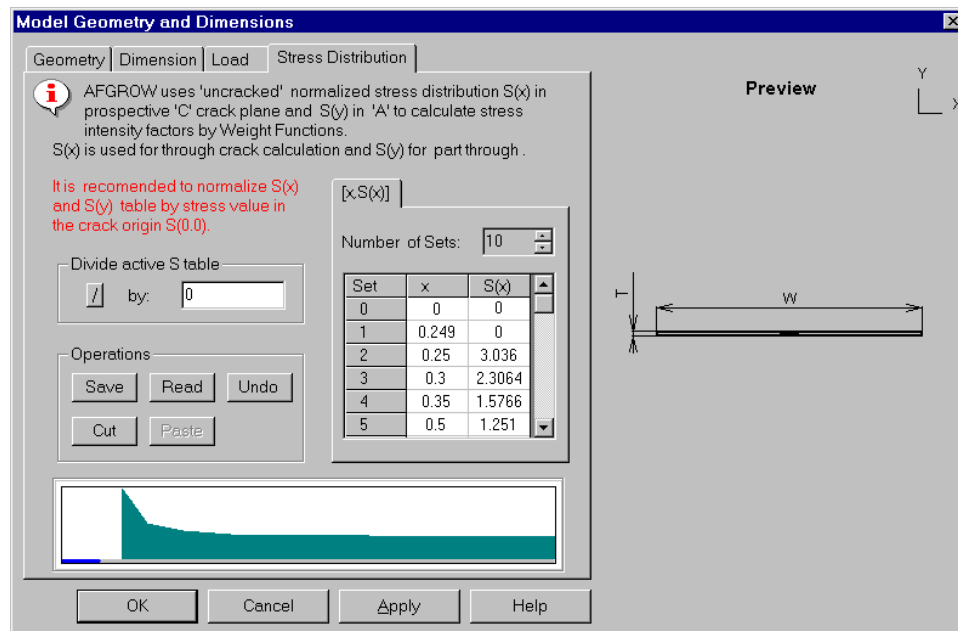


Figure 85: Weight Function Stress Distribution Dialog

AFGROW provides a tool to allow the stress values to be divided by a given number. This is especially helpful in cases where the user wants to normalize the stress values.

Since there is no weight function solution for a double crack at a hole, the center-cracked case was used with a stress free area where the hole would have been located. A total of

ten points were used to characterize the unflawed stress distribution in the crack plane. The input distribution is shown in the stress distribution dialog. AFGROW accepts a maximum of 25 points to define the stress distribution. The points do not have to be equally spaced, but should be spaced such that linear interpolation between points adequately matches the desired distribution.

In order to judge the effectiveness of this approximate solution, a comparison of life prediction analyses was made between this solution and the standard double crack at a hole solution (see Figure 86).

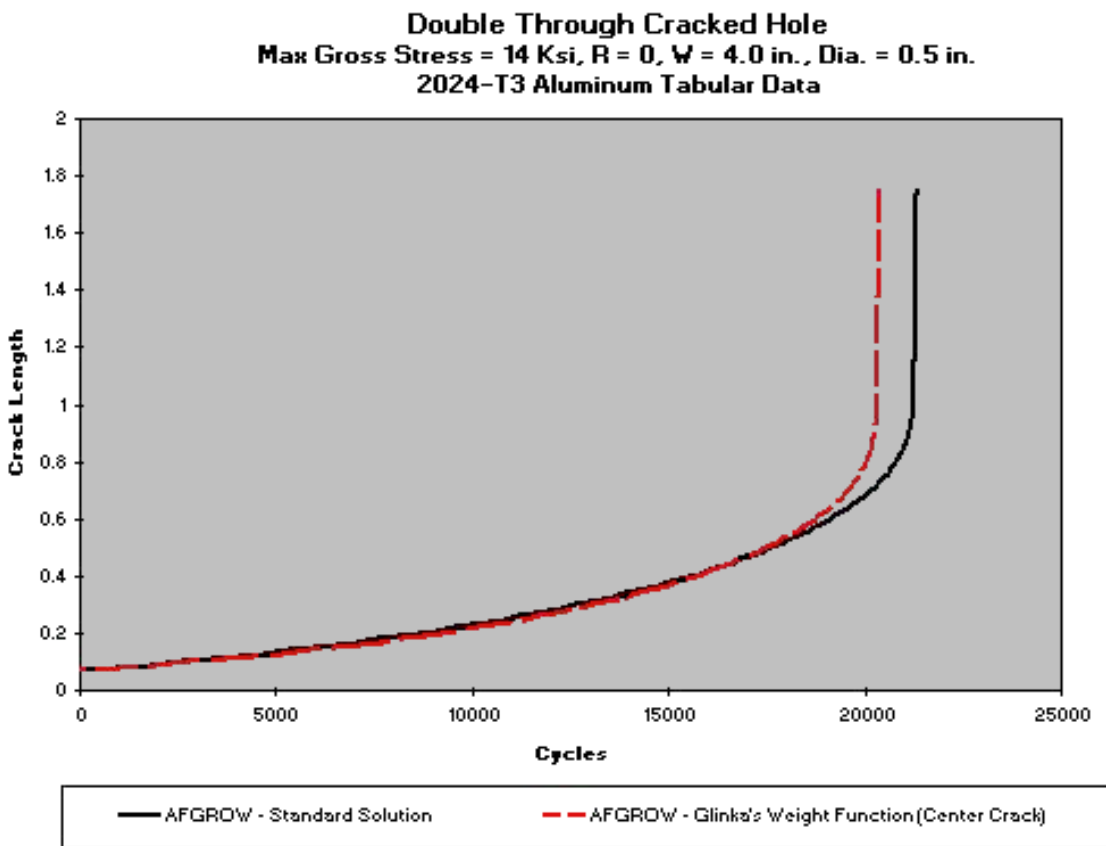


Figure 86: Comparison of Weight Function and Standard Solutions

The weight function solution resulted in an excellent agreement in life to a certain crack length. This comparison is very sensitive to small changes in stress intensity. Hence, this approximation is excellent to a crack length of approximately 0.5 inches.

3.2.3.1.3.2 Weight Function Tips and Tricks

Choose the Appropriate Weight Function Model. Try to choose the model that is geometrically CLOSEST to the problem being approximated.

The current 2-D weight function solutions in AFGROW only permit the input stress distribution to vary in a single direction. Prof. Glinka's solutions for part-through cases can be adjusted to switch dimensions. At this time, only the single corner and surface crack models are available. Future releases are planned which will include additional models.

If a through crack at an edge notched specimen is being modeled, use the edge crack model, determine the unflawed stress distribution, and model the notch depth as a stress free area as was done in the example given above

Additional tips or tricks will be provided as more experience is gained working with the solutions.

3.2.3.1.3.3 Weight Function Verification

Comparisons between weight function and available closed-form stress intensity solutions have been made to aid in the verification of the weight function solutions. Selected weight function stress intensity models, provided by Prof. Glinka, have been compared to existing closed form solutions to demonstrate the accuracy of the weight function solutions. A copy of the stress distribution dialog is provided for each case. The results are shown below in Figure 87 and Figure 88:

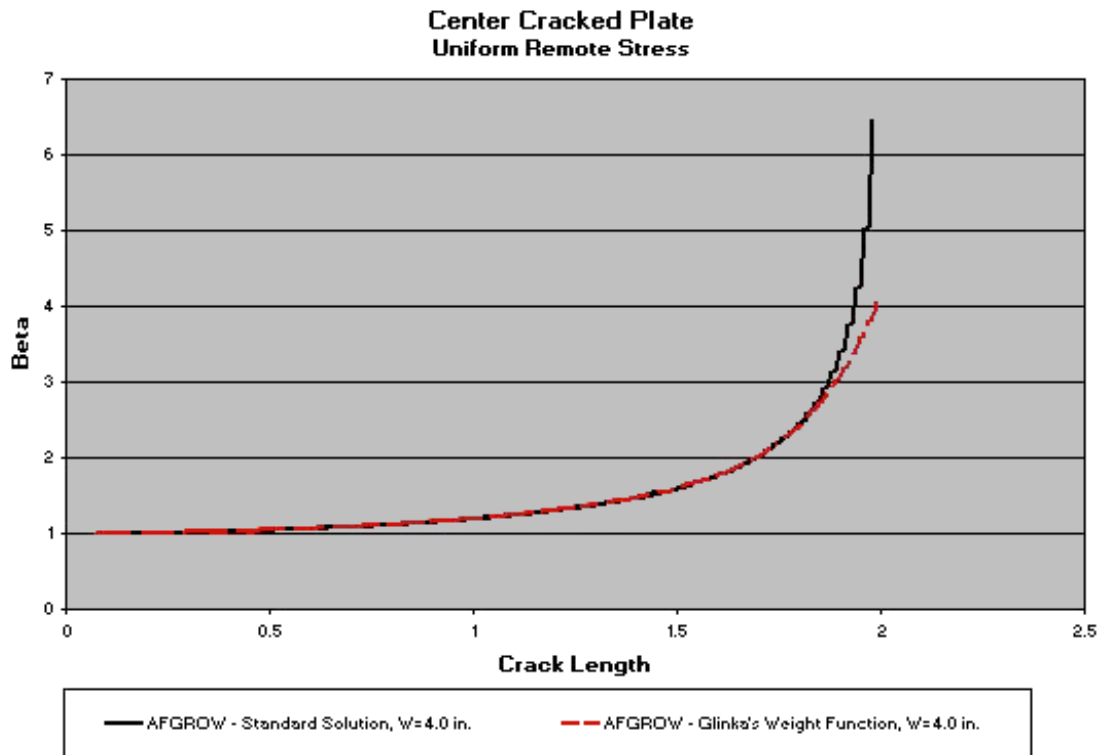


Figure 87: Center Crack with Uniform Axial Loading

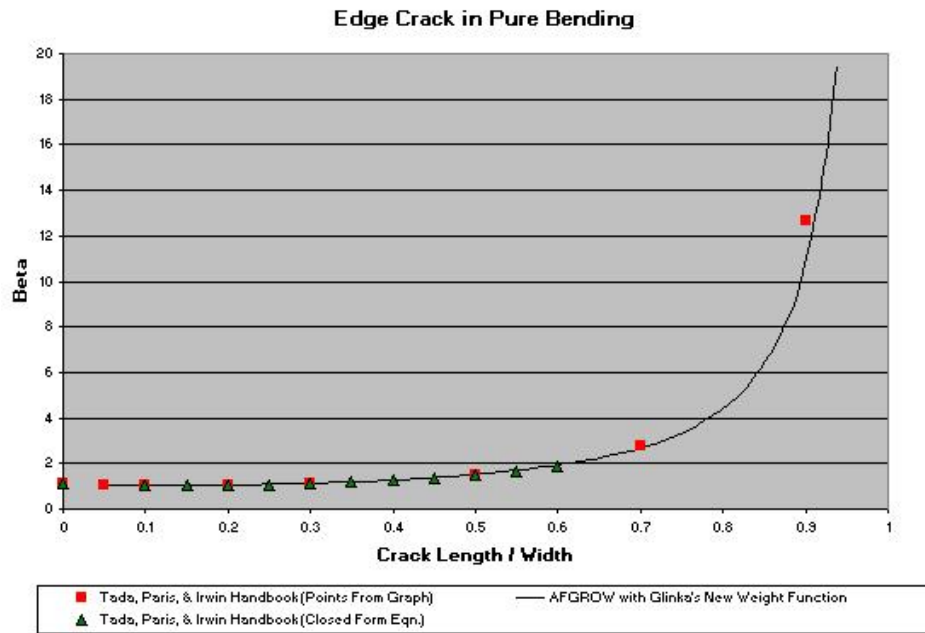


Figure 88: Edge Crack with Pure Bending

The above figures show a comparison between beta values for the weight function case and the corresponding standard stress intensity factor solution. The comparisons show very good agreement to the standard closed-form solutions. There is some divergence at the longer crack lengths. This is expected due to the limits of the weight function solutions. These errors translate to small differences in crack growth life, since the majority of the life is spent at short crack lengths.

3.2.3.1.4 Model Dimensions

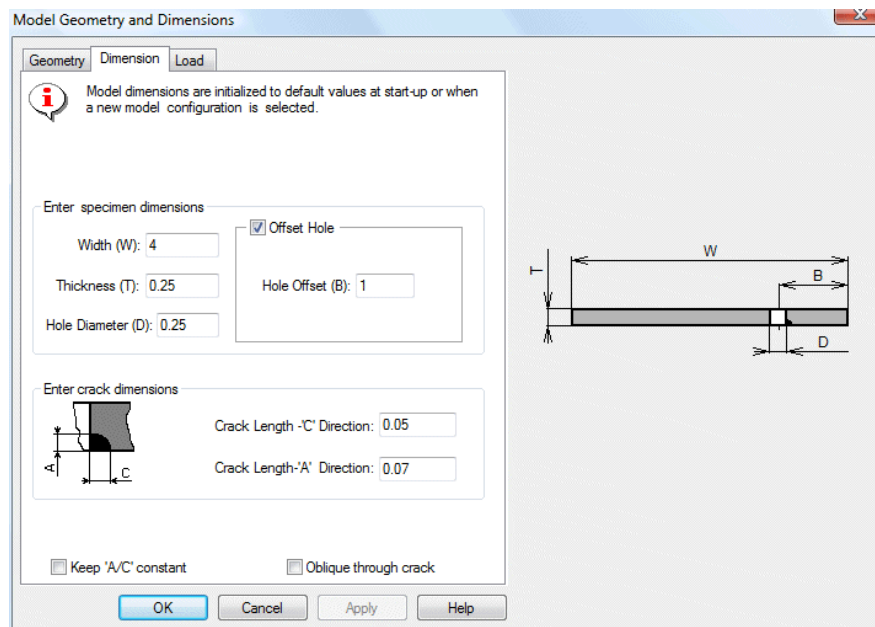


Figure 89: Model Dimension Dialog

The dimensions dialog is used to set the dimensions of the model and the initial crack size. The options in the dimensions dialog reflect the dimensional features of the selected model. In the case of part-through flaws, the user may choose the option for AFGROW to maintain a constant crack shape ($a/c=\text{constant}$). If the option to keep a/c constant is selected, all of the crack growth calculations are based on the c -direction with the a -dimension simply calculated based on the initial crack shape. The preview window will reflect user input dimension changes when the APPLY button is clicked.

3.2.3.1.5 Model Load

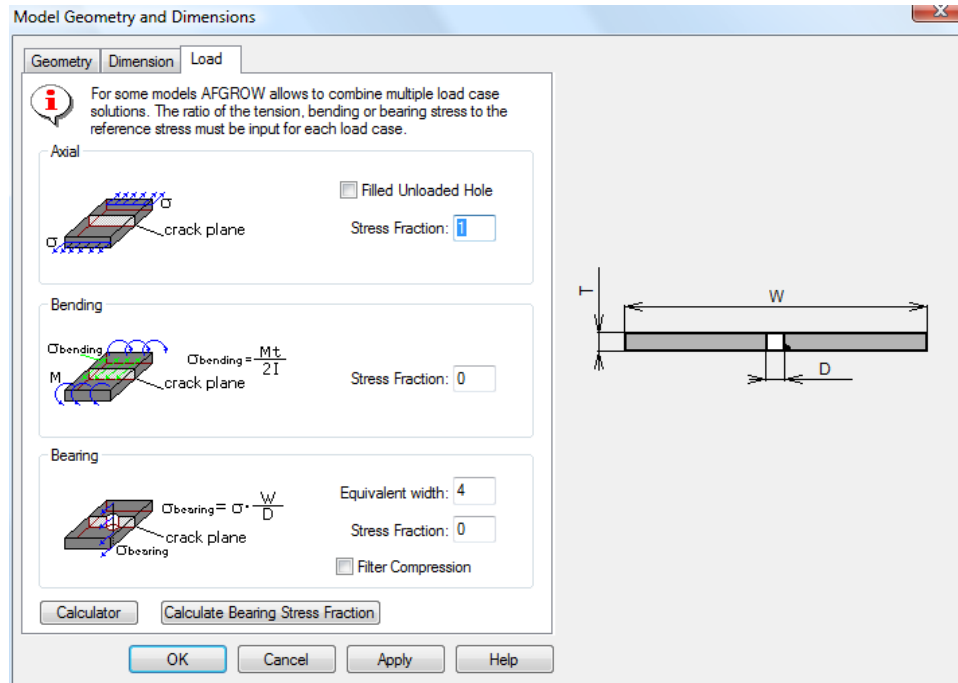


Figure 90: Model Load Dialog

Since some models have multiple load case solutions, AFGROW allows the user to combine these solutions using the superposition method. To use this option, the ratio of the axial, bending, or bearing stress to the reference stress must be input for each load case to be modeled. AFGROW shows the definition of each type of stress in the load tab of the model dialog (see Figure 90). The reference stress is simply the product of the Spectrum Multiplication Factor (SMF) times the current spectrum maximum or minimum value. When using a single channel spectrum, the inherent assumption is that each load case is in-phase and each load case stress to reference stress ratio is constant. Therefore, these ratios should be determined for the reference stress and spectrum condition that the user considers to be appropriate for the given model. This approach allows a user to perform parametric studies for any number of stress levels by simply changing the value of SMF in the spectrum dialog. It is, however, up to the user to be aware of the definitions of the reference stress and the load case stress to correctly use this capability. Every attempt is made to identify the definition of the load case reference stresses.

For all cracked hole models, there is an axial load case option to add a filled hole (zero interference) correction. This correction was originally provided by Dr. Jack Lincoln (~1993), and was most likely developed by Lockheed under contract to the Air Force Research Laboratory in the 1980's. This correction factor (F_{fill}) is given below:

$$F_{fill} = 0.969831 + 0.10763 \log_{10} \left(\frac{c}{r} + 0.01 \right)$$

Where:

c = crack length in the c-direction

r = hole radius

Note: If $F_{fill} > 1$, $F_{fill} = 1$

This correction has more recently been compared to a correction based on finite element model results for a single through cracked hole as shown.

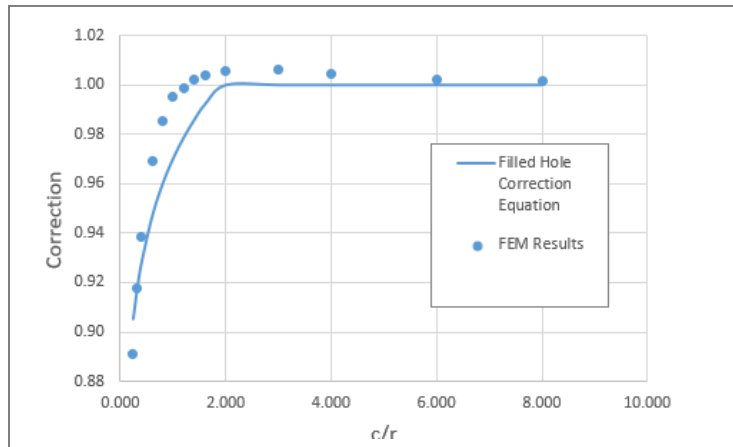


Figure 91: Filled Hole Correction

The original equation is within ~2% of the finite element results, so the equation will remain in use to maintain continuity for legacy purposes. It is possible that the correction may be conservative for part-through cracks. When more information becomes available, we will make necessary modifications.

Combined Loading Example:

For a given reference load case, a 0.25 in. dia. fastener hole in a 0.125 in. thick x 1.0 in. wide plate has a pin load of 200 lbs. The bypass stress is 10 ksi. The bending stress is 5 ksi. If you choose to use the remotely applied gross stress (bypass stress + bending stress + pin load/(width * thickness)) as the reference stress, then the total gross remote reacted stress is:

$$10 \text{ ksi} + 5 \text{ ksi} + 200/(0.125 * 1.0) * 0.001 = 16.6 \text{ ksi}$$

Therefore,

The axial stress ratio is: $10/16.6 = 0.6024$

The bending stress ratio is: $5/16.6 = 0.3012$

The bearing stress ratio is: $(200/(0.25 * 0.125) * 0.001)/16.6 = 0.3855$

These ratios have nothing to do with a "percent load transfer." There is no limitation that these ratios add to 1.0. Depending on the situation, the ratios can easily be much greater than 1.0. The reason the ratios do not add to 1.0 in this case is because the stress intensity solution for the bearing load case is based on a bearing stress reference instead of gross stress. It is necessary to "fool" AFGROW to use a common reference stress. It is generally a good practice to use gross stress as the reference since the majority of models use gross stress and it will usually minimize any necessary conversions.

A calculator option is available to aid the user in making the appropriate bearing stress ratio calculations automatically based on the axial and bending stress ratios using the following relationship:

$$\text{Bearing Stress Ratio} = (1 - \text{Axial Stress Ratio} - \text{Bending Stress Ratio}) * W_{\text{effective}}/D$$

Please Note: This assumes that the input stress spectrum and spectrum multiplication factor are referenced to the remote axial gross stress. In addition, it is important to remember that the bearing stress at the hole should be accounted for with this calculation. For this to be true, the Axial and Bending Stress Ratios must also be applicable over the assumed effective width.

When bearing stress is included in a model, it is also important to remember that local fastener loads have a limited effect for offset holes in wide plates. When modeling a crack at an offset, loaded hole in a wide plate (>10 hole diameters), consider using a total plate width not larger than 10 hole diameters. This will normally allow the user to account for some plate offset and the effect of an adjacent hole. The use of a larger width will tend to underestimate the effect of bearing load. There are a few documents on the AFGROW web page that discuss this issue in more detail.

3.2.3.2 Advanced Crack Models

A major internal code change was made in AFGROW prior to the addition of the advanced crack analysis capability. It was felt that the best way to analyze more complex geometries with multiple cracks was to treat each crack tip as a separate entity (or object). The steps required to predict the growth of each tip are the same and it is much easier to manage the life prediction process if each tip is managed as a separate object. This is NOT to say that each tip has no effect on the other, simply that the same life prediction process must be applied to each tip. The method used to account for the presence of other cracks will be explained in detail in the following section. As a result of the code change, AFGROW contains the basic code infrastructure to handle any number of cracks. The only limitation to this is the logic required to predict geometric changes that occur as cracks grow toward cracks, other holes, or the edge of a specimen.

The Advanced Crack Model interface is illustrated in Figure 92 below:

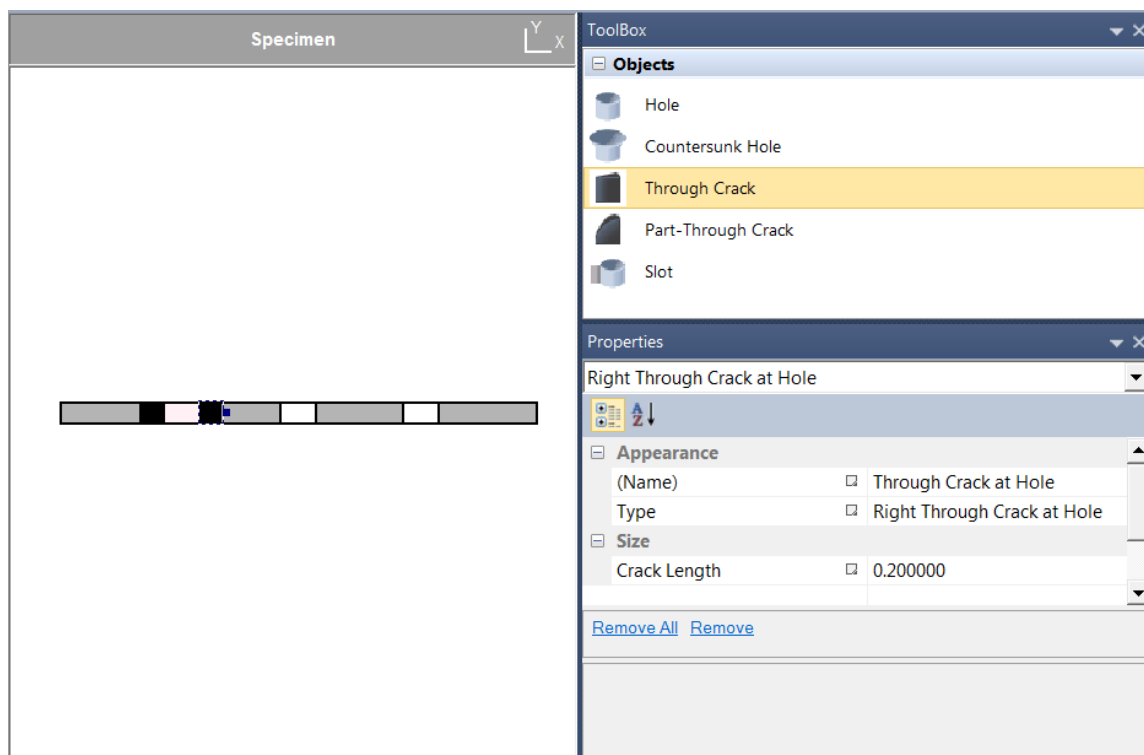


Figure 92: Advanced Model User Interface

The specimen cross-section is shown in the animation frame and objects (Hole, Through Crack, or Part-Through Crack) may be added to the cross-section using the mouse to “drag and drop” the feature on the specimen. These objects are located on the tool box menu bar (see Section 3.3.1.4 for more details) shown in Figure 92.

The specimen and/or any object attached to it may be resized by selecting the desired object with the mouse (single left-click). The object may be resized by dragging it or by entering the appropriate value in the properties menu bar (see Section 3.3.1.3) shown in Figure 92.

The length of a crack shown in the properties bar is consistent with the lengths used in the classic model interface. The crack lengths in the output and status windows reflect the conventions used in AFGROW for all crack lengths. For example, an internal through crack with a total length of 0.2 will have a C-length of 0.1.

The position of an object (crack or hole) is the distance from the left edge of the plate to the center of the object.

User-defined 2 crack geometries may be saved for later use as a *.dax file by using the Save As command in the file menu.

An example of the output for the 2-crack geometry is given as shown in Figure 93.

Crack #1			
Left Tip =	0.076475	Beta= 1.086	R(k)= 0.000 R(final)= 0.000
Right Tip=	0.075922	Beta= 0.968	R(k)= 0.000 R(final)= 0.000
		Delta k=7.436e+000	D()/DN=3.278e-007
		Delta k=6.629e+000	D()/DN=2.091e-007
Crack #2			
Left Tip =	0.45725	Beta= 1.084	R(k)= 0.000 R(final)= 0.000
Right Tip=	0.46735	Beta= 1.134	R(k)= 0.000 R(final)= 0.000
		Delta k=1.829e+001	D()/DN=2.206e-005
		Delta k=1.913e+001	D()/DN=2.590e-005
Max stress 14.000, r = 0.00, 4800 Cycles, Constant amp.: 49, Pass: 49			

Figure 93: Sample Output for a Two-Crack Model

In the case of internal through cracks, crack lengths shown in the output are measured from the initial offset defined by the user. The actual stress intensity values are calculated internally for each crack based on the current offset (this changes as the crack grows). Therefore, remember that the crack lengths printed in the output should not be used to calculate stress intensities by hand (in case you are trying to verify the K or Beta values in the output). It was felt that users would prefer to have the output crack lengths reflect the distance from the initial offset. The only exception to this is the plot file where the crack lengths for internal through cracks are the actual half-lengths. However, the crack length and beta values in the plot file are only given for the left crack tip. This is simply due to limitations of the format for the plot file (for the purposes of plotting in Excel) and the desire to keep the file format consistent. Actual text based output files may be imported to Excel if more detail is required for plotting purposes.

Currently, AFGROW will stop when the first crack fails or grows to a geometric boundary if there is no solution available for the new crack geometry. We continue to make improvements and additions to improve the overall capability of AFGROW.

K-solutions for all possible loading cases are not currently available. Solutions are available for the following loading cases:

Geometry	Loading Case
Non-Symmetric Corner Cracked Hole	Axial/Bending/Bearing
Double, Symmetric Corner Cracked C/S Hole	Axial/Bending
Double, Symmetric Corner Crack at the Base of C/S Hole	Axial
Single Corner Crack at a C/S Hole	Axial/Bending
Single Corner Crack at the Base of a C/S Hole	Axial
Single Corner Crack at a U-Shaped Notch	Axial
Single Through Crack at a U-Shaped Notch	Axial
Non-Symmetric Through Cracks at a Hole	Axial/Bending/Bearing
Non-Symmetric Through Cracks (general case)	Axial/Bending

Please refer to the above list for guidance when using the Advanced Interface. If there are any questions, please contact support@afgrow.net for more information.

Please refer to Section 2.2.3 for a list of general limitations for the Advanced Model Interface.

3.2.3.2.1 Through-the-Thickness Cracks (Axial or Bending Loading Conditions)

The stress intensity solutions²⁹ for two arbitrary through cracks in a plate were determined using finite element models (FEM). The primary finite element code used for this purpose was StressCheck [31]. StressCheck is a P-version finite element code that provides very good information on the convergence of a given case. We also performed MANY verification analyses for geometries with well-established closed-form solutions and compared other FEM codes to provide the highest confidence in the FEM solutions. This work was funded through the U.S. Air Force Aging Aircraft Office (ASC/SMA) and was supported by the Air Force ALCs.

On the surface, solutions for 2 through cracks in a plane may seem like a trivial matter. The fact is that it is a VERY difficult problem simply because of the combination of possible geometries. If you consider an infinite plate, the possibilities are reduced tremendously; however, there are never enough infinite plates around when you need them. There is not enough space in this manual to provide all of the details of the K solutions. The complete details have been published in an Air Force technical report [43]. The general approach and examples are given in the following paragraphs.

First, the K-solution for each crack is determined assuming that it is the only crack in the structure. The classic solutions in AFGROW contain all possible single crack cases for this purpose (including offset internal cracks and cracks on either side of an offset hole).

Then, a 40-inch plate was used to simulate an infinite plate condition (crack lengths were kept short enough so that any finite width effect would be negligible). Combinations of crack lengths and crack spacing were modeled (using FEM methods) to determine a relationship between these variables and the effect of the second crack on the first. An example of this is shown in Figure 94 for two internal through cracks.

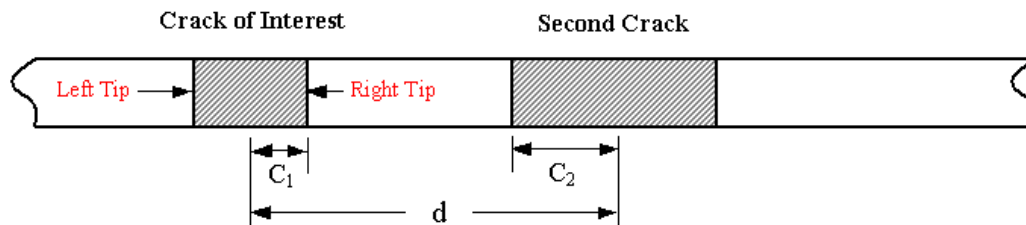


Figure 94: Two Internal Cracks in an Infinite Plate

An example of the beta correction for the left crack tip (caused by the second crack) is shown in Figure 95.

²⁹ These solutions were developed under axial loading conditions, solutions for out-of-plane bending are approximated using a fraction of the axial solution. Refer to Section 3.4.1.8 for an explanation of how this fraction may be changed as a user-defined preference.

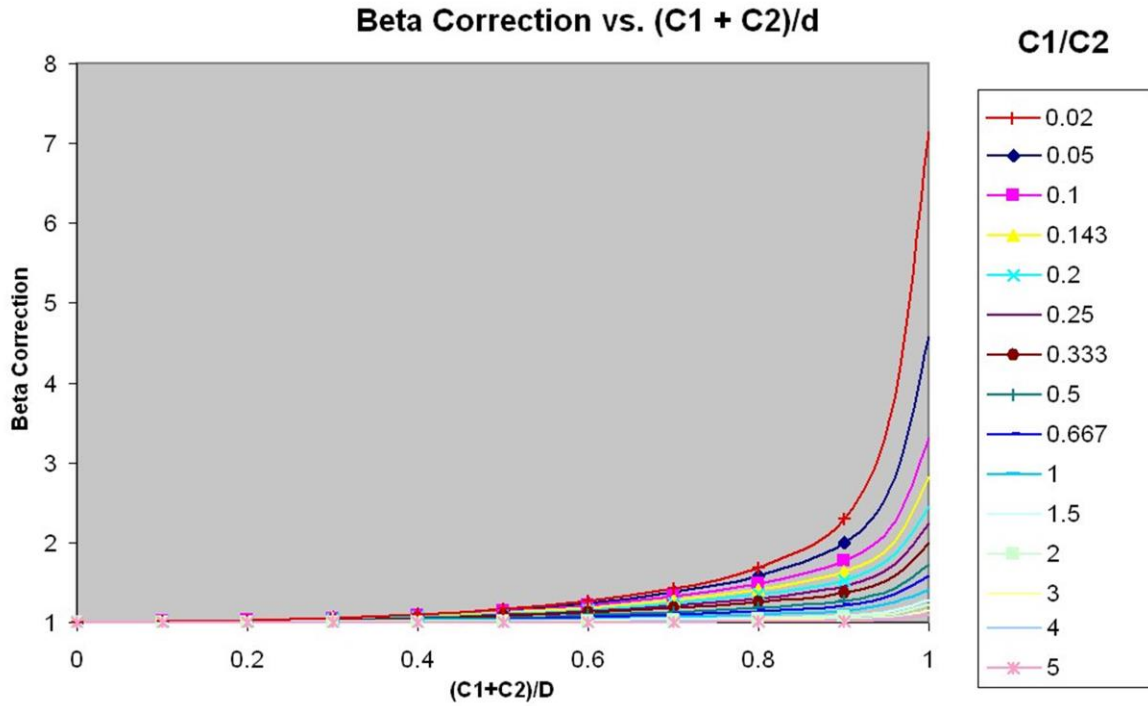


Figure 95: Sample Beta Correction to Account for a Second Crack

The data points were determined by dividing the FEM stress intensity for the left tip (*these FEM models included both cracks*) by the AFGROW stress intensity for the left tip (*assuming the crack in question was the only crack in the plate*). This provided the correction for the left tip for the crack in question (C1). FEM modeling was somewhat problematic for $(C1 + C2)/d > 0.9$, so values were extrapolated using a cubic spline method. In the case of the external crack tips (as shown in Figure 95), the correction for $(C1 + C2)/d = 1$ was determined using the inverse ratio of the infinite plate K-solutions for the crack tip immediately before and after crack coalescence. This ratio simply reduces to:

$$\sqrt{\frac{C_1 + C_2}{C_1}} = \sqrt{1 + \frac{C_2}{C_1}}, \quad \text{for } (C_1 + C_2)/d = 1$$

The actual correction for any given crack length combination is determined in AFGROW using a cubic spline interpolation method. It should be noted that the crack length ratios $(C1/C2)$ above 50 or below 0.02 were never modeled in any of the current solutions. It is expected that this range of values will cover the vast majority of practical problems. No extrapolations are made beyond these limits. In cases, where the correction is less than 1%, no correction is generally applied.

Finally, the effect of the finite plate width must be considered. Hundreds of FEM analyses were performed for numerous crack length combinations for several plate widths (40, 24, 16, 8, and 4 inches). These analyses were performed for several crack combinations including: internal-internal, edge-internal, edge-edge, cracks on each side of a hole, and

cracks growing to holes. The K-value from each FEM analysis (for the crack tip in question) was divided by the K value that had been corrected for the presence of a second crack in an infinite plate (actually in the 40 inch plate – see Figure 95). These ratios are the error in the infinite plate solution caused by the fact that the plate is not really infinite.

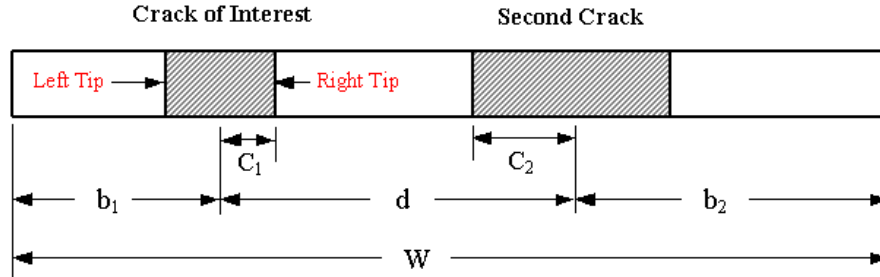


Figure 96: Two Internal Cracks in a Finite Plate

A spreadsheet was used to tabulate the specific parameters of each FEM analysis with the finite plate error. The spreadsheet was then imported to a Microsoft Access database and sorted on increasing error. The resulting table was examined for trends in the parameters and the error correction was curve fit using the most promising parametric trends. This process was VERY time consuming and tedious. The resulting curve fit for the left crack tip for two internal through cracks is given below for the parameters shown in Figure 96.

$$\text{Finite Plate Correction} = \left(1.012 - 0.35 \left(\left(1 - \frac{2b^*}{W} \right) \frac{C_1}{b_1} \right)^2 \right) \left(1.022 - 0.4 \left(\left(1 - \frac{2b^*}{W} \right) \frac{C_2}{b_2} \right)^{2.2} \right)$$

In this particular case, b1 and b2 are the distances between the crack centers and the **NEAREST** plate edge for each crack and may never be greater than W/2. The value, b*, is defined to be the smaller of b1 or b2.

Remember, the finite plate correction is not the same as the finite width effect that is used to account for the free edge in normal stress intensity solutions. The classic finite width effect for each crack is already accounted for in the solution since the first step is to determine K for each crack as if it were alone in the plate. The finite plate effect merely accounts for changes in K caused by the presence of the second crack in a finite plate.

The final result for all cases resulted in solutions that were normally well within 3% of the FEM analyses. As a matter of fact, most are within 1% of the FEM solutions. However, a few extreme cases resulted in errors of approximately 10%. However, considering the complexities involved, it is felt that this effort has been very successful. As a result of the level of complication in the work to develop closed-form K solutions for 2 independent through cracks in finite plates, it is logical to assume that solutions for 3 or more cracks should not be attempted using this approach.

3.2.3.2.2 Through-the-Thickness Cracks at a Hole (Bearing Load)

This solution is based on a closed-form solution based on the methods of compounding and superposition developed at the National Research Council Canada [81].

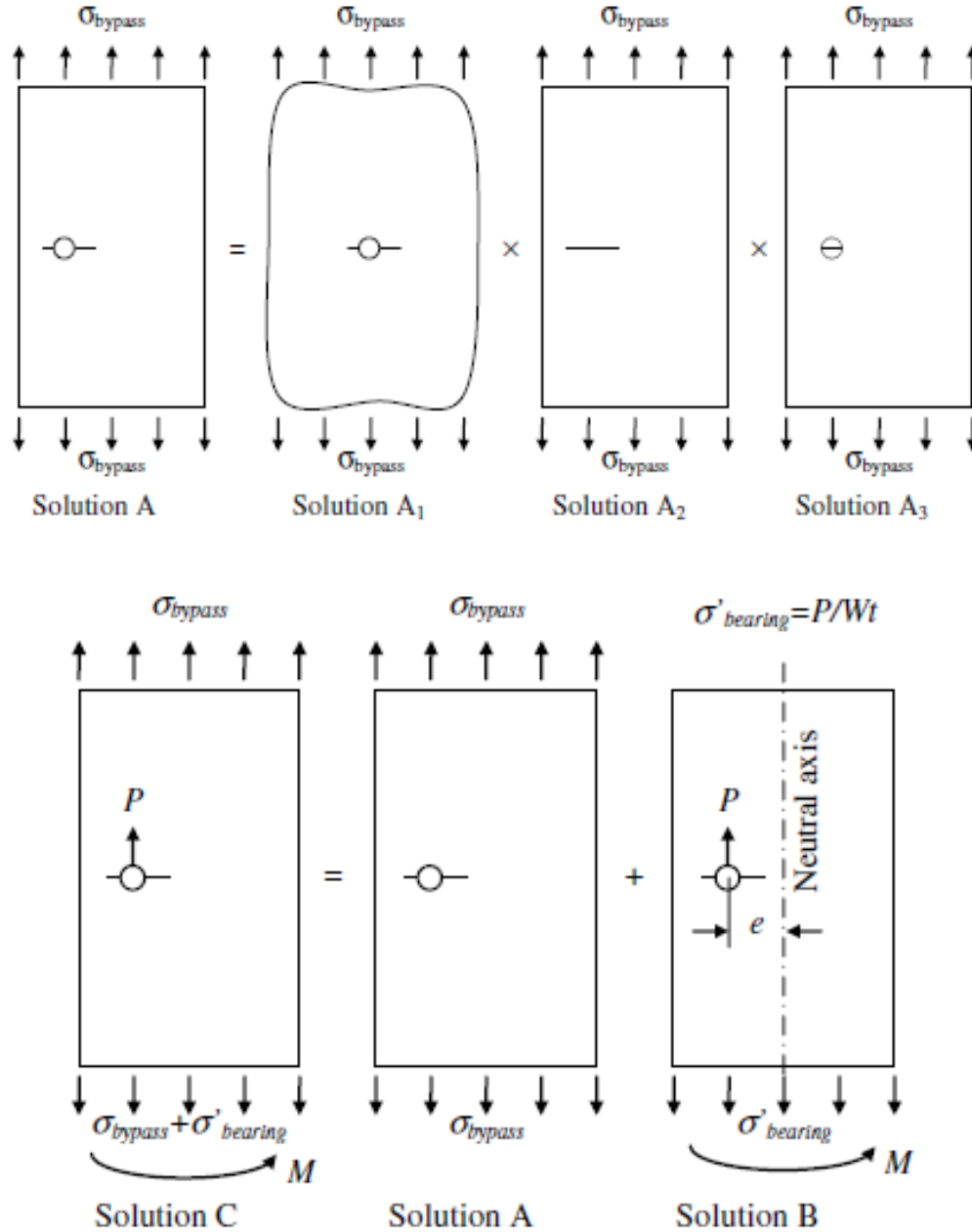


Figure 97: Solution Method for a Through Crack at a Bearing Loaded Hole

The solution was modified by LexTech, Inc. based on FE results available for several combinations of crack length(s) and hole offset³⁰. The original solution assumed no in-plane bending if the hole was located at the center of the plate. Available FE solutions for un-equal crack lengths at a centered hole indicated that crack asymmetry can have a significant effect on in-plane bending. A modified correction for in-plane bending effects was developed using the plate neutral axis as the primary independent variable for the modification. Changes were also made to the solution for crack lengths less than the hole radius. The existing solution assumed the hole was simply part of the total crack and had no other effect on the solution. For the shorter crack lengths, it was necessary to include additional corrections to account for the hole.

This solution appears to be reasonable for Width/Diameter greater than 2 and B/Diameter greater than 1.5 (B is the hole offset measured from the hole center to the near edge of the plate). The new solution is generally within 5% of the FEM results for most of the cases that were tested. For short hole edge distances (B/Diameter < 2), the results showed larger differences. The solution was normalized for each crack length as C/C_{max} , where C_{max} is the longest possible crack length in either direction for a given geometry. When $C/C_{max} > 0.5$, the differences noted with available FEM results increase above 5-10% for some cases. These differences will tend to have a relatively minor effect on total crack growth life since most of the life for a majority of structures is expended at shorter crack lengths.

Suggested Solution Limits:

$W/D > 2$

$B/Diameter > 1.5$

For $C/C_{max} > 0.5$, user's should be aware that the solution is likely to have more significant error

Please note:

When using the Advanced bearing solution, crack(s) may only be placed at the loaded hole. There are currently no solution(s) for cracks growing toward a bearing loaded hole (a very complex case). Any adjacent hole in a model is assumed to be unloaded and open. The adjacent, open hole is generally a conservative assumption, but the effect of any bearing load at the adjacent hole can negate the benefit of hole fill that may otherwise exist.

³⁰ For more information on the modifications, refer to the AFGROW Web Page, 2016 Workshop Presentation: "Stress Intensity Factor Solution for Two, Independent Through-the-Thickness Cracks at an Offset Pin Loaded Hole."

3.2.3.2.3 Double, Non-Symmetric Corner Cracks at a Straight Shank Hole



Figure 98: Double, Unsymmetrical Corner Cracked Hole

Double, unsymmetrical corner cracked hole solutions for multiple load cases (axial, bending, and bearing)³¹ were developed by Fawaz and Andersson (F/A), [44] using an hp-version finite element method (FEM). Solutions were calculated for many combinations of geometric variables as indicated in Figure 99.

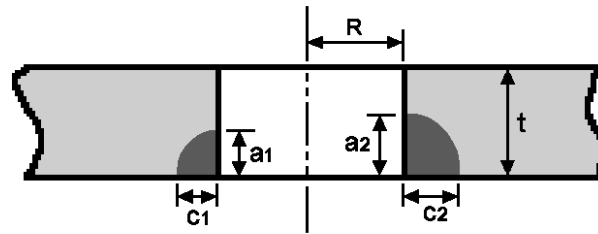


Figure 99: Variables for Corner Cracks at a Straight Shank Hole

All of the cracks were modeled as being part-elliptical in shape. Finite dimension (width and height) effects for the finite element model (FEM) were eliminated using a large plate width ($W/D=100$). The plate height was also very large, relative to the hole diameter ($H/D=100$) to eliminate far field boundary effects at the hole. For a given corner crack configuration, stress intensity solutions were calculated, and are valid³² for the following combinations of geometric parameters:

R/t: 0.1, 0.111, 0.125, 0.1428, 0.1667, 0.2, 0.25, 0.333, 0.5, 0.667, 0.75, 0.8, 1.0, 1.25, 1.33, 1.5, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0

a1/c1, a2/c2: 0.1, 0.111, 0.125, 0.1428, 0.1667, 0.2, 0.25, 0.333, 0.5, 0.667, 0.75, 0.8, 1.0, 1.25, 1.33, 1.5, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0

a1/t, a2/t: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 0.99

Since a general solution for two, non-symmetric cracks was desired; each geometric combination for one crack was modeled for all possible combinations of the above parameters for the second crack – except for R/t since that remains constant for both cracks. The resulting matrix of FEMs is quite large for each load case – $(25 \times 11)^2 = 75,625$ for each R/t (1,890,625 total). Stress intensity values were extracted for as many as seventy-seven points along the crack front for each model. The stress intensity values extracted for

³¹ At transition of the first corner crack, both cracks (C1 and C2) are transitioned to thru-cracks since no solution is available for combinations of corner and thru-cracks.

³² AGROW does not extrapolate beyond the boundaries indicated by the parameter limits.

each FEM are resolved into twenty-five points using curve fitting techniques. An example of this is shown in Figure 100, and the parametric angle, Φ , is defined in Section 3.2.3.1.1.

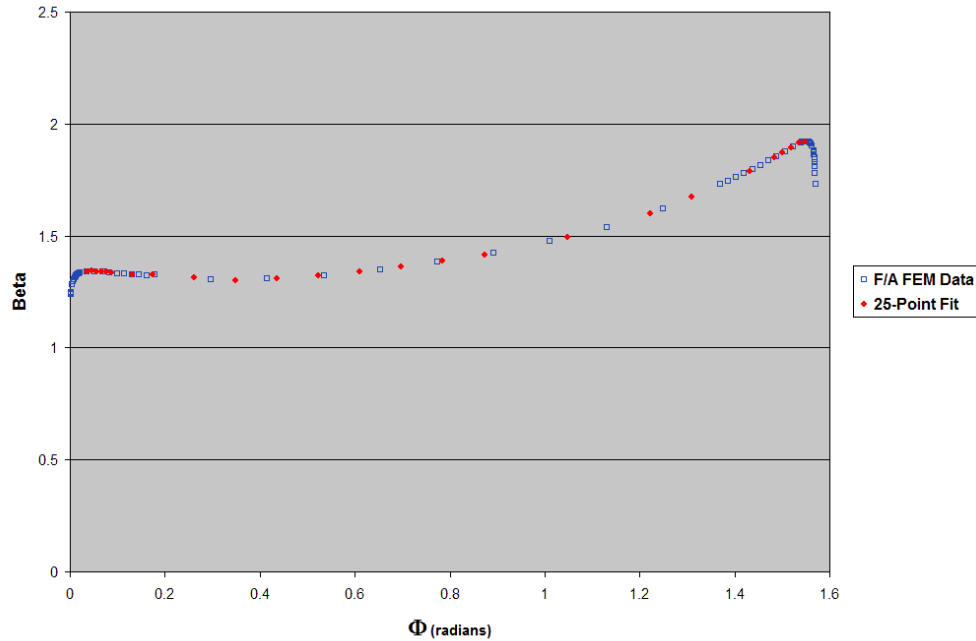


Figure 100: Extraction and Curve Fit Points

The FEM models calculate a stress intensity drop-off as the points approach a free surface³³. The F/A FEMs are finely meshed, and the drop-offs are detected very close to the free edges. The exact position of the effect varies slightly from model to model, and is a function of loading condition. The data in this region is not of interest for crack growth prediction purposes. The data was filtered to find the local maximum values (vertices) to characterize the stress intensity values near each surface. The curve fit points are bounded by the vertices, and twenty-five points³⁴ were found to be the minimum number of points required to interpolate data at any point along the crack front within 1% of the original FEM data. These values are stored in a database for all three load cases and geometric combinations noted above ($3 \times 1,890,625 = 5,671,875$ cases). When this model is initially used, the solution matrix is loaded into memory. This takes some time – depending on the speed of the machine. It may appear that AFGROW is not responding, but the analysis will resume as soon as the data are loaded in memory. These data will remain in memory until AFGROW is closed. The final solution for any given crack geometry is determined using a multi-dimensional interpolation of this matrix. Solutions for this geometry are available for axial, bending, and bearing load cases. Combined load cases are handled as they are in the classic models (Section 3.2.3.1.5). The loading condition is set in the Specimen Properties View as indicated in Figure 101.

³³ The calculated stress intensity values are not valid since the standard square root singularity does not exist at the free surface.

³⁴ In addition to the vertices, $\Phi =$ midpoint to 0.45, 0.45, 1, 3.5, 5, 7.5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 75, 82, 85, 86, 2-points evenly distributed to the vertex (degrees).

Specimen Properties	
Item	Value
(Name)	Specimen
Width	4
Thickness	0.25
Load Tension	1
Load Bending	0
Load Bearing	0
Type	Two Points

Figure 101: Plate Properties

Plate properties are displayed when a user makes a mouse click anywhere on the plate in the animation frame.

Two types of solutions are available for the advanced model, for two corner cracks at a hole, as indicated in Figure 102.

Load Bending	0
Load Bearing	0
Type	Two Points
	Two Points
	Multiple Points

Figure 102: Advanced Solution Types

The two-point solution uses the normalized stress intensity (beta) values at the two vertex points (a and c-directions) for a given crack geometry. As noted earlier, the exact position of these points vary from case to case, but they are the local maximum values that occur very close to each free surface.

Since solutions were extracted along the entire crack front, a multiple-point solution is also available. Using the 25-points available in the F/A database would require significant CPU power. When this solution type is selected, eleven points along the crack front are grown independently for each crack growth increment. After experimenting with several options, it was determined that eleven points could be used to give the same results³⁵ with much shorter run times. Even so, cycle-by-cycle calculations for the 11-point solution can be very slow. Life prediction results using two, ten, and eleven-points are shown in comparison to the comparable twenty-five-point solution in Figure 103 and Figure 104 for the c- and a-dimensions, respectively.

³⁵ The 11-point prediction was within 0.08% of the 25-point prediction in terms of the number of cycles at any crack length. The 2 and 10-point solutions were within 25% and 2.8% of the 25-point solution.

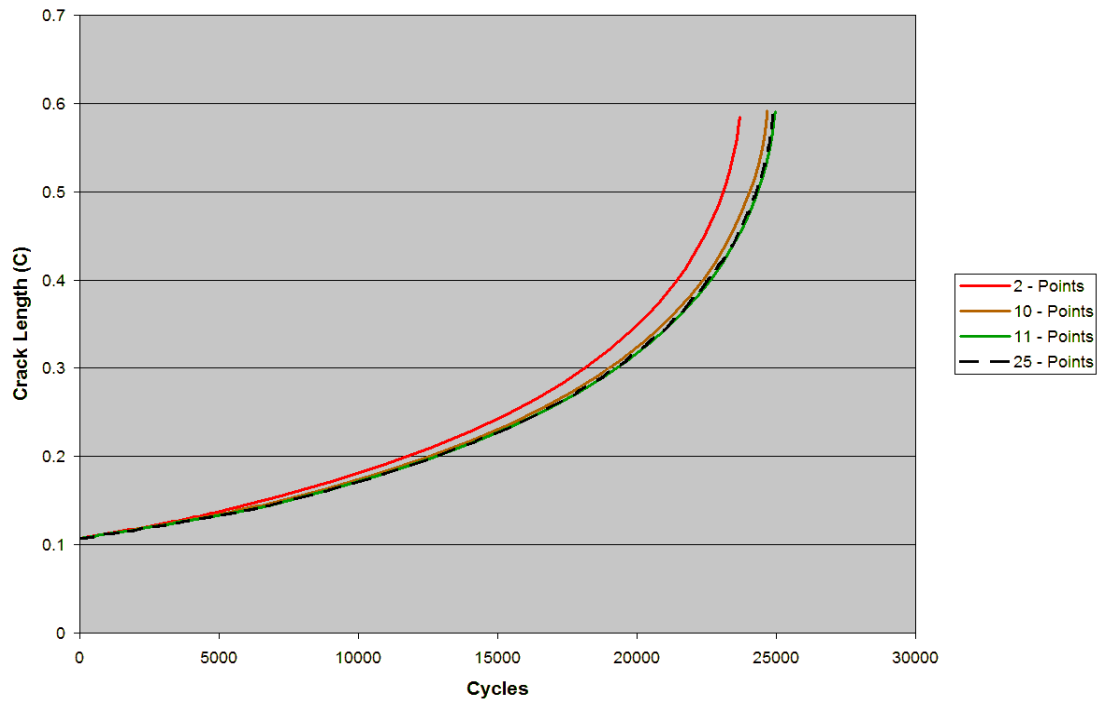


Figure 103: Multi-Point Life Prediction Example (c-direction)

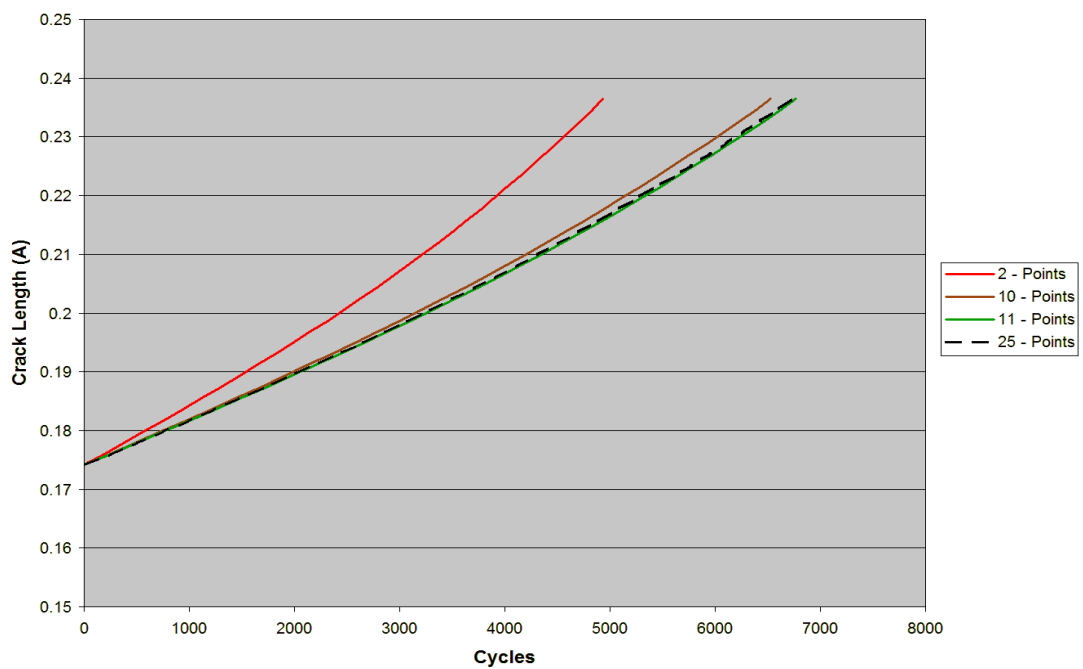


Figure 104: Multi-Point Life Prediction Example (a-direction)

The locations of the points are weighted toward each free surface to better capture the behavior in these regions as shown in Figure 105.

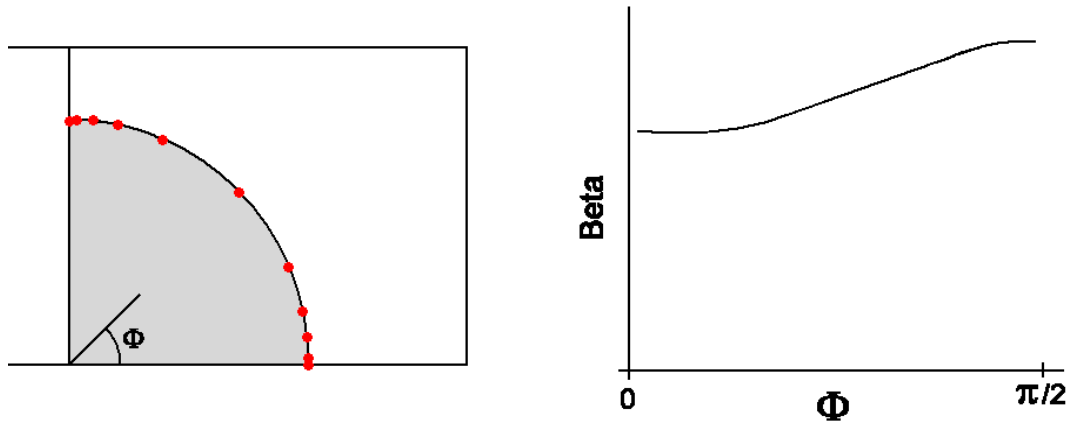


Figure 105: Multiple Point Spacing

The final eleven points³⁶ used in the multiple point option are determined from the twenty-five curve fit points for a given geometric case in the F/A database using multi-dimensional interpolation methods³⁷. Each point is grown based on the local stress intensity value for the user-specified growth increment. At the end of each increment, a quarter-ellipse is fit through all eleven points. This is required to obtain the solution for the next growth increment since the cracks were assumed to be part-elliptical in shape for all of the FEMs.

3.2.3.2.3.1 Finite Width/Geometric Effects

Finite width, hole offset, and effect of an adjacent hole are determined using the method of compound solutions. These additional effects are beta correction factors that are applied (as multipliers) on top of the F/A solutions.

The finite width correction that has become standard for most part-through crack applications [16] is not applicable for bearing load cases. This was discovered when the resulting stress intensity values for bearing were compared to the equivalent remote axial load case for a few relatively narrow plate geometries ($1.5 \leq W/D \leq 4$). The bearing solutions should be higher since the local stress at the hole will be higher under bearing load. When the standard axial finite width correction was applied, the bearing solutions were significantly lower than the corresponding axial case. As a result of this discovery, a large number of FE solutions were required to develop an appropriate bearing width correction.

The double, symmetric crack geometry was selected to allow results to be compared to the F/A wide plate models as well as the classic AFGROW solutions. The verification models included several plate widths ($1.5 \leq W/D \leq 100$) and combinations of crack length. As expected, the axial cases were in reasonable agreement ($\leq 5\%$) with the F/A results (using

³⁶ The 11-points include the vertex points and $\Phi = 1.5, 4, 10, 21, 45, 69, 80, 86$, and 88.5 degrees.

³⁷ This option will run significantly slower than the 2-point method.

the standard finite width correction), but the bearing cases became increasingly un-conservative as the plate width decreased. It was concluded that the standard finite width correction could be applied only to non-bearing load cases (i.e. axial and bending).

The principle of superposition was used to develop a new finite width correction for the bearing load case. This is explained in more detail in Section 3.2.3.2.1.1.5 and illustrated in Figure 72. The F/A bearing solutions were obtained from FEMs for a wide plate ($W/D=100$). It was assumed that this was reasonably close to an infinite plate to allow the use of superposition.

$$\beta_{\text{Bearing}} = \frac{1}{2} \beta_{\text{Axial}} + \beta_{\text{Bearing}(W/D=100)} * F_{\text{WP}} ; \text{ (for any given plate width)}$$

The verification models provided the bearing beta values for a given plate width, and the axial and wide plate bearing values were taken from the F/A database. Of course, the axial result included the standard finite width correction to adjust the F/A wide plate axial results for plate width. The above relationship is easily solved to determine the new width correction for pin loaded holes in plates.

$$F_{\text{WP}} = \frac{\left(\beta_{\text{Bearing}} - \frac{1}{2} \beta_{\text{Tension}} \right)}{\beta_{\text{Bearing}(W/D=100)}}$$

It was logical to assume that F_{WP} would be a function of crack length and plate width, as is the case with the standard open hole (axial) width correction. After examining the data, it appeared that F_{WP} was most directly related to the cracked fraction of the plate as measured from the center of the hole in the c-direction (width direction). This parameter, a/b , varies along the crack front since the crack projection in the c-direction is different at each point. The parametric angle, Φ (Section 3.2.3.1.1), is used to define the location of any point along the crack front.

For example:

$$\text{If } \Phi=0, \frac{a}{b} = \frac{2(R+C)}{W} \text{ (for a crack at a centered hole in a plate)}$$

$$\text{If } \Phi=1.5707, \frac{a}{b} = \frac{2R}{W} = \frac{D}{W} \text{ (for a crack at a centered hole in a plate)}$$

The F/A data for the two-point solution was being used to determine F_{WP} , and the resulting values were curve fit for each point (c and a-directions). The cases shown in Figure 106 and Figure 107 are for $A/C = 1$, and $1.5 \leq W/D \leq 100$.

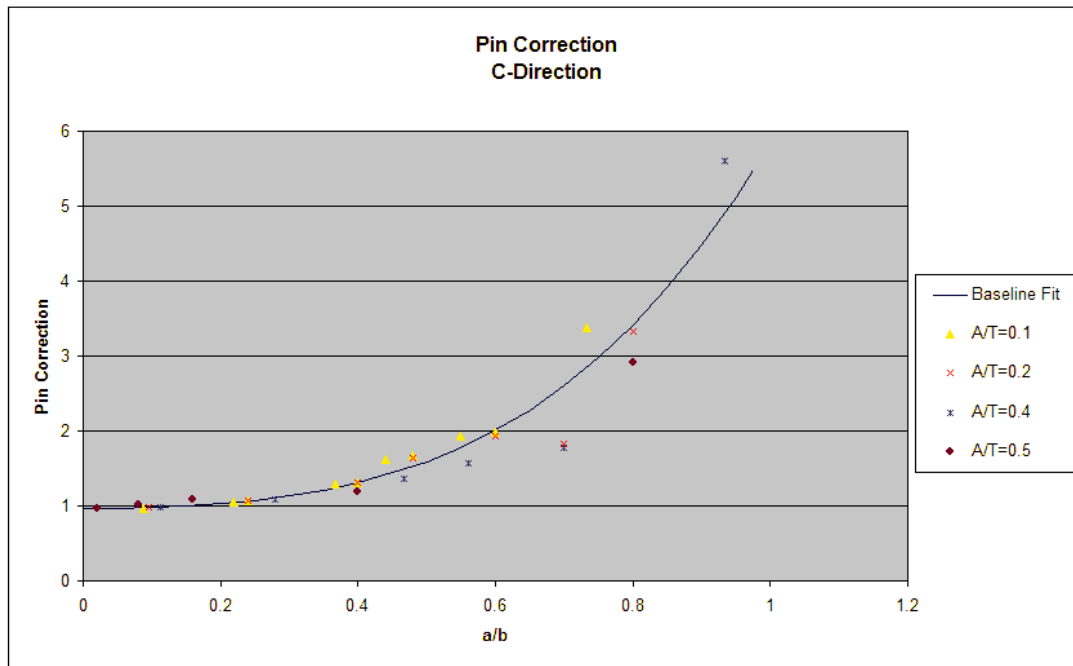


Figure 106: Pin Correction in the C-Direction

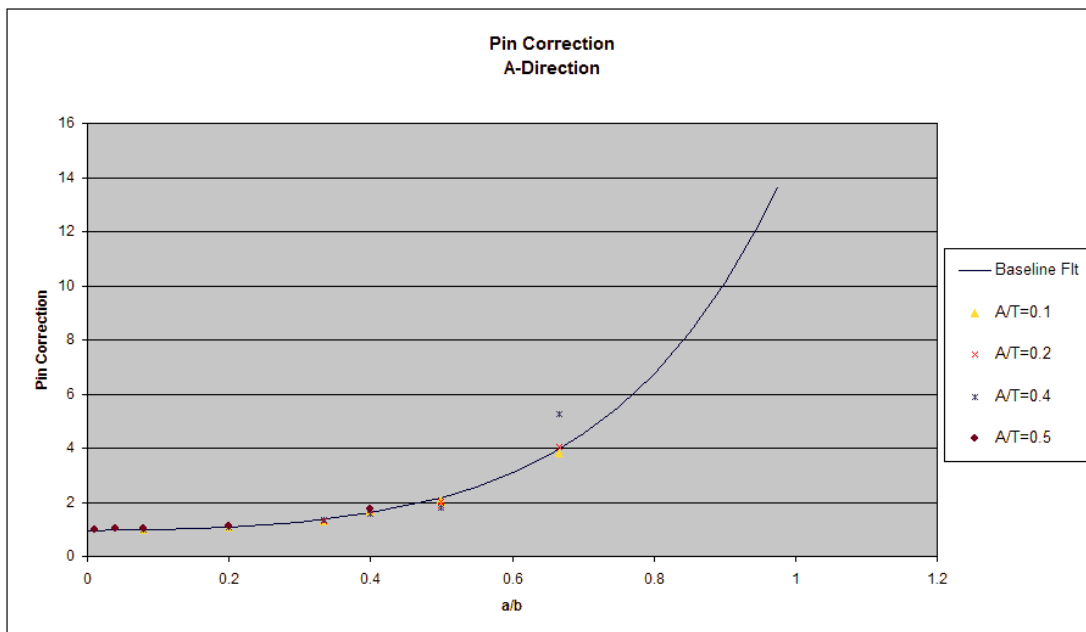


Figure 107: Pin Correction in the A-Direction

For the two-point data, Φ varied from case to case depending on the location of the local maximum stress intensity. However, the distance to each free edge was small in every case, so Φ was assumed as indicated below.

C-Direction: $\Phi = 0$

A-Direction: $\Phi = 1.5707$

The relationship between a/b and F_{WP} was then determined by curve fitting the data with polynomial terms.

$$F_{WP} = \left[0.96 + 0.1 \left(\frac{a}{b} \right) + 0.75 \left(\frac{a}{b} \right)^2 + 2.25 \left(\frac{a}{b} \right)^3 + 1.8 \left(\frac{a}{b} \right)^4 \right] \left[1 + \Phi \left(\frac{a}{b} \right)^{2.1} \right]$$

The curve fit solution for F_{WP} goes to 0.96 instead of 1.0 (as would be expected) as a/b goes to zero. This is most likely due to the fact that the F/A wide plate bearing solution was developed for $W/D=100$, and is somewhat higher than the infinite plate solution. Although the difference between the infinite plate and $W/D=100$ solutions is probably much less than 4%, it should be noted that the total bearing solution also includes one half of the axial solution. The value of F_{WP} is only applied to the F/A wide plate portion of the total bearing solution.

When the above solution is applied to the verification models, the resulting errors are shown in Figure 108 and Figure 109.

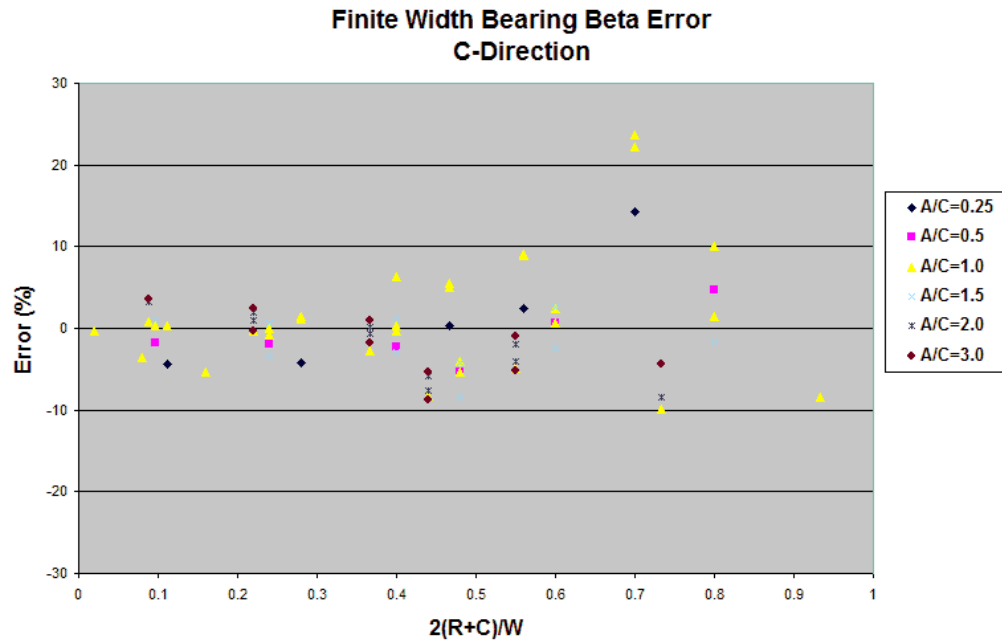


Figure 108: Bearing Beta Solution Error in the C-Direction

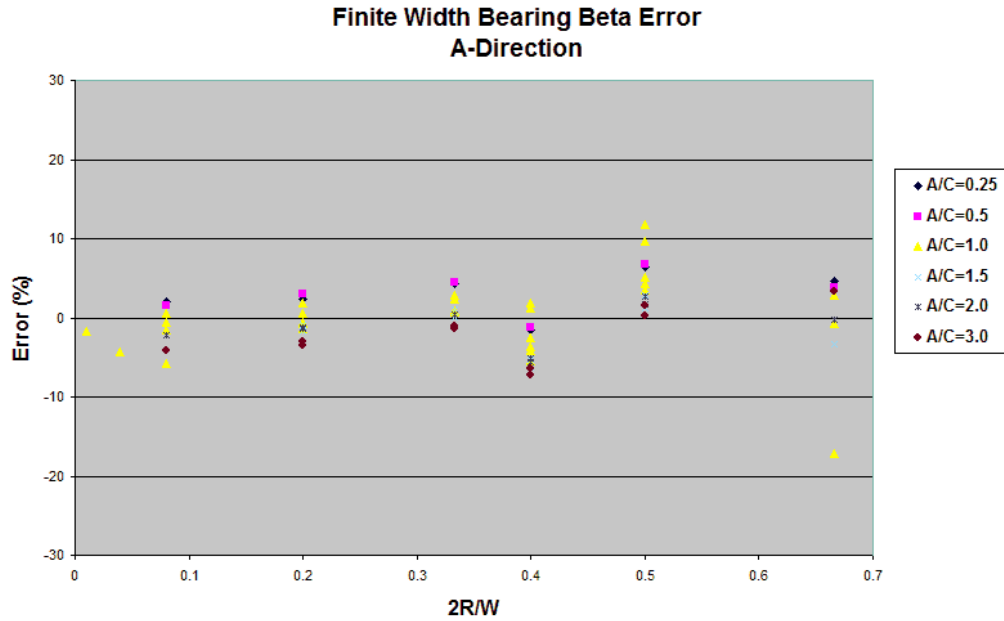


Figure 109: Bearing Beta Solution Error in the A-Direction

The current solution is a great improvement over the use of the standard finite width correction, but there is clearly room for improvement. The data from the verification models shows the error to be within approximately 10% for $a/b \leq 0.6$.

The standard finite width correction is documented to be valid for $a/b \leq 0.5$ [16]. The same correction is applied to both crack dimensions for axial and bending. Data based on other 3-D FE analyses performed for the axial load case, have shown that the magnitude of the required correction is actually different for the two dimensions. This difference increases drastically as W/D becomes small. This is undoubtedly becoming a factor as a/b increases. The limit ($a/b \leq 0.5$) is exceeded at $C=0$ when $W/D < 2.0$, and even higher values of W/D as the crack grows. This means that there are many practical applications for which the current finite width correction is in doubt. In the future, we plan to develop a new finite width correction. When that task is complete, this solution will also be re-examined and changed as required to give the most accurate solution possible.

3.2.3.2.4 Single/Double, Symmetric Corner Cracks at a C/S Hole

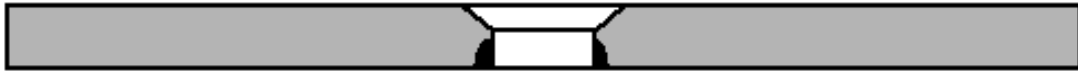


Figure 110: Double, Symmetric Corner Crack(s) at a C/S Hole

The double, symmetrical corner crack at a 100 degree countersunk hole solution for multiple load cases (axial, bending, and bearing) was developed by Reinier de Rijck, [76] using an hp-version finite element method (FEM). Solutions were calculated for many combinations of geometric variables as indicated in Figure 111.

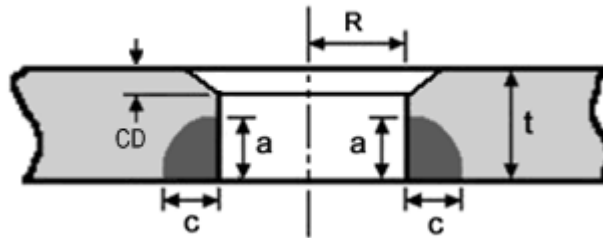


Figure 111: Variables for the C/S Hole Solution

The cracks were modeled with a part-elliptical crack shape. Finite dimension (width and height) effects for the finite element model (FEM) were eliminated using a large plate width ($W/D=100$). The plate height was also very large, relative to the hole diameter ($H/D=100$) to eliminate far field boundary effects at the hole. For a given corner crack configuration, stress intensity solutions were calculated, and are valid³⁸ for the following combinations of geometric parameters:

R/t: 0.1, 0.25, 0.5, 0.667, 1.0, 1.2, 1.5, 2.4

CD/t: 0.5, 0.75, 0.95

a/c: 0.5, 0.667, 0.75, 1.0, 1.5, 2.0, 5.0

a/t: 5 values in the straight shank portion, 10 values in countersunk area (up to $c/R = 5$)

The corner crack(s) may only be placed on either side of a single hole along the faying surface (away from the countersink). A single crack may be placed on either side of the hole, and the solution will be determined from the double cracked results using the Shah correction to convert to a single crack solution. When the crack transitions along the countersink, the a-dimension is defined as the distance through-the-thickness at the point of intersection with the countersink. This provides consistency with the definition of a/t and will not permit the a-dimension to exceed the thickness.

³⁸ AGROW does not extrapolate beyond the boundaries indicated by the parameter limits with an exception for $CD/t < 0.5$ where interpolation is used between the straight shank solution to ensure the solutions converge as CD/t goes to zero.

The user-defined properties for the countersunk hole are shown in Figure 112. Users may set the diameter, offset (from the left edge), and the depth of the countersink. Since no FE solutions are available for countersink depths (CD) less than $t/2$, we have implemented a scheme to estimate the solution based on interpolating between the countersunk hole solution for $CD/t=0.5$ and the straight shank hole solution.

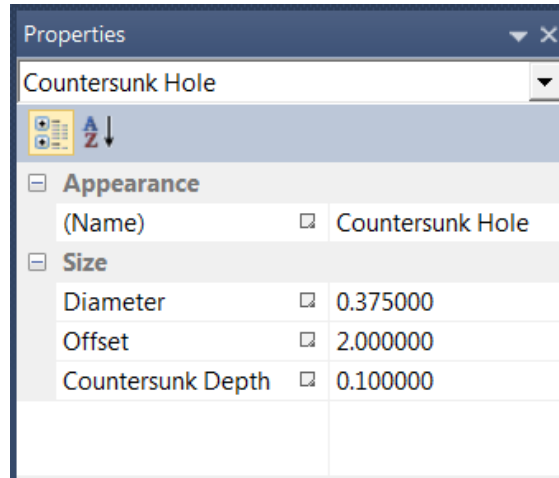


Figure 112: C/S Hole Properties

Due to some difficulties verifying the bearing solutions, we have only implemented the axial and bending load case at this time. We realize that most users want the bearing load case, and we will continue to work to implement it as soon as possible. In addition, please note that the current implementation will not allow other holes to be placed on either side of the cracked hole. This is another unfortunate consequence of complications with the oblique crack and the way the crack is grown along the countersink. We will also continue to work on this issue. In the meantime, we wanted to make this solution available as it is, and we welcome any comments and/or suggestions.

3.2.3.2.5 Single/Double, Symmetric Corner Cracks at the Base of a C/S Hole

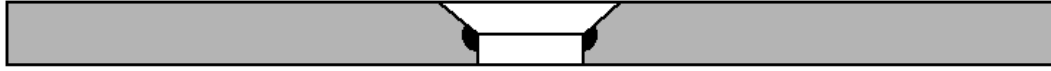


Figure 113: Corner Crack(s) at the Base of a C/S Hole

The Advanced tabular K-solution for a Single corner crack at the base of a 100 degree countersunk hole was developed by Jody Cronenberger for the axial load case [82] using StressCheck®, a p-version finite element analysis program. Solutions were calculated for many combinations of geometric variables as indicated in Figure 114. Solutions for double, symmetric corner cracks are estimated using the Shah correction to the single crack solution.

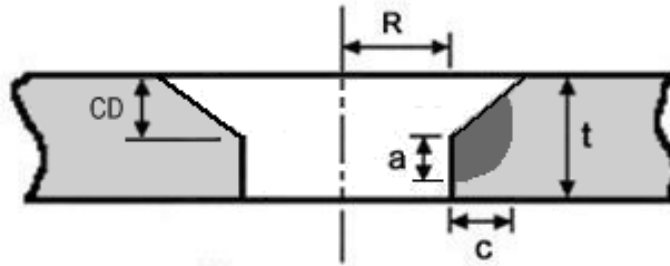


Figure 114: Variables for Corner Crack(s) at the Base of a C/S Hole

All of the cracks were modeled as being elliptical in shape where the crack dimensions (a and c) are the semi minor and major axes of an ellipse centered at the base “knee” of the countersunk hole (crack origin). The solution space used for this solution was based on a survey of aircraft structure.

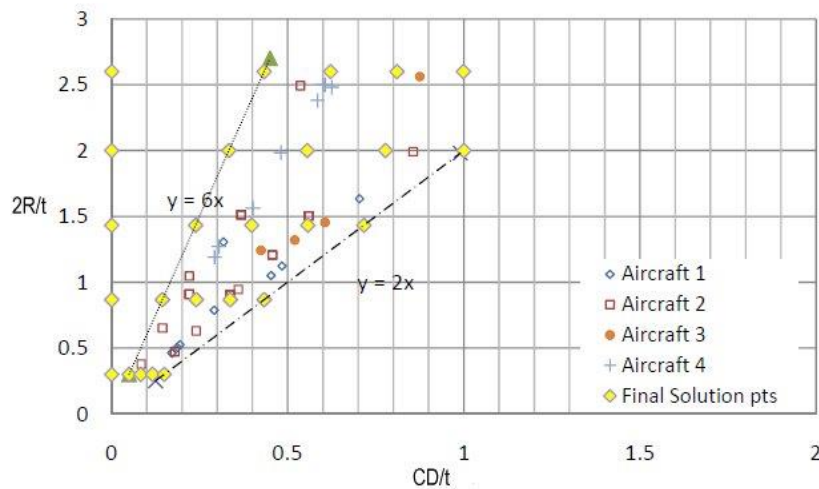


Figure 115: Solution Space from the Aircraft Survey

The limits used for countersink depth (CD), thickness (t), and hole diameter (2R) are given below³⁹:

$$\text{Maximum } 2R/t = 6 * CD/t$$

$$\text{Minimum } 2R/t = 2 * CD/t$$

There is a practical limit on the fastener sizes used for a given thickness, and the fastener size dictates the CD value required. Based on the results of the survey, the values for 2R/t used in the tabular solution are given below.

$$2R/t = 0.3, 0.867, 1.433, 2.0, \text{ and } 2.6$$

The limits used for crack lengths were also based on the aircraft survey so that a minimum crack length for this K-solution in the c-direction was always ≤ 0.005 inches for all practical cases.

$$\text{Minimum } c/t = 0.0125 * 2R/t + 0.002$$

A practical limit for the maximum crack length in the c-direction was also based on the survey and the fact that the effect of the hole will be negligible for cracks beyond 4R.

$$\text{Maximum } c/t = 2.5 * 2R/t$$

The tabular solution is based on 14 crack lengths in the c-direction that are spaced uniformly in logarithmic scale so that the smaller crack lengths are better represented. This improves the interpolation accuracy since the K-solution changes more quickly for shorter crack lengths.

The a-lengths for the tabular solution are based on five, a/c ratios used in the Finite Element (FE) analyses.

$$a/c = 0.5, 1.0, 1.5, 2.0, \text{ and } 4.0$$

Therefore, the crack shape is limited as shown below.

$$0.5 \leq a/c \leq 4.0$$

When the a-dimension extends beyond the physical boundary of the plate, the point of intersection of the ellipse drawn using the a and c-dimensions as the semi-major and minor axes and the plate boundary define the Ct dimension of an oblique crack for this model as indicated in Figure 116.

³⁹ In the aircraft survey, values for 2R/t were in the range ($0.3 \leq 2R/t \leq 2.6$)

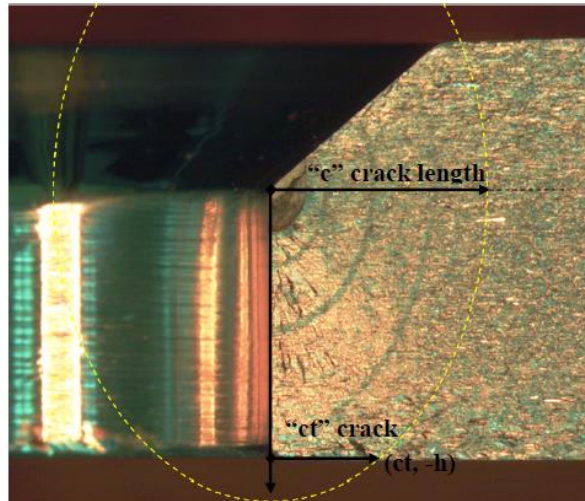


Figure 116: Oblique Crack Dimensions

Transition to a straight through crack at a hole occurs when $a/c > 4$ since no solution exists in the table for higher a/c values. At this point, the solution transitions to the Advanced through crack at a straight shank hole solution (without the C/S).

Please note that solutions for this geometry are only available for the axial load case at this time. We will continue to work to obtain solutions for the bearing and bending load cases as soon as possible.

3.2.3.2.6 Continuing Damage Solution

When a crack grows from a hole to the near edge of a plate, it is often necessary to continue the life prediction process for a crack that may be assumed to exist on the opposite side of the hole.

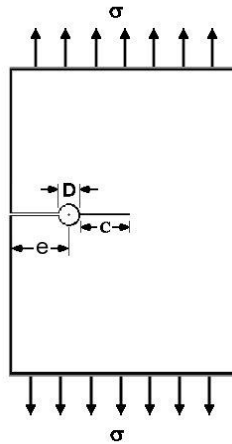


Figure 117: Continuing Damage Geometry

A slot object has been added to the Advanced Model interface so that users may drag it to fill the space between a hole and the edge of a plate as indicated in Figure 118. This simulates a crack that has grown from the hole to that edge.

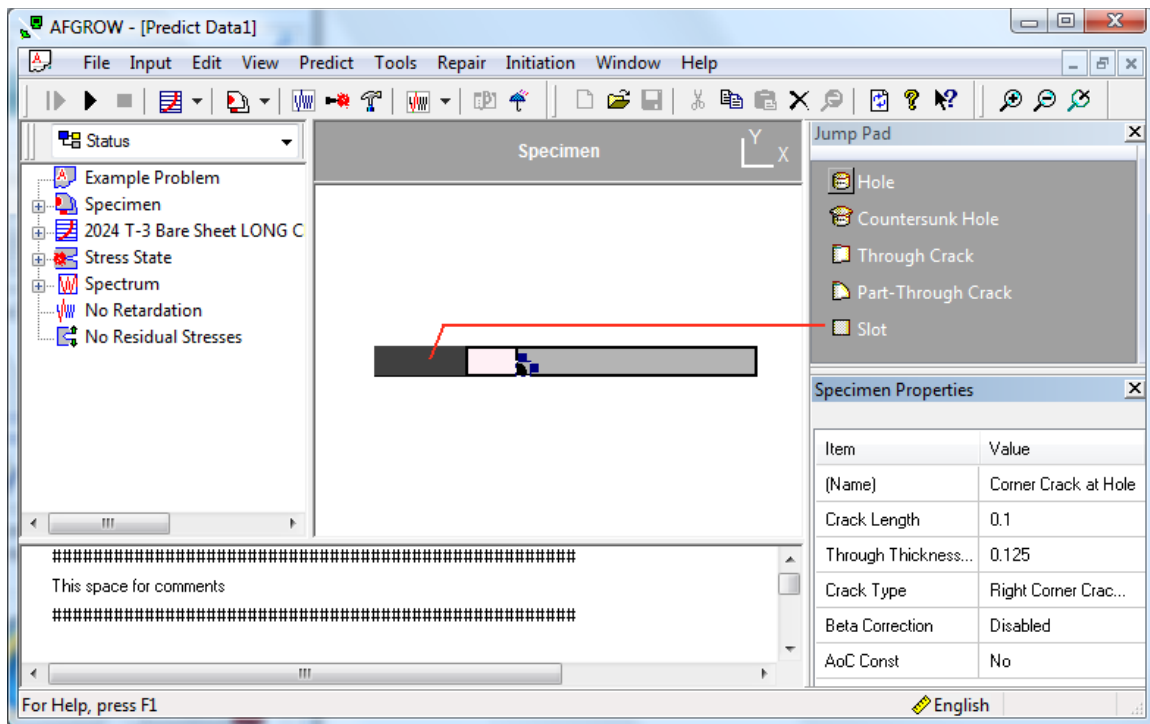


Figure 118: AFGROW Continuing Damage Model

AFGROW uses the superposition principle to account for the effect of additional holes may be placed in the path of the growing continuing damage crack. Once the crack reaches another hole, the prediction stops, and the user may create a new model to resume crack growth from the next hole.

Since many aircraft applications include panels with frames, in-plane bending is constrained by default. However, the interface includes an option to remove the in-plane bending constraint if desired. This option is now at the specimen level of the Properties view as indicated in Figure 119.

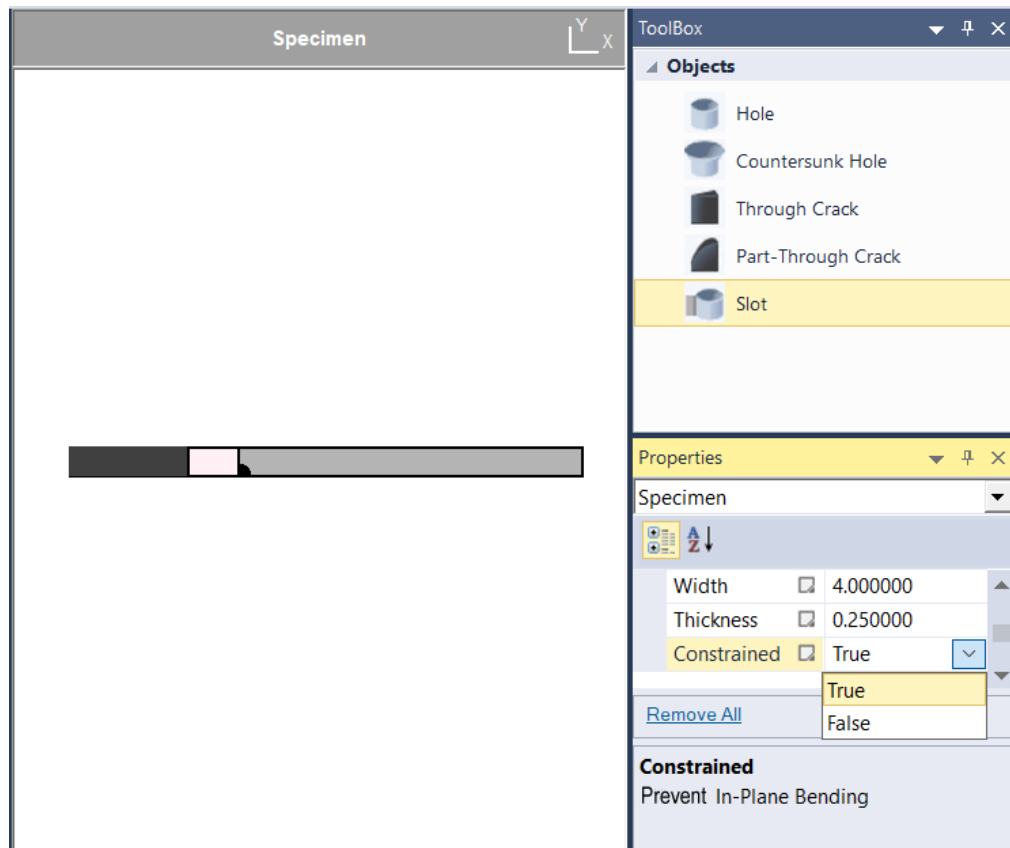


Figure 119: Continuing Damage Bending Constraint

Continuing damage solutions are available for the load cases described in the following sections. Errors above 10% have been noted for some cases outside of the suggested solution limits for each load case. The user notification list will be updated when these limits are exceeded.

3.2.3.2.6.1 Axial Load Case

The continuing damage solution in AFGROW [75] was developed for the axial load case for through-the-thickness and corner cracks at a straight shank hole.

The axial continuing damage solution for the through-the-thickness crack geometry is valid for all combinations of crack length, plate width, notch depth, hole offset and diameter and was shown to be within approximately 5% of all FEM solutions in reference [75]. The corner crack solution demonstrated more variability but remained within 10% of all FEM solutions for $C < 20\%$ of the un-cracked net section (W - notch depth). As stated above, the continuing damage solution is currently only applicable for the axial load case. When used as part of a total life prediction, 100% axial loading will be assumed for this portion of the analysis.

Suggested Limits:

$$0.5 \leq a/c \leq 3.0$$

$$0 < a/t < 0.95$$

$$0.25 \leq r/t \leq 2.0$$

$$c < 0.2 * (W - e - r), \text{ for corner cracks}$$

3.2.3.2.6.2 Transverse Bending Load Case

The transverse (out-of-plane) bending load case is currently unavailable until a solution can be developed for the corner crack geometry.

3.2.3.2.6.3 Bearing Load Case

A proprietary curve fit solution for the bearing load case has been developed for through-the-thickness and corner cracks at a straight shank hole by LexTech, Inc.

The continuing damage solution for the through-the-thickness crack geometry is valid for all combinations of crack length, plate width, hole offset and diameter for notch depths less than one half the total plate width as indicated below. Errors greater than 10% may occur when the notch depth exceeds this limit. The corner crack solution demonstrated more variability but generally remained within 5% for all cases that were tested.

Suggested Limits:

$$0.5 \leq a/c \leq 3.0$$

$$0 < a/t < 0.95$$

$$0.25 \leq r/t \leq 2.0$$

$$e/W < 0.5$$

3.2.4 Input Spectrum

Toolbar Icon:



The spectrum dialog, Figure 120, provides a means of specifying the load/stress spectrum to be used by AFGROW.

Spectrum

Stress Multiplication Factor [SMF] multiplies the stress or load levels found in spectrum files. This allows normalized spectra to be used. If actual stress levels are used in the spectrum files, SMF should be set to 1.

Residual Strength Requirement [Pxx] is the value of stress (or load for models using load input) which must be carried at all crack sizes. It is used to determine the critical crack size - if a non-zero value is entered.

Stress Preload [SPL] is used to account for pre-existing stresses. This value is added to the max and min spectrum stresses after they have been multiplied by SMF.

Enter

Stress Multiplication Factor(SMF): 14

Residual Strength Requirement (Pxx): 0

Stress Preload (SPL): 0

Select

☒ Create new spectrum file

☐ Open spectrum file

☐ Constant amplitude loading

OK

Cancel

Figure 120: Input Spectrum Dialog

3.2.4.1 Spectrum Dialog Options

3.2.4.1.1 Spectrum Multiplication Factor (SMF)

The spectrum multiplication factor is multiplied by each maximum and minimum value in the user input stress spectrum. This allows a user to input spectra, which are normalized (maximum value = 1), and simply use one multiplication factor to predict the life for different stress levels. Of course, this can be done for non-normalized spectra as well, but may be awkward since it requires the user to calculate the appropriate multiplication factor for the actual maximum value in the spectrum.

3.2.4.1.2 Residual Strength Requirement (Pxx)

This value is the source of some confusion, but it is really a simple variable. It is simply the value of stress⁴⁰, which the structure **MUST** be able to carry at all crack sizes. This value is **NOT** multiplied by SMF. It is very useful for cases in which users don't know

⁴⁰ Load is used for certain models – refer to the icon to the right of this variable

when the maximum stress (or load) will occur and wish to check for failure at all times for this condition. If this value is set to zero (default), failure will occur based on the current applied stress (or load for some models). If the input spectrum is large with only one high stress value, the default condition could cause AFGROW to over predict the life depending on the crack length when the high stress was applied.

3.2.4.1.2 Stress Preload (SPL)

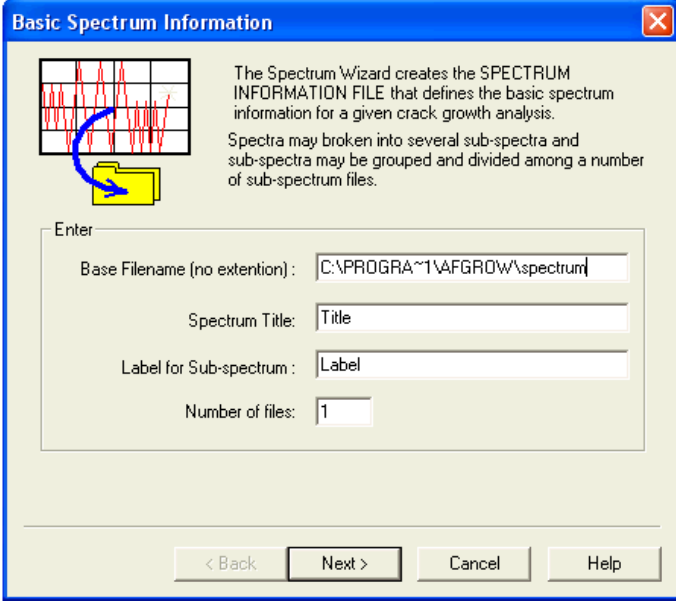
The option allows for a pre-existing stress (or load) to be taken into account in the analysis. Any value entered for SPL will be added to the spectrum maximum and minimum values (after SMF has been applied). The most likely application of this would be for cases where there was some amount of dimensional error in assembly of a component.

3.2.4.1.3 Create New Spectrum File

Opens the Spectrum Wizard that directs the user through several steps to create a loading spectrum:

Step 1: Spectrum Information

At least two files are required to specify any spectrum for AFGROW. The first file is called a spectrum information file that is named [filename].sp3 and the subsequent file(s) contain the actual spectrum data (see Figure 121). The filename convention is [filenameXX].sub, where XX is a two digit file number (from 01 to 99).



Basic Spectrum Information

The Spectrum Wizard creates the SPECTRUM INFORMATION FILE that defines the basic spectrum information for a given crack growth analysis. Spectra may be broken into several sub-spectra and sub-spectra may be grouped and divided among a number of sub-spectrum files.

Enter

Base Filename (no extension): C:\PROGRA~1\AFGROW\spectrum

Spectrum Title: Title

Label for Sub-spectrum: Label

Number of files: 1

< Back Next > Cancel Help

Figure 121: Spectrum Information Dialog

The information entered in this dialog will be saved in the [filename].sp3, which this wizard will create.

Wizard Options:

Base Filename: The filename of the spectrum information file without an extension.

Spectrum Title: Provided for reference or documentation purposes.

Label for Sub-spectrum: Provided to identify what each sub-spectrum represents (flights, hours, blocks, etc.).

Number of Files: Number of files containing the actual spectrum data.

Note: While it is acceptable to use a single file for the actual spectrum data, it may be useful to divide the data into more than one file so it is easier to edit the files if necessary. AFGROW can work with a spectrum file of any size, but no sub-spectrum may exceed 4MB. The number of sub-spectra is unlimited.

Help: Displays the help topic for this step. Users may also press F1 for help.

Cancel: Cancels your previous actions and closes the Wizard.

Back: Disabled in this step.

Next: Move forward to the next step.

Step 2: Type of Spectrum

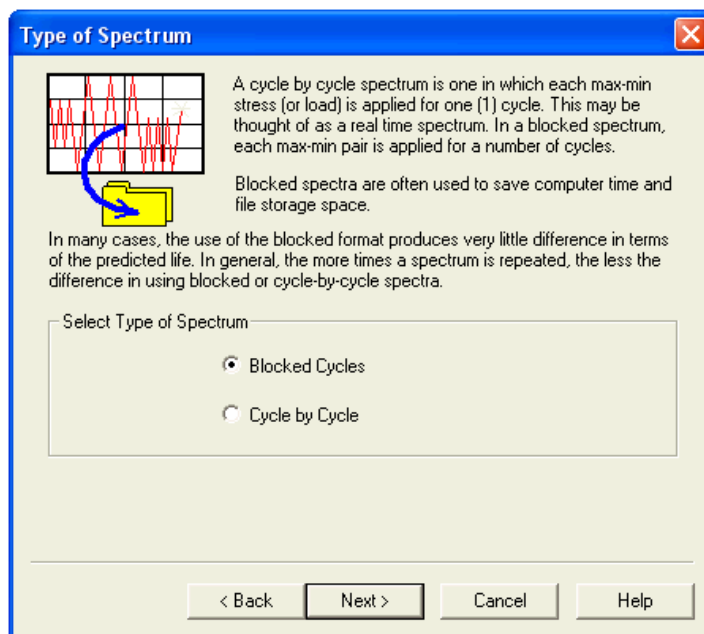


Figure 122: Spectrum Type Dialog

AFGROW uses the term Blocked Cycles to indicate that each Max, Min Stress level may consist of multiple cycles.

The term Cycle by Cycle means that each Max, Min Stress level must only have one cycle.

Note: Although AFGROW expects a Cycle by Cycle spectrum to have one cycle per level, the format of the data must be in the form Max Min 1, where 1 is the number of cycles. In this way, the file format is consistent. AFGROW will not accept a Cycle x Cycle spectrum unless the number of cycles for each stress level is one.

Wizard Options:

Help: Displays the help topic for this step. Users may also press F1 for help.

Cancel: Cancels your previous actions and closes the Wizard.

Back: Move back to the previous step.

Next: Move forward to the next step.

Step 3: Number of Sub-Spectra

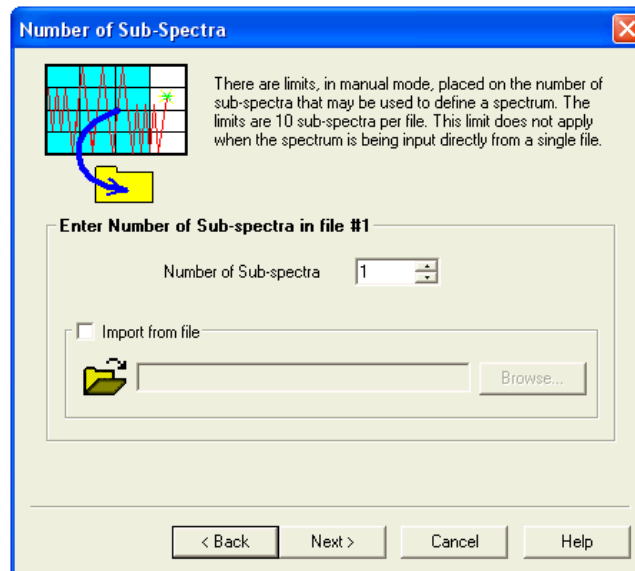


Figure 123: Sub-Spectra Dialog

Wizard Options:

Number of Sub-spectra: This is only used for manual spectrum data entry. This wizard allows users to create small spectra by manual input of up to 10 sub-spectra. This option will be ignored if the spectrum data are being read from a file (Import from file option).

Import from file: The wizard can import a complete spectrum file containing an unlimited number of sub-spectra⁴¹. This file may be a standard AFGROW spectrum file *.sub or a user created ASCII file in the following format:

[*number of stress levels*]

[*max stress*] [*min stress*] [*cycles*] ← Repeat for each stress level

[*number of stress levels*]

[*max stress*] [*min stress*] [*cycles*] ← Repeat for each stress level

(Repeat until data for all sub-spectra have been entered)

Browse: Opens Standard Windows Open File dialog if import from file option is selected.

Help: Displays the help topic for this step. Users may also press F1 for help.

Cancel: Cancels your previous actions and closes the Wizard.

Back: Move back to the previous step.

Next: Move forward to the next step.

Step 4: Number of Stress Levels

Sub-spectra are the smallest unit of the total spectrum that AFGROW can read into memory at once. They are the building blocks of any AFGROW spectrum. If the total spectrum will fit in the allocated memory (currently 4MB), all of the data may be placed in a single sub-spectrum. The minimum size of a sub-spectrum is one stress level.

⁴¹ Refer to section 3.2.4.2 for more information on how sub-spectra are used in AFGROW

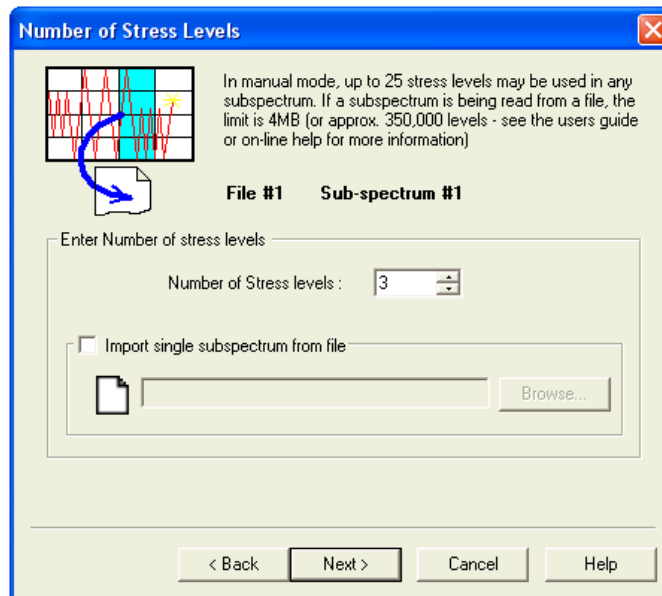


Figure 124: Stress Level Dialog

Wizard Options:

Number of Stress Levels: This is only used for manual spectrum data entry. This wizard allows users to create small spectra by manual input of up to 25 levels of spectrum data for a given sub-spectrum. This option will be ignored if the spectrum data is being read from a file (Import single sub-spectrum from file option).

Import single sub-spectrum from file: The wizard can import a file containing spectrum data for a single sub-spectrum⁴². This file must be an ASCII text file in the following format:

[*max stress*] [*min stress*] [*cycles*]

(Repeat until all of the sub-spectrum data have been entered)

Browse: Opens Standard Windows Open File dialog if import from file option is selected.

Help: Displays the help topic for this step. Users may also press F1 for help.

Cancel: Cancels your previous actions and closes the Wizard.

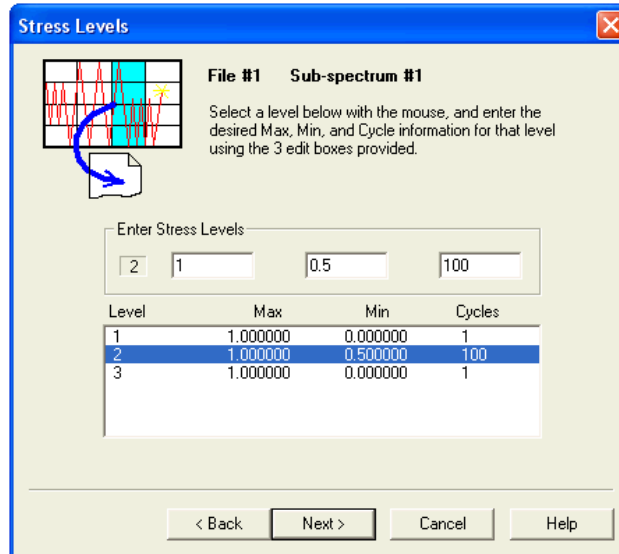
Back: Move back to the previous step.

⁴² Refer to section 3.2.4.2 for more information on how sub-spectra are used in AFGROW

Next: Move forward to the next step.

Step 5: Stress Levels

This page is only used for manual spectrum data entry. Maximum of 25 stress levels may be entered manually. If larger spectra are required, use the read sub-spectrum from file option.



Level	Max	Min	Cycles
1	1.000000	0.000000	1
2	1.000000	0.500000	100
3	1.000000	0.000000	1

Figure 125: Stress Levels

Simply highlight the row you wish to edit in the table (click on it), and enter the values in the appropriate box above the table. Pressing [enter] will cause the change to be registered in the table.

Wizard Options:

Help: Displays the help topic for this step. Users may also press F1 for help.

Cancel: Cancels your previous actions and closes the Wizard.

Back: Move back to the previous step.

Next: Move forward to the next step.

Step 6: Summary

This is the final dialog box for the spectrum creation wizard.

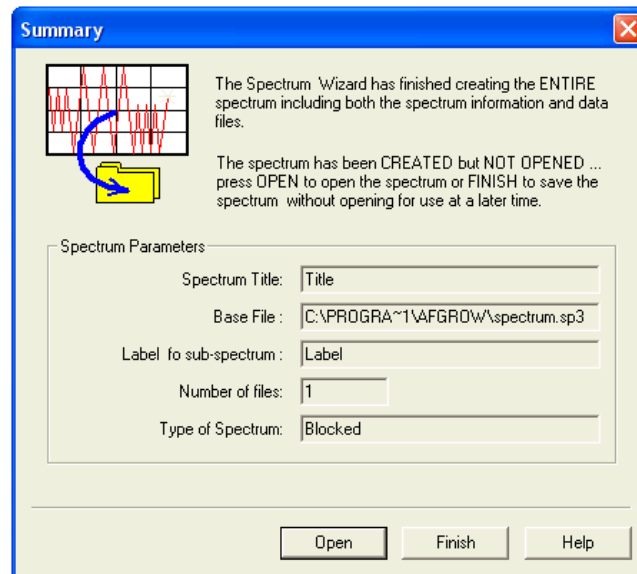


Figure 126: Spectrum Wizard Finish Dialog

The basic spectrum information for the newly created spectrum is shown in this dialog.

Wizard Options:

Open: Saves and Opens the new spectrum in AFGROW and closes the wizard.

Finish: Saves the spectrum file and closes the wizard. Note: The spectrum that was created WILL NOT be opened. The newly created spectrum must be opened before it can be used.

Help: Displays the help topic for this step. You can also press F1 for help.

3.2.4.1.4 Open Spectrum File

Opens the Windows standard Open File Dialog. The Open File Dialog will look in the AFGROW directory by default, but the spectrum files may be in any directory if the user has the appropriate read/write permissions.

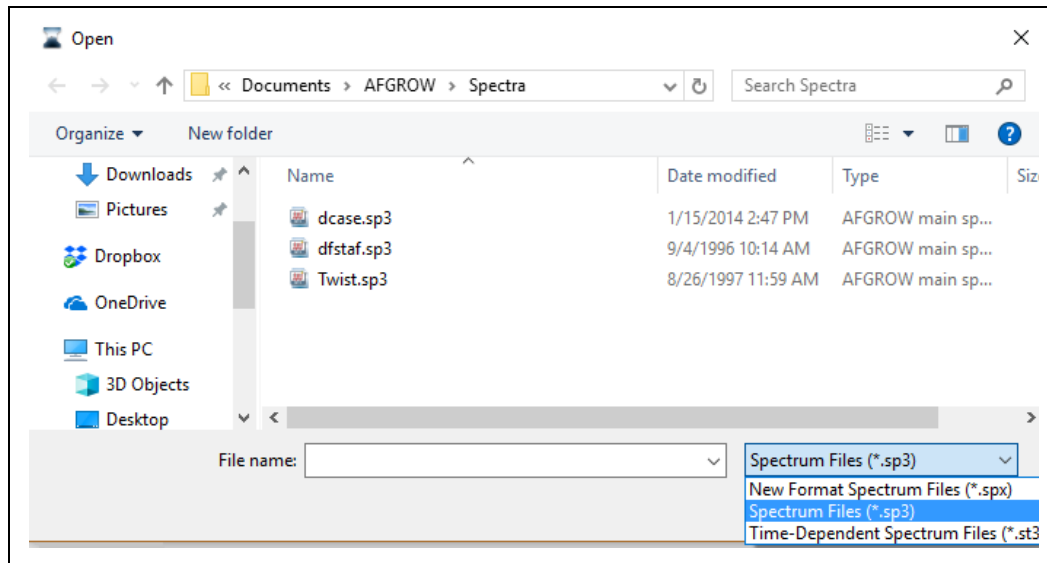


Figure 127: Open a Spectrum File

The user may select a previously created AFGROW spectrum in the desired file format for use in a given life prediction analysis. **All spectra must be cycle counted** (see Section 3.5.3).

3.2.4.1.5 Constant Amplitude Loading

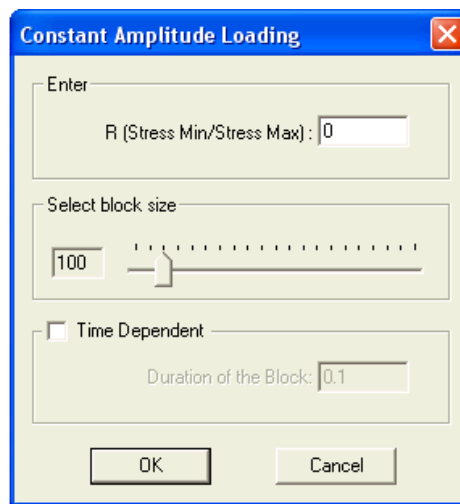


Figure 128: Constant Amplitude Spectrum Dialog

AFGROW provides a method to generate a simple constant amplitude loading spectrum. The stress ratio (R) is the ratio of the minimum to maximum stress level⁴³. The block size is used to determine the number of constant amplitude cycles in one pass of the constant amplitude spectrum. AFGROW uses a Vroman integration scheme to help reduce the time required for life analysis. The use of larger blocks will tend to reduce the time required for analysis but may also reduce the accuracy depending on other user defined software settings.

If the time dependent option is selected (and time dependent crack growth rate data has been entered - see Section 3.5.4), AFGROW will include time dependence in addition to the standard cyclic dependent crack growth calculations. An entry is required to define the duration of the user-defined sub-spectrum block (in **seconds**). Each cycle in the constant amplitude block is assumed to be in the form of a sine wave and each cycle is broken into 100 discrete steps in time where the mean stress for each segment is used to calculate K and the resulting growth rate is used to determine any crack extension over the given time interval. Whenever time dependent spectra are used, AFGROW will automatically determine the time per pass through the spectrum and will over-write any input value in the predict, preferences dialog for the number of hours per pass through the spectrum.

3.2.4.2 General Spectrum Format Information

AFGROW spectra input must represent the COMPLETE cycle counted history of stresses (or loads) to be applied. AFGROW does NOT use partial cycles in user-defined sequences. The input spectra may be as large as required to represent the desired loading sequence. The only limit is the size of a given sub-spectrum. At this time, sub-spectra may not exceed 4MB (~320,000 levels). However, any number of sub-spectra may be used to define a complete spectrum.

A new Spectrum Management Tool is now available for all users with a current license for AFGROW, Version 5.3 and higher. This tool allows users to create/edit spectrum files via manual entry and/or by copy and paste from other Windows applications (e.g. Excel). It includes the ability to clip, truncate, and normalized/de-normalized spectra. This tool may also be used to create three loading channels (axial, bending, and bearing). The Spectrum Management Tool and detailed documentation is available on the AFGROW web page.

Spectra may be represented in any of the formats explained in the following sections that is appropriate for the type of analysis required.

3.2.4.2.1 Standard Spectrum Text Format

⁴³ The maximum value is assigned to be 1.0. The Stress Multiplication Factor (SMF) is required to set the actual maximum stress (or load) value since it is multiplied by each value (maximum and minimum stress) in a given spectrum.

The standard spectrum is used to determine crack growth life on a cyclic basis alone (no time dependence). The first file for the standard spectrum is called a spectrum information file (named [*filename*].sp3) with the format shown below:

[*Title*]

[*sub-spectrum label*] (i.e. Flight, Block , Hour, etc.)

[*type of spectrum*] (Either **BLOCKED** or **CYCLexCYCLe**)

[*number of files to follow*]

The other files associated with the spectrum contain the actual stress (or load) information. Remember that if the spectrum is specified as CYCLexCYCLe, then it MAY NOT have any level (max-min pair) with more than 1 cycle. Also, the spectrum is assumed to have been cycle counted. There are many of cycle counting programs available in the open literature. AFGROW provides an optional cycle counting tool in the tools menu (see Section 3.5.4). In any case, these spectrum data files (**ASCII text**) are named sequentially: [*filename01.sub*], [*filename02.sub*], ..., etc. These files are constructed as follows:

[*Sub-spectrum Number*] [*number of levels*]

[*max*] [*min*] [*cycles*]

.....
.....

The above pattern is simply repeated for as many sub-spectra as desired in each file. Each sub-spectrum is also numbered sequentially, and a given file may have as many sub-spectra as required by the user. The maximum and minimum values are floating point values and the cycles are integer values. A text editor or a simple program may be used to generate these files.

3.2.4.2.2 Time Dependent Spectrum Text Format

The time dependent spectrum format allows BOTH the cyclic and time dependent aspects of crack growth to be considered. If this format is used, user-defined time dependent crack growth rate data (see Section 3.5.4) will be used to determine the time dependent portion of the total crack growth life. The first file for the time dependent spectrum is called a spectrum information file (named [*filename*].st3) with the format shown below:

[*Title*]

[*sub-spectrum label*] (i.e. Flight, Block , Hour, etc.)

[*type of spectrum*] (Either **BLOCKED** or **CYCLexCYCLe**)

[*number of files to follow*]

The other files associated with the spectrum contain the actual stress (or load) information. Remember that if the spectrum is specified as CYCLExCYCLE, then it MAY NOT have any level (max-min pair) with more than 1 cycle. Also, the spectrum is assumed to have been cycle counted. There are a number of cycle-counting programs available in the open literature. AFGROW provides an optional cycle counting tool in the tools menu (see Section 3.5.4). In any case, these spectrum data files (**ASCII text**) are named [filename01.std], [filename02.std], ..., etc. These files are constructed as follows:

[*Sub-spectrum Number*] [*number of levels*] [*seconds in sub-spectrum*] [*loading type*]

[*max*] [*min*] [*cycles*]

.....

The above pattern is simply repeated for as many sub-spectra as desired. If two files are specified in the [.st3] file, there MUST be a [filename02.std] file⁴⁴. The maximum and minimum values are floating point values and the cycles are integer values. A text editor or a simple program may be used to generate these files.

Integer values are used to indicate the type of loading applied:

- 1 – Random Cyclic Sequence** (assumed to be sinusoidal)
- 2 – Ramp Up** (may only have one level describing the ramp up)
- 3 – Ramp Down** (may only have one level describing the ramp down)

In a time dependent spectrum, when a given random cyclic sequence (type 1) sub-spectrum contains a level where the max-min values are set to be equal, the level is treated as a **holding level**. This is useful for cases where a sustained load is applied. Cyclic crack growth calculations are always calculated in addition to any time dependent growth. **In cases where ramp loading is applied, the cyclic growth is ONLY applied during the ramp up to avoid double counting of the cyclic growth.** At this time, AFGROW will not show the hold or ramp type loading in the spectrum plot when using a time dependent spectrum.

3.2.4.2.3 XML Spectrum Format

This new spectrum format for AFGROW is intended to be created/edited by the Spectrum Management Tool (see Section 3.2.4.2). These spectrum files have a “.spx” file extension. The XML spectrum format allows users to develop usage spectra that includes tags to identify environmental conditions for use with the appropriate crack growth rate data and to track damage related to specific levels in a spectrum.

⁴⁴ ALL files associated with a spectrum have the same root name [filename].

3.2.4.2.3.1 Environmental Tags

Environmental tags may be assigned to individual levels in a sub-spectrum using the AFGROW Spectrum Manager as shown in Figure 129. At this time, environment tags are numerical values that are intended to correlate to temperature. AFGROW will use the appropriate tabular crack growth rate model⁴⁵ corresponding to the environment tag in the XML formatted spectrum file. If no tag is assigned to a given spectrum level, the default tabular growth rate data will be used to calculate crack extension for that level. Temperature based environmental tags are easily adapted to other environments (e.g. humidity level) by assigning a numeric tag to crack growth rate data for the desired environment condition.

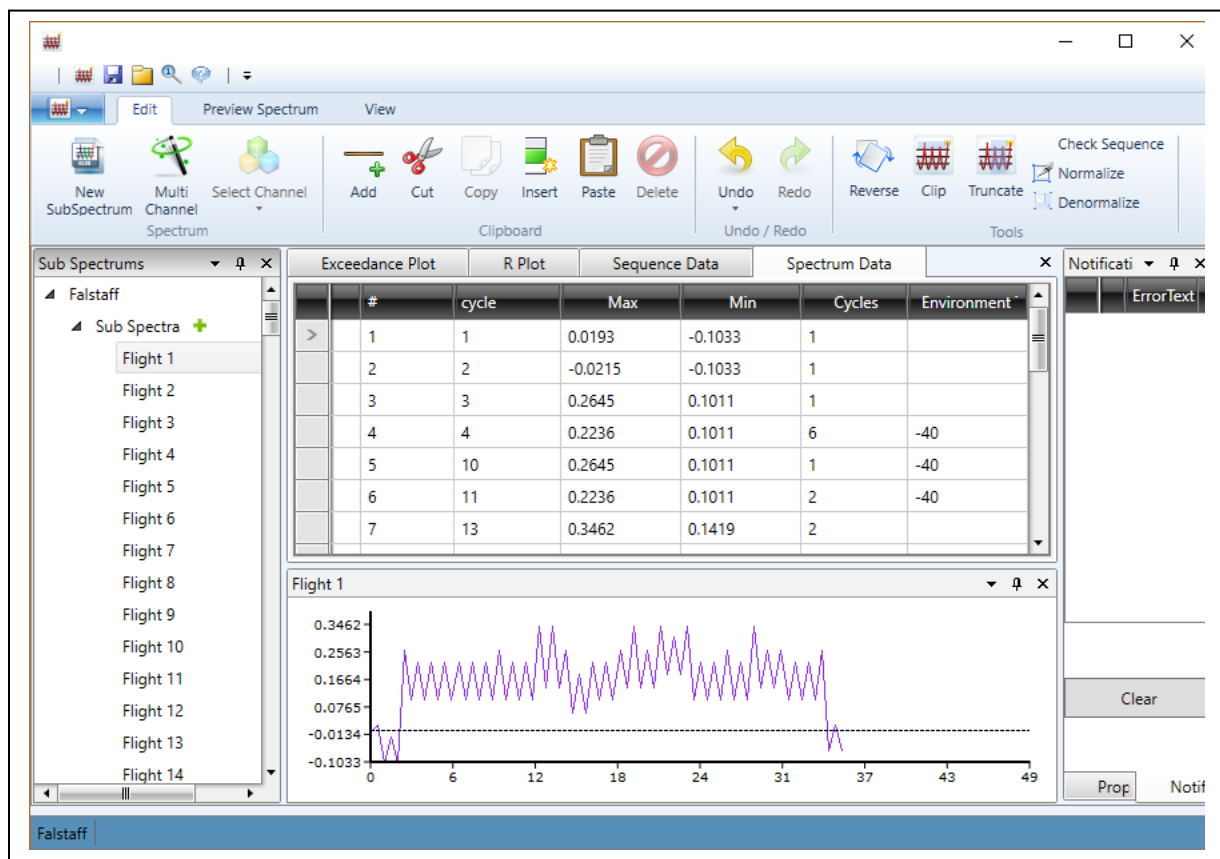


Figure 129: Environment Tags

Please refer to Section 3.2.2.4 for more information about using multiple tabular crack growth rate models. Spectrum Manager is available for download on the AFGROW Web Site (www.afgrow.net)

⁴⁵ The tabular crack growth rate model must be used when using environment tags.

3.2.4.2.3.2 Damage Tags

Damage tags may be assigned to individual levels in a sub-spectrum using the AFGROW Spectrum Manager in the same manner as shown in Figure 129. These tags are alpha-numeric so they can include any combination of text and numeric information. Damage tags are used to track the amount of crack growth resulting from tagged spectrum levels. During a life prediction, AFGROW keeps a running total of the amount of crack growth for each tagged level in the input spectrum. A damage summary is provided at the end of the analysis showing the percentage of total crack growth for each damage tag. Damage accumulated by spectrum levels without tags are reported as the remaining percentage of the total crack growth.

```
Output
Max stress 9.000, r = 0.67, 1.88702e+006 Cycles, Subspectra: 1, Pass: 1217

+++++Kmax Criteria Failure. Edge 1, Crack 1
Crack #1
C Length = 2.1421 Beta Tension= 1.1707 Beta Compression= 1.1707 R(k)= 0.0000 R(final)= 0.0000 Delta k=3.6444e+001 D()/DN=2.7422e-003
Max stress 12.000, r = 0.00, 1.88707e+006 Cycles, Subspectra: 1, Pass: 1217

*****Fracture
Stress State in the 'C' direction (PSC): 3.10287
Fracture has occurred- run time: 0 hour(s) 0 minute(s) 8 second(s)

Damage Summary (By Source)
Percent of total damage due to "": 23.54%
Percent of total damage due to 'GAG': 1.33%
Percent of total damage due to 'Gust': 2.48%
Percent of total damage due to 'Level 1': 36.21%
Percent of total damage due to 'Level 2': 36.44%

Damage Summary (By Sub Spectra)
Percent of total damage due to 'Block 1': 76.46%
Percent of total damage due to 'Block 2': 23.54%
```

Figure 130: Damage Tag Output Information

When damage tags are used, AFGROW also prints out the percentage of crack growth for each sub-spectrum.

Spectrum Manager is available for download on the AFGROW Web Site (www.afgrow.net)

3.2.5 Input Spectrum Filters

The spectrum filtering option allows the input spectrum tension and/or compression values to be adjusted independently to account for internal or external effects.

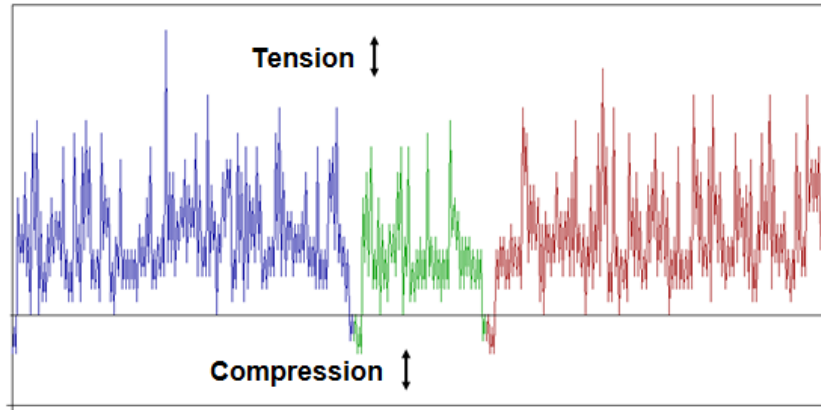


Figure 131: Spectrum Tension and Compression Filters

In the case of a crack at a hole with an interference fit fastener, the fastener creates a pre-load condition at the hole which decreases as a crack length increases (compliance increases with crack length). Currently, the spectrum filtering capability allows the tension and/or compression values to be adjusted by the user as a function of the C-length.

3.2.5.1 Tabular Look-Up Option

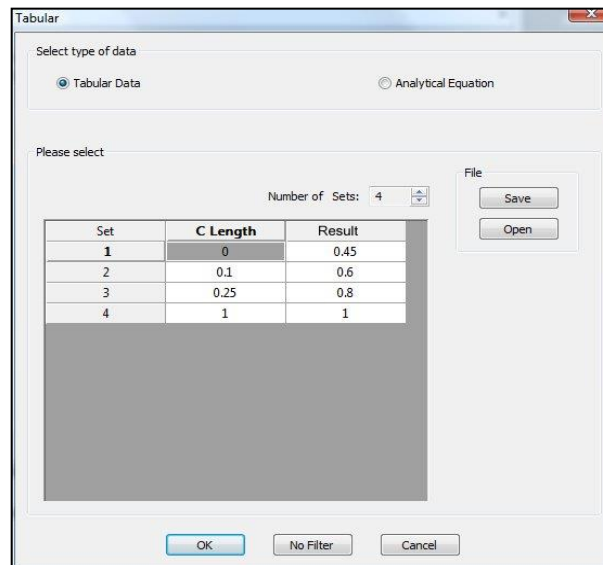


Figure 132: Spectrum Filtering Using Tabular Look-Up

Users may enter up to 100 points to define the spectrum modification as a function of the C-length. The table may be saved to a file which may be opened later for use in another life prediction. In addition, the filtering capability may be turned off by selecting the “No Filter” button at the bottom of the dialog box.

Remember that AFGROW will not extrapolate outside the bounds of the tabular data.

3.2.5.2 Analytical Equation Option

Equation

Select type of data

☐ Tabular Data ☒ Analytical Equation

Please select

```
var result;  
if (crackLengthC < 0.5)  
{  
  result=0.8+0.2*crackLengthC/0.5;  
}  
else  
{  
  result=1.0;  
}  
return result;
```

Test

crackLengthC: 1

width: 4

thickness: 0.25

radius: 0.2

stressMax: 10

stressMin: 0

OK No Filter Cancel

Figure 133: Spectrum Filtering Using the Analytical Equation Option

A java scripting engine is available in AFGROW to allow users to enter the spectrum modification in equation form. Details on writing java script code are readily available on the Internet. Basic model dimensions are available to the user, but the C-length is the only variable that is not constant for any given model.

The “Test” button allows users to verify that the java script returns the expected value for user defined parameters. The desired values for each model parameter are entered in the appropriate box, and the result is displayed in a message box when the user clicks on the test button as shown in Figure 134.

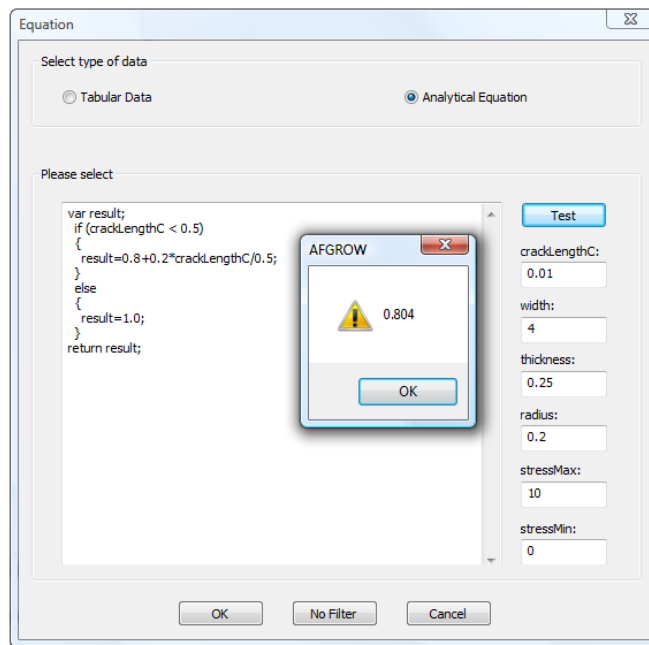


Figure 134: Analytical Equation Test Capability

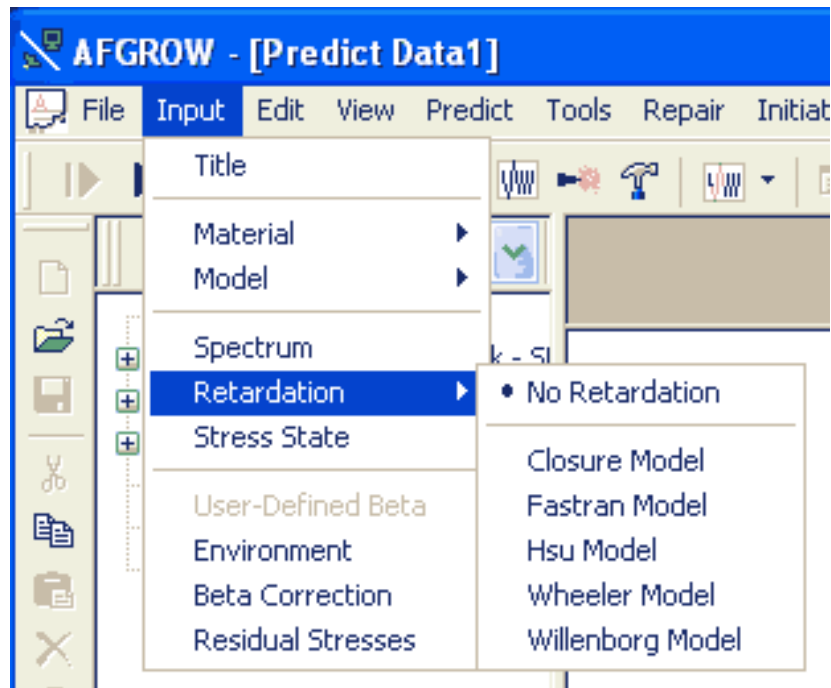


Figure 135: Retardation Model Input Option

There are currently six choices of load interaction, or retardation models in AFGROW. The models can be accessed through either the input menu or by using the pull-down menu located on the toolbar.

Note: Each model has one or more user adjustable parameter(s) that are used to tune the model to fit actual test data. Ideally, a parameter should be a material constant, which is independent of other variables such as spectrum sequence or load level. Some models seem to work better than others in this regard, but there is a need to reproduce results for various types of retardation models. Whenever possible, verification tests should be used to test the models and determine the appropriate parameter(s). These models are provided at the user's discretion and responsibility.

The details of the No Retardation, Closure, FASTRAN, Hsu, Wheeler, and Willenborg models are given in the following sections.

3.2.6.1 No Retardation

This is the default option in AFGROW. When this option is selected, no spectrum load interaction effects are assumed.

3.2.6.2 Closure Model

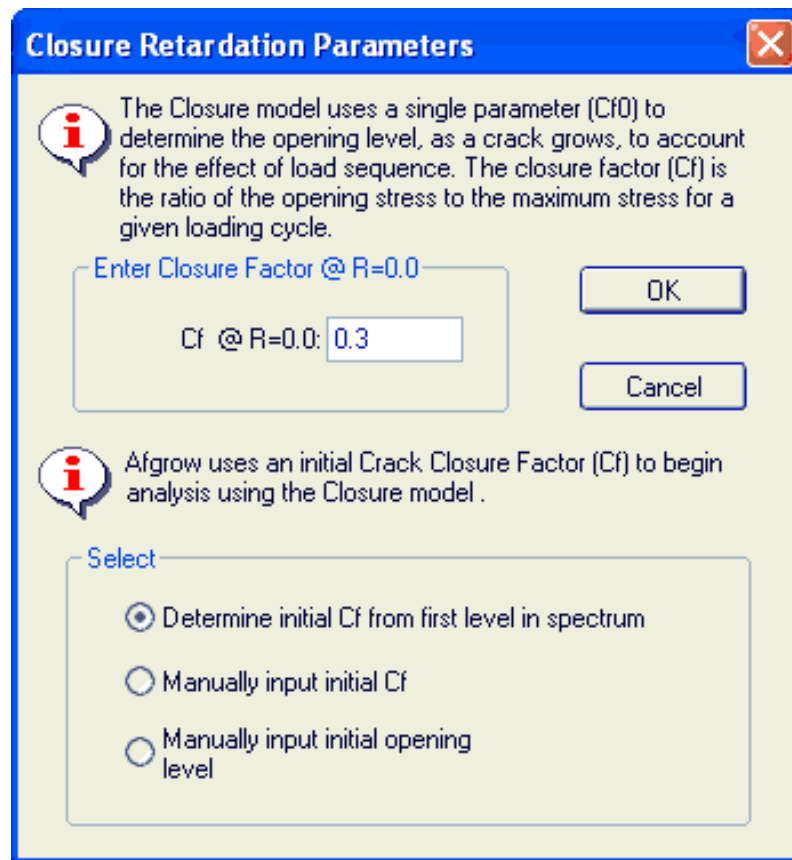


Figure 136: Closure Retardation Model Dialog

The closure model in AFGROW is a fairly simple single-parameter plasticity model. The model is based on early fracture mechanics work by Erdogan and Elber and more recent models proposed by Dr. Matthew Creager and Dr Sunder [2-4]. The model, developed by Mr. James Harter, basically expanded a constant closure model originally developed by Dr. Creager while he and Mr. Harter were involved in performing damage tolerance analyses for the B-2 Bomber in 1982-83.

3.2.6.2.1 Closure Model Overview

It is important to understand that this model is called a closure model because it is based on the idea that the crack is “closed” when no load is applied and a certain load must be applied to “open” the crack tip. Some researchers believe that yielded material in the wake of a growing crack acts as a wedge behind the crack tip [45]. This yielded material is forced to be in compression by the elastic material surrounding it. Other researchers believe that this plastic wake is merely a surface phenomenon caused by the difference between the plane stress state at the surface and an internal plane strain state. They believe that the apparent contact behind the crack tip is merely the result of natural stress equilibrium and plays a very minor role in crack growth behavior [46, 47]. The later researchers believe that there is only a significant compressive residual stress in FRONT of the crack tip. This

compressive residual stress must be overcome by applied tension loading before the crack can extend.

In either case, there is some minimal applied axial load that must be reached before the crack may extend. In AFGROW, this value is referred to as the “opening” load. The early closure work by Elber, et al., [2-4] showed a relationship between the maximum applied stress and this opening stress. The closure factor, C_f , is defined as the ratio of the opening stress to the maximum applied stress and was demonstrated to be a function of stress ratio ($R = \sigma_{\min}/\sigma_{\max}$).

$$C_f = 1.0 - [(1 - C_{f0})(1 + 0.6R)(1 - R)]$$

The closure model uses a single adjustable parameter (C_{f0}) to “tune” the closure model for a given material. Ideally, this parameter would be a true material property and be independent of the applied loading spectrum. The closure model provides reasonable results for most practical cases, but the user is encouraged to verify life predictions with test data whenever possible.

A general description of the use of the closure model in AFGROW is shown schematically in Figure 137.

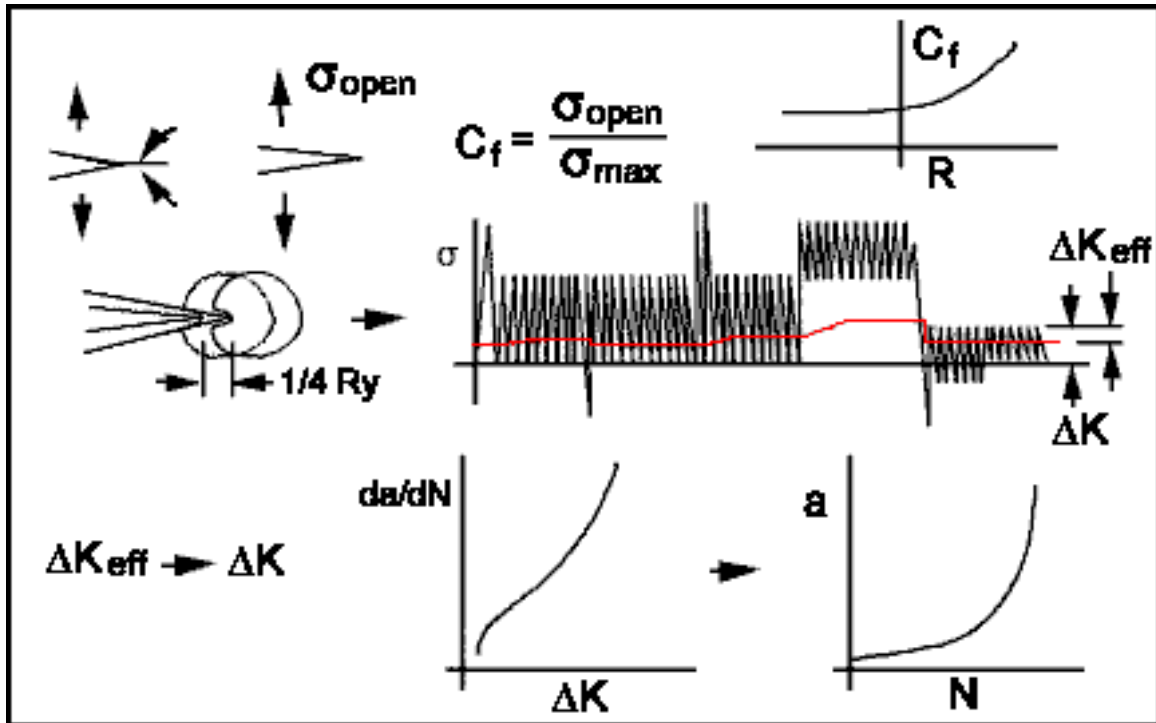


Figure 137: Life Prediction with the Closure Model

An initial opening level is determined based on the option selected by the user (see Figure 136). The opening level changes as a function of the load history as shown in red in Figure 137. Changes in opening level caused by an overload are assumed to vary linearly from the

level at the time the overload was applied to the opening level calculated for the overload when the crack reaches $\frac{1}{4}$ of the distance into the yield zone created by the overload. An overload is defined as any cycle where the crack length plus the yield zone (for the maximum stress) is greater than the previous overload (as indicated in Figure 138).

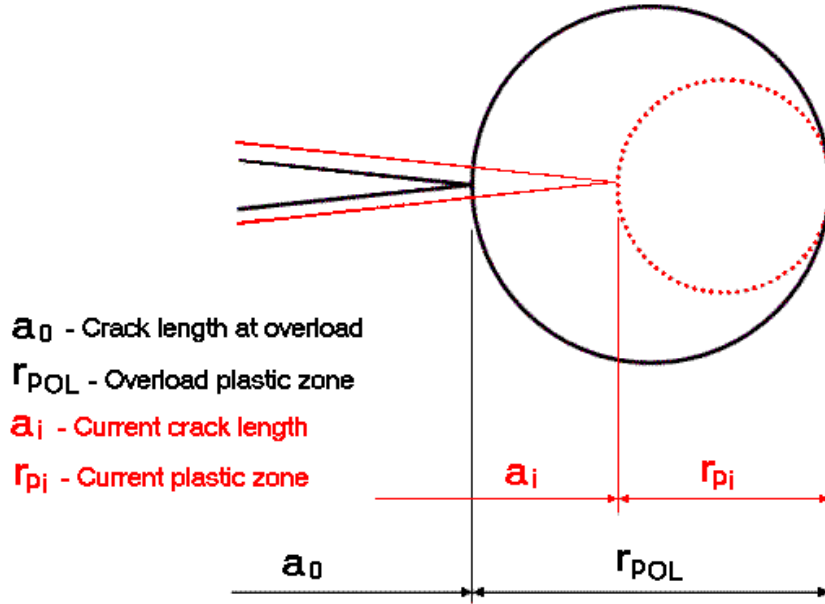


Figure 138: Overload Definition

While tensile overloads generally increase the opening level, compressive stresses tend to lower it (as indicated in Figure 137).

Once the opening level is known, an effective stress intensity factor range (ΔK_{eff}) is determined as follows:

$$\Delta K_{eff} = K_{max} - K_{open}^{46}, \text{ if } K_{open} \geq K_{min}$$

$$\Delta K_{eff} = K_{max} - K_{min}, \text{ if } K_{open} < K_{min}$$

Crack growth rate data available in the open literature are normally given as a function of ΔK and stress ratio (R). The AFGROW closure model converts ΔK_{eff} to an equivalent ΔK based on the relationship between the closure factor (C_f) and stress ratio (R). The details of this conversion are given in Section 3.2.6.2.4. The crack growth rate for a given spectrum cycle is then determined from the user-defined crack growth rate data. Finally, the crack growth life is determined as the sum of the applied cycles required to extend the crack to a critical length.

⁴⁶ $K_{open} = \sigma_{open} \sqrt{\pi x \beta}$, where x is the crack length of interest.

3.2.6.2.2 Closure Factor

As stated in the previous section, the closure factor (Cf) is defined as:

$$C_f = \sigma_{\text{open}} / \sigma_{\text{max}}$$

The general relationship between C_f and R is shown in Figure 139.

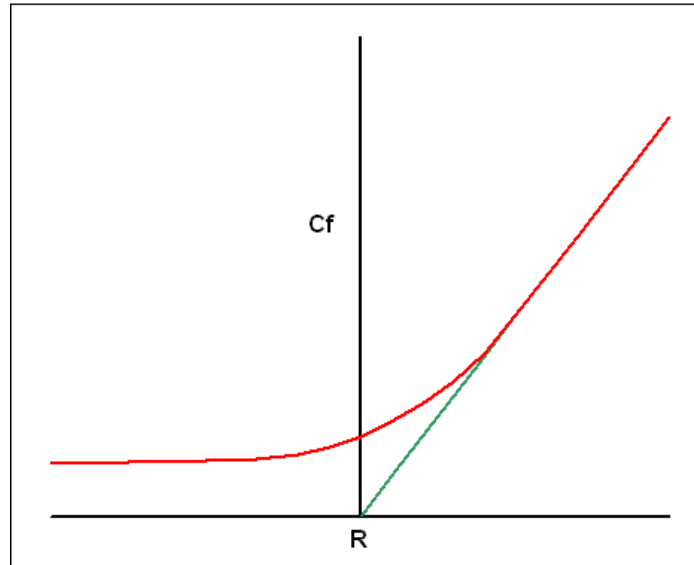


Figure 139: Typical C_f vs. R Relationship

The closure factor (C_f) used in AFGROW is a function of stress ratio (R) as follows:

$$C_f = 1.0 - [(1 - C_{f0})(1 + 0.6R)(1 - R)]$$

$$C_f = R, \text{ for } R > R_{hi} \quad ^{47}$$

$$R = R_{lo}, \text{ for } R < R_{lo} \quad ^{48}$$

Where C_{f0} is the value of C_f for $R = 0$. This is the only user-defined parameter used in the closure model (see Figure 136). Typical values for C_{f0} range from 0.3 to 0.5. Ideally, C_{f0} is a material property and should provide reasonable life predictions for a given material independent of the applied spectrum. However, as is true for most load interaction model parameters, C_{f0} should be thought of as a “tuning” parameter for the closure model.

Since the equation for C_f reaches a minimum at $R = -1/3$, AFGROW ensures that the C_f value will be equal to its minimum when $R < -1/3$. This ONLY affects the C_f calculation to prevent the opening level from increasing as R decreases. As the R -value increases, the

⁴⁷ R_{hi} is defined as the R -value above which the crack is always open (see section 3.2.2)

⁴⁸ R_{lo} is defined as the R -value below which C_f is assumed constant (see section 3.2.2)

Cf-value approaches the line, Cf = R. For all R-values greater than the point where the Cf curve touches the line, the crack is assumed to be fully open.

3.2.6.2.3 Initial Opening Level

The closure model requires an opening level to serve as a starting point in a given life analysis. There are three options to set the initial opening level (see Figure 136):

Determine initial Cf from first level in spectrum,

Manually input initial Cf, or

Manually input initial opening level

The default option is to use the first spectrum cycle to determine the initial closure factor. In this case, AFGROW assumes no previous loading effect (perhaps the previous loading is unknown). The initial Cf is calculated using the R-value for the first cycle in the spectrum. The default value for Cf is set to 0.3, and will generally result in conservative life predictions. Cf values > 0.5 may indicate a problem somewhere else in the model, or that the closure model is not appropriate for the problem.

If the previous loading history is known, the user can choose to manually input an initial value for Cf or enter the opening level directly.

Early versions of AFGROW used Cf instead of a user-defined initial opening level due to internal code structure. Opening level calculations in the closure model are based on the current maximum stress intensity as shown below:

$$\sigma_{open} = \frac{C_f K_{max}}{\sqrt{\pi x \beta}}, \text{ where } x \text{ is the crack length for the dimension of interest}^{49}$$

The initial opening level option requires more effort since the user must first determine an opening level based on the previous load history. The initial closure factor is then determined as follows:

$$C_f = \frac{\sigma_{open} \sqrt{\pi x \beta}}{K_{max}}$$

Where, K_{max} and β are for the initial crack length and first spectrum cycle

This requires the user to run AFGROW once to get the values of Kmax and beta for the first cycle in the spectrum. The user will also need to be able to determine the initial

⁴⁹ σ_{open} will be independent of the dimension chosen

opening level (σ_{open}) caused by a previous load history. Information provided in this guide should provide information required to estimate an initial opening level based on a previous load history.

If a user selects the option to manually input the opening level directly, the input value must match the reference⁵⁰ loading for the model.

$$K = \sigma_{ref} \sqrt{\pi} \times \beta$$

The Wedge Opening Load (WOL) and Compact Tension (CT) classic models use applied load as the reference. In cases where bearing stress is used as a reference stress, the opening level would be a bearing stress. The use of the word, level is used in this option because the value required is stress in all but the CT and WOL cases.

3.2.6.2.4 Closure Calculations

Once the initial opening level has been determined, the closure model keeps track of the current opening level based on the user-defined spectrum. Each time an overload is detected (see Figure 138); a new closure factor is calculated based on the R-value for the applied cycle. The equation for Cf (given in Section 3.2.6.2.2) can result in Cf values less than R in cases where Cf0 is less than 0.375. In those cases, AFGROW will set Cf to the point where the given curve crosses the line: Cf = R.

$$Cf = 1.667/(1.0 - Cf0) - 1.667, \text{ for } R < R_{hi}$$

For any case when $R \geq R_{hi}$, AFGROW sets Cf as follows:

$$Cf = R$$

When $R < R_{lo}$, AFGROW sets Cf as follows⁵¹:

$$Cf = 1.0 - [(1 - Cf_0)(1 + 0.6R_{lo})(1 - R_{lo})]$$

This provides a means of limiting Cf for higher R-values and lower values of Cf0. As Cf0 decreases, these limits also decrease, as would be expected

The opening level changes as a function of the applied stress (or load) history. According to work by Dr. Sunder, the Cf value expected for a given stress ratio will not be reached until the crack grows 1/4 of the way through the yield zone created by the maximum stress

⁵⁰ In cases where load is used as the reference, the beta values printed in the output have been adjusted to include area units. For combined loading cases, the user defines the reference value (see section 3.2.3.1.5).

⁵¹ Since the Cf equation reaches a minimum at $R = -1/3$, the Rlo is never allowed to be less than -1/3 for the purpose of calculating Cf. The R (or Rlo) used to determine the growth rate is not subject to this limitation.

for that cycle. AFGROW assumes that the opening level varies linearly from the current value to the value expected when an overload cycle occurs (as the crack grows through 1/4 of the overload yield zone). AFGROW keeps track of the current overload cycle by defining an overload condition whenever (crack length + yield zone) > previous overload - as is the case in the Willenborg model.

Compressive loads/stresses are treated a bit differently in that the opening level may be INSTANTANEOUSLY shifted to the level determined by the equation above for an R-value equal to the ratio of the compressive minimum load/stress to the current maximum overload load/stress. The INSTANT change in opening level is made IF the maximum value for a given cycle IS an overload (yield zone extends beyond previous overload case) AND the opening level is LOWER than the current opening level OR the maximum value for a given cycle IS NOT an overload AND the opening level (based on the R value determined from the compressive value and the current overload) is LOWER than the current opening level. The idea is that while a crack must grow into the plastic wake of tensile overloads to fully develop a given opening level, a compressive cycle can instantly cause the residual stress field to be changed. If any given compressive load/stress is not low enough to cause the opening level to fall below the current value, then there is no reason to change the opening stress.

When an overload cycle contains a compressive minimum, the overload yield zone size is reduced by 10 percent of the absolute value of the stress ratio for that cycle. This reduction is made to help account for the effect of the compressive minimum. This reduces the effect of the overload since it will take fewer subsequent cycles to grow through a smaller overload yield zone. The quantity, 10 percent, was determined based on actual test data for common aircraft alloys tested in-house at Wright-Patterson AFB and some very helpful data provided by Mr. Kevin Walker [48].

Finally, an effective stress intensity range (ΔK_{eff}) is determined as the difference between K_{max} and the K value for the current opening level. If the opening level is less than the current minimum K , ΔK_{eff} is simply the difference between the maximum and minimum K values. Since the standard crack growth rate data used in AFGROW is NOT based on ΔK_{eff} , a conversion⁵² back to ΔK is made in the crack growth rate module. The steps involved in this conversion are described in the following paragraphs.

First, the stress ratio (based on K_{min} , K_{max} and the C_f value for the current cycle) is determined as follows:

$$R_K = K_{min} / K_{max} \text{ (for the current cycle)}$$

$$\text{If } K_{min} < K_{open}, K_{min} = K_{open}$$

⁵² Determines the ΔK and R that would result in a given ΔK_{eff}

$$Cf = \frac{K_{\min}}{K_{\max}}$$

Then, the equivalent stress ratio⁵³ (R) for the given Cf is determined from the relationship between Cf and R. Since this relationship is a quadratic equation, there are two possible values of R for any Cf. However, since the minimum value of R is $-1/3$, the only R of concern will be the R that is greater than $-1/3$. This solution is shown below:

$$R = \frac{0.4 (Cf_0 - 1) + \sqrt{F(Cf_0, Cf)}}{1.2 (1 - Cf_0)}$$

Where, $F (Cf_0, Cf) = (0.4(1 - Cf_0))^2 - 4.0(0.6(1 - Cf_0)(Cf_0 - Cf))$

Note: When $F (Cf_0, Cf) \leq 0.0$, $R = -1/3$ (footnote 51)

Once the R-value is known, the equivalent ΔK can be determined as follows:

if $((R > R_{hi}) \text{ or } (R > Cf))$, $R = Cf$ (footnote 47)

if $((R_K < 0.0) \text{ and } (R_K \leq R))$, $R = R_K$ (footnote 51)

$$\Delta K = \Delta K_{eff} \frac{(1 - R)}{(1 - Cf)} ; \text{ for } R \geq 0.0 \text{ Since } K_{\max} = \frac{\Delta K_{eff}}{(1 - C_f)}$$

$$\Delta K = \frac{\Delta K_{eff}}{(1 - Cf)} ; \text{ for } R < 0.0 \text{ }^{54}$$

The resulting ΔK value is shown in the AFGROW output and is used to determine the crack growth rate based on the user-define growth rate model. If the current opening level is the same as would have been caused by a given load cycle (if applied by itself), the ΔK -value returned by this conversion will be the same as the original ΔK -value for that cycle. In this way, the result of predictions made using constant amplitude spectra will give the same results as the no retardation case. There may be a slight difference in the closure model vs. no retardation for constant amplitude blocked spectra. This is due to the fact that individual blocks are divided into smaller blocks in the closure model to ensure that a given crack will NOT grow beyond 1 percent of the current overload yield zone. This may be verified by use of a single cycle constant-amplitude spectrum. The results for the closure model will

⁵³ The equivalent R is the R-value that would have caused the current Cf (which is dependent on the load history)

⁵⁴ This relationship is used here because AFGROW uses $K_{\max}$, not ΔK , when $R < 0.0$

match those of the no retardation model for this case - of course; there will be an increase in runtime.

The closure model relies on the use of a single curve representing the relationship between C_f and R . Some researchers [47] have proposed that the actual relationship is much more complex and requires multiple curves. As more data become available, this idea will be explored further.

3.2.6.3 FASTRAN Model

FASTRAN [49] is a crack growth life prediction program that uses a crack closure concept⁵⁵ developed by Dr. James C. Newman, Jr. The FASTRAN closure model is based on the Dugdale yield zone model [50] which was modified to leave plastically deformed material in the wake of a crack tip.

3.2.6.3.1 Overview of the FASTRAN Model

As a cracked specimen is loaded, material just ahead of the crack tip yields and creates a compressive residual stress when the load is removed (due to the elastic material surrounding the yield zone). As a crack grows through the yielded material, the plastically deformed material acts as a wedge behind the crack tip which pre-loads the crack tip. The magnitude of the pre-load is a function of the applied load history. It determines the magnitude of the applied loading required to “open” the crack tip and cause subsequent crack growth. This is illustrated in Figure 140 below.

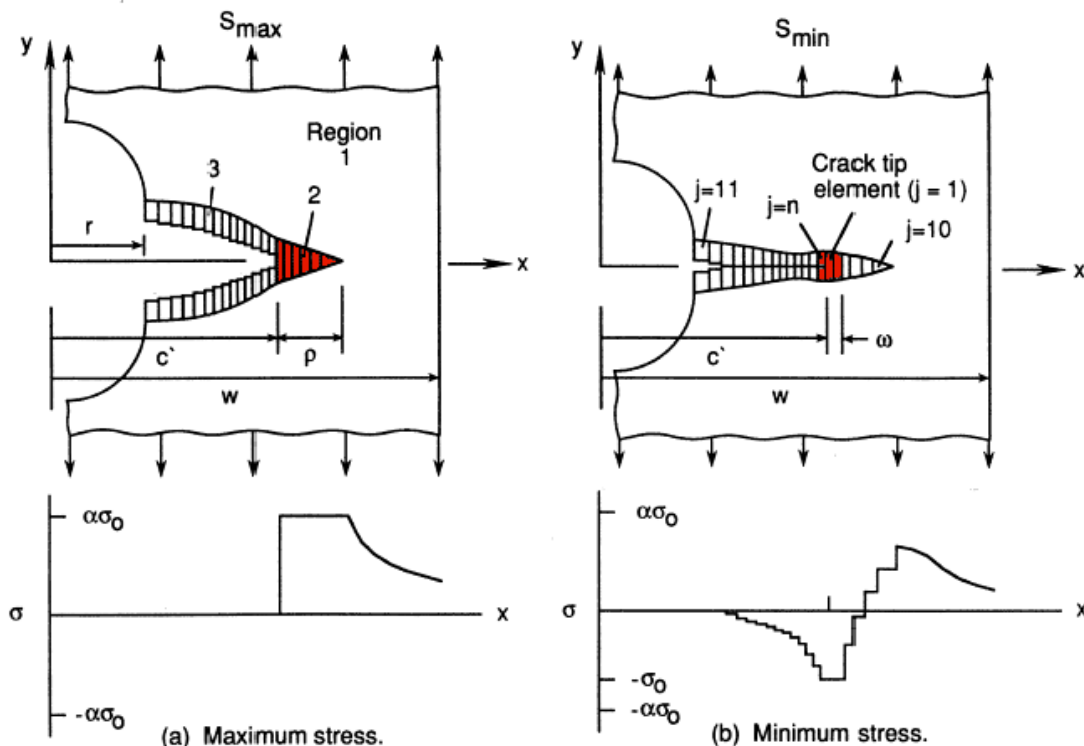


Figure 140: FASTRAN Closure Concept

⁵⁵ Portions of this users guide for the FASTRAN model were taken from reference [49]

The model consists of three regions:

- 1- Linear Elastic
- 2- Plastic Zone
- 3- Residual Plastic Deformation in the Crack Wake

The crack and plastic zone is divided into a number of elements where local stresses and displacements are calculated. The stress and displacement in each element are also a function of stress state. FASTRAN uses a constraint factor (α) to adjust the flow stress (σ_0 - average of the yield and ultimate strength of a given material).

For plane stress: $\alpha = 1$

For plane strain: $\alpha = 3$

The flow stress is the highest stress that can be sustained by a given element and is a first order approximation for strain hardening. Since the elements along the crack face are broken, they can only transfer compressive loads when they are in contact.

For a given applied stress cycle, the length of the plastic zone resulting from the maximum applied stress is shown as ρ , and the length of the compression residual stress ahead of the crack tip at the minimum applied stress is ω (see Figure 140). This information is used to calculate the minimum applied stress required to open the crack tip. This opening stress is used to determine the effective range of stress intensity (ΔK_{eff}) for a given applied stress cycle.

$$\Delta K_{eff} = K_{max} - K_{open}, \text{ if } K_{open} > K_{min}$$

$$\Delta K_{eff} = K_{max} - K_{min}, \text{ if } K_{open} \leq K_{min}$$

The FASTRAN code, written by Dr. Newman, does not allow for multiple cracks to be modeled concurrently. AFGROW includes models for multiple and non-symmetric cracks, where crack growth calculations must be performed for each crack tip. Therefore, the FASTRAN implementation in AFGROW was modified so that these cases may be handled in a consistent manner. When the FASTRAN model is used in AFGROW, an additional internal model is created with an initial crack length equal to the longest initial crack length in the user-defined geometry (even single cracked cases). A very wide plate (10,000 in.) is used for the internal model so that finite width effects do not enter the calculations, and the crack will never reach the free edge before any of the cracks in the user-defined model. The opening load calculated for the internal model is applied to all cracks in the user-defined model. Of course, finite width effects are calculated as appropriate for the user-defined case. The internal model includes a cracked hole of the same size as the largest cracked hole in the user-defined case. If there are no cracked holes in the user defined case, the internal model is a simple, center cracked geometry. Although this method is not ideal, it does allow the FASTRAN model to be used for all cases in AFGROW.

3.2.6.3.2 Using Effective Crack Growth Rate Data for FASTRAN

Crack growth rate data are normally given as da/dN vs. ΔK as shown in

Figure 141 [49].

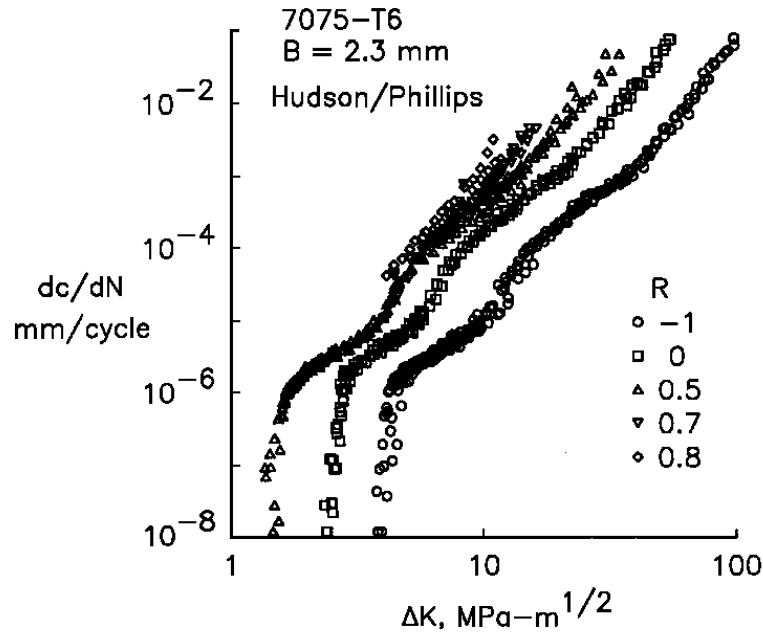


Figure 141: Standard Crack Growth Rate Data

The FASTRAN model requires the use of a single da/dN vs. ΔK_{eff} curve to determine the growth rate at a given value of ΔK_{eff} for any applied spectrum cycle. This single curve must be input in the tabular look-up growth rate model (see Section 3.2.2.4). The conversion from ΔK to ΔK_{eff} is based on the calculation of the stress intensity value corresponding to the opening stress (K_{open}). This calculation is fairly complex, and the user is encouraged to review the information provided in reference [49]. The method used to calculate K_{open} is given below:

$$\frac{K_{open}}{K_{max}} = A_0 + A_1 R + A_2 R^2 + A_3 R^3, \text{ for } R \geq 0$$

$$\frac{K_{open}}{K_{max}} = A_0 + A_1 R, \text{ for } R < 0$$

$$K_{open} = K_{min}, \text{ if } \frac{K_{open}}{K_{max}} < R$$

$$\frac{K_{open}}{K_{max}} = 0, \text{ if } \frac{K_{open}}{K_{max}} < 0$$

$$R = \frac{K_{min}}{K_{max}}$$

$$A_0 = (0.825 - 0.34\alpha + 0.05\alpha^2) \left[\cos\left(\frac{\pi S_{max} F_w}{2\sigma_0}\right) \right]^{\frac{1}{\alpha}}$$

$$A_1 = (0.415 - 0.071\alpha) \frac{S_{max} F_w}{\sigma_0}$$

$$A_2 = 1 - A_0 - A_1 - A_3$$

$$A_3 = 2A_0 + A_1 - 1$$

F_w – Finite width effect for the given specimen geometry

The above equations are considered valid as long as the maximum stress applied for a given crack growth rate test is less than $0.8 \sigma_0$. Based on the definition of ΔK_{eff} given earlier in this section and the above equations, ΔK_{eff} is determined as follows:

$$\Delta K_{eff} = K_{max} - K_{open}$$

As was also noted above, values for the variable, α , range from 1 (plane stress) to 3 (plane strain). The goal is to find a value for α that results in a single da/dN vs. ΔK_{eff} curve for all R-values. This is normally done by trial and error until the data for all R-values collapses to a single curve (or as close as possible). It may be necessary to use multiple α -values for different ranges of crack growth rate. The FASTRAN model includes an option for the use of variable α -values (see Section 3.2.6.3.3). The result of the conversion of the data in

Figure 141 is shown in Figure 142.

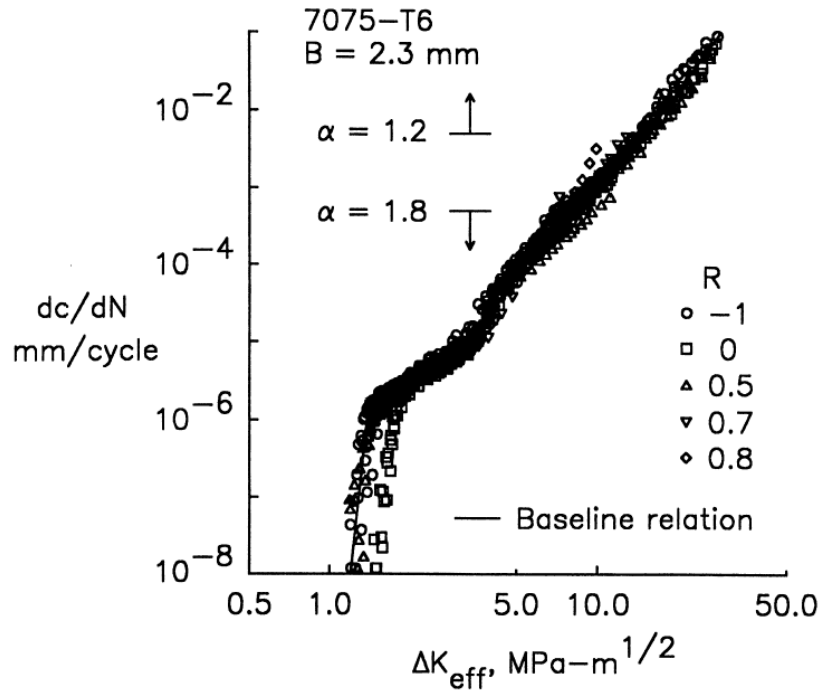


Figure 142: Effective Crack Growth Rate Data

Large crack growth rate data may not collapse in the threshold region due to many possible factors including: oxide or roughness induced closure, and possible load reduction effects. Load reduction is commonly used in crack growth rate tests to obtain threshold data.

3.2.6.3.3 FASTRAN Wizard

Due to the complexity of the FASTRAN model, a wizard is used as a guide through the modeling process.

Geometry:

The first dialog box reminds users that the FASTRAN model requires users to enter a single da/dN vs. ΔK_{eff} curve in the table lookup growth rate model.

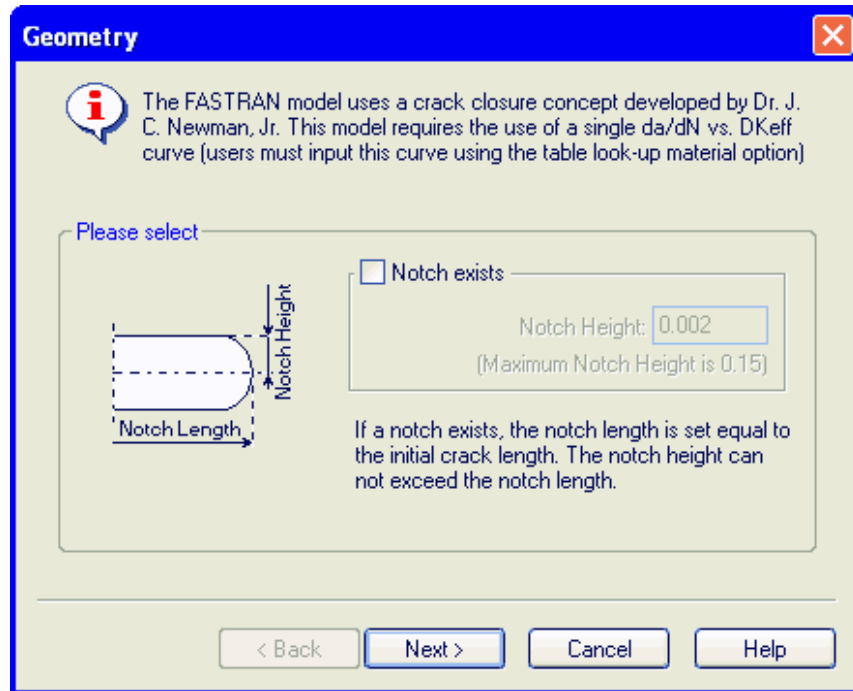


Figure 143: Geometry

If a notch exists in a test specimen (commonly used to initiate a crack in an MT test specimen used for laboratory testing), users can include the notch in this dialog box. The notch can make a significant difference in an analysis since it will change the amount of contact in the wake of the crack.

Currently, AFGROW sets the notch length equal to the initial crack length, so users will have to subtract the cycles required to grow to a longer initial size. There is a plan to allow a notch length less than the initial crack length in a future AFGROW release. The maximum notch height is indicated in the dialog box and is equal to the notch length.

Crack Growth Equation Type:

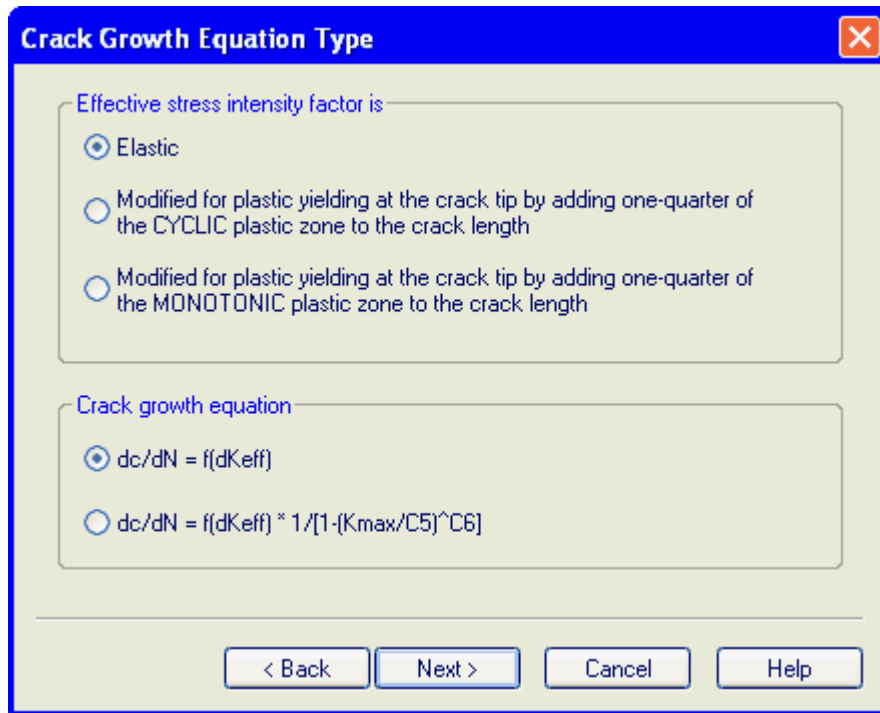


Figure 144: Crack Growth Equation Type

Normally, structures are designed to operate under stress levels far below the material yield stress. In these cases, the stress intensity solutions are assumed to be completely elastic and cracks will grow under standard linear elastic fracture mechanics (LEFM) conditions. However, there are cases when local stresses can be high enough to cause more widespread local yielding and violate LEFM assumptions (i.e. crack length is much smaller than the yield zone ahead of the crack tip). Users have the option to modify the stress intensity calculation by adding a fraction of the cyclic or monotonic plastic zone to the physical crack length. This is done to adjust the K -values to account for the effect of the larger crack tip plastic zone sizes for cracks operating under elastic-plastic fracture mechanics (EPFM) conditions.

The other option in this dialog box is provided to allow more flexibility in regard to the user-defined crack growth rate curve. The default condition simply uses the input growth rate curve as given. The alternative is to make an adjustment to the high end of the user-defined growth rate curve to increase the growth rate as K_{max} approaches the user-defined limit on K_{max} ($C5$). The variable, $C6$, is simply provided to control how quickly this increase occurs as K_{max} increases. Both variables ($C5$ & $C6$) are set in the following dialog in the FASTRAN wizard.

It should be noted that although this dialog shows crack growth rate as dc/dN , ΔK_{eff} values and growth rates for all crack dimensions are determined for the appropriate growth direction. However, the relationship between ΔK_{eff} and growth rate for any crack dimension uses the same user-defined tabular growth rate curve.

Crack Growth Threshold and Fracture Properties:

Crack Growth Threshold and Fracture Properties

These properties make adjustments to the upper and lower end of the user-input crack growth rate curve.

Threshold and fracture properties

Threshold constants

C3: 1 C4: 0

Fracture constants

Cyclic fracture toughness, C5: 1000000

Power on Kmax/C5 term, C6: 0

< Back Next > Cancel Help

Figure 145: Crack Growth Threshold and Fracture Properties

The FASTRAN model allows users to set the threshold value of ΔK_{eff} ($\Delta K_{0\text{eff}}$) using two parameters as follows:

$$\Delta K_{0\text{eff}} = C_3 (1 - C_4 R)$$

Although the FASTRAN model uses a single growth rate curve to describe the relationship between da/dN and ΔK_{eff} , users may wish to allow for changes in the threshold value as a function of stress ratio (R). There may be additional factors that affect the growth rate curve in the threshold region (see Section 3.2.6.3.1). The parameter, C_3 , is the threshold and C_4 is used to adjust the threshold as a function of R .

The fracture constants, C_5 and C_6 control the upper end of the user-defined crack growth rate curve. Cyclic fracture toughness is used as a limiting value on K_{max} . In the previous dialog box (Crack Growth Equation Type), C_5 and C_6 are used to control the increase in growth rate as K_{max} approaches the value of C_5 . However, this only happens if the user selects the option to modify the crack growth equation. If the default growth rate option is selected, the user-defined growth rate curve is used without any adjustment. In this case, C_6 is not used, and the input box for C_6 is disabled (as shown in Figure 145).

Constraint Factors:

The FASTRAN model uses constraint factors to modify the flow stress to account for the local stress state. Constraint values range from 1 (plane stress) to 3 (plane strain). The alpha and beta constraints adjust the tension and compression flow stresses, respectively.

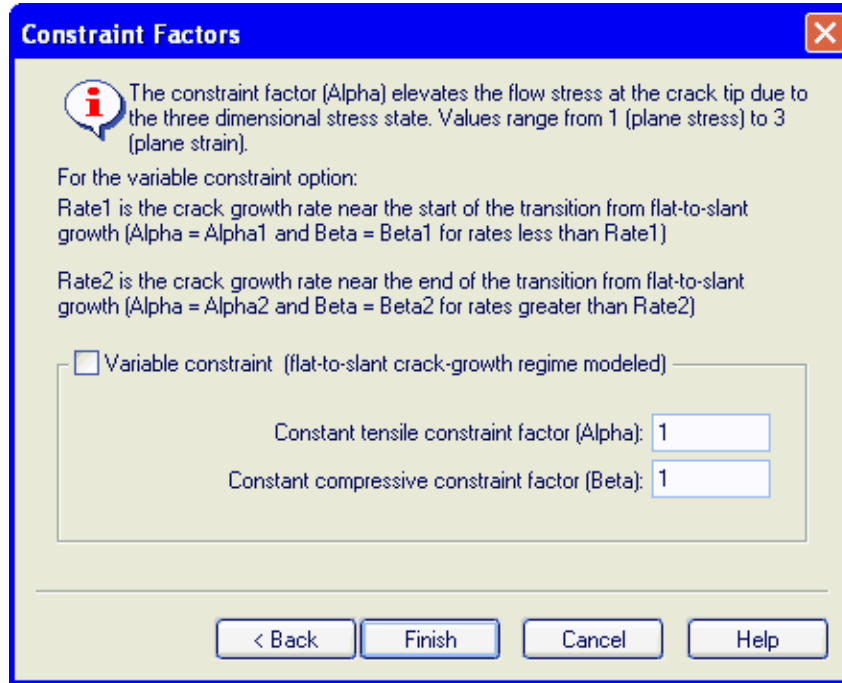


Figure 146: Constant Constraint Factors

The tension and compression flow stress is determined as follows:

$$\sigma_0(\text{tension}) = \text{Alpha}[\sigma_0] = \text{Alpha} \left[\frac{\sigma_{ult} + \sigma_{yield}}{2} \right]$$

$$\sigma_0(\text{compression}) = \text{Beta}[\sigma_0] = \text{Beta} \left[\frac{\sigma_{ult} + \sigma_{yield}}{2} \right]$$

Cracks growing under plane strain conditions exhibit what is often called flat growth (growth in a flat plane normal to the applied loading). As a crack grows and the applied stress intensity increases, the growth tends to transition toward what is called slant growth. The transition begins with the formation of “shear lips” along the free edges. For most common metals, these “shear lips” are slanted at approximately 45 degrees to the plane normal to the applied loading. As the applied stress intensity increases with crack extension, the crack can eventually transition to purely slant growth (no flat growth through the thickness).

Users can select the option to model the flat-to-slant crack growth behavior. If this option is selected, the dialog changes as shown in Figure 147.

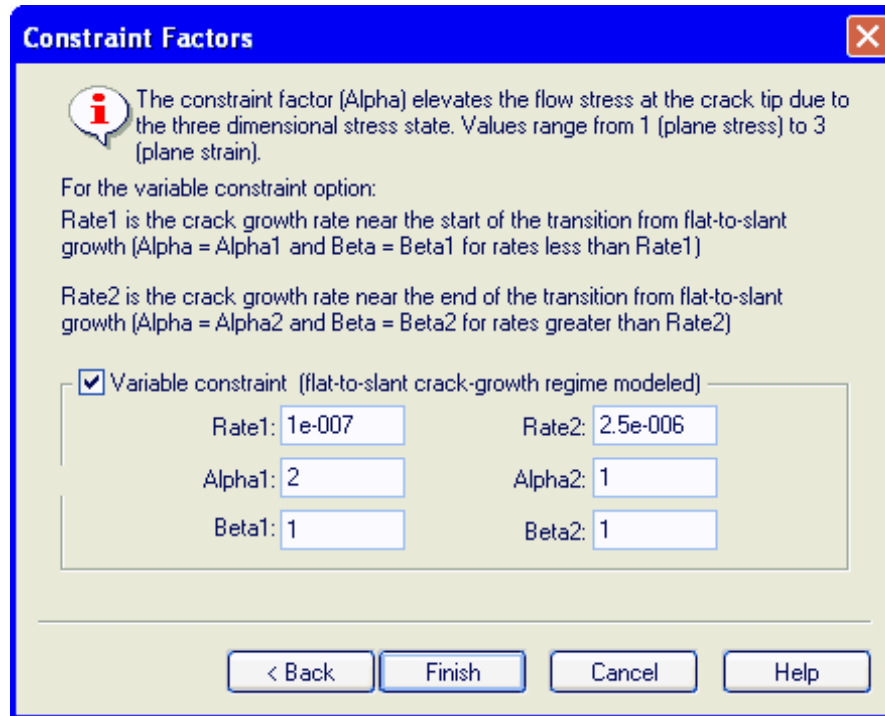


Figure 147: Variable Constraint Factors

In this case, the FASTRAN model applies the user input values for Alpha1 and Beta 1 when the current growth rate is less than Rate1. When the current growth rate is greater than Rate2, the model applies Alpha2 and Beta2. For rates between Rate1 and Rate2, Alpha and Beta values are interpolated (log-linear for rate and constraint) between the two values. Normally, users should use higher constraint values (toward plane strain) at the lower rates. The example shown in Figure 147 is for a case where Alpha is variable and Beta is constant. For most cases, Beta = 1 is recommended.

3.2.6.3.4 Comparison of AFGROW/FASTRAN and FASTRAN Version 3.8e

The FASTRAN source code, by Dr. James C. Newman, Jr., is written in FORTRAN and was translated into C for use in AFGROW. As a result, there are some differences in the results. The differences seen in the current version of AFGROW and relatively minor, but the following examples may be instructive.

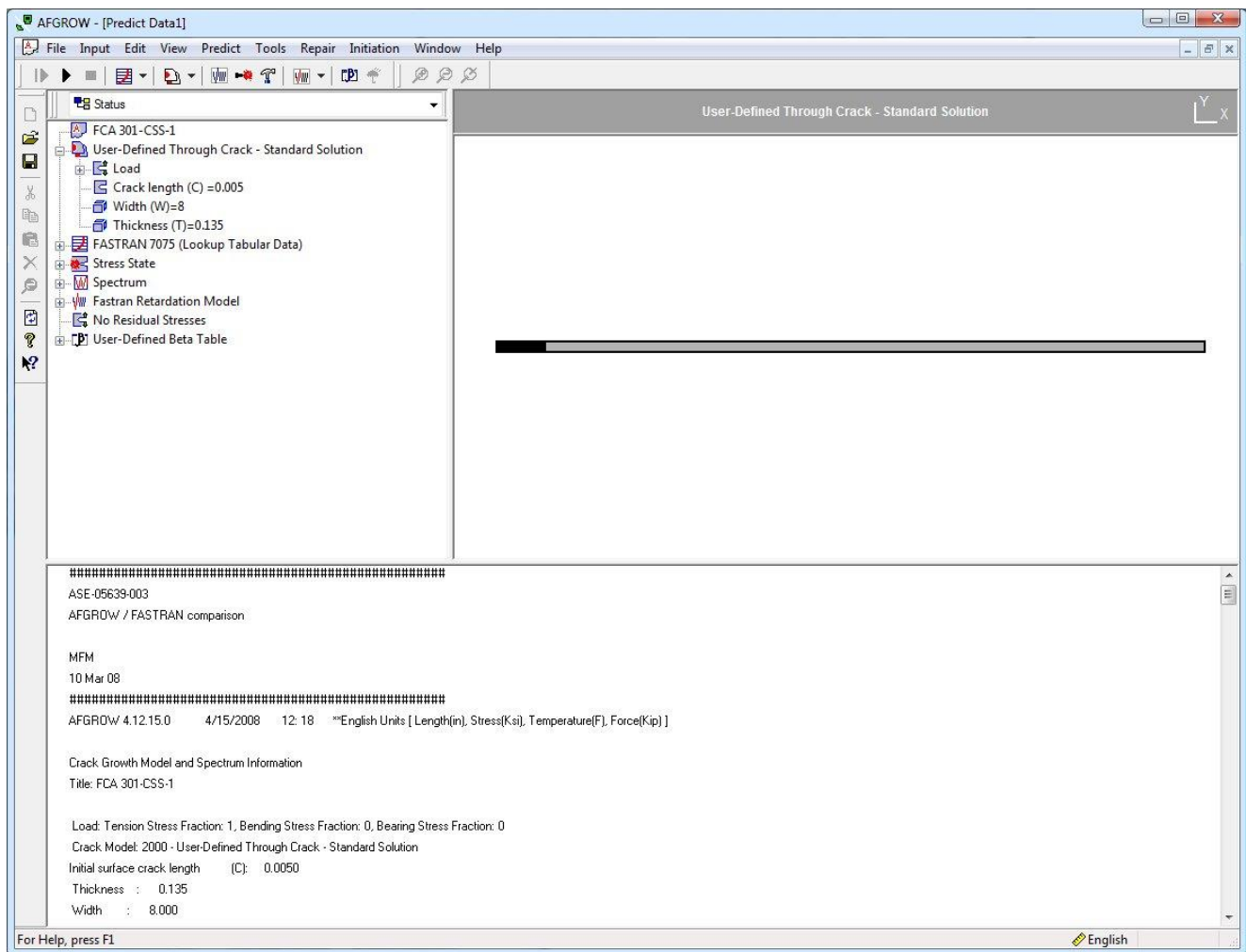


Figure 148: User-Defined FASTRAN Model

The user-defined beta table and effective crack growth rate data are given in Figure 149.

da/DN	Delta K(eff)
1.00E-09	0.75
2.00E-08	1
1.00E-07	1.7
1.00E-05	5
1.00E-04	10.5
1.00E-02	30

C	Beta	C	Beta	C	Beta	C	Beta
0.0000	2.0517	0.0781	1.2211	0.1778	1.1713	0.4571	1.1581
0.0025	2.0009	0.0906	1.1953	0.2214	1.1124	0.4833	1.2068
0.0075	1.9078	0.1094	1.1939	0.2621	1.0825	0.5081	1.2718
0.0125	1.8335	0.1156	1.2242	0.3355	1.0681	0.5750	2.0000
0.0313	1.5499	0.1350	1.3190	0.3688	1.0706		
0.0469	1.3847	0.1351	1.2758	0.4000	1.0893		
0.0625	1.2794	0.1500	1.2309	0.4294	1.1185		

Figure 149: FASTRAN Material and Beta Information

The FASTRAN model parameters for this example are shown in Figure 150.

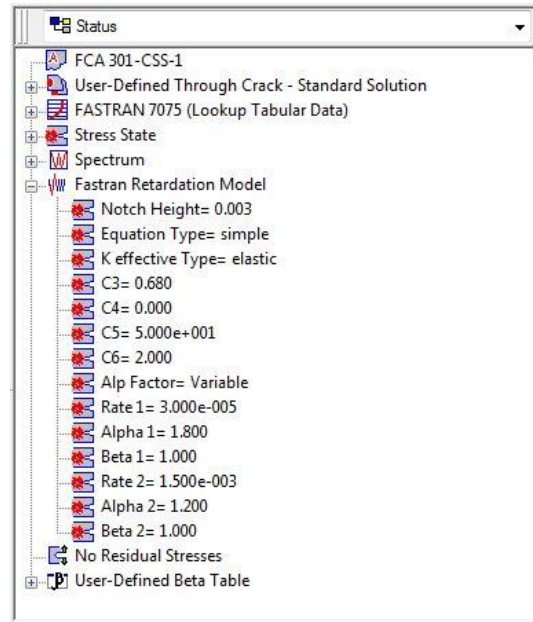


Figure 150: FASTRAN Model Parameters

Finally, two types of loading spectra were used. The first was a relatively large variable amplitude stress spectrum adjusted to a maximum stress level of 31.657 Ksi (SMF = 0.001147). The spectrum is shown in Figure 151.

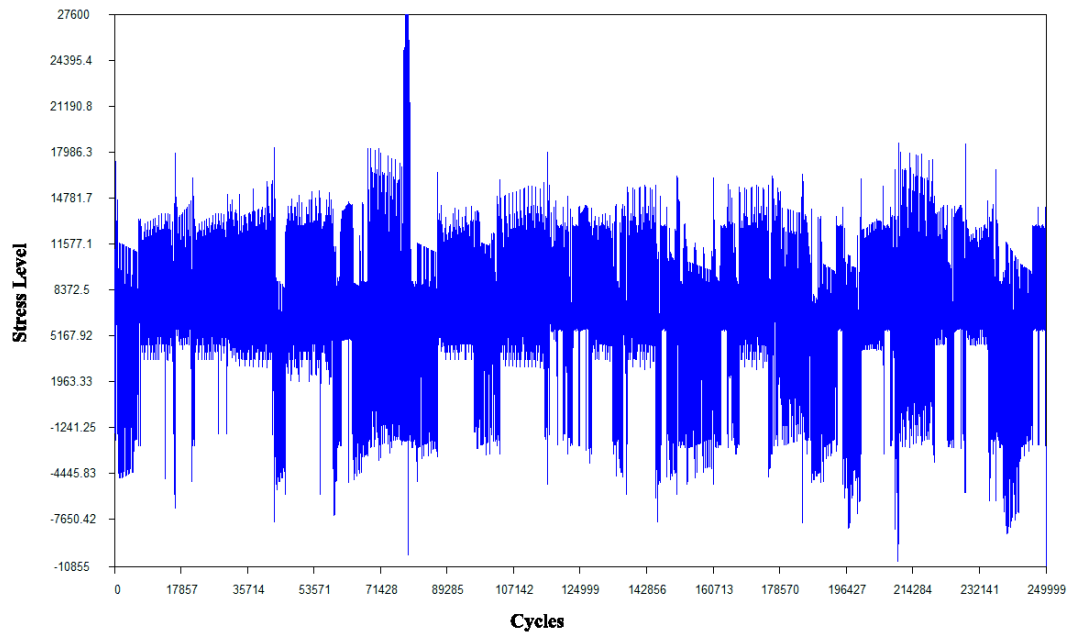


Figure 151: FASTRAN Spectrum

The resulting lives for AFGROW/FASTRAN and FASTRAN, Version 3.8e are shown in Figure 152.

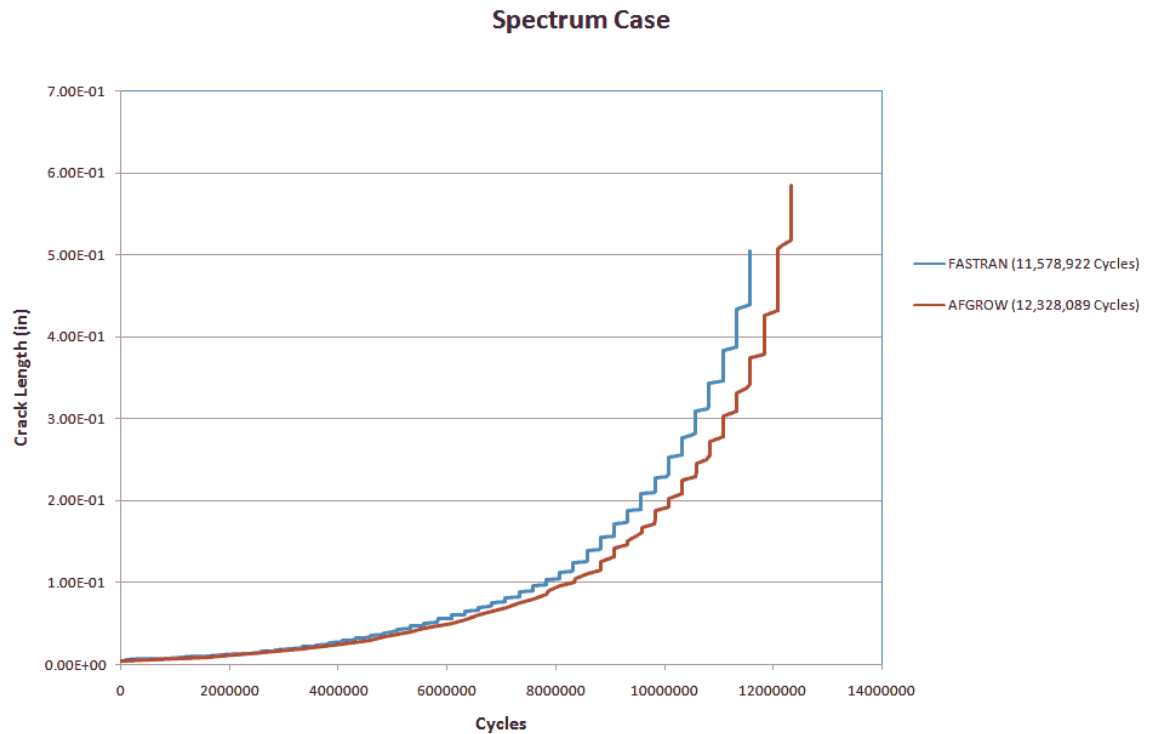


Figure 152: FASTRAN Spectrum Life Comparison

The second spectrum was a simple constant amplitude spectrum ($R=0$) at two stress levels (10, and 31.66 Ksi).

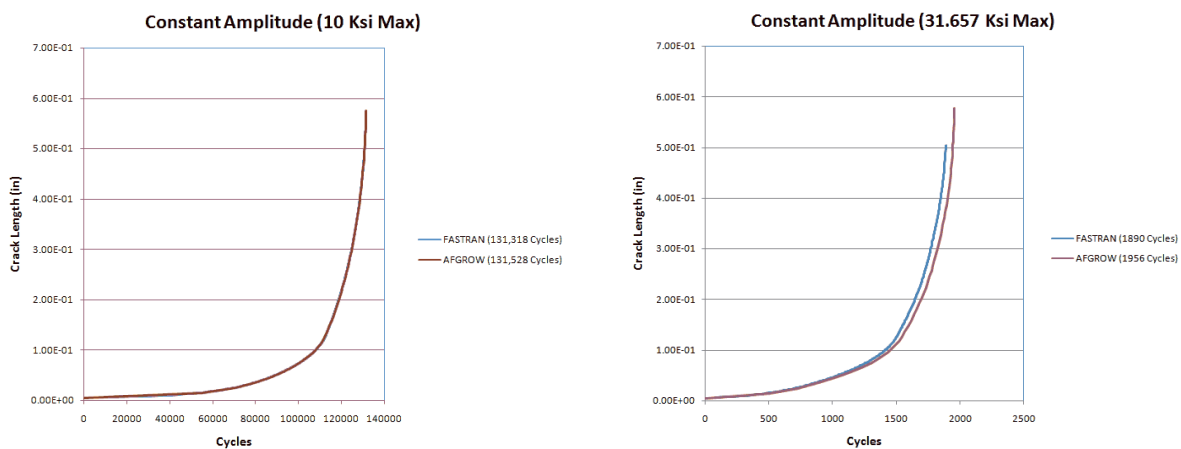


Figure 153: FASTRAN Constant Amplitude Life Comparison

3.2.6.4 Hsu Model

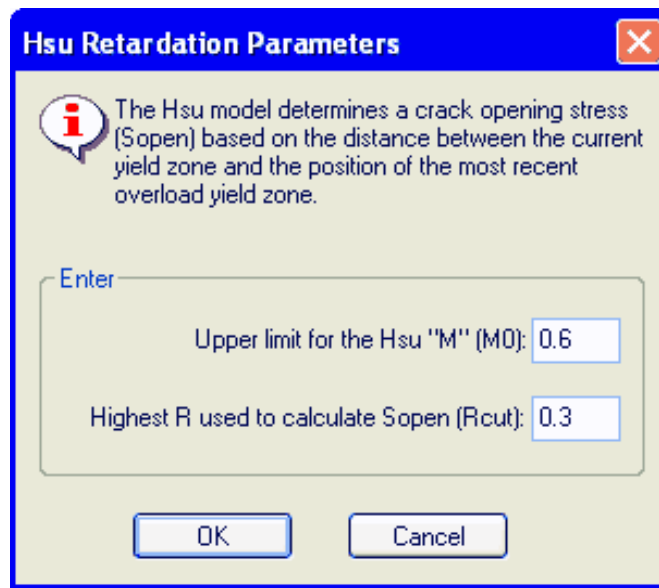


Figure 154: Hsu Model Dialog

The Hsu model⁵⁶ uses an effective stress and closure concept. The model is not only capable of accounting for the retardation effect due to tensile overload, but also accounts for the effect of the compression portions of tension-compression load cycles on the fatigue crack growth rate during subsequent load cycles. The current Hsu model is unable to account for compression-compression cycles. In general, over loads decelerate or retard crack growth, while under loads accelerate crack growth.

3.2.6.4.1 Overview of the Hsu Model

The Hsu process starts by making an innovative assumption by checking both opening stress level and plastic zone size. The spectrum is assumed to start on $\sigma_{\min}$ and growth is calculated for the stress (load) going from $\sigma_{\min}$ to $\sigma_{\max}$ ⁵⁷. Crack growth occurs for the first half of the load cycle - on up ticks. At the instance of the “up tick” of the first cycle, an initial overload opening stress (σ_{oOL}) and effective load interaction zone is calculated. The subsequent half cycle – down tick, does not contribute to crack growth. The ensuing cycles, are processed starting with a check for $\sigma_{\max} > \sigma_{oOL}$. If this test fails, then there will be no crack growth for this cycle since $\sigma_{\max}$ is not high enough to open the crack. If the opening stress check is passed, the plastic zone is calculated at the end of this cycle using $\sigma_{\max}$. Should the current plastic zone be less than the residual effective load interaction zone size, crack growth will be retarded by modifying the minimum stress of the cycle, if not then the residual effective load interaction zone size and opening stress are reset. The minimum

⁵⁶ The AFGROW implementation of the Hsu model was developed by Thomas W. Deiters Engineering, Inc., and this part of the users guide was taken from reference [51]

⁵⁷ Since each cycle has a max and min value, the order used in the AFGROW spectrum input makes no difference

stress of the cycle is checked for compression. If it is compressive, corrections are made to both the residual effective load interaction zone size and the required overload stress (i.e., Willenborg et al required stress) and minimum effective stress (if retarded). Crack growth rate is determined using the minimum effective stress to determine the current minimum stress intensity value for the current cycle. Although the Hsu model uses a closure based concept to determine the opening stress, it still uses the standard user input crack growth rate data to determine growth rate based on delta K (or Kmax if R<0). The Hsu model simply modifies the minimum K-value for a given spectrum cycle to account for load interaction effects.

3.2.6.4.2 Opening Stress

At the instance of the first half load cycle and every overload half cycle thereafter, Hsu calculates an opening stress of overload cycle as follows.

$$\sigma_{oOL} = \frac{\sigma_{max}^2}{\sigma_{Fty}}$$

For subsequent non-overload half cycles or in between half cycles, σ_{oOL} is set to the following.

$$\sigma_{oOL} = \frac{\sigma_{OLeff}^2}{\sigma_{Fty}}$$

Where σ_{OLeff} is the Willenborg et al stress that is required to produce the effective interaction zone, r_{peff} , at the current crack length. It is derived in the next section.

If $\sigma_{max} \geq \sigma_{oOL}$ then the stress cycle is considered for crack growth and the process continues to the check on plastic zone size. If $\sigma_{max} < \sigma_{oOL}$ then this cycle is assumed to produce no crack growth and the process continues to the next half cycle. In both instances, the minimum stress is checked for compression and appropriate corrections are made as covered in the compression effect section. Thus this check is a screening or threshold check. The σ_{max} must be greater than σ_{oOL} or there can be no growth. The initial setting can be explored to gain insight into this check by simple factoring.

$$\sigma_{oOL} = \frac{\sigma_{max}^2}{\sigma_{Fty}} \Rightarrow \frac{\sigma_{oOL}}{\sigma_{max}} = \frac{\sigma_{max}}{\sigma_{Fty}}$$

This equation states that the ratio of opening stress to maximum stress is the same as the maximum stress to yield strength. It can be recognized that the maximum spectra stress for transport aircraft could be around 20 KSI and yield strength could be around 60 KSI, so that the ratio of opening stress to maximum stress could be around 0.333. Therefore, the Hsu process only turns away applied load half cycles whose maximum stress is less than 0.333 times 20 or 6.7 KSI but even this number is reduced during intermediate cycles and so even less cycles are turned away. At the time of its creation, computer time was outlandish costing \$800 per crack run; therefore Hsu implemented this check in an effort to keep processing costs down. If no similar constraint exists today this check step could be eliminated.

3.2.6.4.3 Effective Load Interactive Zone

The Hsu model uses a load interaction zone concept based on the Irwin plastic zone model as a criterion to determine whether the crack growth of the current cycle will be altered from that of constant amplitude. The basic dimensions for the load interactive zone are shown in Figure 155.

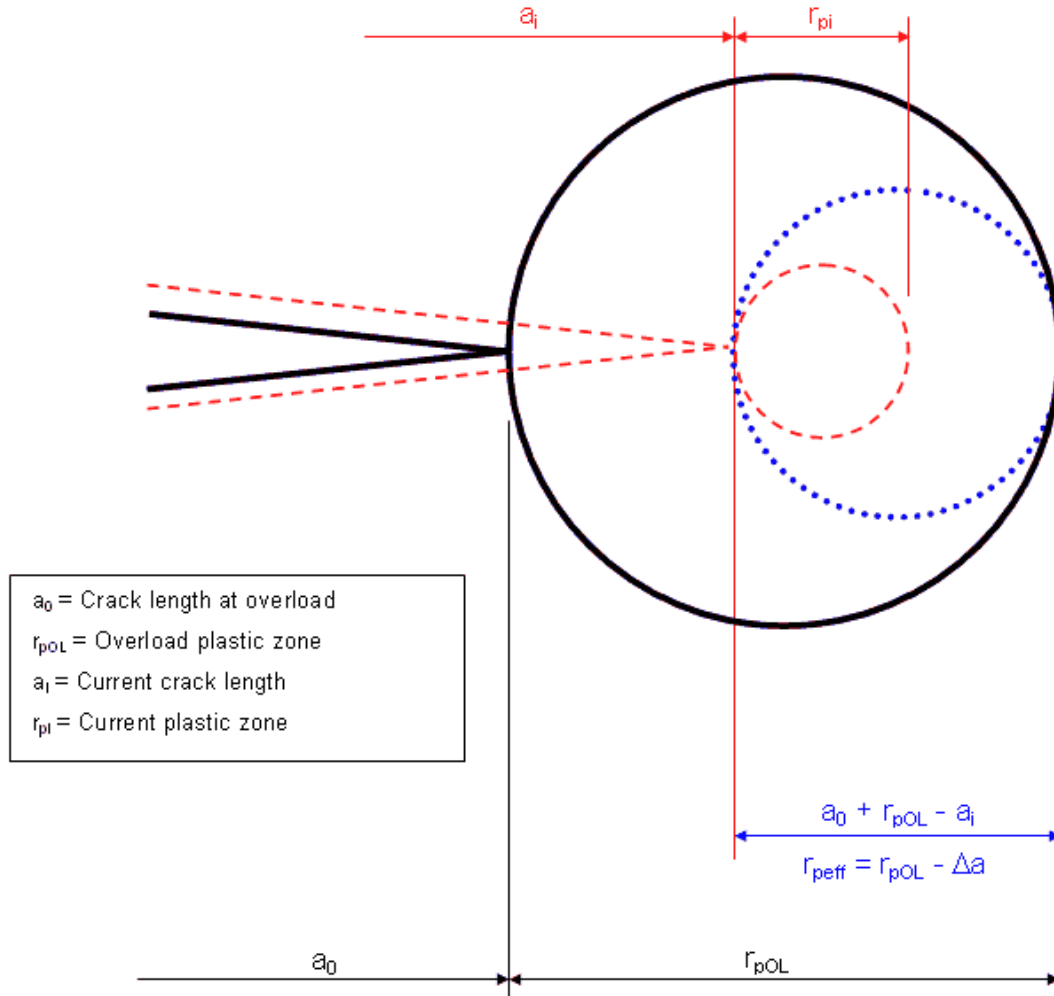


Figure 155: Load Interactive Zone

To start, assume that an over load stress occurred. By definition this will have occurred in Figure 155 at a_0 and produced K_{maxOL} which produced an over load plastic zone equal to the following.

$$r_{pOL} = \frac{1}{\alpha \pi} \left[\frac{(K_{max})_{OL}}{\sigma_{Fty}} \right]^2$$

Next assume that the application of a subsequent half cycle produced growth equal to Δa . Then by definition the effective load interaction zone is determined as follows.

$$r_{peff} = r_{pOL} - \Delta a$$

As the crack grows further away from a_0 , the load interaction zone, r_{peff} decreases. The plastic zone of the current crack, a_i , is.

$$r_p = \frac{1}{\alpha \pi} \left[\frac{K_{max}}{\sigma_{Fty}} \right]^2$$

If $r_p \geq r_{peff}$, there will be no load interaction and the crack growth rate associated with the cycle will be generated as under constant amplitude loading. Conversely, if $r_p < r_{peff}$, then the crack growth rate will be reduced by modifying the minimum stress of the cycle.

At crack length a_i we can associate a stress intensity factor, $K_{max\ eff}$ with the effective interaction zone by solving the following equation.

$$r_{peff} = \frac{1}{\alpha \pi} \left[\frac{K_{max\ eff}}{\sigma_{Fty}} \right]^2 = \frac{1}{\alpha \pi} \left[\frac{K_{OL\ eff}}{\sigma_{Fty}} \right]^2$$

And this stress intensity factor, $K_{max\ eff}$ can be converted into an effective load interaction zone stress, $\sigma_{OL\ eff}$ easily as follows.

$$\sigma_{OL\ eff} = \frac{K_{max\ eff}}{\sqrt{\pi a \beta}} = \frac{K_{OL\ eff}}{\sqrt{\pi a \beta}}$$

This is exactly the same as the required Willenborg et al stress. This is used in the calculation of σ_{oOL} above in the opening stress section.

3.2.6.4.4 Retardation Calculations

As stated in the previous section, the Hsu formulation modifies the minimum stress of the applied cycle to take into account variable amplitude load interaction. Therefore, if the plastic zone size of the current half cycle (see Figure 155) is less than the effective load interaction zone, Hsu redefines the minimum stress to be an effective minimum stress as follows.

$$\sigma_{min\ eff} = \sigma_{min\ i} + \alpha \Delta \sigma$$

Where

$$\Delta \sigma = \sigma_{max\ i} - \sigma_{min\ i}$$

$$\alpha = \left(1 - \lambda_H^{2m} \right) \sqrt{(1 - R)} ; 0 < \alpha < 1.0$$

R must be positive in order to limit α to 1.0

α , λ_H and m

Since the Hsu model predicts load interaction when the current plastic zone is within the effective load interaction zone for a given overload condition, it is important to keep track of the size of this zone as a crack grows. The plastic zone required to fill the interaction zone is:

$$r_{p \text{ req}} = a_0 + r_{pOL} - a_i = r_{peff}$$

This is also the same as the effective interaction zone, r_{peff} as shown in Figure 155. The Hsu model uses the Wheeler model concept (see Section 3.2.6.5) as follows:

$$\phi = \left[\frac{r_{pi}}{r_{p \text{ req}}} \right]^m$$

This can be expressed in terms of stress intensity, K . Since

$$r_{pi} = C \left(\frac{K_{\max i}}{\sigma_{FTY}} \right)^2$$

$$r_{p \text{ req}} = C \left(\frac{K_{\max \text{ req}}}{\sigma_{FTY}} \right)^2$$

Where $C = \frac{1}{\alpha \pi}$ and α = constraint factor

$$\text{Then } \phi = \left[\frac{r_{pi}}{r_{p \text{ req}}} \right]^m = \left[\left(\frac{K_{\max i}}{K_{\max \text{ req}}} \right)^2 \right]^m = \left[\frac{K_{\max i}}{K_{\max \text{ req}}} \right]^{2m}$$

This in turn can be expressed in terms of stress, since

$$K_{\max i} = \sigma_{\max i} \sqrt{\pi a_i} \beta_{(a_i)}$$

and

$$K_{\max \text{ req}} = \sigma_{\max \text{ req}} \sqrt{\pi a_i} \beta_{(a_i)}$$

Finally

$$\phi = \left[\frac{\sigma_{\max i}}{\sigma_{\max \text{ req}}} \right]^{2m} = \left[\left(\frac{\sigma_{\max i}}{\sigma_{\max \text{ req}}} \right)^2 \right]^m$$

Hsu defines,

$$\lambda_H = \frac{\sigma_{\max i}}{\sigma_{\max \text{ req}}}$$

Then $\phi = [\lambda_H^2]^m$

In summary, Wheeler's equation may be expressed in the following forms.

$$\phi = \left[\frac{r_{pi}}{r_{p \text{ req}}} \right]^m = \left[\left(\frac{K_{\max i}}{K_{\max \text{ req}}} \right)^2 \right]^m = \left[\left(\frac{\sigma_{\max i}}{\sigma_{\max \text{ req}}} \right)^2 \right]^m = [\lambda_H^2]^m$$

The exponent, 'm', in Wheeler's equation is empirically derived to give the best fit to test data. In Wheeler's expression it can be seen that 'm' acts as an effectivity constant on the ratios of; plasticity, Ks, or stresses, that is 'm' determines how effective the ratios are. If 'm' equals 1.0, the ratios are unaffected. Hsu formulated an expression in terms of 'm' that does not rely on empirically derived parameters -- except as a limiting case. The Hsu model defines the minimum effective stress as follows.

$$\sigma_{\min \text{ eff}} = \sigma_{\min i} + \alpha \Delta\sigma$$

Where

$$\Delta\sigma = \sigma_{\max i} - \sigma_{\min i}$$

$$\alpha = (1 - \lambda_H^{2m}) \sqrt{(1 - R)}; \quad 0 < \alpha < 1.0$$

R must be positive in order to limit α to 1.0

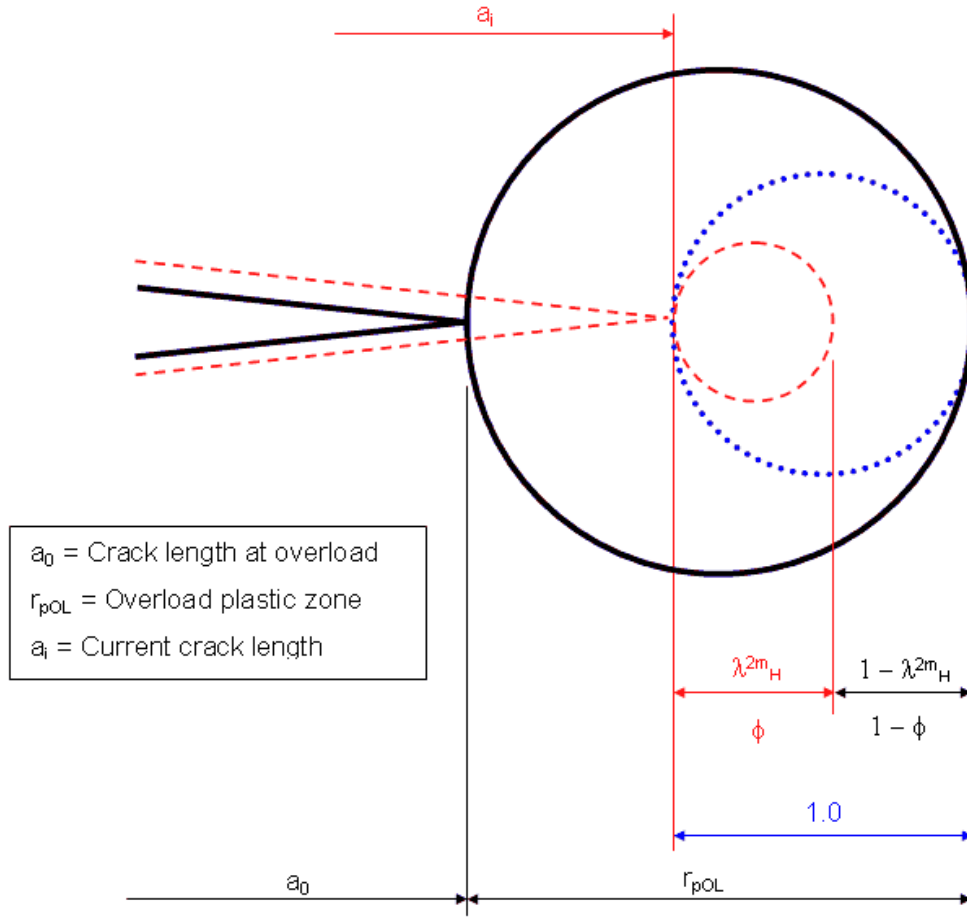


Figure 156: Normalized Load Interaction Zone

To understand the physical significance of this formulation, Figure 156 presents the plastic zone illustration of Figure 155, with the required plastic zone normalized to 1.0 by $a_0 + r_{pOL} - a_i$ or r_p required, and includes the effectivity exponent (m). Load retardation requires that the current plastic zone be less than the required plastic zone or that, $r_{pi} < a_0 + r_{pOL} - a_i$ or r_{peff} which is the same thing as saying that λ^2_H will always be less than one, this is easily seen in Figure 156. The normalized current plastic zone without the effectivity exponent is equal to λ^2_H , and the effective normalized current plastic zone equal to λ^{2m}_H as shown in Figure 156. The distance between the current plastic radius and the overload plastic boundary is equal to Hsu's α without the square root of $(1-R)$ term.

$$\alpha = (1 - \lambda^{2m}_H) \sqrt{(1 - R)}; 0 < \alpha < 1.0$$

The inclusion of the square root term is evidently a correction refinement that provided better correlation to test and suggests that Hsu found that the effect of closure decreased with increasing R-ratio. As R-ratios increase the effect is to reduce α , and reducing α , reduces the effective cyclical stress, which in turn reduces the effective minimum stress, so as R-ratios increase the difference between the effective minimum stress and the minimum stress decreases and in the limit the effective minimum stress equals the minimum stress and there is no load interaction effect. In summary, Hsu bases his formulation on the available plasticity ahead of the current plastic zone to the overload

plastic boundary and modifies its effectivity by ‘m’ and square root of (1-R). Hsu formulates ‘m’ as follows.

$$m = \frac{1}{\lambda_H} - 1 \leq m_0$$

Where m_0 is the limiting value where the delay in crack growth starts to decrease or where the effect of retardation starts being reduced.

It should be obvious that the Hsu ‘m’ is not equal to the Wheeler ‘m’. The Hsu ‘m’ is an expression, which has a shut off value of m_0 . Hsu modifies each $\lambda = (\sigma_{\max i} / \sigma_{\max \text{ req}})$ ratio in the spectrum differently provided ‘ $m < m_0$ ’. The Wheeler ‘m’ modifies every ratio equally. The determination of ‘ m_0 ’ from test data is dependent on the ‘m’ expression as well as the square root of (1-R). The ‘ m_0 ’ value is essentially a tuning factor to adjust the acceleration or deceleration of retardation of the overall spectrum and material. So while Hsu is an improvement over Wheeler and Willenborg et al in that the parameters can be calculated, but it is still to a degree empirically based due to the dependency on ‘ m_0 ’ in the limit.

Rcut

Hsu found that Shih and Wei [52], conducted a study on crack closure in fatigue for Ti-6Al-4V titanium alloy and observed no crack closure for R-values greater than 0.3. The statement that no closure exists above a certain R-value can be interpreted today in terms of the Cf function versus R relationship shown in Figure 139.

$$Cf = \sigma_{\text{opening}} / \sigma_{\max}$$

There is almost universal agreement that σ_{opening} is approximately equal to σ_{closure} . Therefore, the statement that no closure exists means that σ_{opening} is equal to $\sigma_{\min i}$. This can be seen in Figure 139 as the point where the Cf vs. R curve intersects the line (Cf = R).

Based on the Shih Wei study, Hsu decided to set $R = 0.3$ if R is greater than 0.3 in the α equation. Therefore, the Rcut in the Hsu model is taken as the maximum R-value that is used in the retardation calculations

Remembering that Hsu’s minimum effective stress is written as:

$$\sigma_{\min \text{ eff}} = \sigma_{\min i} + \alpha \Delta\sigma$$

and

$$\alpha = (1 - \phi) \sqrt{(1 - R)}$$

When Rcut is used the alpha value will stop decreasing when the applied R-value is greater than Rcut. This means that the opening stress ($\sigma_{\min \text{ eff}}$) will be larger than the

applied minimum stress ($\sigma_{\min i}$). This is not consistent with the idea that the crack is always open above R_{cut} . However, this is how the Hsu model was developed and is implemented in AFGROW.

The R_{cut} value is used as a tuning parameter and provides additional flexibility.

3.2.6.4.5 Compressive Effects

The section presents Hsu model aspects affected by compression. A compression load will accelerate the fatigue crack growth and shorten the life. If the compression load is neglected, the fatigue crack growth life prediction will be un-conservative. Therefore, for the case where the minimum load is compressive, modification of the effective plastic zone and its corresponding effective tensile overload is necessary. The clarity in time history of when and where these modifications are to be made indicates some hurried last minute thinking. During unloading of an overload cycle, the change of stress field and the plastic zone will behave linearly. However, Hsu has stated, should the minimum stress of the subsequent applied load cycle continuously decrease from tension into compression, reverse (or compressive) yielding will start to occur and the benefit of residual strain created by the tensile overload will begin to decrease. Therefore, one may assume that the effect of compressive load on cyclic fatigue growth depends upon the magnitude of the compressive load and compressive yield strength. The compressive correction factor follows.

$$F_c = \left(\frac{\sigma_{Fty} - \sigma_c}{\sigma_{Fty}} \right)^{\frac{1}{2}} = \left(1 - \frac{\sigma_c}{\sigma_{Fty}} \right)^{\frac{1}{2}}$$

The form of F_c is based on the following reasoning.

1. If there is no compressive load then $F_c = 1.0$, i.e., no effect,
2. If the compressive load reaches the compressive yield strength, $F_c = 0$, completely nullifies tension overload,
3. The choice of the exponent $\frac{1}{2}$ is based on the argument that the compressive load effect is proportional to the square root of the plastic zone size, since the plastic zone is proportional to the square of the applied stress. The basis for this was by considering the relations of the terms in the plastic zone equation.

The compressive correction factor is applied to the effective overload plastic zone at the encounter of a compressive minimum stress as follows.

$$(r_{eff})_c = F_c (r_{eff})_t$$

Where Subscripts ‘c’ and ‘t’ are compression and tension, respectively

$(r_{eff})_t$ is the size of the effective tensile plastic zone prior to the encounter with compressive load

$(r_{\text{eff}})_c$ is the size of the effective tensile plastic zone after the encounter with the compressive load

$$r_{\text{eff}} = \frac{1}{\alpha \pi} \left[\frac{(K_{\text{max}})_{\text{eff}}}{\sigma_{\text{Fty}}} \right]^2$$

The effective over load stress following the encounter with a compressive stress will become.

$$(\sigma_{\text{OL}})_{\text{effc}} = F_c^{1/2} (\sigma_{\text{OL}})_{\text{eff}}$$

The effective minimum stress of the half cycle that contains the compressive minimum is to be set as follows

$$(\sigma_{\text{min}})_{\text{effc}} = F_c^{1/2} (\sigma_{\text{min}})_{\text{eff}}$$

This essentially drives the effective opening (minimum effective) stress to a lower value, which reduces the retardation effect – this is the desired compression effect.

In summary, the compressive load effect is developed and applied consistently by modifying the effective residual plastic zone, the minimum effective stress value of the minimum stress half cycle, and the required stress at the current crack length to give the residual plastic zone.

3.2.6.5 Wheeler Model

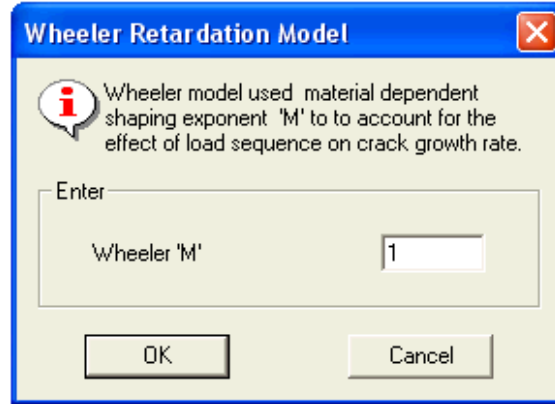


Figure 157: Wheeler Model Dialog

The Wheeler retardation model [53] is one of the most empirical load interaction models in use in Fracture Mechanics. It works by modifying the current crack growth rate with a "knock-down" factor based on the ratio of the current yield zone size to the difference between the effective crack length of an overload condition and the current crack length. Here's how it works:

$$\frac{da}{dN} = C_p \left(\frac{da}{dN} \right)$$

Where:

$$C_p = \left(\frac{R_y}{X_{eff(ol)} - X} \right)^m$$

X is the crack length

X_{eff} is the crack length plus the yield zone size

$$R_y = \left(\frac{K_{max}}{Yield} \right)^2 \left(\frac{1}{\pi PSX} \right)$$

Note: AFGROW uses the Irwin yield zone equation (and the current stress state) to determine the yield zone size. The subscript (ol) refers to an overload condition. It is changed each time that an applied maximum stress (or load) exceeds a previous maximum, or when the current yield zone size (R_y) grows beyond the yield zone created by an overload ($R_y(ol)$). PSX is the stress state for the given crack length (2 – Plane Stress, 6 – Plane Strain).

Retardation Parameter:

m : Wheeler exponent

The value of the Wheeler exponent, m , is determined from test data for a given material, spectrum, stress level, etc. As mentioned above, this model is extremely empirical and the m value, which gives good correlation to test data, has been known to be dependent on MANY test parameters. Users should use this model with caution.

3.2.6.6 Generalized Willenborg Model

The Generalized Willenborg model [54] is one of the most commonly used load interaction models in crack growth life prediction programs. The model is based on early fracture mechanics work performed at Wright-Patterson AFB, OH and was named after a student who worked on the model. The model uses an "effective" stress intensity factor based on the size of the yield zone in front of the crack tip. The formulation of the Willenborg retardation model used in AFGROW is given below:

$$K_{\max}(\text{eff}) = K_{\max} - K_{\text{red}}$$

$$K_{\min}(\text{eff}) = K_{\min} - K_{\text{red}}$$

$$R(\text{eff}) = K_{\min}(\text{eff})/K_{\max}(\text{eff})$$

$$K_{\text{red}} = \phi \left(K_{\max}(\text{ol}) \sqrt{1 - \frac{(x - x(\text{ol}))}{R_y(\text{ol})}} - K_{\max} \right)$$

$$\phi = (1 - \Delta K_{\text{Threshold}}/K_{\max})/(\text{SOLR} - 1)$$

$$R_y(\text{ol}) = \left(\frac{K_{\max}(\text{ol})}{Yield} \right)^2 \left(\frac{1}{PSX \pi} \right)$$

Where:

x : Crack Length

x(ol) : Crack Length at Overload

$\Delta K_{\text{Threshold}}$: Threshold value of ΔK at $R = 0$

SOLR: Shutoff Overload Ratio (Ratio of the overload to nominal load required to effectively stop further growth under nominal loading)

Yield : Material yield strength

PSX : Stress State in a Given Crack Growth Direction (2.0 (Plane Stress) - 6.0 (Plane Strain))

The subscript (ol) refers to an overload condition. It is changed each time that an applied maximum stress (or load) exceeds a previous maximum, or when the current yield zone size (R_y) grows beyond the yield zone created by an overload ($R_y(\text{ol})$). The value, ϕ , is simply a parameter used in the Generalized Willenborg model. The $\Delta K_{\text{Threshold}}$ is the lowest

value of ΔK that will cause a crack to grow for $R = 0$. The value is based on user input for the crack growth rate model being used in a given prediction.

The generalized Willenborg model uses the shutoff overload ratio (SOLR) as a material property to control the effect of load history on the predicted life. This parameter is input by the user when this model is selected (as shown in Figure 158)

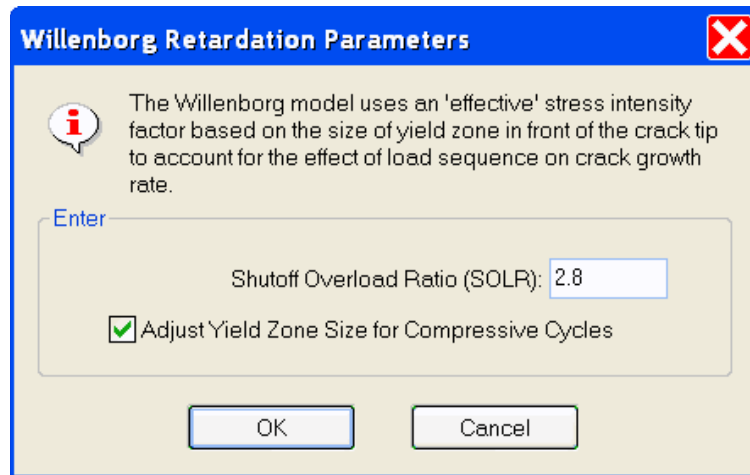


Figure 158: Willenborg Retardation Parameter Dialog

SOLR : Shut-off Ratio - Ratio of overload maximum stress to the subsequent maximum stress required to arrest crack growth

The exact value of the SOR is varied to adjust the life prediction to match test results. Ideally, the SOLR should be a material parameter, which is insensitive to spectrum or stress level. However, this does not always work out. The following is a list of common SOLR values for some materials:

Aluminum: SOLR = 3.0

Titanium: SOLR = 2.7

Steel: SOLR = 2.0

Many crack growth programs use the Chang acceleration model [55] with the Willenborg retardation model to account for the effect of compressive stress (or load) cycles. The Chang model requires the use of negative stress intensity values. AFGROW does not consider negative stress intensity factors to be valid (in general). In place of the Chang acceleration model, AFGROW uses the following method to account for the effect of compressive stresses (or loads):

$$Ry(ol) = \left(1 - 0.9ABS\left(\frac{\sigma_{compression}}{\sigma_{overload}}\right) \right) Ry(ol)$$

This method is used by default in AFGROW, but users may turn this option off by de-selecting the option: “Adjust Yield Zone Size for Compressive Cycles” in the Willenborg Retardation Parameters dialog box (see Figure 158).

Using the absolute value of the ratio of the compressive stress (or load) to the overload stress (or load) reduces the size of the current overload yield zone. This method will NOT increase the effective stress intensity; it will merely reduce the retarding effect of a previous overload. Therefore, the Willenborg model used in AFGROW can NEVER result in a life prediction that is less than the life prediction with no retardation.

The Generalized Willenborg model is a yield zone based empirical model that has been used for many years. It is quite different from crack closure based models since it applies a reduction to Kmax and Kmin instead of a reduction in ΔK_{eff} .

3.2.6 Input Stress State

Toolbar Icon:

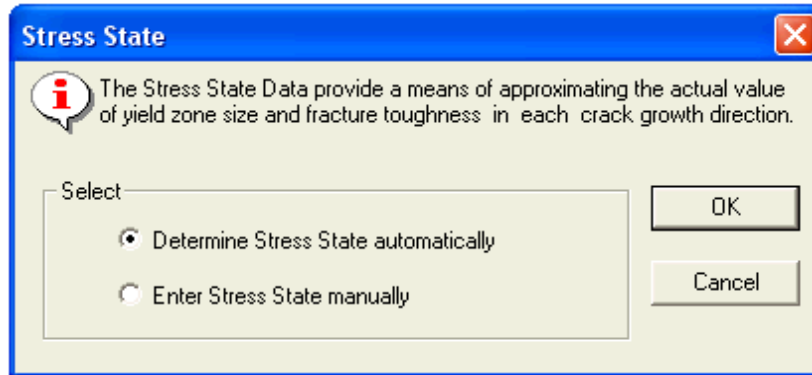


Figure 159: Stress State Dialog

There are currently two choices in AFGROW for Stress State: Automatic Stress State Determination and User Specified. AFGROW uses a stress state index of real numbers that range from 2 to 6. The range was chosen because of the relationship between stress state and the Irwin yield zone size.

$$\text{Plane Stress: Yield Zone Size} = \left(\frac{K_{\max}}{Yield} \right)^2 \frac{1}{2\pi}$$

$$\text{Plane Strain: Yield Zone Size} = \left(\frac{K_{\max}}{Yield} \right)^2 \frac{1}{6\pi}$$

AFGROW uses the stress state index to determine the yield zone size, which is required for the load iteration models, AND to determine the appropriate value of fracture toughness. The yield zone size is determined as follows:

$$\text{Yield Zone Size} = \left(\frac{K_{\max}}{Yield} \right)^2 \frac{1}{(index)\pi}$$

The actual value of fracture toughness that defines the stress intensity failure limit for a given geometry is often called the apparent fracture toughness since it is determined by the given geometry and applied failure stress. The highest possible value of fracture toughness is the plane stress fracture toughness and the lowest possible value is the plane strain fracture toughness. The plane stress and strain fracture toughness values are material properties. The apparent fracture toughness value is determined by a linear interpolation between the plane strain (K_{IC}) and plane stress (K_C) fracture toughness values input by the user.

$$\text{Apparent Fracture Toughness} = K_{IC} + \frac{(6.0 - index)}{4.0} (K_C - K_{IC})$$

The stress state index is a function of the specimen thickness and maximum applied stress intensity. Specimens that are relatively thin are generally operate under plane stress conditions (index = 2.0) and thick specimens are generally plane strain (index = 6.0).

3.2.6.1 Automatic Stress State Determination

The default choice for stress state determination in AFGROW is to automatically determine the stress state index based on K_{max} and specimen thickness for each applied load/stress cycle. The relationship between K_{max} , thickness (t), and stress state index [56] is:

$$\text{Index} = 6.7037 - \frac{1.4972}{t} \left(\frac{K_{max}}{Yield} \right)^2$$

If Index > 6.0, Index = 6.0 (Plane Strain)

If Index < 2.0, Index = 2.0 (Plane Stress)

The above relationship has been verified with fracture test data for several metal alloys. The complete details will be published at a later date. The test results are shown in Figure 160.

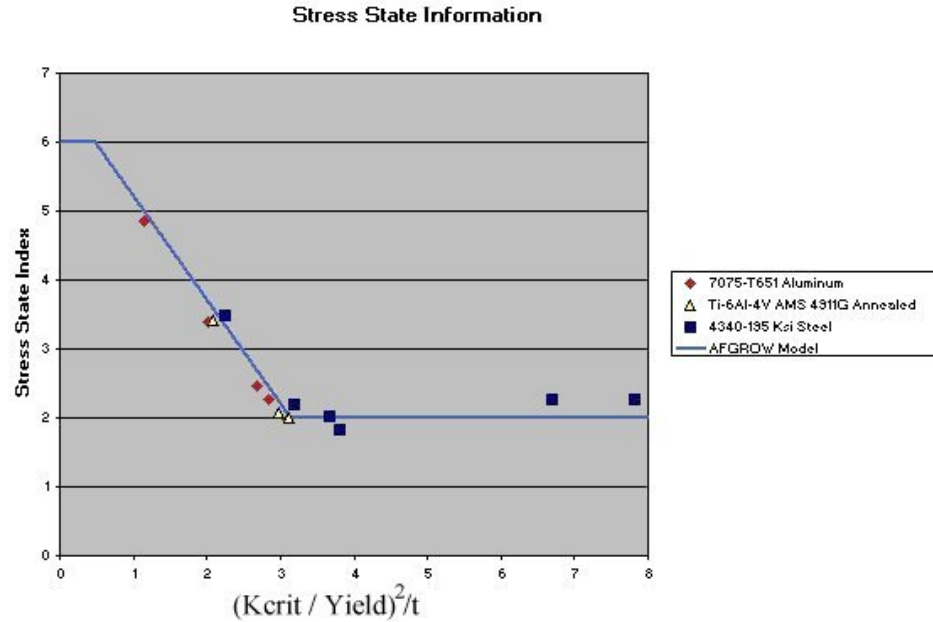


Figure 160: Stress State Information

According to David Broek [57], the plane strain condition is:

$$\text{Plane Strain (index = 6.0)} : \left(\frac{K_{crit}}{Yield} \right)^2 \leq 0.47 t$$

The ASTM standard for a plane strain condition [58] is:

$$t \geq 2.5 \left(\frac{K_{crit}}{Yield} \right)^2$$

The ASTM standard is slightly more conservative, but it meets Dr Broek's plane strain condition.

There is no definitive reference for the plane stress condition. The test data shown in Figure 160 for three common aircraft alloys (aluminum, titanium, and steel) led Mr. Harter [56] to believe that the following plane stress condition may be applied for these alloys:

$$\text{Plane Stress (index=2.0)} : \left(\frac{K_{crit}}{Yield} \right)^2 \geq \pi t$$

A linear equation was used to determine intermediate stress state indices for conditions between the plane stress and plane strain limits. Plane stress and plane strain fracture toughness values were known for the alloys used in the test program. Each center cracked (MT) specimen was pre-cracked to various crack lengths and loaded monotonically to failure. The failure stress and crack length was used to determine the critical stress intensity factor. The stress state index was determined by linear interpolation based on the plane stress and plane strain fracture toughness values for each material.

3.2.6.2 User Specified Stress State

Users have an option to input stress state index values. If this option is selected, AFGROW will use a constant value for stress state index in a given crack growth dimension. The index range is a real number from 2 to 6, where 2 is used for Plane Stress and 6 for Plane Strain. The user-specified value(s) will remain constant during the life prediction calculations and will be used to determine the apparent fracture toughness.

3.2.7 Input User-Defined Beta

Toolbar Icon:



Users can input their own solutions through the user-defined beta option. However, to use this option, the user must first select either the 1-D or 2-D user defined geometry from the Standard Solutions dialog (see Section 3.2.3). Beta factors are defined as follows:

$$\beta = \frac{K}{\sigma_{ref} \sqrt{\pi x}} ; \text{Where } x \text{ is the appropriate crack length}$$

The crack length dimension in the thickness direction is the a-dimension and the crack length in the width direction is the c-dimension. Application and user defined solutions are identified in the beta solution column in the geometry tab of the model dialog (see Figure 69, Section 3.2.3). There are only two user-defined models among the standard solutions since AFGROW only handles 1-D or 2-D cracks. These geometries are simply generic

models, which depict either a 2-D crack (2 crack dimensions) or a 1-D crack (1 crack dimension). Since the user inputs the beta values, the actual geometry is taken into account by the beta values themselves. The image in the animation frame is merely showing a generic view since it is difficult to show all possible user-defined geometries.

3.2.7.1 One-Dimensional User Defined Betas



The one-dimensional user-defined beta option is used when a user has an existing stress intensity factor solution (in the form of a beta table) for any crack that may be described with one length dimension (1-D) to input in AFGROW.

The geometric beta values are NOT calculated by AFGROW, but are merely interpolated from a one-dimensional user defined table of beta values. Users must supply beta values for a given reference stress at various crack lengths so that the appropriate value at a given crack length may be interpolated. This model is shown as an edge cracked plate in the animation frame. The representation of the model is merely meant to indicate the one-dimensional nature of the crack. It was not possible to create representations of all possible geometries that may be modeled using user defined beta factors.

For the [c] crack length dimension: $K = \sigma_{ref} \sqrt{\pi c} \beta(c)$

When the user-defined beta option is selected for a through crack case, the dialog box shown in Figure 161 appears:

Enter crack length and Beta factors. If the current model geometry is 1000 (user - defined part-through crack), this table is used after the transition to a Beta factors for any crack length are obtained by linear interpolation from this table. For crack lengths outside the range of the table, the nearest value in the table will be used.

Enter up to 25 sets
Number of 'C' Sets: 6

Set	C	Beta
1	0.1	1.001
2	0.2	1.005
3	0.5	1.1
4	1	1.2
5	1.5	1.32
6	2	1.55

File
Open
Save

Finish Cancel Help

Figure 161: Through Crack User-Defined Beta Table Dialog

The initial crack length should be the same or less than the initial part through crack length in the C direction. This is because it may be difficult to know what the crack length will be when transition to a through crack occurs, and it is important that the input data cover the entire range of possibilities. AFGROW will NOT extrapolate user-defined betas and will simply use the nearest data in the event the data are out of range.

If the user-defined through crack input data are saved, AFGROW will give the file a .bet extension which will be visible the next time this dialog is opened (clicking on the read button will open it again). Just remember which directory the data are in if you decide to save to some directory other than the default.

3.2.7.2 Two-Dimensional User Defined Betas



This option is used when a user has an existing stress intensity factor solution (in the form of a beta table) for any crack, which may be described with two length dimensions (2-D) to input in AFGROW. Some users have mistakenly assumed that only corner cracks may be modeled using this option. A corner-cracked plate is merely used to illustrate any two-crack dimensions. The width and thickness dimensions should be appropriate for the actual geometry being modeled.

The geometric beta values are NOT calculated by AFGROW, but are merely interpolated from a two-dimensional user defined table of beta values. Users must supply beta values at various crack lengths so that the appropriate value at a given crack length may be interpolated. This model is shown as a corner cracked plate in the animation frame. The representation of the model is merely meant to indicate the two dimensional nature of the crack. It was not possible to create representations of all possible geometries that may be modeled using user defined beta factors.

For the [a] crack length dimension: $K = \sigma_{ref} \sqrt{\pi a} \beta(a)$

For the [c] crack length dimension: $K = \sigma_{ref} \sqrt{\pi c} \beta(c)$

Please note: The reference stress for the user-defined solution is determined by the user to normalize the beta values, so the same reference (gross stress, net stress, etc.) must be used in the input spectrum.

When the beta icon (or User-Defined Beta menu option) is selected, the dialog shown in Figure 162 will appear:

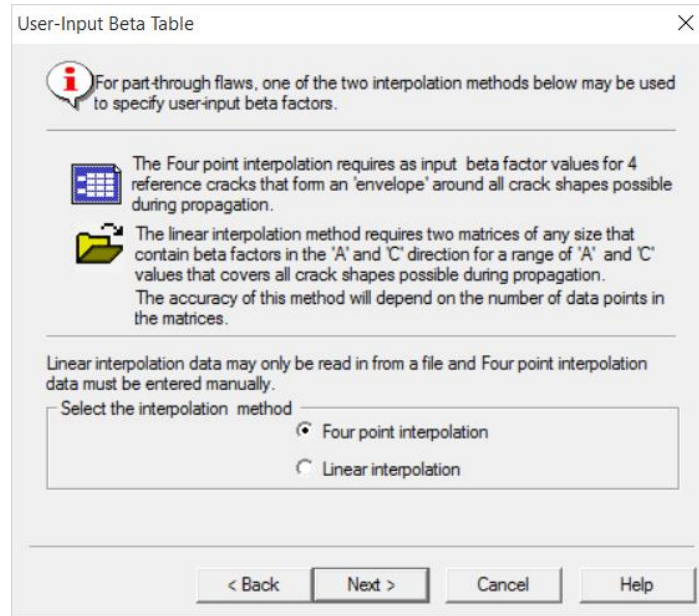


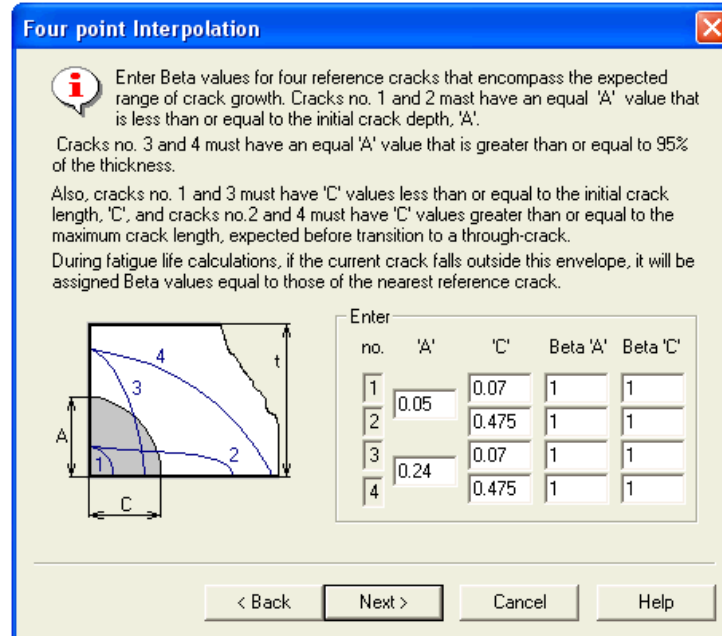
Figure 162: 2-D User Input Beta Dialog

Since beta values for a 2-D crack are dependent on crack shape (a/c), a matrix of beta values are required to determine the appropriate stress intensity factors for each dimension (assuming the dimensions are independent). The two dimensions of crack growth are allowed to grow based on the stress intensity for each dimension. It is not possible to anticipate the changes in crack shape that are possible as the 2-D crack grows under any arbitrary loading.

There are currently two choices in AFGROW to input User Defined Betas for Part-Through Cracks: Four Point and Linear. Actually both use linear interpolation. The Four Point method attempts to provide a simpler method of interpolation, which is based on Schijve's weighted interpolation [59] (without weights, which are model specific). The Linear method is a straightforward double table look-up that must be read from a file.

3.2.7.2.1 Four-Point User-Defined Beta Values

AFGROW will produce the following dialog with crack length suggestions (which may be changed as desired - within the guidelines noted) if the four-point method is selected (see Figure 163).



The dialog box titled "Four point Interpolation" contains an information icon and text explaining the requirements for entering Beta values for four reference cracks. It includes a diagram of a crack in a plate with dimensions A, C, and t, and a table for entering the values.

Enter Beta values for four reference cracks that encompass the expected range of crack growth. Cracks no. 1 and 2 must have an equal 'A' value that is less than or equal to the initial crack depth, 'A'.
Cracks no. 3 and 4 must have an equal 'A' value that is greater than or equal to 95% of the thickness.

Also, cracks no. 1 and 3 must have 'C' values less than or equal to the initial crack length, 'C', and cracks no. 2 and 4 must have 'C' values greater than or equal to the maximum crack length, expected before transition to a through-crack.
During fatigue life calculations, if the current crack falls outside this envelope, it will be assigned Beta values equal to those of the nearest reference crack.

Enter

no.	'A'	'C'	Beta 'A'	Beta 'C'
1	0.05	0.07	1	1
2		0.475	1	1
3	0.24	0.07	1	1
4		0.475	1	1

< Back Next > Cancel Help

Figure 163: Four-Point Beta Interpolation Dialog

This method is offered to allow users to input part through crack beta information for a minimum number of crack lengths. This method is not expected to be terribly accurate, but may be sufficient for cases where there is limited time or resources available for detailed analyses. AFGROW will suggest crack lengths expected to cover the range of lengths in both crack growth directions. The user may also change these values. In either case, beta values for any arbitrary crack shape (a/c) will be determined by linear interpolation on these data. Data will NOT be extrapolated – the nearest point in either direction will be used. It is important for users to know this and enter data that covers the expected range of crack lengths in both directions.

The crack lengths for the c-direction are repeated for each a-dimension (see the dialog box above). It is important to maintain this for the purpose of interpolation.

If the option to keep a/c constant is selected in the model dimensions dialog (Figure 89), all of the crack growth calculations are based on the c-direction. The beta values for the a-direction will not be used in this case. However, the beta values for the a-direction must be filled nonetheless (with 1's if desired). Also, if the betas for the c-direction are not considered to be a function of the a-direction, the data for the c-direction may be simply repeated for the second a-direction (see the dialog box above).

3.2.7.2.2 Linearly Interpolated User-Defined Beta Values

When the linear option is selected, AFGROW will open the standard file open dialog, Figure 164, and will show any files of this type exist in a given directory.

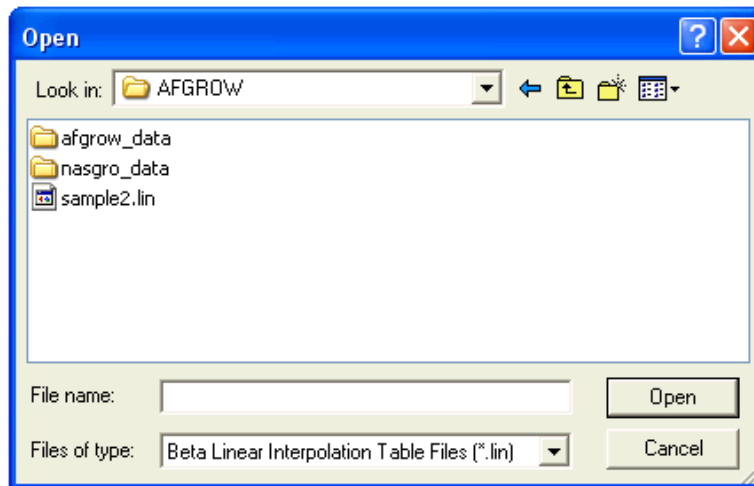


Figure 164: Linear Interpolation Dialog

This information MUST be read from a file since it will probably include a relatively large amount of data. The file format is set up in a table format that will make it easy to export from a spreadsheet. In addition, it should be noted that there are TWO tables, the first is for the betas in the A-dimension and the second is for the betas in the C-dimension. These data are required to allow AFGROW to interpolate in both crack growth dimensions to find the appropriate beta values (for both dimensions). Also, remember:

- The matrices must be square and both must be the same size.
- The crack lengths for which the Beta values are specified must be the same for each table. The A and C lengths do not have to be the same, just the C's and the A's must match in both tables.
- More data provides more accuracy.

The matrices are square because it is easier to work with square matrices. In addition, the interpolation accuracy is generally better if there are an equal number of crack lengths in both directions. This arrangement handles the most general case, where the crack shape is not known in advance and is allowed to change based on the local growth rate. It may seem excessive in cases where a user may want to keep the crack shape (a/c) constant, but it is easy to copy columns or rows of data in a spreadsheet. If the option to keep a/c constant is selected, all of the crack growth calculations are based on the c-direction. The beta values for the a-direction will not be used in this case. However, the table for the a-direction must be filled nonetheless (with 1s if desired). Also, if the betas for the c-direction are not considered to be a function of the a-direction, the data for each column may be copied to fill the table for all the a-dimensions.

The second point above merely states that the dimensions used for each matrix (for the a-direction and the c-direction) must match. It should make sense that the dimensions for both tables are the same. Any redundancy is just for the purpose of readability. Again, the lengths used for each dimension do not have to be the same, but the c-lengths used for the a-direction table must match the c-lengths used for the c-direction table. The same goes for the a-lengths for both tables.

The final point mentions accuracy. This is obvious, but more data will yield better accuracy. This is the reason for this option. Users are in control over the amount of data used in this method. AFGROW will merely linearly interpolate in both crack growth directions to determine the beta value used in the life prediction. Data will NOT be extrapolated – the nearest point in either direction will be used. It is important for users to know this and enter data that covers the expected range of crack lengths in both directions. The final line in the file is reserved to let AFGROW know the desired units for the input crack lengths. The enumerated values are 0 for English and 1 for Metric units (see Section 4.0 for more information about units). The word (units) should be capitalized in the file.

The [filename].lin file format is as follows ([Blank Spaces] allow the columns to align):

```
[Matrix Order (N)] (Maximum is currently 100)

[Blank Spaces] [ 1st A Length ] [ 2nd A Length ] ... [ Nth A Length ]

[1st C Length] [Beta in A dir.] [Beta in A dir.] .... [Beta in A dir.]

[2nd C Length] [Beta in A dir.] [Beta in A dir.] .... [Beta in A dir.]

.....<data pattern is continued>.....

[Nth C Length] [Beta in A dir.] [Beta in A dir.] .... [Beta in A dir.]

[Blank Spaces] [ 1st A Length ] [ 2nd A Length ] ... [ Nth A Length ]

[1st C Length] [Beta in C dir.] [Beta in C dir.] .... [Beta in C dir.]

[2nd C Length] [Beta in C dir.] [Beta in C dir.] .... [Beta in C dir.]

.....<data pattern is continued>.....

[Nth C Length] [Beta in C dir.] [Beta in C dir.] .... [Beta in C dir.]
[UNITS=0]
```


3.2.8 Input Environment

Toolbar Icon:



Currently, AFGROW allows cyclic environmental effects to be determined based on crack growth rate data obtained for the environment of interest. In the case of some environmental effects (i.e. corrosion (material loss)), the effect may be merely limited to local increased stress levels. However, some environmental effects can have a direct effect on the crack growth rate behavior of a given material. In the latter case, the cyclic environmental effect model may be used to more accurately predict crack growth life. AFGROW allows as many as six separate applications of the same or different environments as indicated in Figure 165.

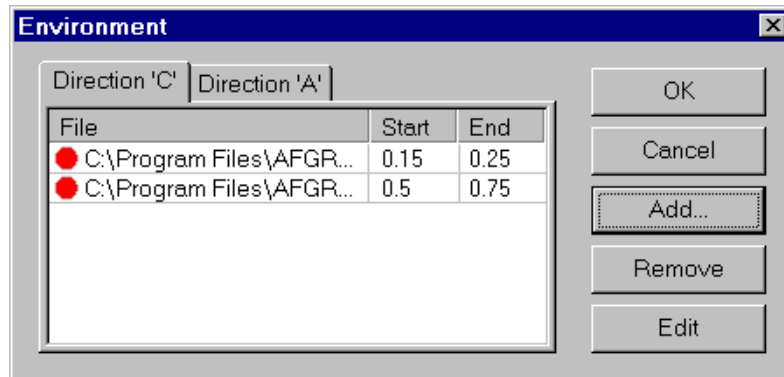


Figure 165: Environment Dialog

For the example above, the depiction of the applied environments is indicated on the model as shown in Figure 166.



Figure 166: Environmental Depiction in the Animation Frame

Each application of a given environment is shown (color-coded) on the specimen in the animation frame. For now, the environmental capability is only available when the "Harter T-Method" (Section 3.2.2.2) for crack growth rate data representation is used. The reason for this is because separate crack growth rate data files must be created for the environmental data. The tabular data format was considered to be the most accurate means of representing actual crack growth rate data. The initial capability was designed prior to the development of the tabular look-up crack growth rate model (Section 3.2.2.4). The material title used for each material in the material data file in the "Harter T-Method" is compared to the title lines in the environment data file to ensure that data for the same material are being used.

When a user selects the “Add” button in the environmental dialog (Figure 165), the dialog shown in Figure 167 appears.

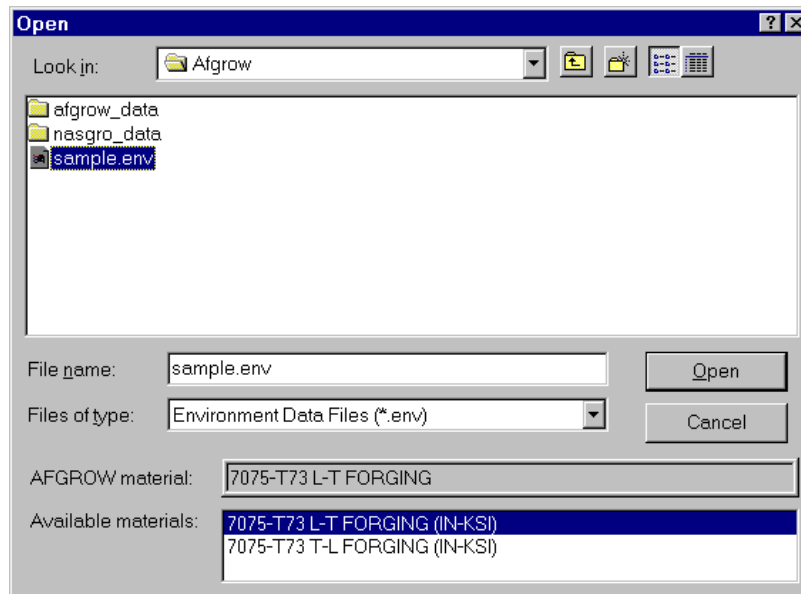


Figure 167: Environmental File Open Dialog

The current baseline material is displayed in the lower portion of the dialog box. Once a file containing the desired environmental crack growth rate data is selected (single left-click), the dialog box displays the titles for the material data available in that file. The environmental data files are simply text files containing much of the same information that is contained in the material files used with the “Harter T-Method” (Section 3.2.2.2). There are a few additional parameters that control the transition from the baseline data to the environmental data. The default file extension is [.env].

The format of the file is:

[*Material Title*] (Must match the baseline material title)

[*da/dN*] [*Delta K @R=0*] [*m*] (25 lines of this data – EXACTLY 25 lines)

[*Dist*] [*A1*] [*A2*] [*A3*]

[*Rlo*] [*Rhi*] [*KIC*] [*Yield*]

[*END*] (after the last material)

Refer to Section 3.2.2.2 (Harter T-Method) for more information on the variables listed above. Note, that the additional variables: Dist, A1, A2, and A3 are required to characterize the transition behavior as described in the next section.

3.2.8.1 Modeling Environmental Crack Growth Rate Transition Behavior

AFGROW uses a third order polynomial function to interpolate the appropriate crack growth rate as a crack grows through a transition region between two different environments. This is illustrated in Figure 168.

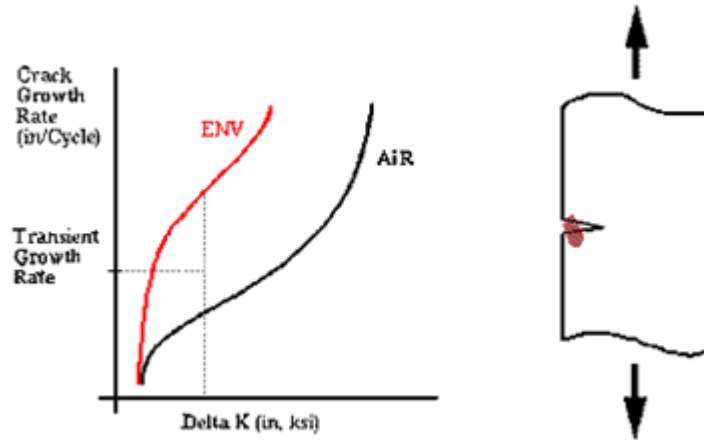


Figure 168: AFGROW Environmental Rate Transition Model

The form of the transition relationship that has been implemented in the current version of AFGROW is as follows:

$$\text{Rate} = \text{Rate1} + \text{Factor} (\text{Rate2} - \text{Rate1})$$

Where:

$$\text{Factor} = A1 (\text{Trans}) + A2 (\text{Trans})^2 + A3 (\text{Trans})^3$$

$$\text{Trans: Fraction of Transition Distance Penetrated } (0 - 1) = \frac{x}{\text{Dist}}$$

x: Relative crack tip position (0 - Dist)

Dist: Maximum distance from environment boundary where crack growth rate is affected

Rate1: Rate curve from which the crack tip is growing

Rate2: Rate curve toward which the crack tip is growing

$$A1 + A2 + A3 = 1.0^{58}$$

⁵⁸ This is due to the boundary conditions: when Trans=1.0, Rate must be equal to Rate2, which means that Factor must also be equal to 1.0

3.2.9 Input Beta Correction

AFGROW includes an ability to estimate stress intensity factors for cases, which may not be an EXACT match for one of the stress intensity solutions in the AFGROW library (Section 3.2.3). For example, a case is being modeled with a high stress gradient. It is unlikely that an exact solution would be available, and the creation of a boundary or finite element model would be time consuming. AFGROW offers a method to approximate the solution using a beta correction technique (see Figure 169).

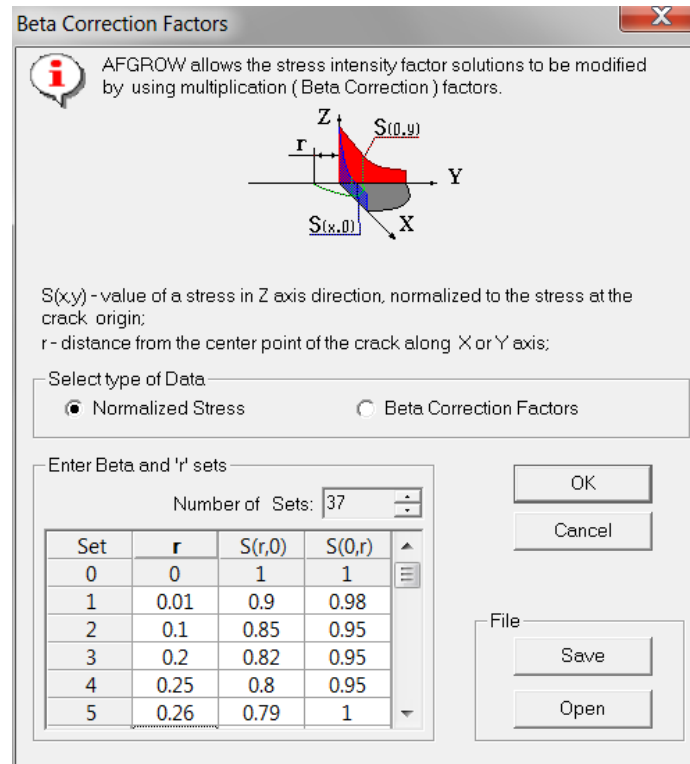


Figure 169: Beta Correction Factor Dialog

Users have the option of entering normalized stress values in the crack plane and allow AFGROW to calculate beta correction factors or enter pre-determined beta correction values. Please note that when the option to keep a/c constant is selected in the baseline model, all crack growth calculations are based on the c-direction.

When using the Advanced Model Interface, the beta correction item in the input menu will be disabled. This is because the beta correction is applied to a selected crack, so the beta correction capability is enabled by clicking on the box to the right of the beta correction item in the Properties window for the selected crack object as indicated in Figure 170.

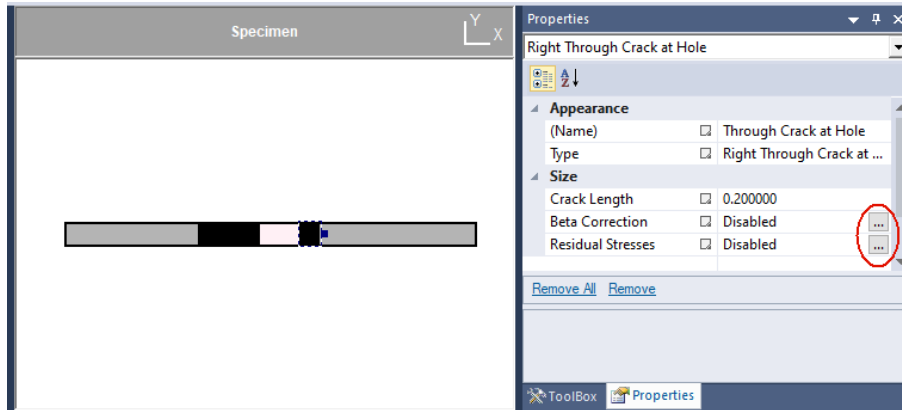


Figure 170: Activating Crack Corrections for Advanced Models

3.2.9.1 Determine Beta Correction Factors Using Normalized Stresses

AFGROW employs a Gaussian integration method, which uses a point load stress intensity solution from the Tada, Paris, and Irwin Stress Intensity Handbook [60] to integrate a given 2-D unflawed stress field (in the crack plane) to estimate stress intensity values at user defined crack length increments.

Users should choose the standard model with a stress field that is as close as possible to the stress field of interest. Then determine the ratio of the unflawed stress field of interest (S2) to the unflawed stress field for the chosen geometry (S1) at sufficient crack length intervals to represent the stress distribution. The intervals should be selected such that linear interpolation would provide a reasonable curve fit between the points. These intervals do NOT have to be uniform, but there should not be a large change in the slope between adjacent intervals. AFGROW uses a Newton interpolation scheme to determine the Gaussian integration points. If the slope change between intervals is large ($> |600|$), the code can generate erroneous integration points.

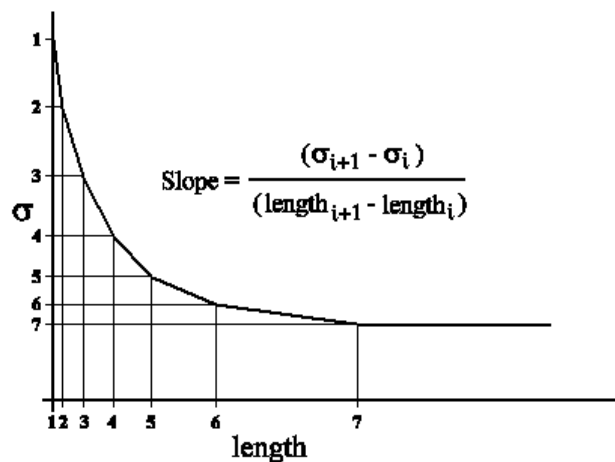


Figure 171: Curve Slope Between Input Data Points

Dividing each stress ratio by the stress ratio at the crack origin normalizes the stress values as shown below.

$$\frac{S_2(x)}{S_1(x)} / \frac{S_2(0)}{S_1(0)}$$

This provides a reference for the actual stress at the crack origin. Therefore, the value of the spectrum multiplication factor multiplied by the spectrum stress (or load) values MUST now be the appropriate value at the crack origin (based on the reference stress for the standard model being used).

For example, if you have a notch case with a K_t of 4.0 and the standard model is the notch case ($K_t = 3.17$). The spectrum SMF value would then be equal to: $4.0/3.17$, or 1.262.

The normalized stress distribution is integrated using the point load solution as shown in Figure 172.

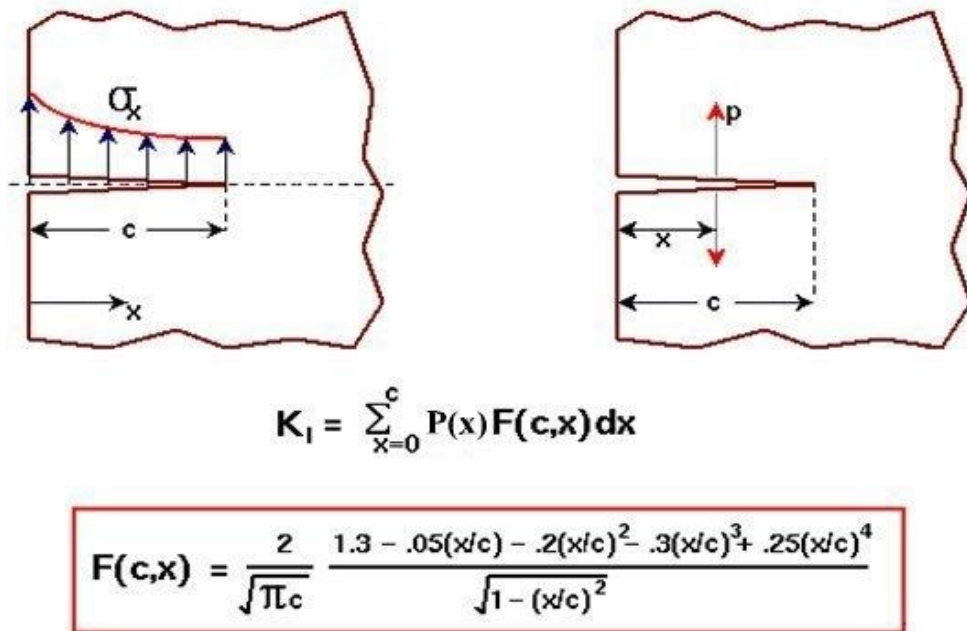


Figure 172: Point Load Stress Intensity Solution

The beta correction factor is calculated by dividing the stress intensity determined by integrating the input stress field by the stress intensity value for a unit stress distribution at each crack length increment. Obviously, this method is not exact since it can't account for

stress field changes as the crack grows, but it is fairly good - especially at shorter crack lengths where most of the life is spent.

AFGROW multiplies the resulting beta correction factor by the beta factor for the user-selected model at a given crack length. These corrected beta values are printed in the output list in AFGROW in the beta column.

For the [a] crack length dimension: $K_a = \sigma \sqrt{\pi a} \beta_a$

For the [c] crack length dimension: $K_c = \sigma \sqrt{\pi c} \beta_c$

Where, in this case: $\beta = \beta_{\text{model}} * \beta_{\text{correction}}$

In two-dimensional cases, users must refer to the x and y dimensions in the animation frame and the dimensions shown in the beta correction dialog (see Figure 169). The length dimension, r, shown in the dialog box is the radial distance from the crack origin. The input stress ratio values are shown for (r, 0) – along the y = 0 axis (c-direction) and (0, r) – along the x = 0 axis (a-direction).

There are a few important points to remember:

- The stress field is normalized to the stress at the crack origin
- The spectrum stresses (or loads) MUST be adjusted to account for the fact that the stress field has been normalized (i.e. multiply the SMF by the normalization factor at the crack origin)
- Choose crack length intervals such that linear interpolation on stress ratio is adequate between points (slope change between points < |600|)
- When entering stress ratio data for 1-D, values of 1.0 should be input for the other dimensions
- If there is a stress gradient in only 1-D, enter values of 1.0 for all points in the other dimensions
- Accuracy increases with the number of points
- Enter two points beyond the longest crack length expected in the analysis

There are additional documents on the AFGROW web page that provide additional help. To access these documents, users must register and login to the web page.

3.2.9.2 Enter Beta Correction Factors Manually

Users have the option to enter beta correction factors directly instead of allowing AFGROW to calculate them. There may be cases where a user simply wants to apply beta correction factors that have been obtained from some external source. To enter beta correction factors manually, simply select “Beta Correction Factors” in the “Select Type of Data” section of the beta correction dialog box (see Figure 169).

The beta correction at the crack origin is set equal to 1.0 by default only because the values are required to be normalized at the crack origin when stress values are input. The beta correction value at the crack origin can only be used as an interpolation limit since all cracks must have a finite length. The first user supplied beta value should be entered for a crack length less than the initial crack size for interpolation purposes. In two-dimensional cases, users must refer to the x and y dimensions in the animation frame and the dimensions shown in the beta correction dialog (see Figure 169). The length dimension, r , shown in the dialog box is the radial distance from the crack origin. The input stress ratio values are shown for $(r, 0)$ along the $y = 0$ axis (for the width direction) and $(0, r)$ along the $x = 0$ axis (for the thickness direction). AFGROW will NOT extrapolate beta correction values for crack lengths extending past the input table limits.

3.2.10 Input Residual Stresses

AFGROW can account for the existence of residual stresses by calculating additive residual stress intensities at user defined crack length increments. The dialog shown in Figure 173 will appear when the residual stress option is selected.

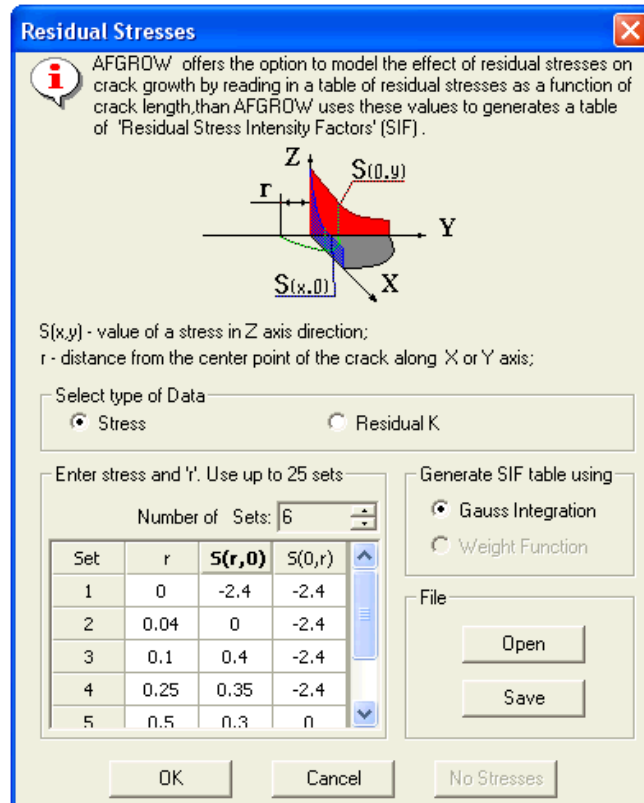


Figure 173: Residual Stress Dialog

AFGROW does not use negative values of stress intensity (K) directly⁵⁹, since K is not defined for compression. However, in this application, negative stress intensities can be used since the residual Ks are merely added to the stress intensities (both maximum and minimum) caused by the applied loads. This will not change ΔK , but will change the stress ratio, which will result in a change in the crack growth rate.

When you use this option, AFGROW will print out the residual K value each time it prints out the standard crack growth information. It is important that you input stress information for the entire range of crack growth lengths since AFGROW WILL NOT extrapolate and will just use the last or nearest applicable value.

⁵⁹ Negative K's are used with the Forman crack growth rate equation, and when calculating the final stress ratio for crack growth rate calculations for the other growth rate models.

Users have the option of entering residual stress values in the crack plane and allow AFGROW to calculate residual stress intensity factors or enter pre-determined residual stress intensity values. When residual stress values are entered, the residual K values may be determined using either the Gaussian integration technique or the weight function method.

When using the Advanced Model Interface, the residual stress item in the input menu will be disabled. This is because the residual stress correction is applied to a selected crack, so this capability is enabled by clicking on the box to the right of the beta correction item in the Properties window for the selected crack object as indicated in Figure 170.

3.2.10.1 Determine Residual Stress Intensity Values for Known Residual Stresses

There are two methods available in AFGROW to calculate residual stress intensity values. The first is the Gaussian integration method which uses a point load stress intensity solution from the Tada, Paris, and Irwin Stress Intensity Handbook [60] to integrate a given 2-D unflawed stress field (in the crack plane) to estimate residual K values at user defined crack length increments. The second method uses the weight function stress intensity solutions provided by Prof. Glinka [7].

3.2.10.1.1 Gaussian Integration Method

The Gaussian integration method is the same stress intensity approximation method used to calculate beta correction factors discussed in Section 3.2.9. The only difference is that actual stress intensity (K) values are being calculated instead of a beta correction factor. Stress ratios are NOT used or normalized, since a K value is being determined. Users should enter the actual residual stress distribution starting at the crack origin. Users may enter as many points as required to allow linear interpolation to provide a reasonable curve fit between each point. These intervals do NOT have to be uniform, but there should not be a large change in the slope for adjacent intervals. AFGROW uses a Newton interpolation scheme to determine the Gaussian integration points. If the slope change between intervals is large, the code can generate erroneous integration points. AFGROW will provide a warning message if a large slope change is detected (see Figure 171). Also, the interpolation method looks two points ahead of any given point in the table, so it is important to include two points beyond the longest crack expected in the analysis. Failure to do this can result in large interpolation error.

AFGROW will perform two integration procedures for part-through crack models. The first integration will be performed for the 2-D case, and a second integration will be performed for the 1-D case starting with the radial distance (r) equal to the plate thickness. This eliminates the need to include several transition points in the table to reduce the stresses in the a-direction to zero for $r > \text{thickness}$ to avoid a large stress gradient. However, an input point for $r = \text{thickness}$ is still required. The results of these integrations are provided in the AFGROW output.

There are a few important points to remember for the Gaussian integration method when used to calculate residual K values:

- Choose crack length intervals such that linear interpolation on stress is adequate between points (slope change between points $< |600|$)
- When entering stress data for a through crack (1-D), or for radial distance greater than thickness, input values of 0.0 should be input for stress in the a-direction
- If users only want to use a stress gradient in 1-D for a 2-D case, enter the stress at the crack origin for the a-dimension (up to a radial distance equal to the plate thickness) and values of 0.0 for all points in the second dimension beyond the thickness as shown in Figure 173
- Include at least four points in the table
- Enter two points beyond the longest crack length⁶⁰ expected in the analysis

3.2.10.1.2 Weight Function Method

The weight function solutions were provided through the effort of Prof. Glinka (University of Waterloo, CA). This method will only be possible IF a weight function solution is available for the geometry being analyzed. The currently available weight function solutions are given in Section 3.2.3.1.2. The current weight function solutions ONLY use a stress distribution in a single crack growth dimension. For part-through cracks, the distribution is in the thickness direction. In the case of through cracks, the distribution in the width direction is used. In cases where a part-through crack is used, AFGROW will use the distribution in the thickness direction until the crack becomes a through crack and will then switch to use the distribution in the width direction. This makes it less desirable to use the weight function method to determine residual K values for most practical 2-D cases.

3.2.10.2 Enter Residual Stress Intensity Factors Manually

Users have the option to enter residual stress intensity factors directly instead of allowing AFGROW to calculate them. There may be cases where a user simply wants to apply residual stress intensities that have been obtained from some external source. To enter these values manually, simply select “Residual K” in the “Select Type of Data” section of the residual stress dialog box (see Figure 173).

In two-dimensional cases, users must refer to the x and y dimensions in the animation frame and the dimensions shown in the residual stress dialog (see Figure 173). The length dimension, r, shown in the dialog box is the radial distance from the crack origin. The input stress ratio values are shown for (r, 0) along the y = 0 axis (for the width direction) and (0, r) along the x = 0 axis (for the thickness direction). AFGROW will NOT extrapolate residual K values for crack lengths extending past the input table limits.

⁶⁰ This applies to each growth direction

3.2.11 Input K-Solution Filters

The K-solution filtering capability allows users to modify stress intensity solutions for the axial⁶¹, bending, and/or bearing load case independently for applied tensile and/or compressive spectrum loading.

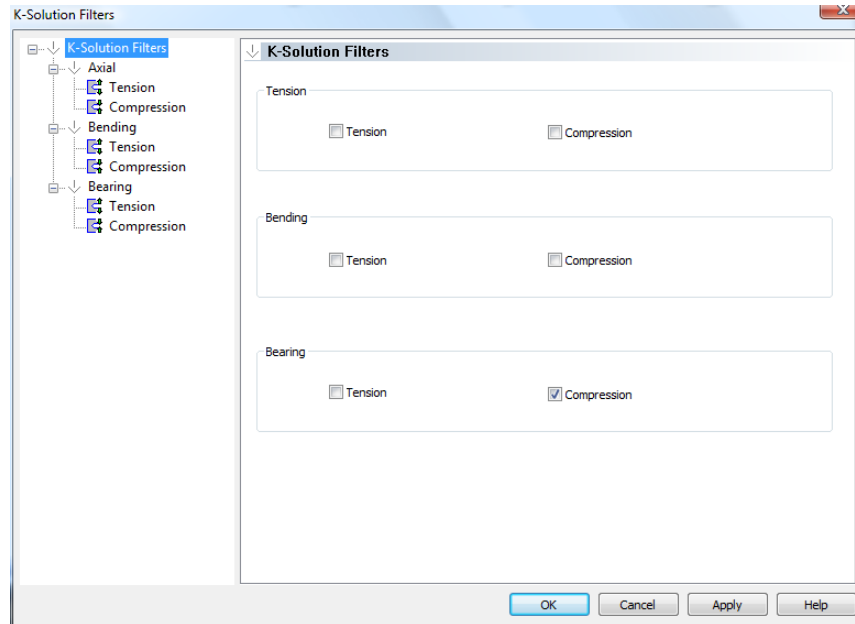


Figure 174: Stress Intensity Solution Filter Dialog

The lug or plate bearing K-solution is perhaps the most practical example of how this capability may be used.

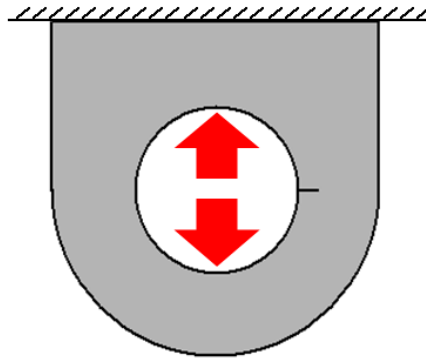


Figure 175: K-Filtering Example for Lug Bearing Load

⁶¹ This was called the “tension” load case in the past, but “axial” is a more descriptive and less confusing label in light of the filtering capability for tensile and compressive spectrum loading.

When the applied spectrum stress value is positive, the resulting bearing stress will open a crack at the edge of the hole. However, under compressive spectrum stresses, the crack will not be affected at all since the resulting compressive bearing stress is carried above the crack plane. In this case, the K-Filtering tool can be used to set the solution to zero when a compression spectrum stress is applied. Without K-Filtering, AFGROW would assume that tensile and compressive spectrum stresses apply equally when calculating the stress intensity value at the crack tip.

While it is true that stress intensity has no real significance under compressive loading, it is important to remember that the resulting crack growth rate is also dependent on the stress ratio (R) that is applied at the crack tip. This stress ratio must include the effect of any applied residual stress and/or load sequence (retardation). Therefore, AFGROW internally keeps track of both positive and negative Kmax and Kmin values in order to determine the appropriate stress ratio for crack growth rate calculations.

In the case of a lug, there is a single load case (bearing), so the above problem could be resolved using the new spectrum filtering tool (Section 3.2.5). However, other models (i.e. plates with holes) allow for combined loading (axial, bending, and bearing). The spectrum filtering tool applies equally to all load cases of a given model. The K-Filtering tool is designed to allow each load case to be modified independently as a function of the current crack length using the tabular or equation form dialog as explained below.

After the user chooses the load case(s) for which K-filtering is desired, the appropriate compression and/or tension filter is selected as indicated in Figure 176.

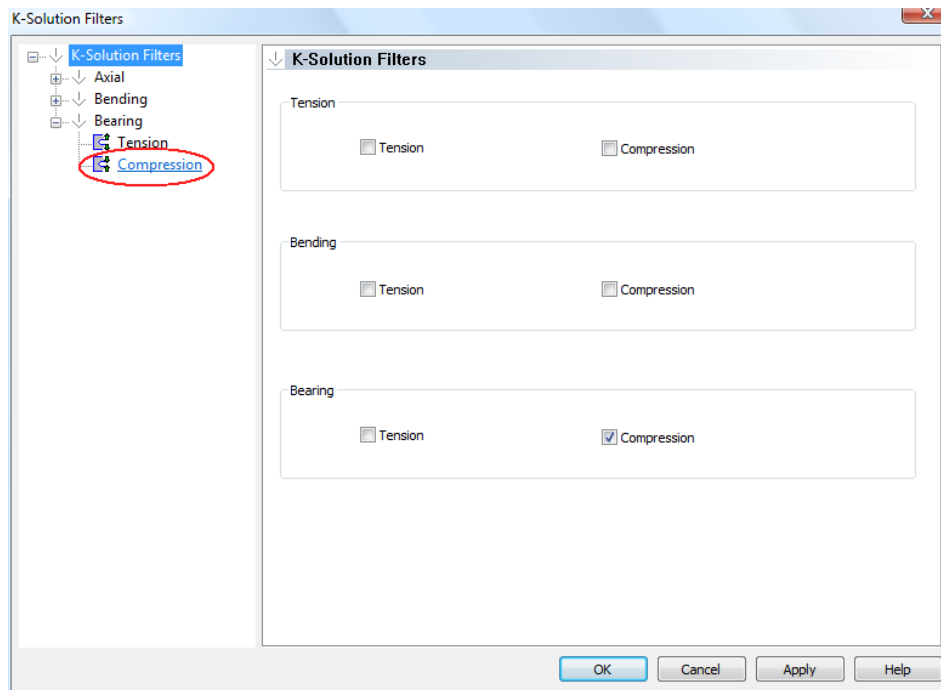


Figure 176: Using the K-Filter Dialog

When the user clicks on the desired filter, another dialog appears to allow the filtering information to be entered. The user may select either tabular or equation form input by selecting the appropriate type at the top of the dialog as shown in Figure 177.

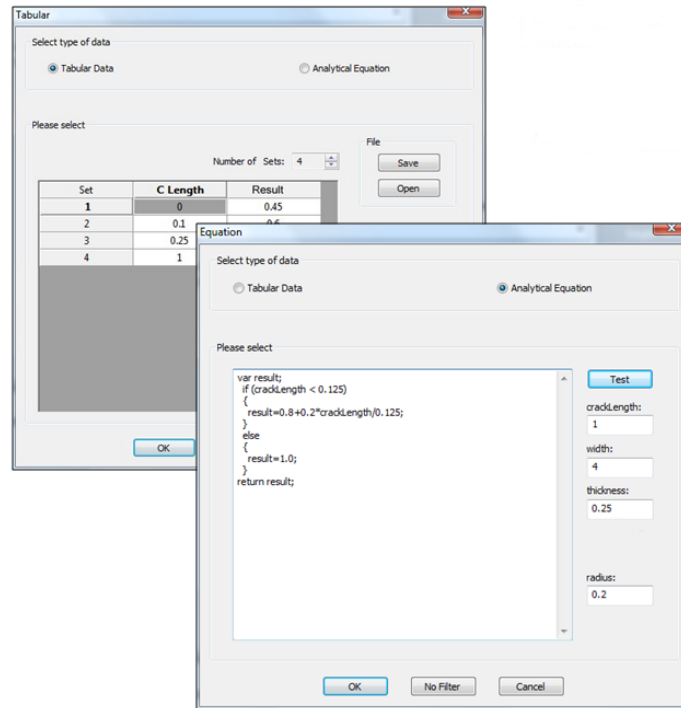


Figure 177: Tabular and Equation Form K-Filter Dialogs

Up to 100 points may be entered in the form of tabular data, and java script is used for the analytical equation. These dialogs are used in the same manner as the equivalent dialogs used for the Spectrum Filtering Tool (see Section 3.2.5).

Remember that AFGROW will not extrapolate outside the bounds of the tabular data. For the example of the lug/bearing case noted above, the bearing compression result is easily set to 0.0 using a single tabular point with a result of 0.0. Also, an option is also available in the bearing load section of the model dialog to set the bearing compression K-Filter to 0.0 since this is expected to be a commonly used feature.

These dialogs are used as explained in Sections 3.2.5.1 and 3.2.5.2, with the exception that the K-filter is a function of the current crack length. The result of this is that the filtering relationship specific to each crack length defined for a given model.

As users gain experience with this capability, and provide feedback, further modifications may be made to improve its functionality.

3.3 View Menu

The view menu, Figure 178, provides control over what is displayed in the various AFGROW frames.

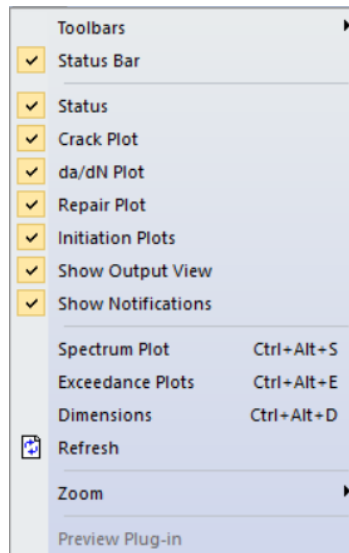


Figure 178: View Menu

The view menu is divided into four sections. The first section controls the display of the toolbars and status bar. The second section controls the display of the main frame. The third section controls additional special features displayed in the animation frame. Finally, the fourth section controls the magnification of the view in the animation frame.

3.3.1 View Toolbars

AFGROW currently uses four toolbars to aid users in performing various tasks as indicated in Figure 179.



Figure 179: AFGROW Toolbars

3.3.1.1 Predict Toolbar

The predict toolbar, Figure 180, allows a user to use shortcuts to perform many common operations required to perform crack growth life predictions:



Figure 180: Predict Toolbar

When selected, the predict toolbar appears in the AFGROW window, and a checkmark appears by this item in the view menu. This toolbar is dockable and can be relocated at the top, sides, or bottom of the AFGROW window. It may also be placed as a floating toolbar anywhere on the desktop. The toolbar is moved by placing the mouse pointer in a blank area between two icons, holding down the left mouse button, and dragging the toolbar to the desired location.

The function of each icon is displayed through the standard Windows help when you move the mouse over the icon.

3.3.1.2 Standard Toolbar

The standard toolbar, Figure 181, allows a user to perform many common Microsoft Windows® operations:



Figure 181: Standard Toolbar

When selected, the standard toolbar appears in the AFGROW window, and a checkmark appears by this item in the view menu. This toolbar is dockable and can be relocated at the top, sides, or bottom of the AFGROW window. It may also be placed as a floating toolbar anywhere on the desktop. The toolbar is moved by placing the mouse pointer in a blank area between two icons, holding down the left mouse button, and dragging the toolbar to the desired location.

The function of each icon is displayed through the standard Windows help when you move the mouse over the icon.

3.3.1.3 Specimen Design (Properties) Toolbar

This toolbar is ONLY used with the advanced model option in AFGROW and is provided to allow users to view or manually edit specific specimen dimensions. This item is similar to the properties window used in Microsoft Visual Basic to view and edit properties.

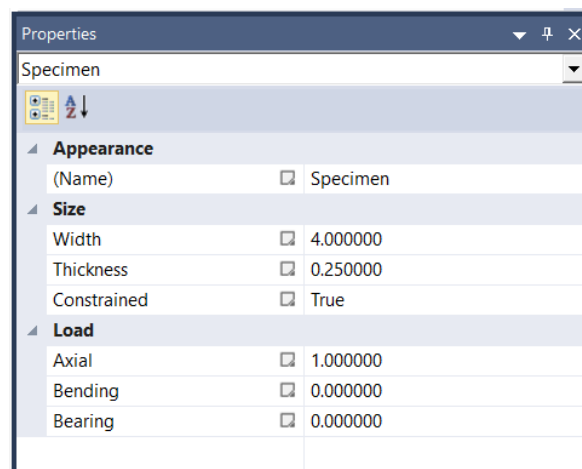


Figure 182: Specimen Design (Properties) Toolbar

Properties are shown for the selected specimen object (crack, hole, or specimen cross-section). This toolbar may also be resized by dragging any edge with your mouse.

The constraint and load options have limited applicability. The constraint option refers to in-plane bending resulting from crack asymmetry across the specimen width and currently applies for a single edge through crack and any model using a slot object. It has

no effect on any other model at this time. These two geometries have the greatest potential for significant in-plane bending effects, and since most practical applications include natural bending constraint (frames and stiffeners), it was important to include an option to model constraint on or off for those cases. This option will be applied for more geometries as solutions become available and/or interest increases. Unless otherwise stated, the other geometric cases in the Advanced Model Interface were modeled without in-plane bending constraint. Axial, Bending, and Bearing load cases only apply to select crack geometries. These geometries are being expanded as quickly as solutions are available.

This is a **dockable toolbar**. Dockable toolbars may be moved to other areas on your screen or **docked** on any border of the AFGROW window.

3.3.1.4 Tool Box Dialog

The Tool Box dialog is shown below in Figure 183.

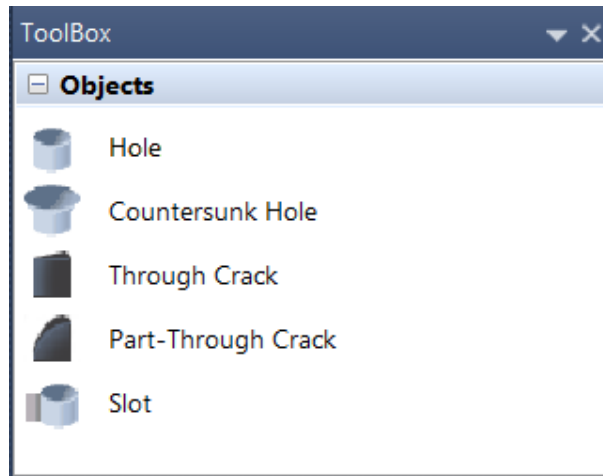


Figure 183: Tool Box

This tool box displays the objects that may be added to any User defined advanced model. These objects currently include:

- Hole**
- Countersunk Hole**
- Through Crack**
- Part-Through Crack**
- Slot** (between a hole and the edge of the plate)

Simply drag and drop the desired object on the specimen view in the animation frame using the mouse.

This toolbar includes a window below the area containing the available specimen objects. This portion of the toolbar is not used at this time but may be used in the future as a location to store user-defined two crack configurations as icons. These icons could then be dragged into the specimen view to save time in cases where a user has a library of crack configurations.

This toolbar may also be resized by dragging any edge with your mouse.

This is a **dockable toolbar**. Dockable toolbars may be moved to other areas on your screen or **docked** on any border of the AFGROW window.

3.3.2 View Status Bar

The status toolbar is found in the margin at the bottom of the output frame (see Figure 29). The purpose of the status toolbar is to provide additional information related to a given analysis. The status bar is a tool that Microsoft provides and should not be confused with the "status view" which is used by AFGROW to display the current input data in the upper left window. For more details, refer to Section 2.1.6.

3.3.3 View Status

The status window is one of the optional "windows" which may be displayed in the upper left-hand window (main frame) in the AFGROW main window (see Figure 8 and Figure 9). The window may be displayed by selecting the status tab. If the status view does not appear in any window, make sure that this option has been selected in the View Menu (Figure 178). For more details, refer to Section 2.1.1.1.

3.3.4 View Crack Plot

The crack length plotting capability allows a user to graphically examine the crack growth life predictions being performed by AFGROW in real time (see Figure 11). When selected, a new menu item (Plots) will appear in the menu bar. This menu item provides access to the same features given in the rebar tool in the main frame. If the crack plot view does not appear in any window, make sure that this option has been selected in the View Menu (Figure 178). For more details, refer to Section 2.1.1.2.

3.3.5 View da/dN Plot

The da/dN plotting capability allows a user to graphically examine the crack growth rate properties to be used by AFGROW (see Figure 18). When selected, a new menu item (da/dN Plots) will appear in the menu bar. This menu item provides access to the same features given in the rebar tool in the main frame. If the da/dN plot view does not appear in any window, make sure that this option has been selected in the View Menu (Figure 178). For more details, refer to Section 2.1.1.3.

3.3.6 View Repair Plot

The repair plotting capability allows a user to graphically examine the beta correction plots for up to eight repair designs (see Figure 22). If the repair plot view does not appear in any window, make sure that this option has been selected in the View Menu (Figure 178). For more details, refer to Section 2.1.1.4.

3.3.7 View Initiation Plots

The initiation plotting capability allows a user to graphically examine the cyclic stress-strain and strain-life plots for the current input data (see Figure 23). When selected, a new menu item (Initiation Plots) will appear in the menu bar. This menu item provides access

to the same features given in the rebar tool in the main frame. If the status view does not appear in any window, make sure that this option has been selected in the View Menu (Figure 178). For more details, refer to Section 2.1.1.5.

3.3.8 View Show Output

This view is normally located as a tabbed view in the bottom half of the AFGROW user interface, and output data is sent to this view by default. As is the case with all of the view options, this view may be moved anywhere on any display. If the output view does not appear in any window or display, make sure that this view option has been selected in the View Menu (Figure 178). For more details, refer to Section 2.1.3.1.

3.3.8 View Notifications

This view is also normally located as a tabbed view in the bottom half of the AFGROW user interface, and notifications related to the applicability of certain model parameters are sent to this view. This view may be moved anywhere on any display. If the output view does not appear in any window or display, make sure that this view option has been selected in the View Menu (Figure 178). For more details, refer to Section 2.1.3.2.

3.3.8 View Spectrum Plot

The da/dN plotting capability allows a user to graphically examine the spectrum being used by AFGROW (see Figure 184).

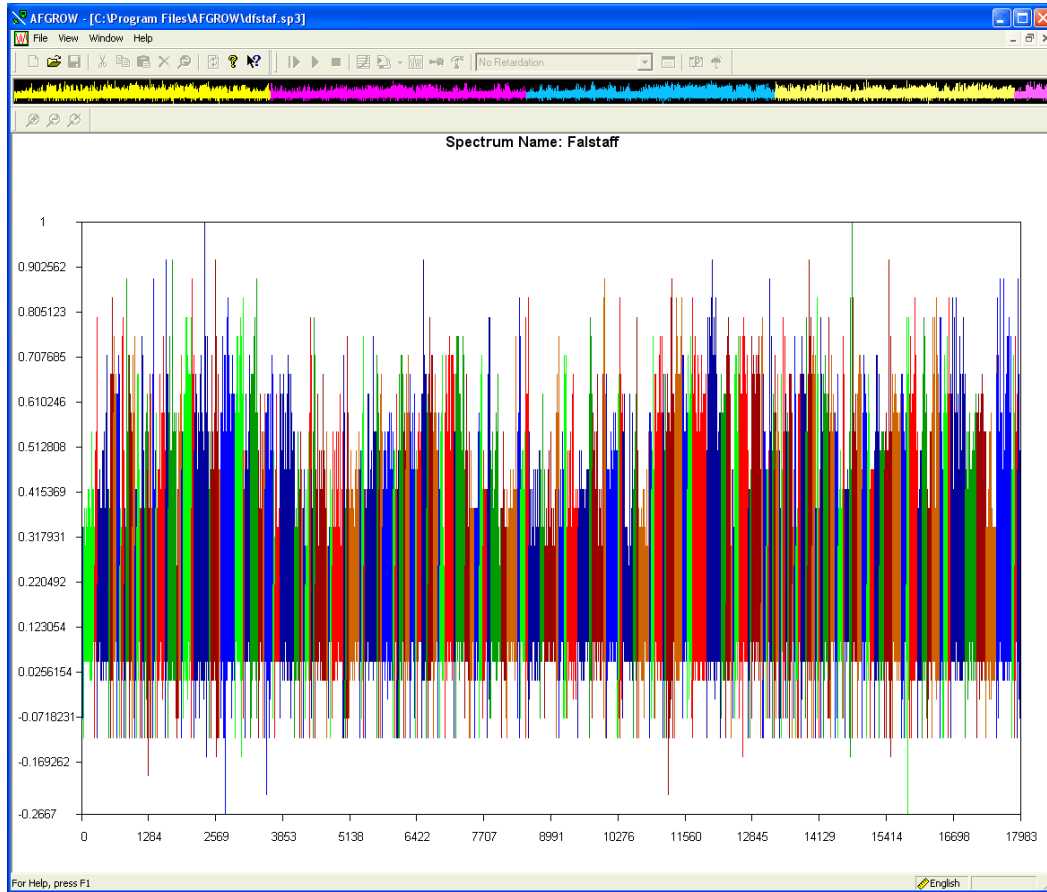


Figure 184: Spectrum Plot

When this option is selected, a new spectrum window will be created. While in this view, several tools in the toolbar are now grayed out since they serve no purpose in the spectrum view window. The maximum and minimum values in the spectrum are indicated on the plot. The color of the data plotted is changed for each sub-spectrum.

Users can zoom-in the spectrum view by using the mouse and dragging out the area to view on the spectrum plot. The entire spectrum is always displayed in the upper area of this view. Users can also adjust the view in the upper area of the window by using the mouse and dragging the highlighted box.

Currently, the only way to close this view is to click on the black "x" below the red "x" that will close the entire AFGROW session (upper right hand corner of the AFGROW interface).

3.3.9 View Exceedance Plots

The exceedance plotting capability allows a user to graphically examine the exceedance information for the current spectrum to be used by AFGROW (see Figure 185).

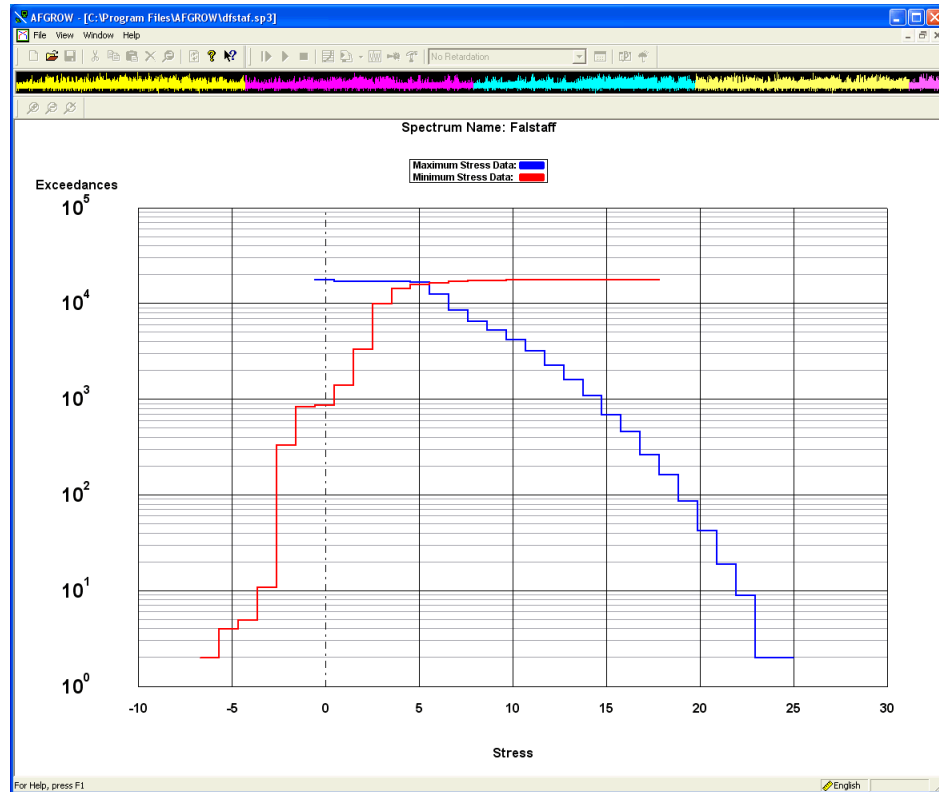


Figure 185: Exceedance Plot

When this option is selected, an exceedance window will be created. While in this view, several tools are now grayed out since they serve no purpose in the exceedance view window. You can switch back to the "normal" window by using the window menu. You can choose to cascade, tile, or simply switch views.

This view⁶² shows the number of cumulative exceedances for each maximum and minimum value in the current spectrum. This is ONLY a view of the current spectrum. AFGROW does not allow users to input exceedance information in lieu of actual spectrum data.

Currently, the only way to close this view is to click on the black "x" below the red "x" that will close the entire AFGROW session (upper right hand corner of the AFGROW interface).

⁶² Note: The exceedance plot shows the spectrum information after being multiplied by the spectrum multiplication factor (SMF) that is input by the user.

3.3.10 View Dimensions

The Dimensions option in the View menu simply shows the definition of the basic geometric dimensions for the current model being analyzed. For example, the corner cracked hole dimensions are shown in Figure 186.

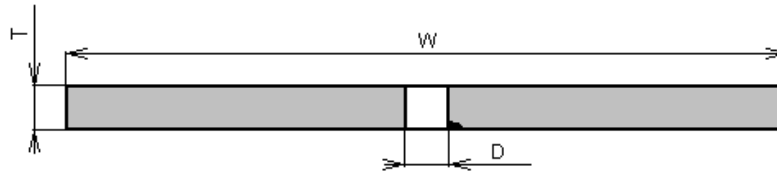


Figure 186: Specimen Dimensions

The specimen dimension display is turned on and off by selecting this menu item. A check mark is displayed beside this menu item when the dimensions are being displayed.

3.3.11 View Refresh

Toolbar Icon:



The Refresh option in the View menu resets the initial crack dimensions in the model being analyzed. Once an analysis is performed, the final crack size is shown in the upper right window. The refresh option will reset the image to show the initial crack dimension(s).

3.3.12 View Zoom

The zoom option in the view menu allows a user to control the magnification of the specimen view in the animation frame. The options are shown in Figure 187.

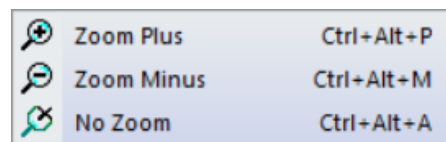


Figure 187: Magnification Options for the Animation Frame

In addition to the magnification options listed above, a specific area is magnified by holding down the left mouse button and dragging out the desired viewing area within the animation frame. For more details, refer to Section 2.1.2.

3.4 Predict Menu

The predict menu, Figure 188, controls options related to life prediction.

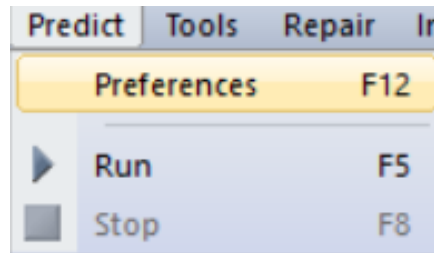


Figure 188: Predict Menu

3.4.1 Predict Preferences

The preferences menu selection is one of the most important menu items in AFGROW. There are several optional settings which may be changed to suit the various requirements of a given life prediction. The preferences are divided into eight categories and are accessible by selecting any of these categories as shown in Figure 189.

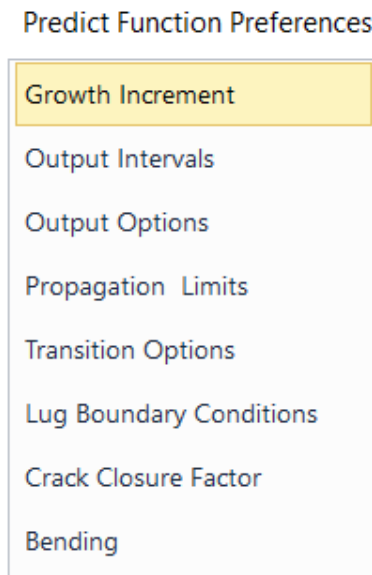


Figure 189: Preference Categories

The preferences dialog is accessible through the AFGROW menu OR by right clicking anywhere in the output frame. The user sets the preference options with the buttons shown in Figure 190.



Figure 190: Saving and Restoring Preferences

Use the Save button to save all parameter settings. These settings will be retained until changed by the user. The Default button will return the original AFGROW preference settings.

All current preference settings are saved in the AFGROW input and output files.

3.4.1.1 Growth Increment

AFGROW allows users to set the crack growth increment for use in calculating the current stress intensity as indicated in Figure 191.

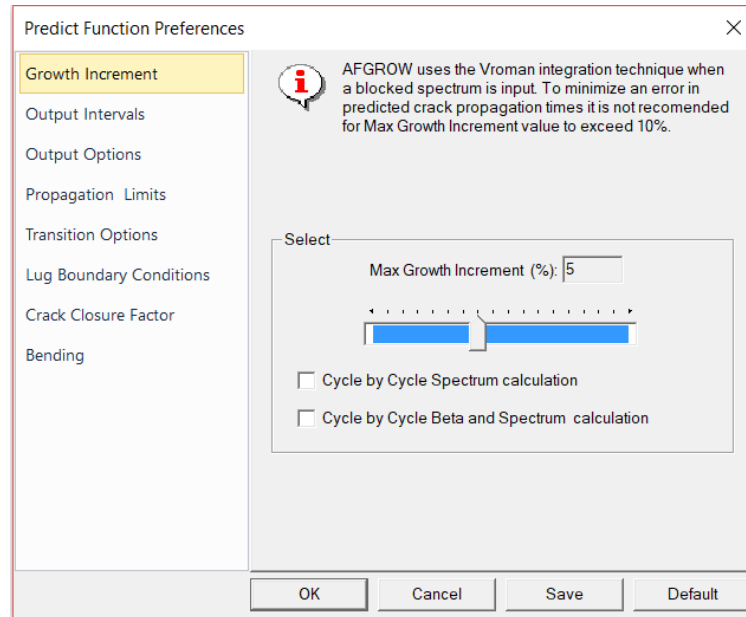


Figure 191: Growth Increment Dialog

The increment is used to determine the maximum number of cycles in a given spectrum level which may be used before the stress intensity values must be recalculated (**Vroman integration method**). A “blocked” spectrum is a spectrum that has been simplified to consist of stress (or load) levels, which may have more than one cycle. Since crack growth per cycle is NOT linear, stress intensity and crack growth rates MUST be recalculated at some crack length increment. This option is designed to give the user more control over an analysis. There is a direct trade-off between speed of calculation and accuracy. Higher increments reduce runtimes, but also decrease accuracy.

The increment value is also important when a “cycle-by-cycle” (1 cycle per stress level) spectrum is used. The increment ALSO controls how often AFGROW runs the internal routine to determine the alpha (α) values that are used to determine stress intensity. These alpha routines can be very CPU intensive and this control also provides the same kind of trade-off of speed and accuracy noted above. The following definitions are important for a good understanding of how this works in AFGROW:

$$K = \sigma_{\text{ref}} \alpha$$

$$\alpha = \beta \sqrt{\pi x} ; \text{ Where } x = \text{crack length}$$

In addition to controlling how blocked spectra are analyzed, AFGROW currently allows users to control how often beta factors are calculated based on a percentage of crack length. The limits are from 0.25 to 15 percent of a given crack length. Increasing this percentage may reduce run times; however, the speed is traded for life prediction accuracy.

The cycle-by-cycle spectrum option allows the increment to be adjusted from 0.25 to 5 percent. The alpha values are calculated based on the selected increment, but the betas are adjusted (from the alphas) for crack length on a cycle-by-cycle basis. The current cycle-by-cycle beta option is a TRUE cycle-by-cycle alpha, beta, and spectrum calculation. Run times may be significantly increased when using this option. If neither option is selected, the allowed increment range will be from 0.25 to 15 percent.

One question that is sometimes asked is "Why does a crack growth plot sometimes appear somewhat jagged even when a constant amplitude spectrum is used?" This is caused when an increment is used which is too large to give an accurate answer. This "jagged" plot will be smoothed by reducing the increment or essentially eliminated by using the cycle-by-cycle beta option. However, it should be noted that a random stress spectrum would tend to produce a "jagged" crack growth curve due to the fact that the stress (or load) levels are changing.

3.4.1.2 Output Intervals

The printing interval for output data is controlled by the Output Interval dialog (see Figure 192).

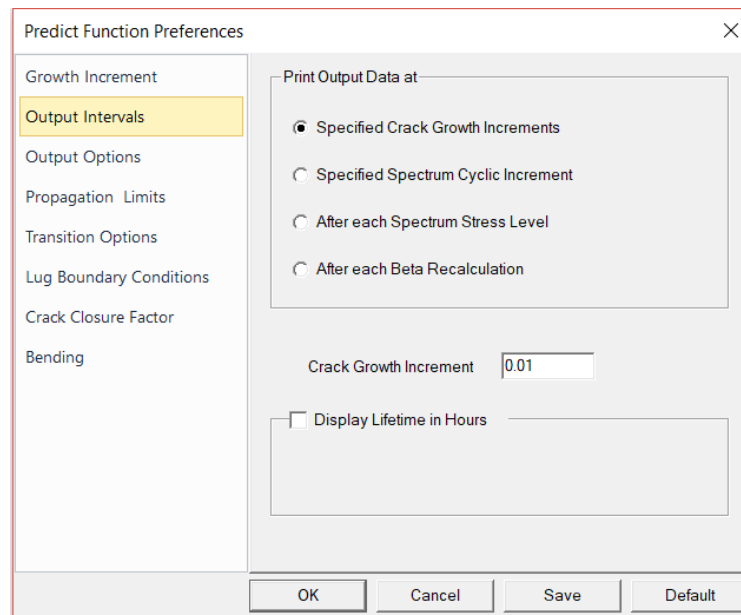


Figure 192: Output Interval Dialog

The crack growth or cyclic options will prompt the user to input the numeric value for the appropriate interval. The option to print after each Spectrum Stress Level is provided for debugging or error checking purposes and can result in a LARGE amount of data.

The option to display the lifetime in hours is merely a conversion from spectrum passes to hours, which is printed at the end of the output file. If this option is selected, a text box will appear so that the number of hours per spectrum pass may be entered. The plot file will have a column that will be converted to hours for plotting purposes. Whenever time dependent spectra are used, AFGROW will activate this option and automatically determine the time per pass through the spectrum.

3.4.1.3 Output Options

Users may select different output file options as indicated in Figure 193.

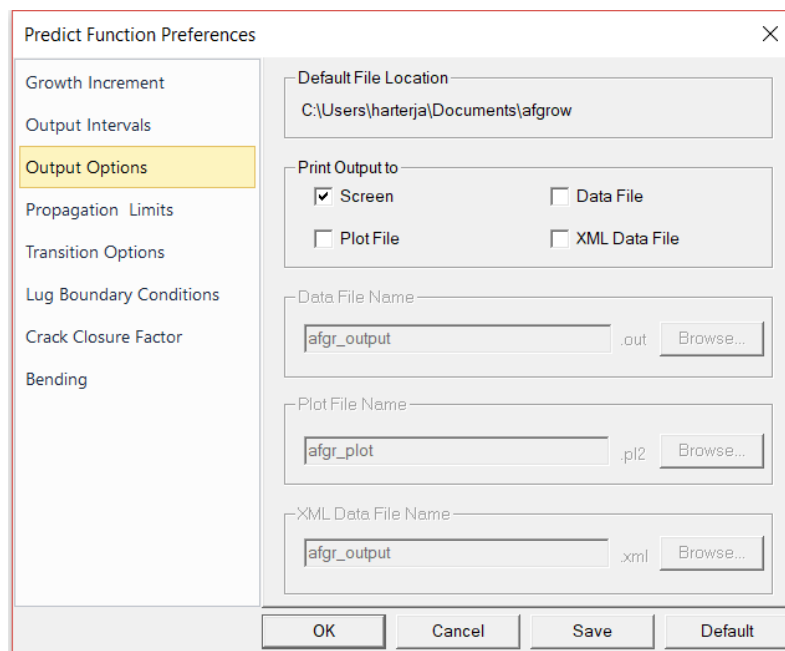


Figure 193: Output Options Dialog

The default option is the Screen, which prints the output data to the output frame in the AFGROW window. The Data File option allows a user to write the output data to a user specified file. The file may be printed or merely saved as a record of a given analysis. The Plot File option is used to create a file containing crack length, beta, and spectrum cycle data that may be plotted in Excel or other plotting software. It should be noted that the Plot File option MUST be selected in order to use the option in the Tools menu to export plot data to Excel.

When the Data File or Plot File options are selected, a default filename will appear in the appropriate text window. It is important to use the Browse button to select a location for the output file. The location must be accessible to the user (not a system directory). If the

default filename is not changed, AFGROW will OVERWRITE any existing default file. If a user types any other filename, AFGROW will display a WARNING dialog BEFORE OVERWRITING an existing file with the same name.

AFGROW prints three different R-values in the Screen and Data File output. An example of the screen output is given in Figure 194.

```

C Crack size= 0.15 Beta Tension= 1.0033 Beta Compression= 1.0033 R(k)= 0.0000 R(final)= 0.0000 Delta k=1.2398e+001 D(I)/DN=6.1298e-006
Max stress = 18.000 r = 0.00 0 Cycles Constant amp.: 1 Pass: 1

C Crack size= 0.16028 Beta Tension= 1.0037 Beta Compression= 1.0037 R(k)= 0.0000 R(final)= 0.0000 Delta k=1.2820e+001 D(I)/DN=6.7761e-006
Max stress = 18.000 r = 0.00 1600 Cycles Constant amp.: 17 Pass: 17

C Crack size= 0.17092 Beta Tension= 1.0041 Beta Compression= 1.0041 R(k)= 0.0000 R(final)= 0.0000 Delta k=1.3244e+001 D(I)/DN=7.4690e-006
Max stress = 18.000 r = 0.00 3100 Cycles Constant amp.: 32 Pass: 32

C Crack size= 0.18103 Beta Tension= 1.0045 Beta Compression= 1.0045 R(k)= 0.0000 R(final)= 0.0000 Delta k=1.3636e+001 D(I)/DN=8.1510e-006
Max stress = 18.000 r = 0.00 4400 Cycles Constant amp.: 45 Pass: 45

C Crack size= 0.1912 Beta Tension= 1.0050 Beta Compression= 1.0050 R(k)= 0.0000 R(final)= 0.0000 Delta k=1.4020e+001 D(I)/DN=8.8581e-006
Max stress = 18.000 r = 0.00 5600 Cycles Constant amp.: 57 Pass: 57

```

Figure 194: Sample Output Data

The value (r), which is printed next to the Max stress value, is the ratio of the minimum to maximum applied stress (or load) for the current spectrum cycle. The value (R(k)) is the ratio of the minimum to maximum stress intensity values which are calculated for the current cycle AFTER the load interaction model is applied. This value ALSO includes the effect of residual (additive) K values caused by residual or thermal stress effects. The value (R(final)) is the ratio of the minimum to maximum stress intensities determined in the crack growth rate routine. This value is used with the printed value of Delta K to determine the appropriate crack growth rate. These values are printed to provide the information needed for a user to verify that the appropriate crack growth rate is being calculated for the options selected for a given analysis.

If a user chooses a value for Rlo or Rhi that does not cover the actual applied stress ratios in a given spectrum, the value of R(final) will show this limitation when compared to “r” and R(k).

Note: Since the definition of Delta K for $R(\text{final}) < 0.0$ depends on the crack growth rate model, the definition of Delta K is printed in the output data in the section where the crack growth rate model is printed. Also, if the Wheeler retardation model is used, remember that this model uses a "knock-down" factor on crack growth rate to affect the retardation.

3.4.1.4 Propagation Limits

AFGROW allows the user to set crack growth propagation limits on life analysis as indicated in Figure 195. The propagation limit is currently only applied to the c-dimension since final failure of a given model is assumed to occur when a crack has transitioned to a through-the-thickness flow.

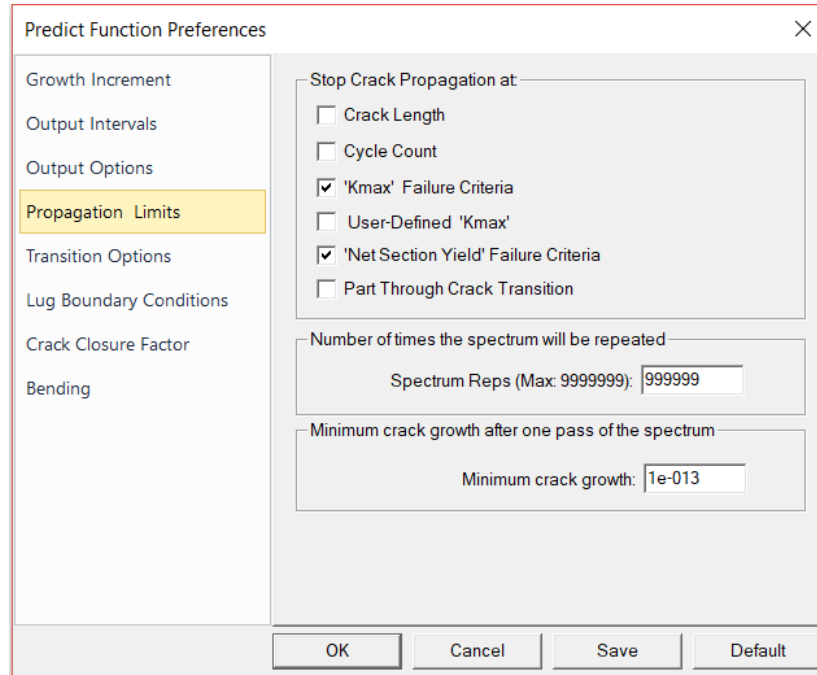


Figure 195: Propagation Limits Dialog

When selected, AFGROW will terminate at the first instance of any selected limit. When a limit is selected which requires additional input, AFGROW will open the appropriate input box for data entry.

Note: In the case of the net section yield criteria, the net section stress is based on the remote tensile load and the net area in the crack plane (minus the yield zone).

Bearing load is assumed to be uniformly distributed through the net section and is determined as:

$$\text{Bearing Load} = \text{Bearing Stress} * \text{Hole Diameter} * \text{Thickness}$$

For cases that include out-of-plane bending, it would be far too conservative to use the bending stress value (taken at the plate surface). In this case, the tensile stress due to the bending at 1/6 of the thickness is substituted to approximate the equivalent axial stress acting at the centroid of the tensile portion of the bending distribution for the un-cracked condition.

AFGROW does not include any contributions of crack asymmetry to in-plane bending contributions to the net section stress. Although it is possible for in-plane bending to play a role in the true net section yielding, there are usually geometric constraints that will prevent or mitigate this effect.

In addition to the failure criteria, users may also set limits on the number of passes of the input spectrum, and the minimum crack growth per pass that is required to continue an analysis. These limits will stop an analysis in case no (or very little) crack growth is predicted for a given problem.

3.4.1.5 Transition Options

AFGROW allows the user to set the part-through to through-the-thickness crack transition criteria as indicated in Figure 196.

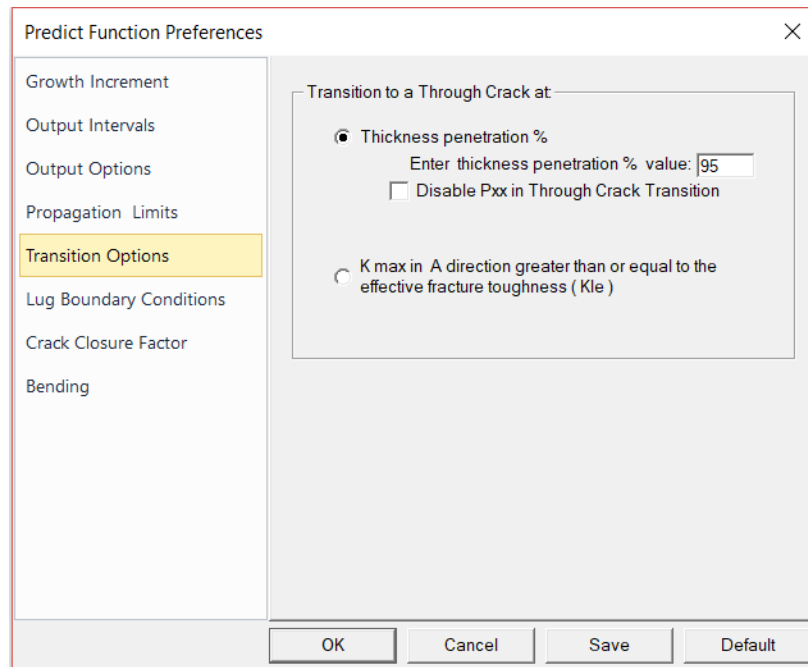


Figure 196: Transition Options Dialog

These criteria set the maximum a-dimension when a part-through crack becomes a through-the-thickness crack. Many K-solutions are not extremely accurate as the crack grows close to the free edge. The K-solution is not defined when the a-dimension touches the free edge, so it is important to set an upper bound to define the transition to a through-the-thickness crack. Transition may occur at a shorter crack length if the K-value in the a-direction exceeds the appropriate fracture toughness for a given stress state or if net section yielding is detected. These checks are done independently from the final failure criteria (see Section 3.4.1.4) since ultimate failure is assumed to occur after the crack has transitioned.

The default transition criterion is 95 percent thickness penetration. When the a-dimension reaches 95% of the thickness (or 2a for surface or fully embedded cracks), the crack is

assumed to become a through-the-thickness crack. This criterion may be adjusted by the user as indicated in Figure 196.

An additional feature has been added to allow Users to disable the Pxx criterion for transition to a through-the-thickness crack. By default, this option is not checked and a part-thru crack will be transitioned if failure is detected based on the current value of Pxx. If the box is checked, the Pxx value will not be used for the Kmax failure criteria but will still be used for net section yield failure.

The alternative criterion is the K_{Ie} method. This has been used in NASGRO based on observations that transition may occur if the maximum stress intensity value in the a-direction exceeds a prescribed value. This value is called K_{Ie} (equivalent fracture toughness for a part-thru crack). Values of K_{Ie} are included in the NASGRO material database. Typically, K_{Ie} may be estimated as:

$$K_{Ie} = 1.4(K_{Ic})$$

K_{Ic} is (of course) the plane strain fracture toughness for a given material. Therefore, if the NASGRO material database is NOT used, K_{Ie} will be estimated as shown above.

3.4.1.6 Lug Boundary Conditions

Lug pin loading boundary conditions may be adjusted (not recommended for novice users) as indicated in Figure 197.

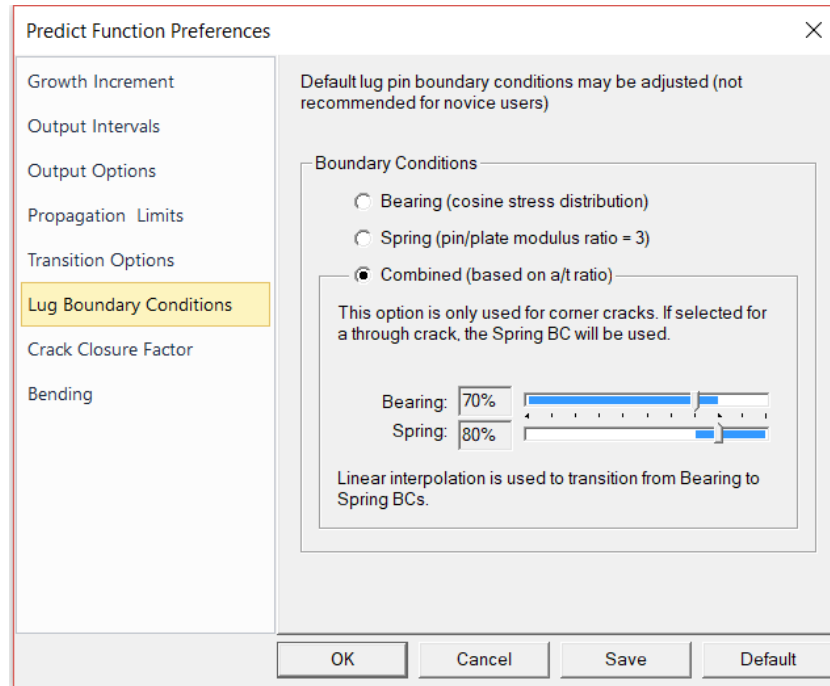


Figure 197: Lug Boundary Condition Dialog

The stress intensity solution for the lug geometry is a tabular look-up solution that was generated using the p-version finite element program, StressCheck[®]. Two different pin loading boundary conditions (BC) were used to obtain the solutions – bearing (cosine stress distribution) and a distributed spring (pin/plate modulus ratio = 3)⁶³. The major difference in the two boundary conditions is the fact that the bearing condition allows the hole to deform, and the spring condition constrains the hole. Verification testing (performed at Purdue University on aluminum lugs with steel fasteners) indicated that the spring BC matched the results for through-the-thickness cracks, and the bearing BC worked best for most corner cracks. As the corner cracks grew larger, the stress intensity values generally transitioned toward the finite element solutions using the spring BC. According to the work at Purdue, this was seen when the a-dimension reached approximately 75-80 percent of the specimen thickness. As noted above, the bearing BC allowed the hole to deform in the FEM, and the corner cracked tests at Purdue [61] had an average pin clearance of 0.002 inches. Subsequent FEM modeling [62] for clearance fit pins showed that a clearance in the range (0.0005 – 0.001) resulted in ~40% increase in K over the results for the spring BC, for $a/t < 0.1$. This may explain why the bearing BC worked better for the corner cracked lug tests. While much more work is required to be certain, the current release default BC is set to begin transition from the bearing to the spring BC at 70% of the

⁶³ Ratios as low as 1.0 showed very little change in the K-solution

specimen thickness. Between 70 and 80 percent of the thickness, the K-solution is determined using a linear interpolation of both BCs. Once the corner crack has reached 80 percent of the thickness, the spring BC is used. For through-the-thickness cracks, the default condition uses the spring BC.

There is a significant difference between the two loading conditions. The FEM results for the FEMs using the bearing distribution were approximately 40% higher than the results for the spring condition. No data were used for model verification other than the testing performed at Purdue. It is left to the user to determine which BC is more appropriate for any given life prediction. In cases where the user is certain that there is no measurable pin clearance, the spring BC may be a good option for longer predicted lives. However, as noted above, this flexibility is intended for experienced users.

3.4.1.7 Crack Closure Factor

AFGROW allows for the use of a crack closure factor (β_r) for corner and surface cracks as indicated in Figure 198.

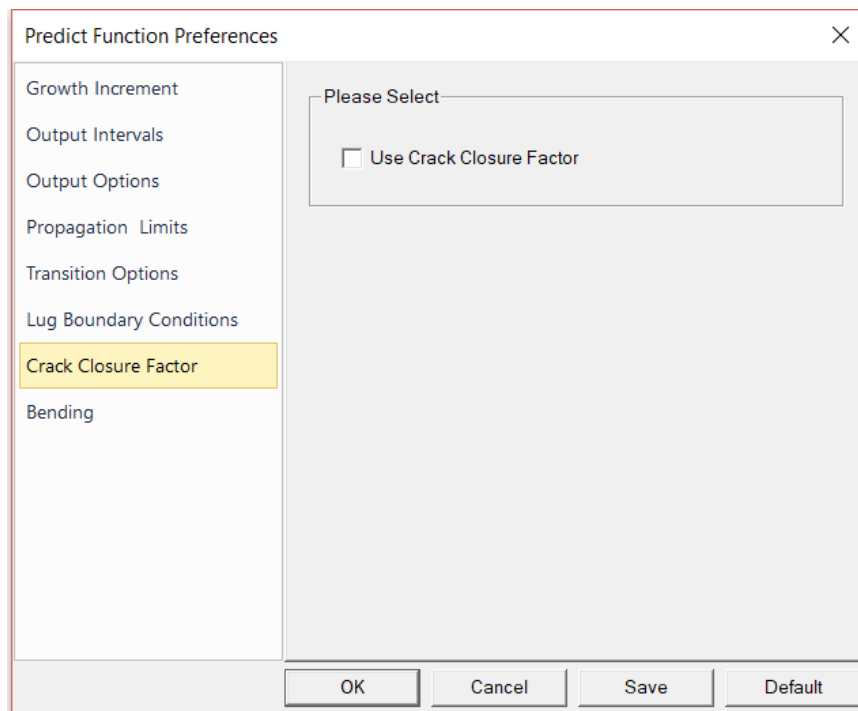


Figure 198: Crack Closure Factor

The crack closure factor [80] is a concept that is used to modify the stress intensity solution for part-through cracks when either crack tip intersects a free surface. The idea is that crack tips at a free surface are growing under less constraint (plane stress) and tend to grow more slowly since more of the applied energy is absorbed by the larger plastic zone around the crack tip. This is used as a “knock-down” factor for the K-solution as a function of stress

ratio (R) as shown below.

$$\beta_r = 0.9 + 0.2 R^2 - 0.1 R^4, \text{ for } R \geq 0$$

$$\beta_r = 0.9, \text{ for } R < 0$$

It is difficult to accurately predict crack shapes that occur as a crack grows through the thickness of a plate. When crack growth shape changes for part-through cracks are predicted using two crack tips, the assumption is that the crack growth increments for each tip are representative of the crack driving force that occurs over the entire crack front. Moreover, most K-solutions assume that the cracks are always elliptical in shape. As a result, this correction is fairly empirical in nature, and should not be interpreted as part of the actual K-solution. This is why this option is located in the Preferences Menu.

3.4.1.8 Bending

When out-of-plane (transverse) bending is modeled for a straight, through-the-thickness crack, there is a basic geometric incompatibility since this loading condition will result in an oblique crack shape. There are limited solutions available for oblique cracks, so for straight, internal and edge through cracks, AFGROW uses the existing axial load solution for a given geometry and applies a “knock-down” factor in an attempt to account for the net through-the-thickness effect. This does not apply to through cracks at holes since AFGROW uses a referenced solution for those cases.

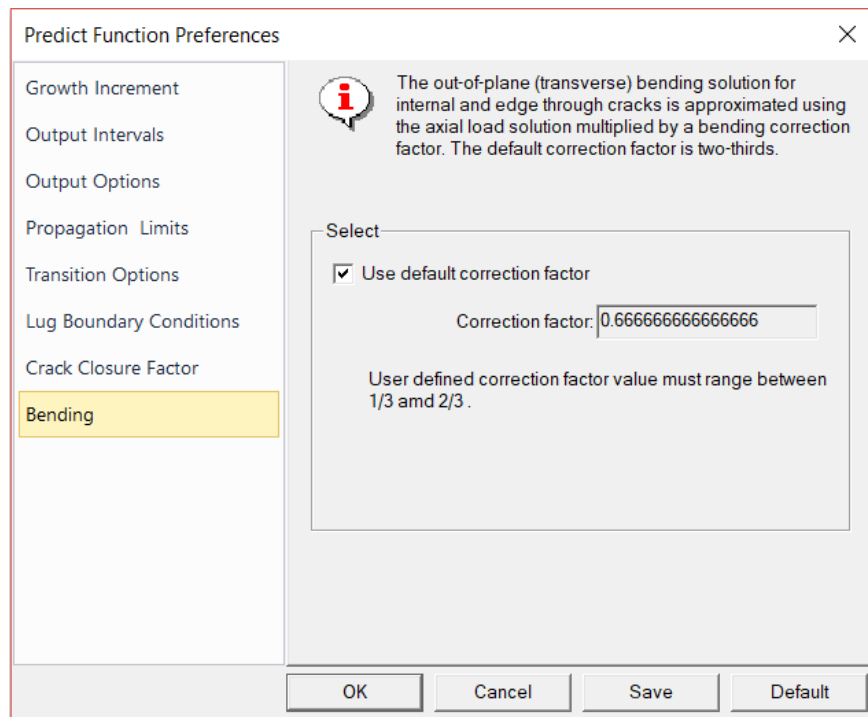


Figure 199: Bending Solution Preference

As indicated in Figure 199, the default factor is set at 2/3, but users can change this to any desired value between 1/3 and 2/3.

3.4.2 Predict Run

Toolbar Icon:



This option will start the AFGROW life prediction process.

3.4.3 Predict Stop

Toolbar Icon:



This option will stop the AFGROW life prediction process.

3.5 Tools Menu

Access to other software tools is available through the tools menu. These tools enhance the capabilities of AFGROW in several areas: viewing plots in Excel, spectrum translation, and interfacing with an aging aircraft structures database. The current tool options are shown in Figure 200.

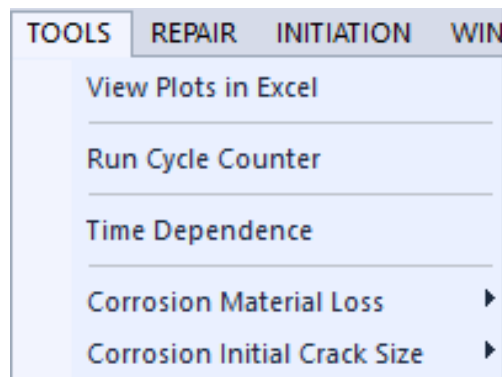


Figure 200: AFGROW Tools

3.5.1 View Plots in Excel

AFGROW allows plot files to be written directly to Microsoft Excel. When this option is selected, the Open Excel dialog appears (see Figure 201).

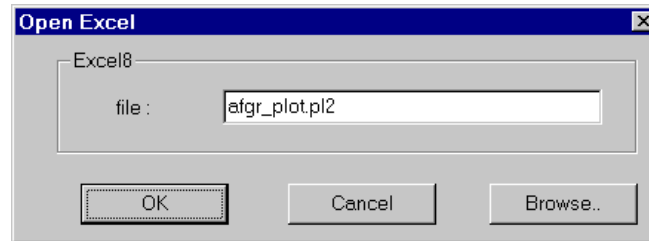


Figure 201: Dialog Box to View Plots in Excel

The default plot file name will appear in the dialog box. The user may enter the desired file name manually by clicking inside the text box or may browse the computer to find the desired plot file. Use the Browse button and choose the location of the plot file to make sure that AFGROW is saving the file to the desired file folder.

Once started, AFGROW opens Excel on the user's PC and writes the data to Excel. The crack length vs. cycle data will be plotted on separate worksheet(s). The speed at which this happens will, of course, be dependent on the PC. Once this is complete, users can work with the Excel file as desired. For more details on creating a plot file, see Section 3.4.1.3.

AFGROW is updated periodically to maintain compatibility with the latest version of MS Excel.

3.5.2 Run Cycle Counter

A cycle is defined as shown below in Figure 202. A cycle begins at a certain stress (or load) level, moves to a different level, and returns to the starting level.

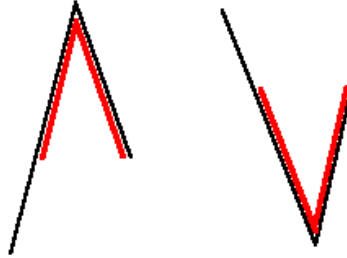


Figure 202: Cycle Definition

Many people have submitted questions related to how AFGROW uses the input spectrum data. Each line in an AFGROW spectrum consists of one or more cycles. A cycle is described by any two of the following parameters: minimum value, maximum value, or stress ratio (R).

Real structures are loaded and unloaded periodically so that the peak-valley sequence of applied stresses is unlikely to form true cycles. The actual peak or valley points are often referred to as reversals since the loading direction (increasing or decreasing) is reversed at each point (see Figure 203).

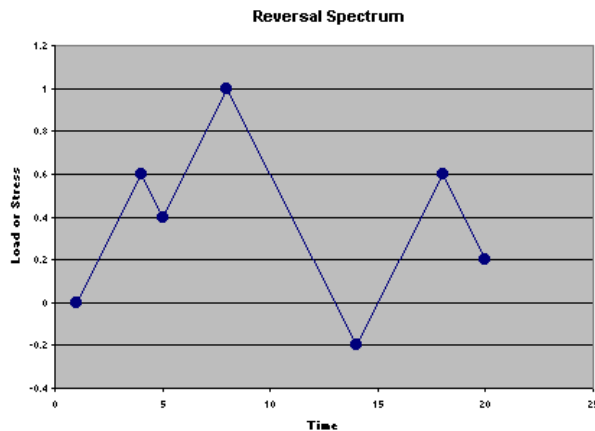


Figure 203: Sample Uncounted Stress Sequence

In any case, the important fact is that AFGROW assumes that the input spectrum is given in the form of cycles, not simply an uncounted sequence.

AFGROW provides a cycle counting program [63] than can be used to convert uncounted sequences to cycles (see Figure 204). This tool is provided for the convenience of our users,

but there are other cycle counting methods in the open literature that may be used as desired.

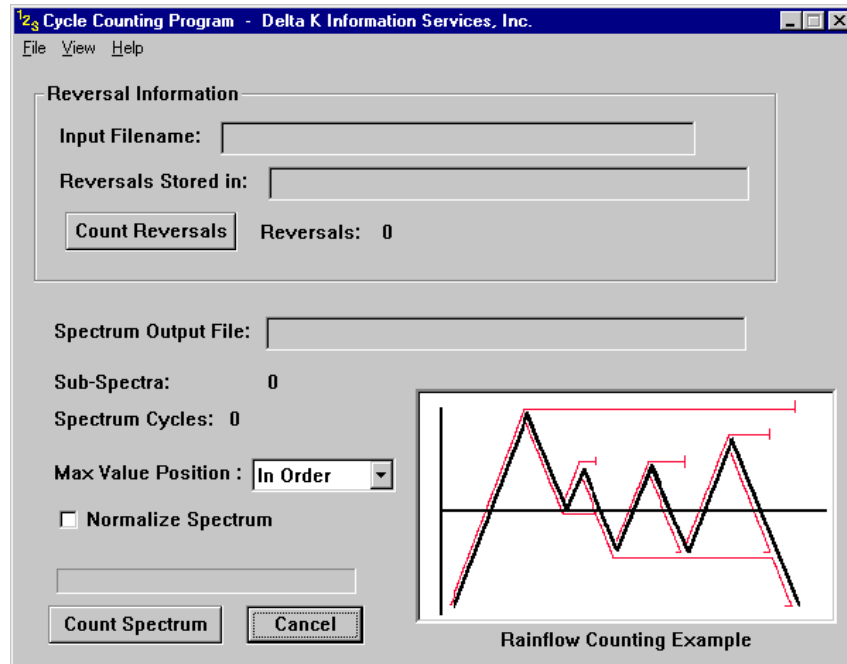


Figure 204: Cycle Counting Software Interface

This program has many features, which are described in detail in its own on-line help. The program will convert stress values to reversals and reversals to cycles. If reversals are available, they can be used as is to create the counted spectrum.

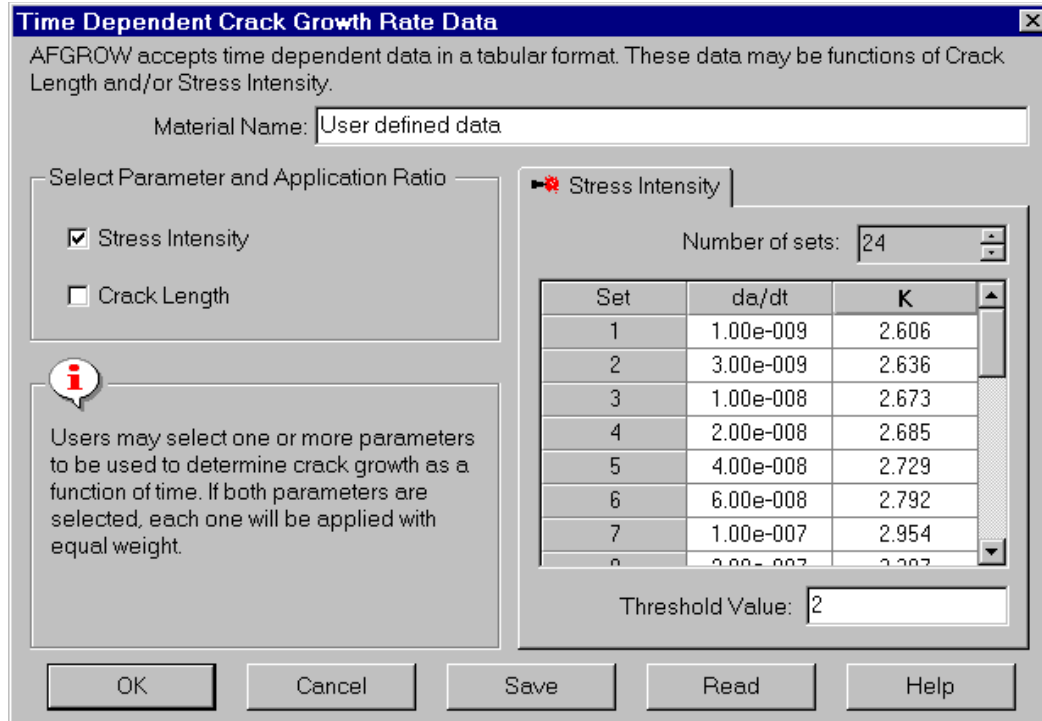
An important point to note is the placement of the maximum value in the spectrum. The cycle counting program will place counted cycles in the order determined by the order of the peak points. Some fracture mechanics experts prefer to place the overall maximum value at the end of the spectrum to minimize the effect on crack growth retardation (more conservative result). Others may prefer to place the maximum peak in the order that the peak occurs in the original sequence. There is an option to place the maximum value at the beginning, end, or in the original order of the peak values.

The resulting spectrum may be normalized so that the maximum value is 1.0. This is very common and allows users to scale the spectrum values based on the overall maximum value - without using a calculator.

Finally, it is important to know whether a spectrum has already been cycle counted. Generally, spectra created for crack growth life prediction will be counted. It is not easy to tell, so it is important to find out. If a counted spectrum is counted twice, it will be altered (unless it is a constant amplitude sequence). The initiation module in AFGROW assumes the spectrum is also counted (see Section 3.7). It would have been very difficult to manage both counted and uncounted spectra in AFGROW. In short, if you are using a spectrum in AFGROW, it should be cycle counted.

3.5.3 Time Dependence

Time dependent crack growth rate data **MUST** be entered as a function of stress intensity and/or crack length (as indicated in Figure 205 below). Users may enter data as a function of both parameters. If this is done, the effect of time will be determined for both parameters. This could have the effect of doubling the effect of time – of course; this depends on the magnitude of the input data.



The dialog box is titled "Time Dependent Crack Growth Rate Data". It contains a text area stating: "AFGROW accepts time dependent data in a tabular format. These data may be functions of Crack Length and/or Stress Intensity." Below this is a "Material Name:" field with the text "User defined data".

Under "Select Parameter and Application Ratio", there are two checkboxes: "Stress Intensity" (checked) and "Crack Length" (unchecked).

An information icon (i) is followed by a text box: "Users may select one or more parameters to be used to determine crack growth as a function of time. If both parameters are selected, each one will be applied with equal weight."

On the right, the "Stress Intensity" tab is active. It shows "Number of sets:" as 24. Below is a table with 3 columns: "Set", "da/dt", and "K".

Set	da/dt	K
1	1.00e-009	2.606
2	3.00e-009	2.636
3	1.00e-008	2.673
4	2.00e-008	2.685
5	4.00e-008	2.729
6	6.00e-008	2.792
7	1.00e-007	2.954
8	2.00e-007	3.202

Below the table is a "Threshold Value:" field with the value 2.

At the bottom are buttons for "OK", "Cancel", "Save", "Read", and "Help".

Figure 205: Time Dependent Rate Data Dialog

This option **MUST** be used in conjunction with a time dependent stress (or load) spectrum (see Section 3.2.4.2.2). If the time dependence option is selected, and a time dependent spectrum is not used, there will be no effect of time on the resulting life analysis. There are currently no tools in AFGROW to develop random, time dependent spectra. The user must do this with a separate program or any text editor.

3.5.3.1 Using Time Dependent Data as a Function of Stress Intensity

If da/dt data are entered as a function of stress intensity, the input spectrum values are used, along with the current crack geometry, to determine the appropriate stress intensities required to calculate crack extension as a function of time. Currently, AFGROW allows the following four types of time dependent cycles in an input spectrum: [Ramp Up](#), [Ramp Down](#), [Hold](#), and [Random Cycles](#).

In the case of the ramped or hold cycles, the method used to determine crack extension is shown in Figure 206.

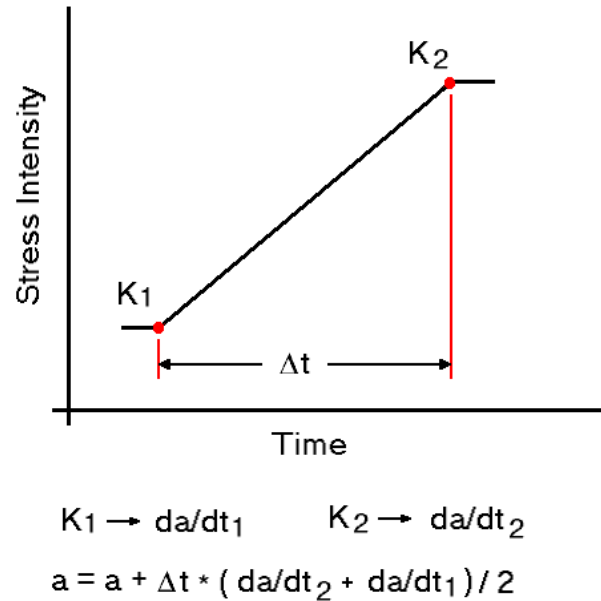


Figure 206: Crack Extension From a Ramped Cycle

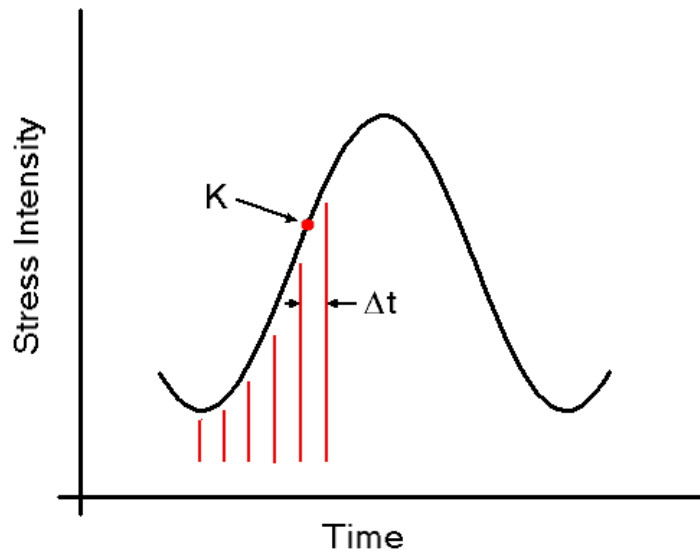
Users should be aware that AFGROW uses the ENTIRE cycle (regardless of how much time is assigned) to determine time dependent crack extension for ramped or hold cycles. The reason for this is because the logic already existed in the code for cyclic crack growth, and the smallest interval in that case is a single cycle. The crack extension for each time dependent cycle is added to the crack extension calculated using the standard cyclic dependent data.

$$\Delta a = da/dN * \Delta N + da/dt * \Delta t$$

AFGROW will NOT recalculate stress intensity values within a single cycle for these cases. **If a ramp or hold cycle occurs over a relatively long period of time, it is recommended that the cycle be divided into multiple cycles** so that changes in stress intensity, due to crack extension, may be accounted for more accurately.

In addition to the time dependent crack growth, cyclic dependent crack extension is calculated for [Ramp Up](#), [Hold](#), and [Random Cycles](#). Cyclic dependent crack extension is NOT calculated for [Ramp Down Cycles](#) because it is assumed that each [Ramp Up](#) would be followed by an equivalent [Ramp Down](#) at some point in the spectrum. Twice as much crack extension would result if these calculations were performed for each case.

In the case of random cyclic loading, the load cycles are assumed to be sinusoidal and each cycle is divided into 100 segments as indicated in Figure 207.



$$K \rightarrow da/dt$$

$$a = a + \Delta t * da/dt$$

Figure 207: Crack Extension from a Random Cycle

3.5.3.2 Using Time Dependent Data as a Function of Crack Length

If time dependent da/dt data are entered as a function of crack length, crack extension is determined based on the current crack length and the time associated with the given Vroman increment (see Section 3.4.1.1).

AFGROW determines the growth rate for a given crack length and then determines the amount of time required to grow to a size consistent with the Vroman increment (within a given spectrum stress level). If there is more time available in a given stress level, the rate is recalculated for the new crack length, and the process is repeated until the time in that stress level is used. In any case, crack extension and rate values are always calculated at least once for each stress level in a spectrum.

3.5.4 Corrosion Material Loss

Two types of material loss models (exfoliation and intergranular corrosion) have been added and may be selected from the Corrosion Material Loss menu.

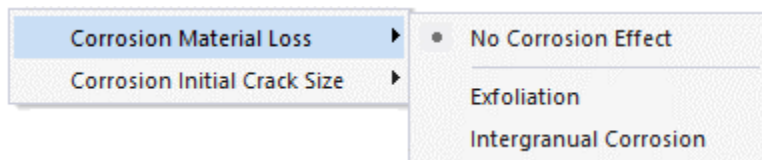
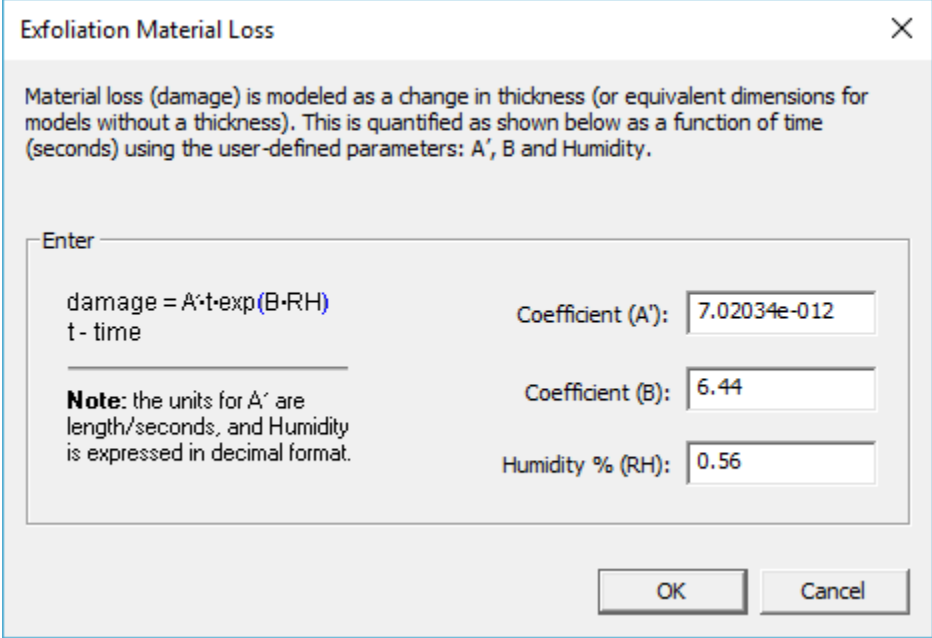


Figure 208: Corrosion Material Loss Menu

These models are time dependent and must be used with a time dependent spectrum which provides the relationship between applied loading and total elapsed time (seconds). If a time dependent spectrum is not used, the material loss will not be calculated and will have no effect on life prediction.

A general description of each material loss model and associated parameters is provided in the following sections

3.5.4.1 Exfoliation



Exfoliation Material Loss

Material loss (damage) is modeled as a change in thickness (or equivalent dimensions for models without a thickness). This is quantified as shown below as a function of time (seconds) using the user-defined parameters: A', B and Humidity.

Enter

damage = $A' \cdot t \cdot \exp(B \cdot RH)$
t - time

Note: the units for A' are length/seconds, and Humidity is expressed in decimal format.

Coefficient (A'): 7.02034e-012

Coefficient (B): 6.44

Humidity % (RH): 0.56

OK Cancel

Figure 209: Exfoliation

Exfoliation is a particular type of intergranular corrosion that occurs on elongated grain boundaries. Corrosion products set up internal stresses which separate the metal along these boundaries, giving rise to a layered appearance. This exfoliation model uses a linear rate law. The empirical model is based on experimental exfoliation corrosion data on an L-T surface of 7178-T6 aluminum recovered from wing skin of a legacy aircraft performed by X. Zhao and G.S. Frankel [64]. When the exfoliation corrosion model is selected in AFGROW, the material loss rate and the remaining thickness are calculated and are documented for each output interval. Since plate thickness has a direct effect on crack growth, the predicted life will be affected by the use of this model. To avoid the possibility that thickness loss may be predicted to outpace crack growth for corner cracks (effectively removing the crack), thickness loss is modeled on the side of the plate away from the corner crack origin.

The input parameters are defined below:

A' : Material constant based on exposure and condition (length/sec)

t : Exposure time (seconds)

B : Material constant

RH : Relative humidity at room temperature (decimal format)

Note: Experimental data showed that exfoliation corrosion rate propagates above a critical humidity of 56%. It is also important to note that the model is more accurate at longer times and higher humidity.

3.5.4.2 Intergranular Corrosion

Intergranular Corrosion [X]

Intergranular corrosion (damage) is modeled as a length from a user-defined initial defect size. It is quantified as a function of time (seconds) using the parameters: A and N as indicated below.

This is not used in crack growth predictions, but is reported in the output for reference purposes.

Enter

damage = $A t^N$
t - time

Note: the units for A are length/seconds

Coefficient (A): 8.60656e-008

Coefficient (N): 0.5

Initial defect size: 0

OK Cancel

Figure 210: Intergranular Corrosion

During the solidification process of engineered metallic alloys, crystals nucleate and grow together to form a solid. In some cases, the crystals or grains chemical composition and the properties of the boundary region can differ significantly causing the rate of corrosion for the alloy to increase. This is referred to as intergranular corrosion. This intergranular corrosion (IGC) model is based on the work of Weilong Zhang and G.S. Frankel, Ohio State University [65]. The model is a power rate law that allows the user to input equation coefficients (A & N) and initial defect size. AFGROW calculates the IGC crack length and the IGC rate for times up to 100 hours.

When the intergranular corrosion model is selected in AFGROW, the predicted IGC growth rate is calculated, added to the initial defect size, and are documented for each output interval.

The input parameters are defined below:

A: Material constant based on exposure and condition (length/sec)

t: Exposure time

N: Material constant based on exposure and condition

Initial defect size: User defined initial flaw size (length)

Note: AFGROW does not use the IGC calculations as part of the life prediction process. The intergranular corrosion rate and total IGC length are provided to users so that they may compare the potential effect of IGC to a baseline crack growth prediction and/or calculate the loss of mechanical strength due to intergranular corrosion.

3.5.5 Corrosion Initial Crack Size

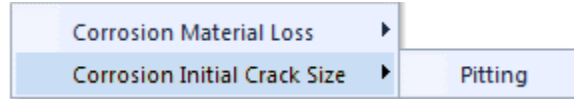


Figure 211: Corrosion Initial Crack Size

3.5.5.1 Pitting

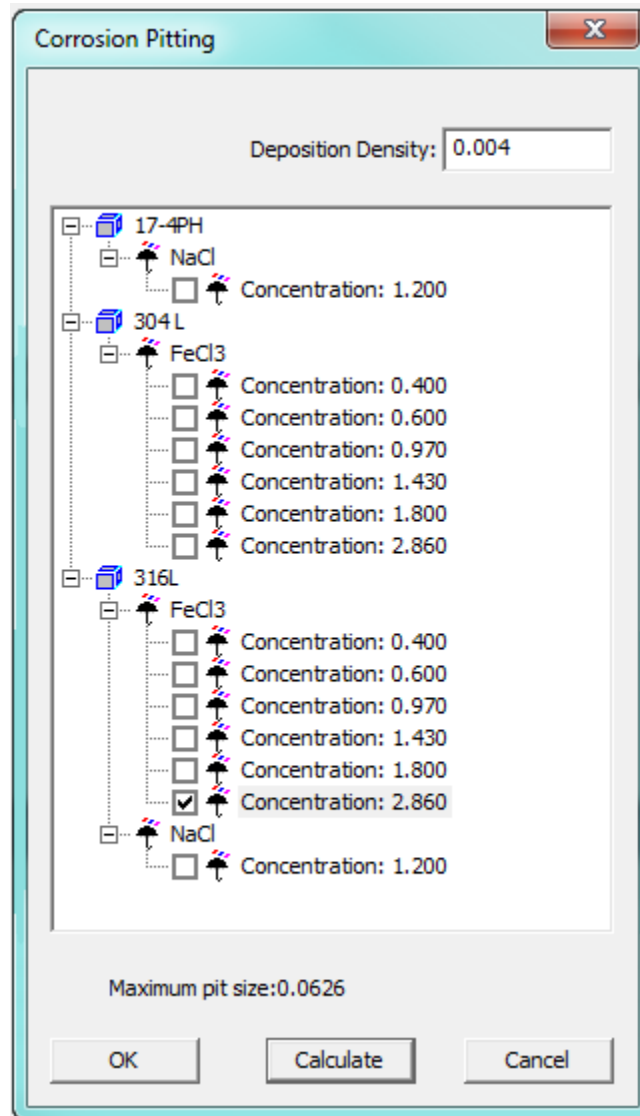


Figure 212: Corrosion Pitting

Pitting Corrosion commonly occurs in localized areas where a material loses its corrosion resistant coating. Another cause of pitting corrosion is the presence of aggressive halide ions such as chloride. This pitting model is based on the work of Prof. Robert G. Kelly,

University of Virginia. The code was developed and validated based on the experimental results conducted by, M.T Woldemedhin, M.E. Shedd and R.G. Kelly [66].

This model predicts the maximum pit size possible for 17-4PH, 304L, and 316 stainless steel surfaces under thin film electrolyte conditions for the available test data and assumes a hemispherical pit. The maximum pit size possible is predicted by coupling the maximum cathode current (I_{cath}) available from the thin film electrolyte with the minimum current (I_{anode}) needed to grow a hemispherical pit. Users have the option to choose from five different concentrations for FeCl₃ and one concentration for NaCl for the thin film conditions as indicated above.

Select the Calculate button to determine the maximum pit size for the input conditions.

The single, user-defined input parameter is defined below:

Deposition Density: Enter in units of weight/area (lbm/in² – English, Kg/m² – Metric)

Note: The maximum pit size may be used to define an initial crack length for a subsequent life prediction. This will enable users to account for the total damage accumulated due to pitting corrosion for a given material and environmental conditions. The upper bound of the maximum pit radius is set to 0.0394 inches. This is believed to be the largest pit that one would ever expect to see on a surface. If the solution is outside the set boundary, the dialog will state “cannot find solution”.

3.6 Repair Menu

AFGROW includes an option to account for the effect of a bonded repair patch on crack growth life. This analysis is based on a Green's function method and was developed by Dr. Mohan Ratwani [7]. Currently, this method is only valid for the following conditions:

- Through-the-thickness cracks
- Thin structure (< 0.125 in.)
- Non-stiffened panels
- Crack remains under the patch

The stress intensity solution is determined by integrating the 2-D adhesive shear stresses in an area surrounding a centered through crack in an infinite plate. This area is simulated using a telescopic grid with a fine mesh covering the crack and a course mesh extending a distance of one half of the total crack length on either side. The height of the mesh extends to one and a half of the total crack length above and below the crack. Due to symmetry conditions, a quarter of the panel is analyzed with a total of 144 nodes. A unit stress is applied to the cracked panel and stress intensity values are determined for approximately 20 crack lengths (crack intervals are calculated using an algorithm in the model). The initial crack length is the same as the initial crack length specified by the user and the final crack size (c) does not exceed 2 inches. A beta correction table is generated by dividing the stress intensity for the patched case by the stress intensity for the same case without a patch. The correction for cracks exceeding 2 inches is assumed to be constant⁶⁴. The assumption is that a centered through crack solution is used to determine the beta correction due to the bonded repair at various crack lengths and is applied to the actual geometry selected by the user. The 2-inch limit on the beta correction values is based on analysis and test verification data. These data indicate that the ratio of the patched to non-patched stress intensity values tend to be nearly constant above a half crack length (c) of 2 inches for the center cracked case using typical patch materials and adhesives.

AFGROW will store up to eight repair designs and their beta correction tables. The most current design is active by default, but the user may change the active design through the repair plot option in the view window or menu selection.

The repair menu options are described in the following sections.

3.6.1 Repair Design

When the repair design is selected, AFGROW will not allow certain values to be changed for the given crack model. The reason for this is that material properties and model dimensions are required for the repair analysis. If any of these values were changed, the

⁶⁴ The stress intensity value will NOT be constant since the applied K value for the non-patched case will increase with crack length. The beta correction value, which is multiplied by the K value for the non-patched case, will be assumed constant.

repair analysis would have to be redone. A dialog will appear informing users of this situation.

A wizard is used to guide users through the repair design process as described in the following sections.

3.6.1.1 Ply Design and Lay-up

Ply Design and Layup

Material Properties

Name: GRAPHITE/EPOXY

Id	Value	Description
EL	25000	Youngs ...
ET	1500	Youngs ...
GLT	700	Shear M...
PNU	0.3	Poisson ...
ALFL	-4e-007	Coef. of ...
ALFT	2e-005	Coef. of ...

Change Material Database File

Number of Plies: 13

Ply Thickness: 0.0052

Delta T : 0

Ply Layup:

Orient...	Ply #
45	1
-45	2
0	3
0	4
90	5
0	6
0	7
0	8
90	9
0	10
0	11
-45	12
45	13

Patch Type

Number of Plies is half the total numbers of plies for symmetric patches

☐ Symmetric

☐ Double_Sided

☐ No Bending

Patch Stiffness Indicator(% of Plate Stiffness)

70% 100% 120% 150%

Out of bounds Allowed Ideal Allowed Out of bounds

Auto Design < Back Next > Cancel Help

Figure 213: Ply Design and Lay-up Dialog

This dialog contains the information for the repair patch including whether to consider out of plane bending and an option to consider thermal residual stresses in the stress intensity solution. The repair design is performed automatically using an internal algorithm based on the maximum applied stress and the modulus of the cracked plate. The automatically generated ply lay-up includes cross plies to provide delamination resistance. In addition, Dr Ratwani's method tends to produce errors if the patch moduli in the x and y directions differ greatly. For these reasons, cross ply lay-ups are preferred. The load direction is assumed to be normal to the crack plane. Users can make any desired changes to the design by making changes to the material properties, ply lay-up, or type of patch.

3.6.1.1.1 Material Properties

The material properties are given for the appropriate material in the default database file. The user MAY NOT change these values since they are interrelated. The use of invalid

composite material properties will cause the analysis to crash. Users may create their own material database files⁶⁵, but must be sure to input valid property values.

The user has control over the number of plies, ply thickness, and Delta T. The Delta T parameter (degrees F) is included to provide a means to account for the residual thermal stressed caused by the differences between the thermal expansion of the cracked plate and composite patch. Some believe that Delta T should be the difference between the patch curing temperature and the operating temperature. Others think that there may be some relaxation in the adhesive after curing which results in a lower effective Delta T. In any case, the user is free to use judgment in setting this value.

3.6.1.1.2 Ply Lay-up

The ply lay-up is initially determined by AFGROW based on a criterion to include cross plies for some biaxial strength, symmetry, and a target value of patch stiffness of 110 percent of the cracked plate stiffness. The user may change the lay-up⁶⁶ by using the mouse to either drag a ply to a new location or selecting a ply (single click) and touching the control key (or a second, single mouse click after a few second pause). AFGROW also includes an option to auto design the ply orientation (left click in the Orient... button) and an auto design option for both the orientation and number of plies (left click in the Ply # button). Cross ply lay-ups are desirable to help prevent the patch from delaminating during normal use. Also, it should be noted that Dr. Ratwani's method has been known to have problems if the patch E_x and E_y values differ by large amounts (i.e.: uniaxial lay-up).

3.6.1.1.3 Patch Type

The three options for patch types are:

Symmetric: The ply lay-up shown is doubled and the lay-up is therefore symmetric with respect to the center of the patch.

Double Sided: The patch is applied on both sides of the cracked plate (eliminates out of plane bending for symmetric patches).

No Bending: Do not account for out of plane bending in the calculations. The plate may be constrained to prevent bending or the user may wish to compare the results with and without out of plane bending.

⁶⁵ The data in the material database file must be in English units. AFGROW will make the appropriate conversion based on the current units being used.

⁶⁶ The maximum number of allowable plies in the current version is 32 (16 if the symmetric option is active).

3.6.1.1.4 Patch Stiffness Indicator

The patch stiffness indicator allows the user instant feedback on the patch design. The target is 110 percent of the plate stiffness to provide strength to help keep the crack closed, but not so stiff to attract too excessive load to the patch. Remember that this is calculated based on thickness and is independent on the patch width. Where possible, it is recommended that the patch width be twice the width of the crack over the projected life of the repair.

3.6.1.2 Patch Dimensions and Adhesive Properties

This dialog, Figure 214, contains the information for the repair patch dimensions and adhesive properties. This includes modeling the local disbond in the adhesive, which tends to occur around a cyclically loaded crack. There is also an option to control whether the patch is considered when using the critical stress intensity factor failure criterion.

Sample C-Scan image of a repair:

Adhesive Properties:

Name: FM-73

Shear Modulus (GXY): 60

Thickness: 0.006

Disbond (Dh/C): 0

Patch Dimensions:

Width (Wp): 4

Length (Lp): 4

Critical SIF based on:

☒ Patched Structure

☐ Unpatched Structure

< Back Next > Cancel Help

Figure 214: Patch Dimensions and Adhesive Properties Dialog

3.6.1.2.1 Sample C-Scan Image of a Repair

The sample C-Scan image (see Figure 214) is provided to show the patch dimensions and explain the concept of the adhesive disbond, which normally occurs around the crack tip under cyclic loading. The method, proposed by Dr Ratwani, assumes the disbond follows the crack tip and is elliptical in shape.

3.6.1.2.2 Adhesive Properties

The adhesive properties consist of the following:

Name: Adhesive name (for documentation purposes)

Shear Modulus (GXY): Adhesive shear modulus

Thickness: Thickness of adhesive layer

Disbond (Dh/C): Ratio of minor to major axis of an assumed elliptical disbond at the crack. Of course, zero indicates that there is no disbond.

3.6.1.2.3 Patch Dimensions

The current solution provided by Dr Ratwani assumes the patch to be twice the width of the crack. However, verification tests have shown that the solution provides reasonable results for cracks extending to the edge of the patch⁶⁷. The only purpose for the user input patch width is for the out of plane bending calculations.

Width (Wp): Patch Width (in.)

Length (Lp): Patch Length (in.) - At this time this variable is not used in the analysis; however, the patch length is assumed to be infinite in the analysis at this time.

3.6.1.2.4 Critical SIF

The critical stress intensity factor may be based on either of the following:

Patched Structure: The critical stress intensity factor calculation includes the patch beta correction factor.

Unpatched Structure: The critical stress intensity factor calculation DOES NOT include the patch beta correction factor. All this does is allow a user to be a bit more conservative in the life prediction. Of course, this conservatism is only valid if you assume the patch would fall off AFTER the crack is EQUAL to or LARGER than the critical crack size without a patch.

⁶⁷ In cases where the crack is longer than one half the patch width, AFGROW sets the adhesive shear stress values to zero for nodes that fall outside the patch boundary when calculating the beta correction values.

3.6.1.3 Designed Patch Properties

This dialog, Figure 215, shows the ply lay-up and resulting laminate structural properties.

The dialog box is titled "Designed Patch Properties". It is divided into two main sections: "Ply Orientations" on the left and "Patch Properties" on the right. The "Ply Orientations" section displays a stack of 10 horizontal bars representing the ply lay-up. The orientations are listed as follows: -45.0, +0.0, +0.0, +0.0, +0.0, +90.0, +90.0, +0.0, +0.0, and +45.0. The "Patch Properties" section contains a list of material and structural properties with their corresponding values in input fields. At the bottom of the dialog are five buttons: "Save", "< Back", "Next >", "Cancel", and "Help".

Patch Properties	
Name:	GRAPHITE/EPOXY
Youngs Modulus in X Dir. (EX):	15400.3886718
Youngs Modulus in Y Dir. (EY):	6708.38
Shear Modulus (GXY):	2464.54
Thickness (THICK):	0.0676
XY Poisson Ratio (NUXY):	0.316096
YX Poisson Ratio (NUYX):	0.137691
Coef. of Thermal Exp. in Y:	1.3414e-007
Coef. of Thermal Exp. in X:	2.87297e-006
Width (Wp):	4
Length (Lp):	4

Figure 215: Patch Dimensions and Adhesive Properties Dialog

Ply Orientations: This window simply shows the ply orientation of the complete patch.

Patch Properties: The patch properties are given for the total patch laminate.

Save Button: The complete repair design may be saved to a file for later use.

At this point, clicking on the NEXT button will start the repair analysis. This can take a few minutes (depending on the computer) and a progress bar will appear to give an indication of the expected run time.

Once the analysis is complete, the repair beta correction vs. crack length plot is displayed, as shown in Figure 216.

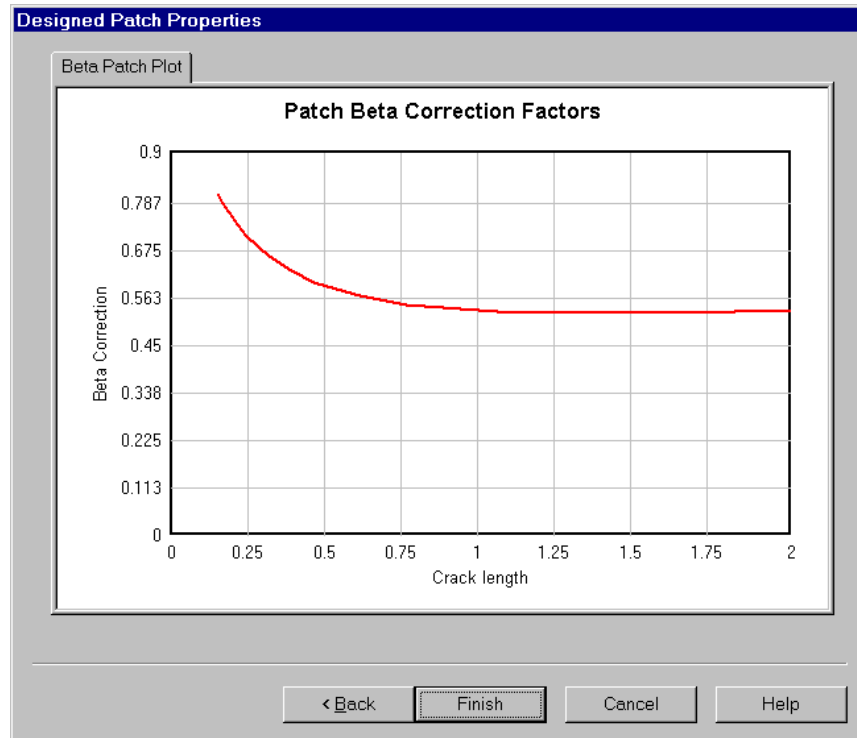


Figure 216: Repair Beta Correction vs. Crack Length

Users may accept the design by clicking on the “Finish” button or return to the repair design wizard by clicking on the “Back” button. If the “Finish” button is selected, the specimen cross-sectional view in the animation frame is shown with a depiction of the bonded repair (see Figure 217).



Figure 217: Specimen Cross-Sectional View with a Bonded Repair

3.6.2 Read Design Data

This option opens a file containing data for a previously saved repair design (see Section 3.6.1.3). The file dialog is shown in Figure 218.

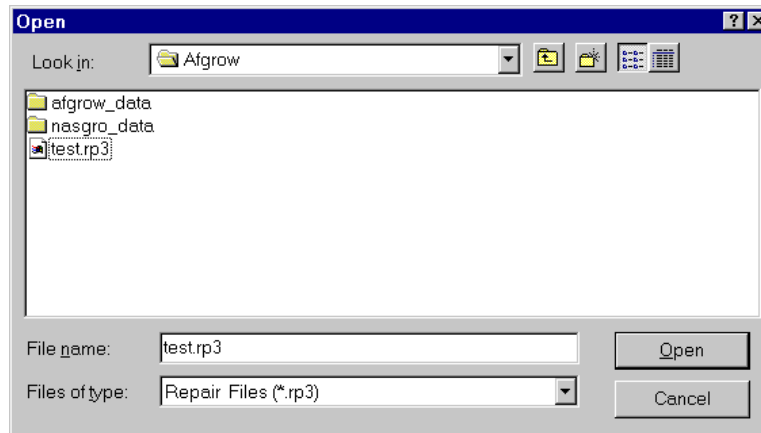


Figure 218: Opening a Repair Design File

3.6.3 Repair/No Repair

This option simply activates/deactivates the repair so a user may perform a crack growth analysis for the same case with or without the effect of the bonded repair. If a repair is active, this menu item is shown as “No Repair” and will deactivate the current repair design if selected. If the repair is not active, this menu item is shown as “Repair” and will activate the current repair. This may be useful when comparing the analytical results with and without the effect of the repair patch.

This option will NOT delete the patch.

3.6.4 Delete Repair

This option WILL delete the patch. This is required if you wish to change the material properties or geometry of the repaired structure⁶⁸.

3.7 Crack Initiation Menu

Eric Tuegel (AP/ES, INC.) initially provided the strain-life based fatigue crack initiation module used in AFGROW [8]. The original module was written in Visual Basic for Applications (Excel Macro). This code was converted to the C/C++ language and a visual interface was added to make the code easier to use.

⁶⁸ AFGROW will not allow a user to change certain properties while a repair beta correction table is being used.

In addition, it should be noted that the original module assumed the input stress spectrum was a peak/valley, uncounted spectrum. Uncounted stress spectra consist of peaks and valleys that are not arranged (counted) such that each peak/valley pair defines a closed hysteresis loop (see Figure 226). Since counted spectra are required for crack growth life prediction, this module was modified to accept cycle counted spectra. Each cycle is assumed to lie on the tension side of the overall hysteresis loop for the maximum and minimum values in the spectrum. This should provide conservative results since the mean stress for any cycle will be greater than or equal to the corresponding case for an uncounted input spectrum.

3.7.1 Strain-Life Initiation Methodology

The module uses standard strain-life methods including:

- Neuber's Rule
- Smith-Watson-Topper Equivalent Strain
- Fatigue Notch Factor (K_t)

The first important point to make about this implementation is that it was designed to work in conjunction with the rest of AFGROW as an additional capability. When used, it will provide an initiation prediction (cycles), which will be added to the cycles calculated for subsequent crack growth life. The flaw size after initiation is assumed to be equal to the initial crack size that was input in the model dimensions dialog (see Figure 89, Section 3.2.3.1.4). This provides additional flexibility since a user can use any initial crack length, which is felt to be best for the given input crack initiation data. Note that AFGROW will simply determine the initiation life based on the input data provided and add the initiation life to the crack growth life from the initial input crack size.

It should also be noted that the initiation module should ONLY be used in cases where there is a notch or hole. Since the code uses Neuber's rule, input data obtained using smooth bar specimens will not return accurate results if K_t is set equal to 1.0. It is possible to model a notch case using an un-notched model as long as the appropriate K_t , notch radius, and fatigue notch constant are used.

Another item worth noting is the fact that Young's modulus (E) is part of the material data associated with the crack growth rate data. Young's modulus is required for the initiation module, but it would be a bad idea to have the same parameter in two different dialog boxes. It is important to be sure that the modulus is correct for the given model when any changes are made to the initiation parameters. This will show up graphically in the cyclic stress-strain curve in the initiation plot option in the main frame (see Figure 23, Section 2.1.1.5).

3.7.1.1 Neuber's Rule

Neuber's equation [67] may be expressed in the following form:

$$\frac{(K_f \Delta S)^2}{4E} = \frac{\Delta \sigma}{2} \frac{\Delta \varepsilon}{2}$$

Where S is the applied stress and σ and ε are the resulting local stress and strain values corrected for the notch effect.

Since the local corrected stress and strain values are two unknown values, the input material cyclic stress strain curve is used in conjunction with Neuber's equation to determine these values as indicated in Figure 219.

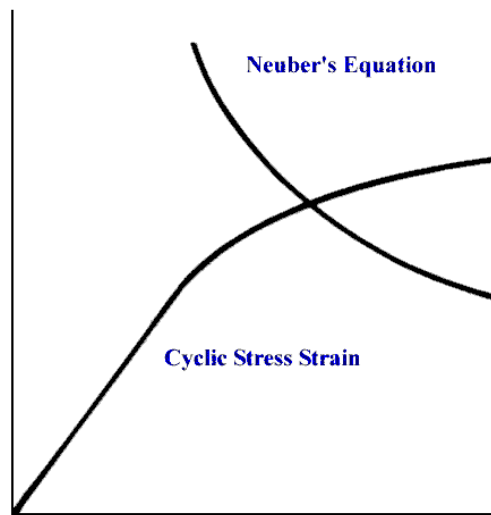


Figure 219: Neuber's Rule

3.7.1.2 Smith-Watson-Topper Equivalent Strain

Normally, strain-life data are available for the case of fully reversed loading ($R = -1.0$). To account for the effect of load cycles that are not fully reversed, an equivalent applied strain must be determined for each cycle in the applied spectrum. The Smith, Watson, and Topper equivalent strain equation [68] is probably the most common method used to convert the strain amplitude for a given load cycle to the equivalent fully reversed strain amplitude. The equation may be expressed in the following form for equivalent strain amplitude:

$$\left[\frac{\Delta \varepsilon}{2} \right]_{eq} = \sqrt{\left(\frac{S_{max}}{E} \right) \left(\frac{\Delta \varepsilon}{2} \right)}$$

Where, S_{max} is the maximum applied stress, ε is the applied strain, and E is Young's Modulus for the material

It should be noted that this form of the SWT equation for equivalent strain amplitude is not defined for $S_{max} \leq 0$. As stated in Section 3.7, AFGROW makes the conservative assumption that each cycle falls on the tension side of the hysteresis curve. For $S_{max} \leq 0$, AFGROW also assumes the loading is fully reversed with a mean stress of zero and a local strain amplitude determined from the applied local stress amplitude. This avoids any problem with the SWT equation shown above and yields a conservative result.

3.7.1.3 Fatigue Notch Factor

For a given notched specimen geometry, the effect of the notch on the fatigue life is not simply a matter of determining the local stress from the stress concentration factor (K_t) and applying the strain-life data. There is an effect of the notch for the given material and notch radius. This effect is commonly known as the fatigue notch sensitivity (q).

The Fatigue Notch Factor, (K_f), is essentially the K_t value corrected to account for the notch sensitivity for the given material [69]. It is determined as follows:

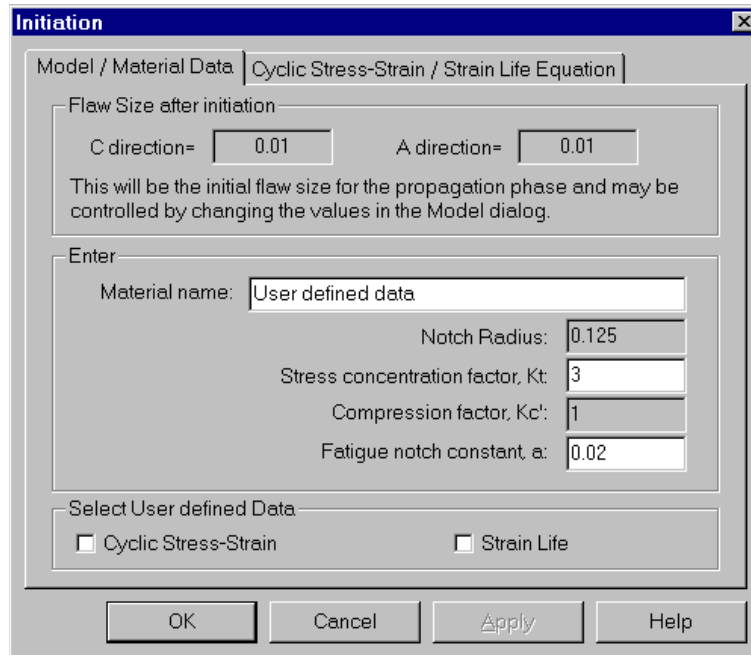
$$K_f = 1.0 + \left(\frac{K_t - 1.0}{1.0 + \left(\frac{a}{r} \right)} \right)$$

Where, a is an empirically determined material constant⁶⁹, and r is the notch root radius

⁶⁹ Values of $[a]$ for some common materials may be found in sources like "Stress Concentration Factors," by R.E. Peterson [66]

3.7.2 Initiation Parameters

When the initiation parameters menu item is selected, the dialog shown in Figure 220 appears.



The dialog box is titled "Initiation" and has two tabs: "Model / Material Data" and "Cyclic Stress-Strain / Strain Life Equation". The "Cyclic Stress-Strain / Strain Life Equation" tab is currently selected. It contains the following fields and controls:

- Flaw Size after initiation:** A section with two input fields: "C direction=" (value: 0.01) and "A direction=" (value: 0.01). Below these is a text note: "This will be the initial flaw size for the propagation phase and may be controlled by changing the values in the Model dialog."
- Enter:** A section with a "Material name:" label and a text input field containing "User defined data".
- Notch Radius:** An input field with the value 0.125.
- Stress concentration factor, Kt:** An input field with the value 3.
- Compression factor, Kc:** An input field with the value 1.
- Fatigue notch constant, a:** An input field with the value 0.02.
- Select User defined Data:** A section with two checkboxes: "Cyclic Stress-Strain" (unchecked) and "Strain Life" (unchecked).

At the bottom of the dialog are four buttons: "OK", "Cancel", "Apply", and "Help".

Figure 220: Initiation Parameters Dialog

The parameter dialog is divided into two categories:

- Model/Material Data
- Cyclic Stress-Strain / Strain-Life Equation

AFGROW also includes an option to enter tabular stress-strain or strain-life data. These data are described in more detail in the following sections.

3.7.2.1 Model/Material Data

The model/material data dialog is shown in Figure 220.

Notch Radius (r): Physical radius of local notch (or hole) which is causing a local stress concentration.

Stress Concentration Factor (Kt): Stress concentration factor $\sigma_{\text{local}}/\sigma_{\text{ref}}$.

Compression Factor (Kc): Determines the amount of the applied compressive stress (fraction of applied tension) to be used in the initiation analysis - not currently active.

Fatigue Notch Constant (a): Material constant used to determine the Fatigue Notch Factor, K_f .

3.7.2.2 Cyclic Stress-Strain / Strain-Life Equation

Initiation

Model / Material Data | Cyclic Stress-Strain / Strain Life Equation

Enter

Cyclic Stress-Strain Equation: $\frac{\Delta \epsilon}{2} = \frac{\Delta \sigma}{2E} + \left(\frac{\Delta \sigma}{2K'}\right)^{1/n'}$

Strain Life Equation: $\frac{\Delta \epsilon}{2} = \left(\frac{SIGF'}{E}\right) (2N_f)^b + EPSF' (2N_f)^c$

Cyclic strength coefficient, K' : 74

Cyclic strain hardening exponent, n' : 0.032

Fatigue strength coefficient, $SIGF'$: 116

Fatigue strength exponent, b : -0.098

Fatigue ductility coefficient, $EPSF'$: 0.26

Fatigue ductility exponent, c : -0.73

OK Cancel Apply Help

Figure 221: Cyclic Stress-Strain / Strain-Life Equation Dialog

Engineers have used the stress-strain and strain-life equations shown in Figure 221 for decades to estimate the fatigue initiation lives. The equations are curve fits to actual fatigue test data. The parameters for various materials are available in the open literature from several sources such as the ASM Handbook® [70]. The parameters are defined below:

Cyclic Strength Coefficient (K'): Stress Value at $\Delta \epsilon_p/2 = 1$ on a log plot of $\Delta \sigma/2$ vs. $\Delta \epsilon_p/2$

Cyclic Strain Hardening Exponent (n'): Slope of the log ($\Delta \sigma/2$) vs. log ($\Delta \epsilon_p/2$)

Fatigue Strength Coefficient ($SIGF'$): Stress Value at $2N_f = 1$ on a log plot of $\Delta \sigma/2$ vs. $2N_f$

Fatigue Strength Exponent (b): Slope of log ($\Delta \epsilon_e/2$) vs. log ($2N_f$)

Fatigue Ductility Coefficient ($EPSF'$): Plastic Strain Value at $2N_f = 1$ on a log plot of $\Delta \epsilon_p/2$ vs. $2N_f$

Fatigue Ductility Exponent (c): Slope of log ($\Delta \epsilon_p/2$) vs. log ($2N_f$)

Note: The subscripts e and p denote elastic and plastic values, respectively. The value $2N_f$ refers to cyclic reversals to failure (1 cycle = 2 reversals).

AFGROW includes a limited amount of strain-life data for a few common materials. These data are available by clicking on the “home” button on the initiation dialog as indicated in Figure 222.

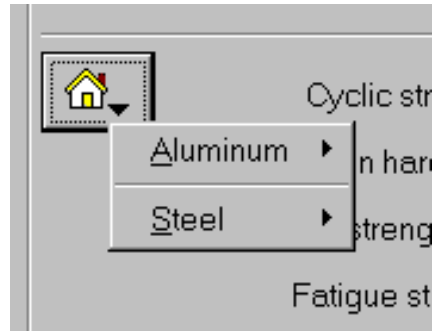


Figure 222: Using Default Initiation Parameters for Common Materials

These data are provided for users who may not have access to their own initiation data and want to use some generic aluminum or steel data.

3.7.3 User-Defined Cyclic Stress-Strain / Strain-Life Data

A user may choose to enter the Cyclic Stress-Strain data and/or the Strain-Life data in tabular form. The choice is controlled by the following check box controls in the initiation parameter dialog shown in Figure 223.

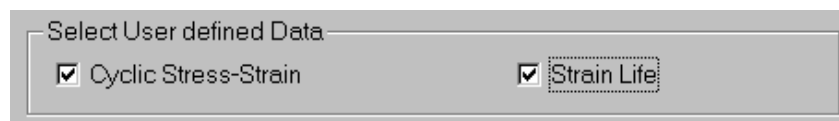


Figure 223: Options for User Defined Initiation Data

A new tab will appear for each box that is selected as shown in Figure 224.

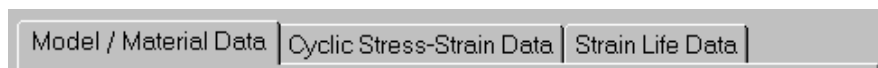


Figure 224: Options for Stress-Strain and Strain-Life Input Data

The data must be entered in tabular format by selecting the appropriate tab. These data may be pasted from Excel or entered in the grid control by hand.

3.7.3.1 Cyclic Stress-Strain Data

The cyclic stress-strain curve is determined by connecting the positive stress tips of the stable hysteresis curve for a given material. The initial slope of the curve should be equal to Young's modulus.

Enter up to 25 values: Number of Sets: 20

Set	Strain	Stress
0	0	0
1	0.0006	6.48
2	0.00123	12.96
3	0.00185	19.44
4	0.0025	25.92
5	0.0031	32.4
6	0.0037	38.88
7	0.0043	45.36

Figure 225: User-Defined Cyclic Stress-Strain Data

Cyclic stress-strain data are obtained from fully reversed cyclic load tests. These tests are conducted at several load levels where stress vs. strain data are obtained and monitored until the hysteresis curve (map of stress vs. strain for each cycle) becomes stabilized. The cyclic stress-strain curve is the locus of the tips of the stable hysteresis curves in the positive stress and strain quadrant of the plot (see Figure 226).

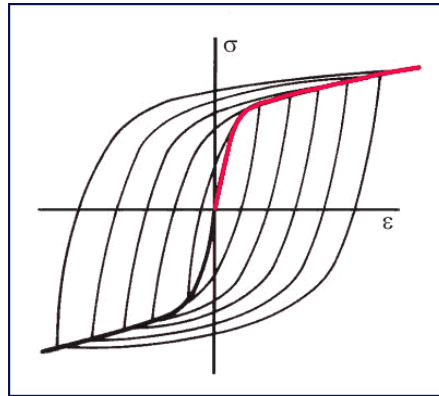


Figure 226: Stable Hysteresis Curves

The first point is defined at zero stress and zero strain. There is a linear range of stress vs. strain whose slope is equal to the Young's modulus of the material (by definition). AFGROW uses linear interpolation and extrapolation to determine the values between input points and beyond the last input point. This is the reason for requesting data for the linear range in addition to data that describes the non-linear behavior. It is a good idea to look at a plot of the initiation data in the main frame view (see Figure 23 in Section 2.1.1.5). This option will permit the input data, the current Young's modulus, and any desired test

data to be overlaid on the same plot. As may be imagined, the resulting crack initiation life is sensitive to the degree to which the input data match the actual test data.

3.7.3.2 Strain-Life Data

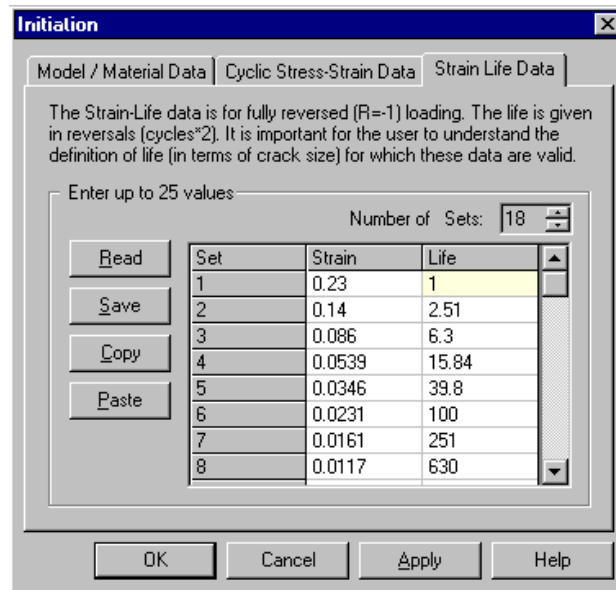


Figure 227: User-Defined Strain-Life Data

The strain-life data are in terms of reversals instead of cycles. A reversal refers to a change in the loading direction during cyclic loading. A complete cycle consists of two load reversals. The first input point must be the strain to initiation (or perhaps failure) for one reversal (monotonic loading). AFGROW uses logarithmic interpolation and extrapolation to determine the values between input points and beyond the last input point. The reason for this is to avoid any case where a negative strain value could result from an interpolation or extrapolation. It was also determined that logarithmic interpolation results in most accurate results. The resulting crack initiation life tends to be VERY sensitive to the degree in which the input data matches the actual test data. It is a good idea to look at a plot of the initiation data in the main frame view (see Figure 23 in Section 2.1.1.5). This option will overlay the input data and any desired test data. As noted in the dialog, **IT IS VERY IMPORTANT** for the user to know the definition of life⁷⁰ for the input strain-life data. This definition should be used in the initial crack length, which is input by the user for subsequent crack growth analysis. AFGROW will determine an initiation life from the input data and proceed with a crack growth analysis from the initial crack length(s) entered for the given problem.

⁷⁰ Cycles to obtain a crack length assumed as the definition of crack initiation (normally 0.01 inch)

3.7.4 Miners Rule and Spectrum Handling

The crack initiation module uses Miners rule to determine life based on the fully reversed strain-life information provided in either equation or tabular form.

The crack initiation module also uses a binning process to divide the user-defined spectrum into 820 bins of strain amplitude, maximum stress values. Crack initiation life is calculated for each of these bins and Miners rule is applied to determine the initiation life. This process introduces some error in the crack initiation calculations since the applied cycles are discretized into bins but allows very large spectra to be managed more efficiently.

The crack initiation module reads the input spectrum, finds the overall maximum and minimum values (including the SMF and local stress concentration effect, K_f), and defines a fully reversed maximum operating stress range based on the maximum absolute value of the overall maximum or minimum local spectrum stress. For example, if the overall local maximum and minimum stress values for a given input spectrum are 30 and -15, then the module defines a maximum operating stress range of ± 30 . The resulting operating stress range is divided in 40 stress steps from -30 to +30 Ksi. In the given example, this would be in stress steps of 1.5 $((30 \times 2)/40)$. To account for all possible combinations of local strain amplitude and maximum local stress, the 40 stress steps are divided into 820 strain amplitude and maximum stress bins in the following manner. The strain amplitude corresponding to each stress step is determined from the cyclic stress-strain curve. The first strain amplitude is paired with all 40 max stress steps, the second strain amplitude is paired with the last 39 maximum stress steps, and this continues until the final strain amplitude is paired with the final maximum stress. This results in 820 combinations of strain amplitude and maximum stress bins $(40 + 39 + 38 + \dots + 1 = 820)$. AFGROW calculates the crack initiation life for every bin. Each cycle in the input spectrum is placed in the corresponding bin so that a cycle count for each bin may be used to determine the total crack initiation life for the input spectrum.

3.7.5 Initiation/No Initiation

This option simply activates/deactivates the initiation analysis, so a user may perform a total life analysis for the same case with or without the crack initiation. If the initiation option is active, this menu item is shown as “No Initiation” and will deactivate the initiation analysis if selected. If the initiation option is not active, this menu item is shown as “Initiation” and will activate the initiation analysis. This may be useful when comparing results with and without including the time to crack initiation.

3.8 Window Menu

The three frames, discussed in detail in Sections 2.1.1, 2.1.2, and 2.1.3, make up the views for the life prediction analysis. To allow the largest view of the spectrum when the view, spectrum plot option is selected (see Figure 184, Section 3.3.8), the entire AFGROW window is used to display the spectrum. The window menu is used to control the display of the spectrum and the three AFGROW frames.

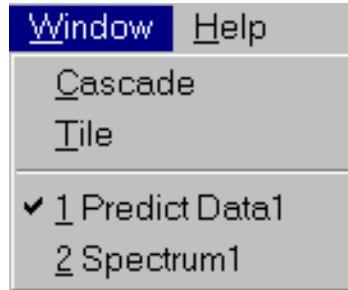


Figure 228: Window Menu

3.8.1 Window Cascade

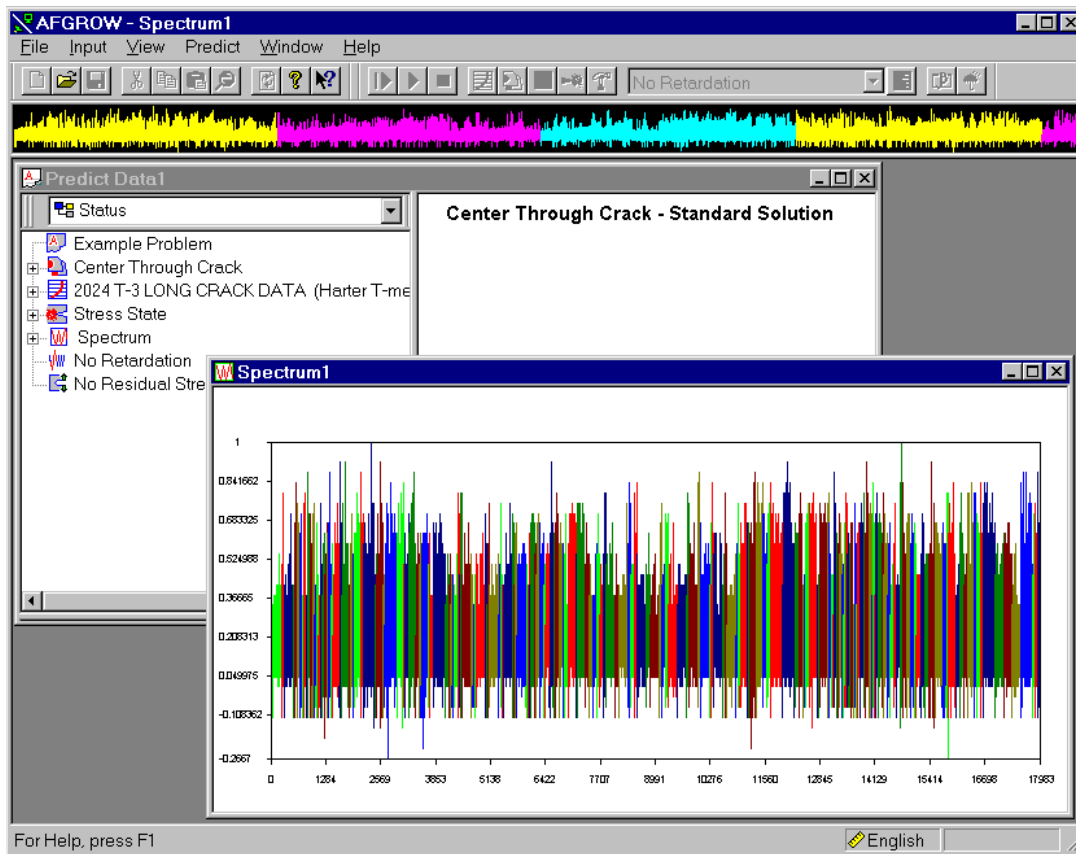


Figure 229: Cascade Window View

The cascade option shows both views overlapping each other. When the spectrum view is active (blue title bar), the menu options related to the prediction data are either grayed out or removed. Activating the prediction view returns the menu to normal. Users can switch between prediction data and spectrum views by clicking on the appropriate title bar or selecting the desired view in the Window menu (Figure 228).

Both views will be reduced in size to fit within the AFGROW window. The three frames of the prediction view will be automatically reduced in size. The output frame may not be visible. The frames can be resized by dragging the frame boundaries with the mouse as desired. The views can also be minimized, restored, or maximized using the standard Windows tools in the upper right hand corner of either view.

3.8.2 Window Tile

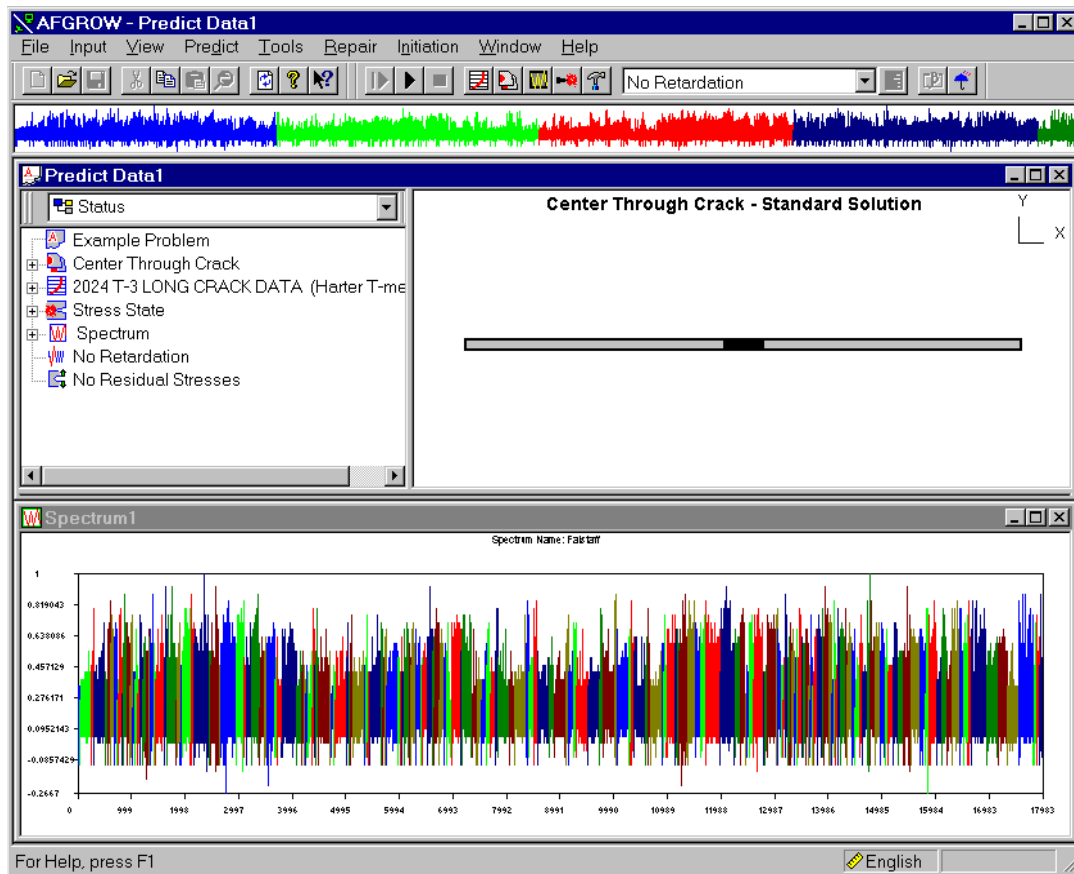


Figure 230: Tile Window View

The tile option shows both views above and below each other. When the spectrum view is active (blue title bar), the menu options related to the prediction data are either grayed out or removed. Activating the prediction view returns the menu to normal. Users can switch between prediction data and spectrum views by clicking on the appropriate title bar or selecting the desired view in the Window menu (Figure 228).

Both views will be reduced in size to fit within the AFGROW window. The three frames of the prediction view will be automatically reduced in size. The output frame may not be visible. The frames can be resized by dragging the frame boundaries with the mouse as desired. The views can also be minimized, restored, or maximized using the standard Windows tools in the upper right hand corner of either view.

3.9 Help Menu

As with most windows programs, AFGROW includes a help menu, which includes extensive on-line help (Help Topics) and version information (About AFGROW).

3.9.1 Help Topics

This action allows you to access the on-line help that is available for the VISTA/Windows 7/Windows 8, and Windows 10 version of AFGROW. The Help Topics dialog is shown in Figure 231.

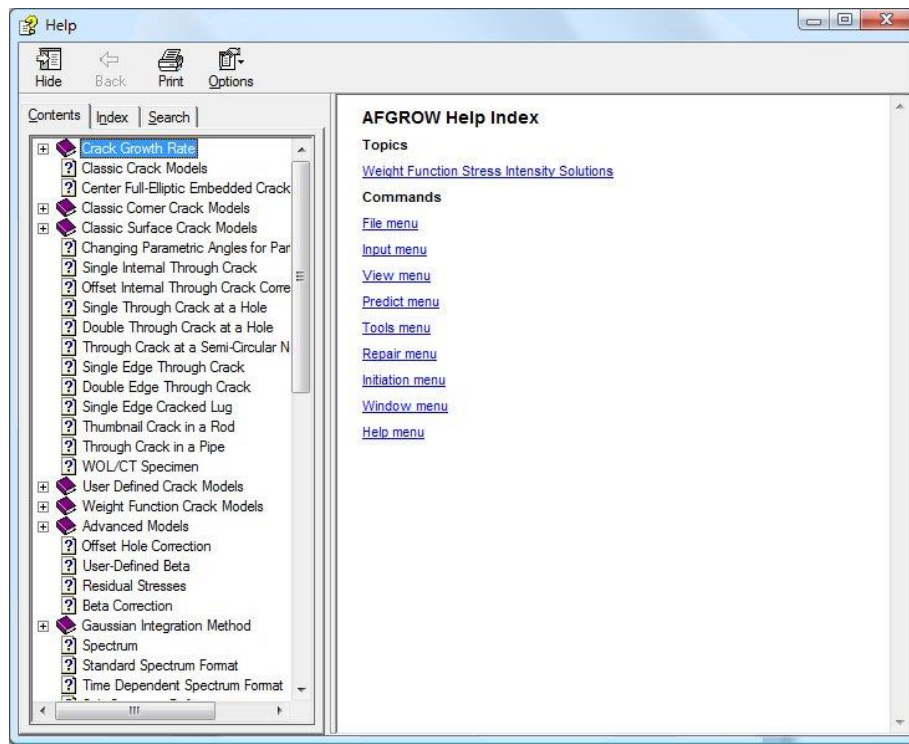


Figure 231: AFGROW Help Topics

This help is the standard Windows help where users can select a topic, view the index, or search for a keyword. Help is available directly from the keyboard for any open dialog by using the F1 function key. There is also a question mark tool in the AFGROW standard toolbar (see Figure 181, Section 3.3.1.2) that may be used to select help for any item in the menus or any other toolbar shortcut. You just click on the question mark (the cursor becomes a question mark) and click again on the item of interest in the AFGROW main window.

3.9.2 About AFGROW

This action allows users to view information about the version of AFGROW being used. The About AFGROW dialog is shown in Figure 232.

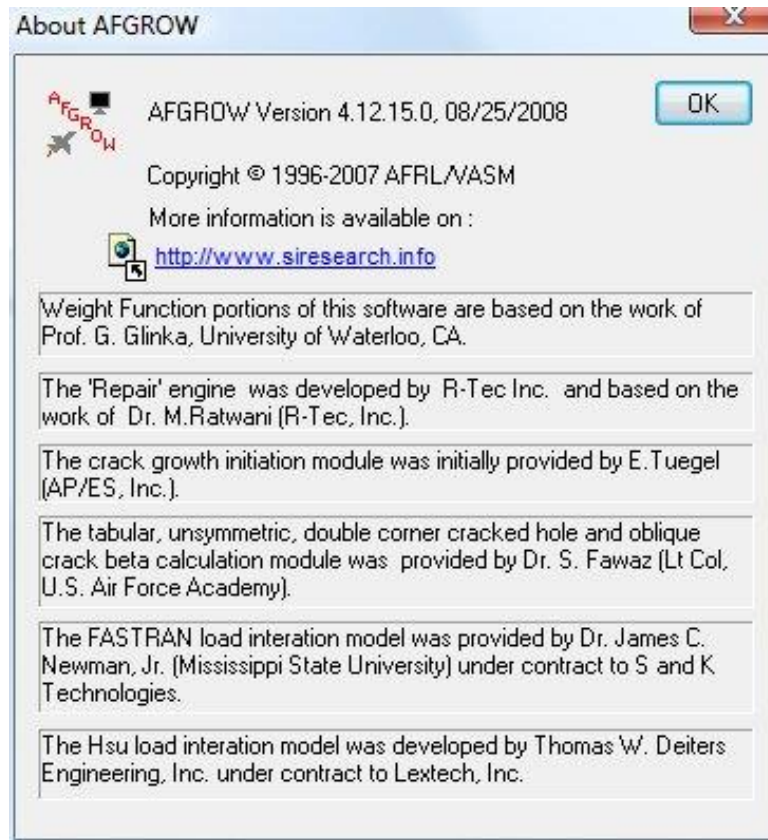


Figure 232: Help about AFGROW

4.0 ENGLISH AND METRIC UNITS

AFGROW uses either English⁷¹ or Metric⁷² units of measurement.

The units used in AFGROW are controlled by the choice of the units displayed on the status bar (see Figure 29, Section 2.1.6). Users may switch between English and Metric units by clicking on the small ruler on the status bar and selecting the units of choice as shown in Figure 233.

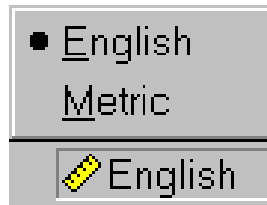


Figure 233: Switching Between English and Metric Units

The current system of units may be changed by clicking (right or left) on the units icon on the status bar and selecting the units of choice. The units may be changed at any time and all input parameters will be converted accordingly. AFGROW uses ASTM Standard Metric Practices [71] for all internal conversions. Care has been taken to prevent loss of data precision after multiple conversions. However, some rounding may be experienced for some small numbers relative to standard values. However, Values that are known to be small (i.e., Paris C, Coefficient of Thermal Expansion, etc.) are handled correctly internally.

For example:

If users prefer to work in Metric units and certain data are available in English units, users can switch to English units, enter these data, and switch back to Metric units (or vice versa).

AFGROW users MUST remember to be consistent in the use of units within the English or Metric Systems. The AFGROW output data will be consistent with the units selected by the user.

Some users may ask why the metric units of length are meters. The reason is consistency. It was felt that since the standard metric units for stress intensity is $MPa\sqrt{m}$, the length units should be in meters. This consistency is important in AFGROW for internal calculations.

⁷¹ Length – inches, Force – Kpounds, Stress – ksi, Temperature – Degrees Fahrenheit

⁷² Length – meters, Force – MNewtons, Stress- MPa, Temperature – Degrees Centigrade

5.0 COMPONENT OBJECT MODEL SERVER

AFGROW for Windows[®] operates in two different modes; first, as a normal interactive Windows program, and second, as a Component Object Model (COM) Server [9]. The COM server technology is an outgrowth of the Object Linking and Embedding technology used by Microsoft for many years. A COM Server may be called from other Windows software and the results from the server can be sent back to the calling program. In the case of AFGROW, users can write Windows programs or macros to generate input data, and call AFGROW to perform structural life analyses. AFGROW can perform the life analyses and return the results directly to the calling software. The most commonly used application of this capability is seen in the following example, Figure 234, using Microsoft Excel[®].

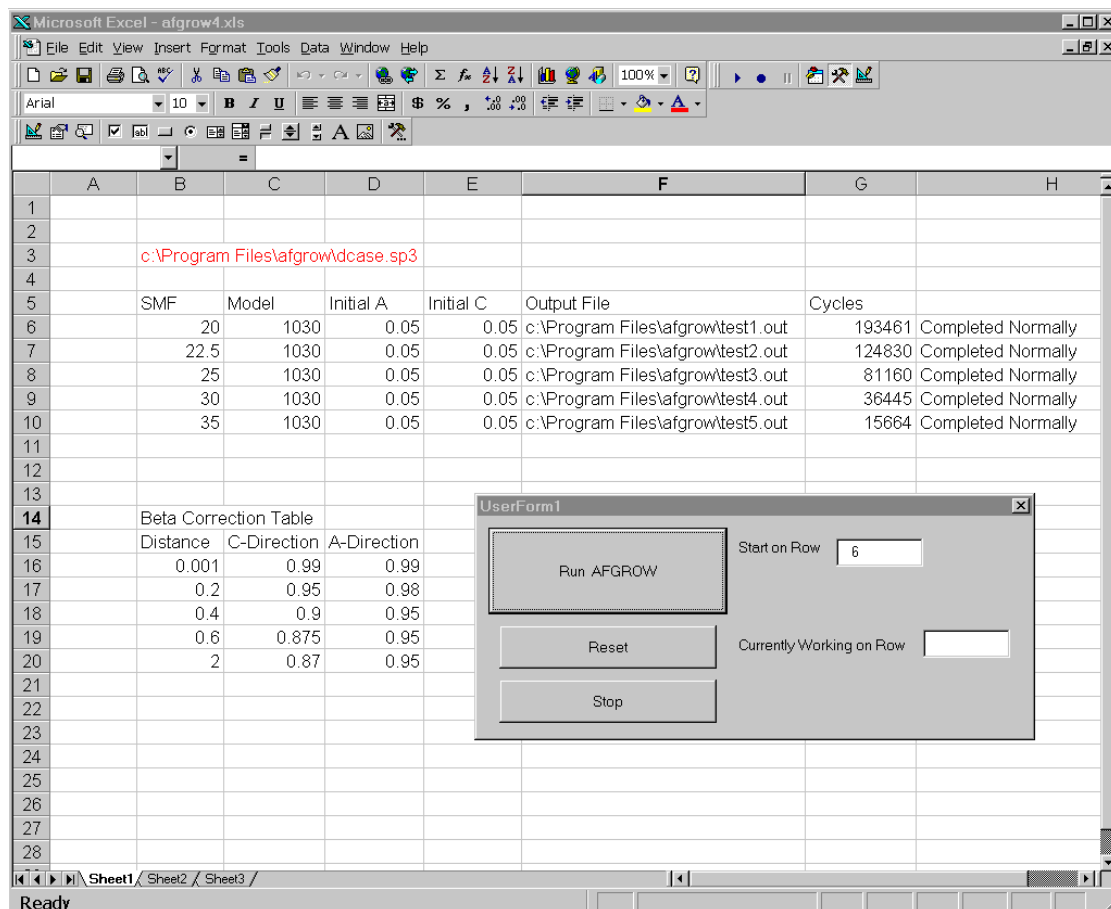


Figure 234: Microsoft Excel Macro Using AFGROW

The above example is a fairly simple application of the COM capabilities in AFGROW, but is intended to show how the technology may be used to perform multiple life analyses. Other uses of this capability can extend as far as a user's imagination can carry it in terms of application to structural life prediction. This capability has already been used to estimate the crack growth life of specimens subjected to a corrosive environment [72].

The latest extensive manual on the use of AFGROW as a COM server is available on the AFGROW Web Site, [73] and the manual for the last Air Force version of AFGROW (version 4.12.15) is also available as an Air Force Technical report [74]. An excerpt from the manual is given below:

General Instructions

Before using the server version from another windows program, AFGROW MUST be run at least once (with administrator privileges⁷³) as a stand-alone program. When the server version is executed for the first time, Windows will recognize that it is a COM server and will look for a Type Library Binary (TLB) file (*afgrow.tlb*) and register AFGROW as a COM object on the local machine⁷⁴. Once this is complete, the AFGROW server will be available for use by other COM compatible software.

The TLB file contains detailed information that other programs use to determine which variables and sub-routines are available in AFGROW. Whenever the AFGROW server is updated and a new version is downloaded, all references to the previous server version MUST be updated.

Documentation and examples of several COM applications are available on the AFGROW web site (<http://afgrow.net/downloads/ddownload.asp>). Simply download the Component Object Model manual. The manual and several excel spreadsheet examples are contained in a zipped file.

⁷³ Please note that AFGROW should not be set to always run as administrator when using the COM interface. When running as administrator, a dialog box will appear and wait for manual confirmation that the code will be run in administrator mode. This dialog will not appear when the COM client code is executed, and AFGROW will not run. There should be no reason to run AFGROW as administrator once the TLB library is registered.

⁷⁴ Many Government and Corporate users will need to contact IT support to install and run AFGROW for the first time.

6.0 TUTORIALS

This section will take users through a few sample problems to show how to use many of the features described in previous sections of this manual.

6.1 Corner Cracked Offset Hole with Residual Stress

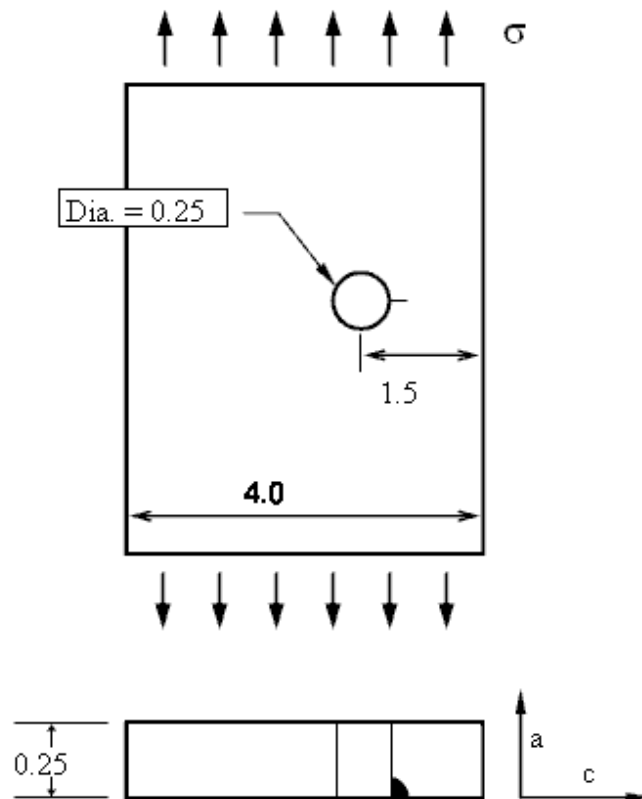


Figure 235: Corner Cracked Hole Problem Geometry

Specimen Geometry:	Corner Crack at an Offset Hole in a Plate
Dimensions:	$W = 4.0$ in., $T = 0.25$ in., $\text{Dia.} = 0.25$ in.
Hole Offset:	$B = 1.5$ in.
Initial Crack Size:	$c = 0.05$ in., $a = 0.05$ in.
Material:	7050-T74 Plate (from matfile.da3 – Harter T-Method)
Stress Spectrum:	16 ksi to 0 ksi 1 Cycle 12 ksi to 8 ksi 1000 Cycles
Retardation Model:	Generalized Willenborg Model, $\text{SOLR} = 2.8$
Stress State:	Automatic
Beta Correction:	None
Environment:	N/A

Residual Stresses:

r	Residual Stress (r,0)	Residual Stress (0,r)
0	-2.4	-2.4
0.025	-1.75	-2.4
0.05	-1.125	-2.4
0.1	0	-2.4
0.15	0.2	-2.4
0.2	0.25	-2.4
0.249	0.23	-2.4
0.25	0.23	0
0.5	0.175	0
0.75	0.15	0
1	0.135	0
1.25	0.125	0
1.374	0.115	0
1.375	0.115	0

Note: The table includes two points beyond the largest crack length expected in both directions as discussed in Section 3.2.10.1.1.

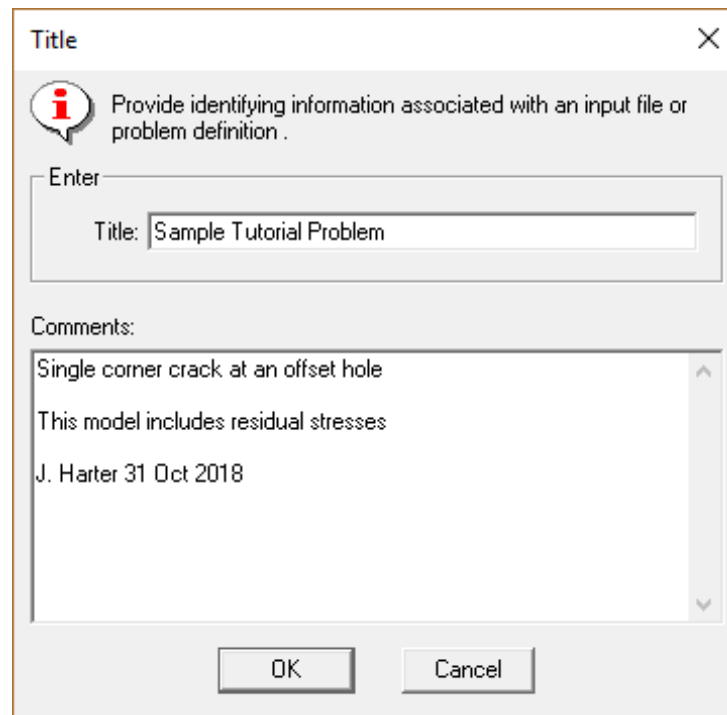
Predict Preferences:

Use defaults except set the growth increment to cycle-by-cycle beta and spectrum calculations, and the print interval to 0.05 inches.

6.1.1 Entering Data

The AFGROW interactive interface is written so that the user may enter data in any order. The philosophy is that the user should control the software; the software shouldn't control the user. The only exception to this general philosophy occurs in the case of the bonded repair analysis option. In the bonded repair case, the effect of the repair is dependent on the applied stress level, specimen dimensions, and material properties. The order in which the data are entered in the following section is simply the preference of the author.

6.1.1.1 Input Title



Title [X]

 Provide identifying information associated with an input file or problem definition .

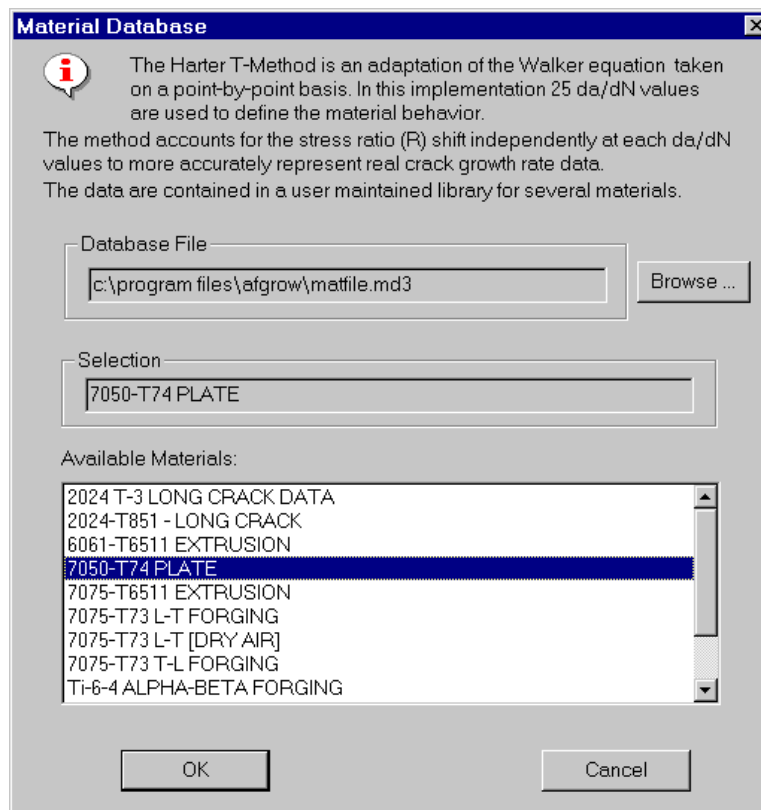
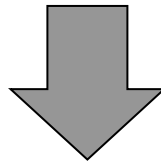
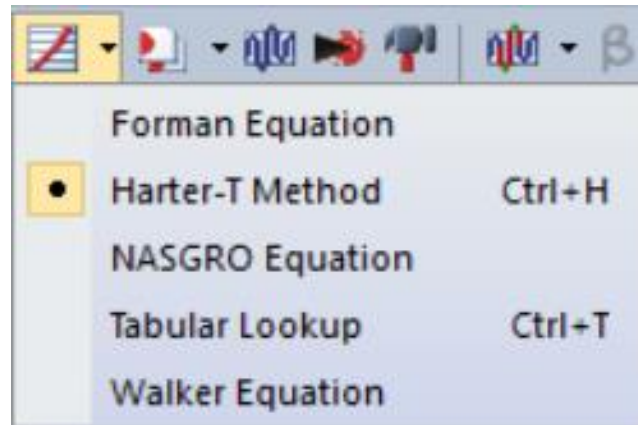
Enter

Title:

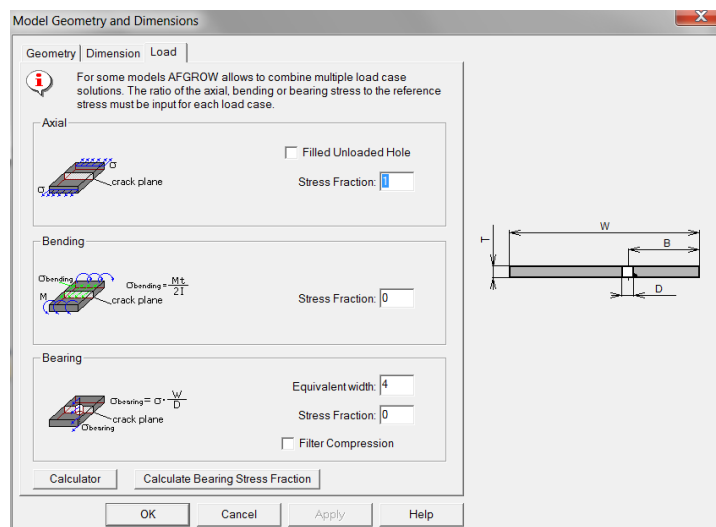
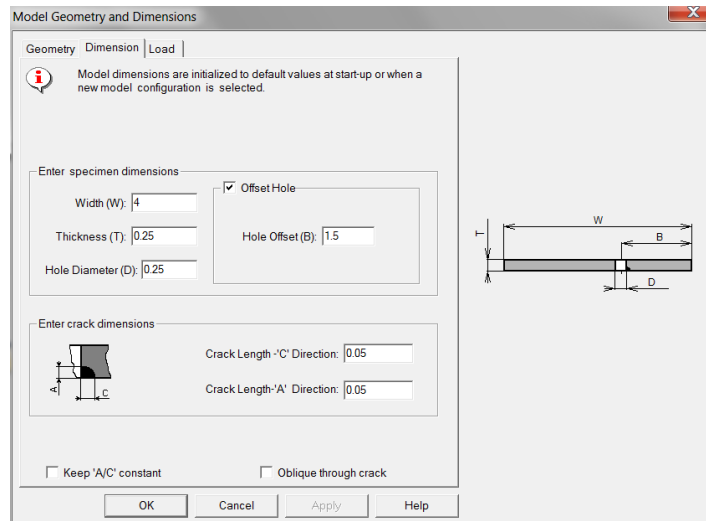
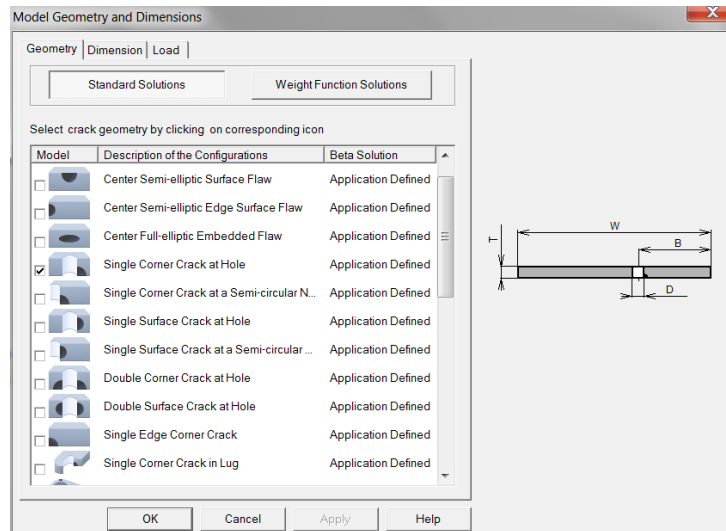
Comments:

This model includes residual stresses
J. Harter 31 Oct 2018

6.1.1.2 Input Material



6.1.1.3 Input Model (Classic Models)



6.1.1.4 Input Spectrum

Spectrum

 Stress Multiplication Factor [SMF] multiplies the stress or load levels found in spectrum files. This allows normalized spectra to be used. If actual stress levels are used in the spectrum files, SMF should be set to 1.

Residual Strength Requirement [Pxx] is the value of stress (or load for models using load input) which must be carried at all crack sizes. It is used to determine the critical crack size - if a non-zero value is entered.

Stress Preload [SPL] is used to account for pre-existing stresses. This value is added to the max and min spectrum stresses after they have been multiplied by SMF.

Enter

Stress Multiplication Factor(SMF):

Residual Strength Requirement (Pxx):

Stress Preload (SPL):

Select

 ☒ Create new spectrum file

 ☐ Open spectrum file

 ☐ Constant amplitude loading

OK Cancel

Basic Spectrum Information

 The Spectrum Wizard creates the SPECTRUM INFORMATION FILE that defines the basic spectrum information for a given crack growth analysis.

Spectra may be broken into several sub-spectra and sub-spectra may be grouped and divided among a number of sub-spectrum files.

Enter

Base Filename (no extension):

Spectrum Title:

Label for Sub-spectrum:

Number of files:

< Back Next > Cancel Help

Type of Spectrum

 A cycle by cycle spectrum is one in which each max-min stress (or load) is applied for one (1) cycle. This may be thought of as a real time spectrum. In a blocked spectrum, each max-min pair is applied for a number of cycles.

Blocked spectra are often used to save computer time and file storage space.

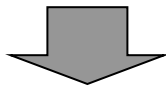
In many cases, the use of the blocked format produces very little difference in terms of the predicted life. In general, the more times a spectrum is repeated, the less the difference in using blocked or cycle-by-cycle spectra.

Select Type of Spectrum

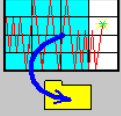
☒ Blocked Cycles

☐ Cycle by Cycle

< Back Next > Cancel Help



Number of Sub-spectra



There are limits, in manual mode, placed on the number of sub-spectra that can make up a spectrum. The limits are 10 sub-spectra per file.

Enter Number of Sub-spectra in file #1

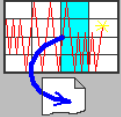
Number of Sub-spectra :

☐ Import from file





Number of Stress levels



There are limits, in manual mode, placed on the number of stress levels that can make up a spectrum. The limits are 25 levels per spectrum.

File #1 Sub-spectrum #1

Enter Number of stress levels

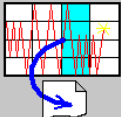
Number of Stress levels :

☐ Import single subspectrum from file





Stress levels

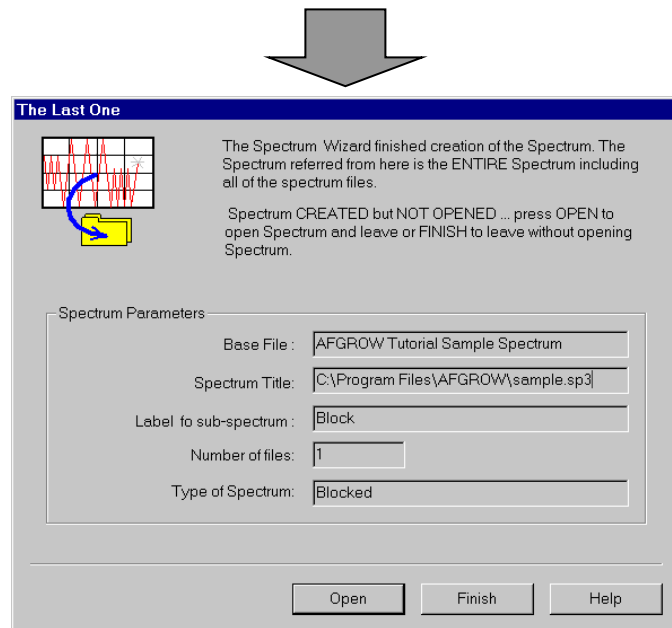


File #1 Sub-spectrum #1

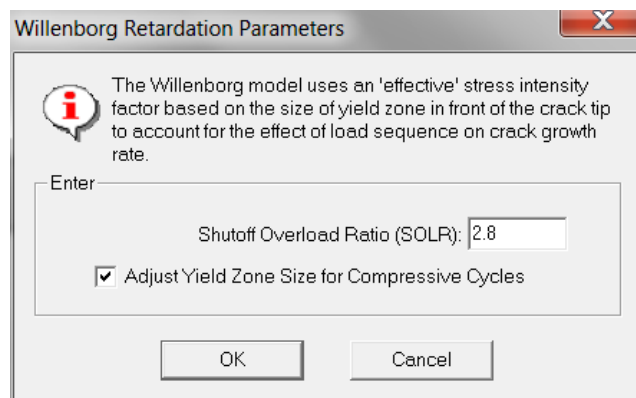
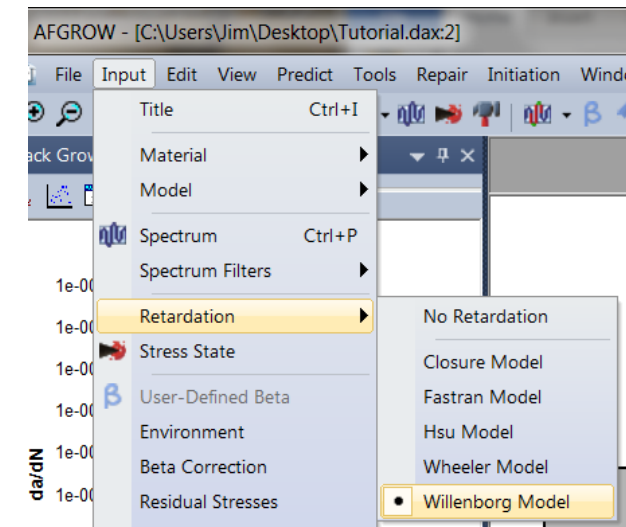
Input Spectrum Stress /Load Levels information in the following format.

Enter Stress Levels

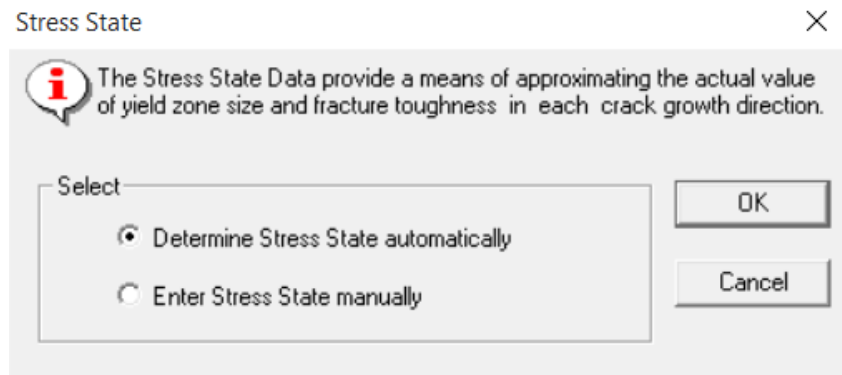
Set	Max	Min	Cycles
1	16.000000	0.000000	1
2	12.000000	8.000000	1000



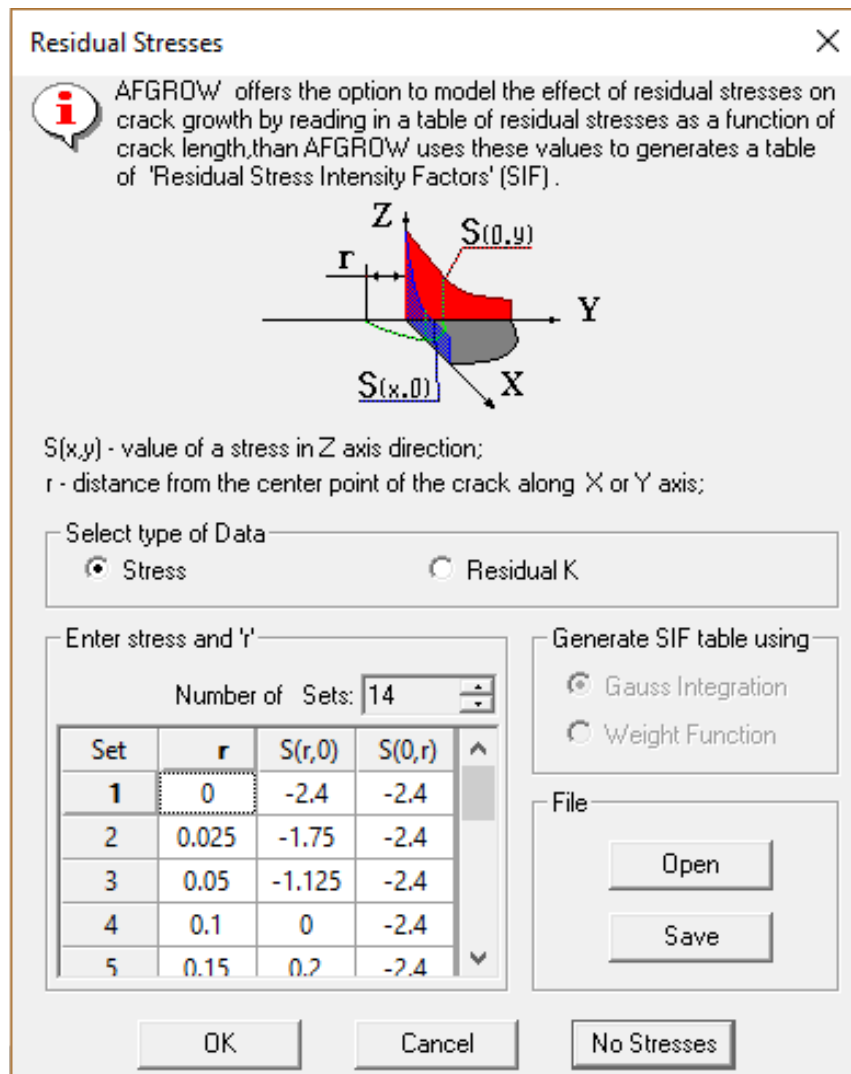
6.1.1.5 Input Retardation



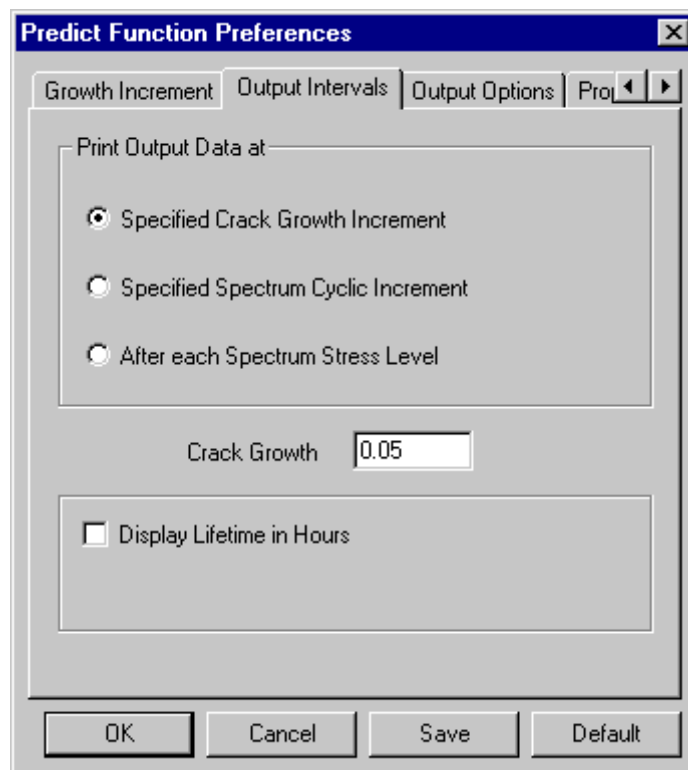
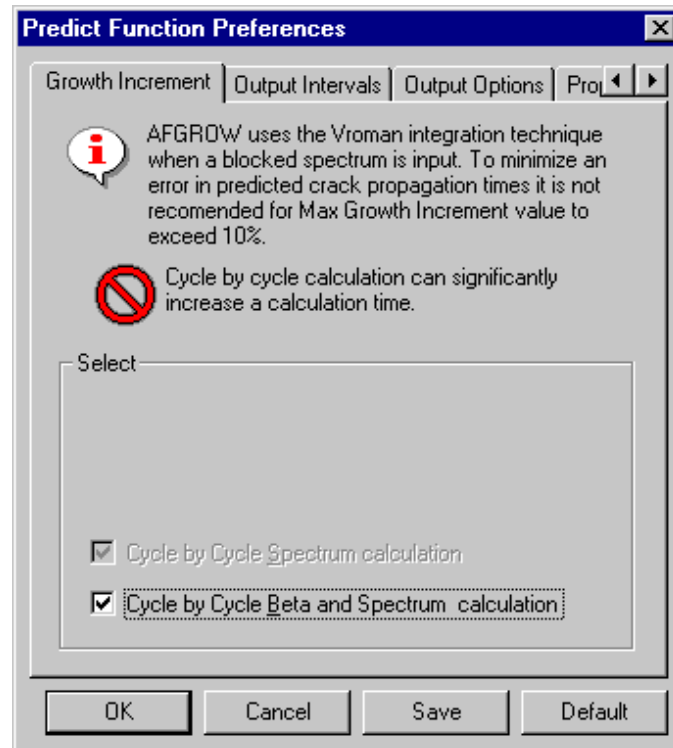
6.1.1.6 Stress State



6.1.1.7 Residual Stresses



6.1.1.8 Predict Preferences



6.1.2 AFGROW Output

The results of the AFGROW analysis are given below.

#####

Single corner crack at an offset hole

This model includes residual stresses

J. Harter 31 Oct 2018

#####

AFGROW 5.3.3.23 10/31/2018 12: 38 **English Units [Length(in), Stress(Ksi),
Temperature(F), Force(Kip)]

Crack Growth Model and Spectrum Information

Title: Sample Tutorial Problem

Load: Axial Stress Fraction: 1, Bending Stress Fraction: 0, Bearing Stress Fraction: 0

Crack Model: 1030 - Single Corner Crack at Hole - Standard Solution

Parametric Angle for the Newman and Raju Solution: C-Direction = 5.00, A-Direction = 80.00

Initial crack depth (A): 0.0500

Initial surface crack length (C): 0.0500

Thickness : 0.250

Width : 4.000

Hole Diameter : 0.250

Hole Offset : 1.500

Young's Modulus =10400 , Poisson's Ratio =0.33 , Coeff. of Thermal Expan. =1.34e-005

Retardation: WILLENBORG

Shut-off ratio : 2.800

Adjust Yield Zone Size for Compressive Cycles = Yes

Determine Stress State automatically (2 = Plane stress, 6 = Plane strain)

The stress intensity factors are being adjusted for a 2d residual stress field as follows:

Length	Stress C	Residual K C	Stress A	Residual K A
0.0000000	-2.40000e+000	-4.53752e-002	-2.40000e+000	-4.53654e-002
0.0250000	-1.75000e+000	-3.69510e-001	-2.40000e+000	-4.41523e-001
0.0500000	-1.12500e+000	-4.19094e-001	-2.40000e+000	-6.11363e-001
0.1000000	0.00000e+000	-2.21060e-001	-2.40000e+000	-7.92839e-001
0.1500000	2.00000e-001	-1.63537e-001	-2.40000e+000	-9.45649e-001
0.2000000	2.50000e-001	2.86416e+000	-2.40000e+000	1.69851e+001
0.2490000	2.30000e-001	1.13313e+000	-2.40000e+000	-6.85607e+000
0.2500000	2.30000e-001	1.11552e+000	0.00000e+000	-6.37854e+000
0.5000000	1.75000e-001	2.34175e-001	0.00000e+000	-8.02175e-002
0.7500000	1.50000e-001	1.81629e-001	0.00000e+000	2.63229e-002
1.0000000	1.35000e-001	1.67749e-001	0.00000e+000	4.01691e-002
1.2500000	1.25000e-001	1.69568e-001	0.00000e+000	4.30248e-002
1.3740000	1.15000e-001	1.64807e-001	0.00000e+000	4.33644e-002
1.3750000	1.15000e-001	1.64767e-001	0.00000e+000	4.33261e-002

The stress intensity factors are being adjusted for a 1d residual stress field as follows:

Length	Stress C	Residual K C
0.000000	-2.40000e+000	-7.06346e-002
0.0250000	-1.75000e+000	-5.46591e-001
0.0500000	-1.12500e+000	-5.80857e-001
0.1000000	0.00000e+000	-1.29291e-001
0.1500000	2.00000e-001	4.33442e-002
0.2000000	2.50000e-001	1.49622e-001
0.2490000	2.30000e-001	1.78044e-001
0.2500000	2.30000e-001	1.78695e-001
0.5000000	1.75000e-001	2.34603e-001
0.7500000	1.50000e-001	2.55180e-001
1.0000000	1.35000e-001	2.63670e-001
1.2500000	1.25000e-001	2.82598e-001
1.3740000	1.15000e-001	2.78449e-001
1.3750000	1.15000e-001	2.78505e-001

No K-Solution Filters

Harter T-Method crack growth rate approach is being used

For $R_{eff} < 0.0$, K_{max} is used in place of ΔK

Material: 7050-T74 PLATE

Lower 'R' value boundary: -0.33

Upper 'R' value boundary: 0.8

Plane strain fracture toughness: 33

Yield stress: 65

Failure is based on the current load in the applied spectrum

Cycle by cycle Beta and Spectrum calculation

Spectrum: Class Example Spectrum

Spectrum multiplication factor: 1.000

SPL: 0.000

The spectrum will be repeated up to 999999 times

Total Cycles: 1001

Levels: 2

Subspectra: 1

Max Value: 16

Min Value: 0

No Spectrum Filters

Critical Crack Length is Based on the Maximum Spectrum Stress

Critical crack size in 'C' direction=1.32995, Stress State=2 (Based on K_{max} criteria)

Transition will be based on K_{max} or 95% thickness penetration Criteria

C Crack size= 0.05 Beta Tension= 1.3224 Beta Compression= 1.3224 $R(k)=-0.0526$ $R(\text{final})=-0.0526$
Delta k=7.9666e+000 $D()/DN=1.9152e-006$ Residual K=-0.419

A Crack size= 0.05 Beta Tension= 1.6429 Beta Compression= 1.6429 $R(k)=-0.0623$ $R(\text{final})=-0.0623$
Delta k=9.8069e+000 $D()/DN=4.8317e-006$ Residual K=-0.6114

A/t ratio= 0.2 A/C ratio= 1

Max stress = 16.000 $r = 0.00$ 0 Cycles Subspectra: 0 Pass: 1

C Crack size= 0.079766 Beta Tension= 1.3177 Beta Compression= 1.3177 R(k)= 0.6108 R(final)= 0.6108
 Delta k=2.6386e+000 D()/DN=2.4518e-007 Residual K=-0.301
 A Crack size= 0.1 Beta Tension= 1.3251 Beta Compression= 1.3251 R(k)= 0.5877 R(final)= 0.5877
 Delta k=2.9708e+000 D()/DN=3.5440e-007 Residual K=-0.7928
 A/t ratio= 0.4 A/C ratio= 1.2537
 Max stress = 12.000 r = 0.67 178808 Cycles Subspectra: 0 Pass: 179

C Crack size= 0.1 Beta Tension= 1.2607 Beta Compression= 1.2607 R(k)= 0.6149 R(final)= 0.6149
 Delta k=2.8265e+000 D()/DN=3.3981e-007 Residual K=-0.221
 A Crack size= 0.12709 Beta Tension= 1.2329 Beta Compression= 1.2329 R(k)= 0.5829 R(final)= 0.5829
 Delta k=3.1161e+000 D()/DN=4.2486e-007 Residual K=-0.8756
 A/t ratio= 0.50834 A/C ratio= 1.2709
 Max stress = 12.000 r = 0.67 247781 Cycles Subspectra: 0 Pass: 248

C Crack size= 0.11909 Beta Tension= 1.2159 Beta Compression= 1.2159 R(k)= 0.6280 R(final)= 0.6280
 Delta k=2.9750e+000 D()/DN=4.9711e-007 Residual K=-0.199
 A Crack size= 0.15 Beta Tension= 1.1727 Beta Compression= 1.1727 R(k)= 0.5943 R(final)= 0.5943
 Delta k=3.2201e+000 D()/DN=5.6398e-007 Residual K=-0.9455
 A/t ratio= 0.6 A/C ratio= 1.2595
 Max stress = 12.000 r = 0.67 296108 Cycles Subspectra: 0 Pass: 296

C Crack size= 0.13354 Beta Tension= 1.2874 Beta Compression= 1.2874 R(k)= 0.6473 R(final)= 0.6473
 Delta k=3.3353e+000 D()/DN=1.0671e-006 Residual K=-0.182
 A Crack size= 0.2 Beta Tension= 1.0099 Beta Compression= 1.0099 R(k)= 0.8796 R(final)= 0.8000
 Delta k=3.2022e+000 D()/DN=3.2469e-006 Residual K=16.9841
 A/t ratio= 0.80001 A/C ratio= 1.4977
 Max stress = 12.000 r = 0.67 319155 Cycles Subspectra: 0 Pass: 319

C Crack size= 0.15 Beta Tension= 1.2961 Beta Compression= 1.2961 R(k)= 0.6458 R(final)= 0.6458
 Delta k=3.5590e+000 D()/DN=1.3119e-006 Residual K=-0.163
 A Crack size= 0.23509 Beta Tension= 0.9368 Beta Compression= 0.9368 R(k)= 0.6353 R(final)= 0.6353
 Delta k=3.2203e+000 D()/DN=8.3124e-007 Residual K=-0.0887
 A/t ratio= 0.94036 A/C ratio= 1.5673
 Max stress = 12.000 r = 0.67 334051 Cycles Subspectra: 0 Pass: 334

Transitioned to a thru-crack at 95% thickness penetration

C Crack size= 0.1553 Beta Tension= 1.2802 Beta Compression= 1.2802 R(k)= 0.6674 R(final)= 0.6674
 Delta k=3.5769e+000 D()/DN=1.5484e-006 Residual K= 0.157
 A Crack size= 0.2375 Beta Tension= 0.9398 Beta Compression= 0.9398 R(k)= 0.5715 R(final)= 0.5715
 Delta k=3.2470e+000 D()/DN=4.8353e-007 Residual K=-1.2608
 A/t ratio= 0.95 A/C ratio= 1.5293
 Max stress = 12.000 r = 0.67 338146 Cycles Subspectra: 0 Pass: 338

C Crack size= 0.1553 Beta Tension= 1.2245 Beta Compression= 1.2245 R(k)= 0.6554 R(final)= 0.6554
 Delta k=3.4211e+000 D()/DN=1.2285e-006 Residual K= 0.055
 Max stress = 12.000 r = 0.67 338146 Cycles Subspectra: 0 Pass: 338

C Crack size= 0.2 Beta Tension= 1.1261 Beta Compression= 1.1261 R(k)= 0.6238 R(final)= 0.6238
 Delta k=3.5706e+000 D()/DN=1.1463e-006 Residual K= 0.150
 Max stress = 12.000 r = 0.67 374550 Cycles Subspectra: 0 Pass: 375

C Crack size= 0.25 Beta Tension= 1.0564 Beta Compression= 1.0564 R(k)= 0.6334 R(final)= 0.6334
 Delta k=3.7449e+000 D()/DN=1.4319e-006 Residual K= 0.179
 Max stress = 12.000 r = 0.67 408715 Cycles Subspectra: 0 Pass: 409

C Crack size= 0.3 Beta Tension= 1.0104 Beta Compression= 1.0104 R(k)= 0.6186 R(final)= 0.6186
 Delta k=3.9235e+000 D()/DN=1.5234e-006 Residual K= 0.190
 Max stress = 12.000 r = 0.67 437518 Cycles Subspectra: 0 Pass: 438

C Crack size= 0.35001 Beta Tension= 0.9786 Beta Compression= 0.9786 R(k)= 0.6660 R(final)= 0.6660
 Delta k=4.1049e+000 D()/DN=2.4074e-006 Residual K= 0.201
 Max stress = 12.000 r = 0.67 462126 Cycles Subspectra: 0 Pass: 462

C Crack size= 0.40001 Beta Tension= 0.9565 Beta Compression= 0.9565 R(k)= 0.6721 R(final)= 0.6721
 Delta k=4.2888e+000 D()/DN=2.8862e-006 Residual K= 0.212
 Max stress = 12.000 r = 0.67 483194 Cycles Subspectra: 0 Pass: 483

C Crack size= 0.45001 Beta Tension= 0.9411 Beta Compression= 0.9411 R(k)= 0.6721 R(final)= 0.6721
 Delta k=4.4760e+000 D()/DN=3.3115e-006 Residual K= 0.223
 Max stress = 12.000 r = 0.67 501356 Cycles Subspectra: 0 Pass: 501

C Crack size= 0.50001 Beta Tension= 0.9310 Beta Compression= 0.9310 R(k)= 0.6715 R(final)= 0.6715
 Delta k=4.6674e+000 D()/DN=3.7728e-006 Residual K= 0.235
 Max stress = 12.000 r = 0.67 517130 Cycles Subspectra: 0 Pass: 517

C Crack size= 0.55001 Beta Tension= 0.9252 Beta Compression= 0.9252 R(k)= 0.6376 R(final)= 0.6376
 Delta k=4.8647e+000 D()/DN=3.5008e-006 Residual K= 0.239
 Max stress = 12.000 r = 0.67 530811 Cycles Subspectra: 0 Pass: 531

C Crack size= 0.60002 Beta Tension= 0.9231 Beta Compression= 0.9231 R(k)= 0.6250 R(final)= 0.6250
 Delta k=5.0697e+000 D()/DN=3.7245e-006 Residual K= 0.243
 Max stress = 12.000 r = 0.67 542693 Cycles Subspectra: 0 Pass: 543

C Crack size= 0.65002 Beta Tension= 0.9246 Beta Compression= 0.9246 R(k)= 0.6683 R(final)= 0.6683
 Delta k=5.2850e+000 D()/DN=5.5477e-006 Residual K= 0.247
 Max stress = 12.000 r = 0.67 553056 Cycles Subspectra: 0 Pass: 553

C Crack size= 0.70002 Beta Tension= 0.9295 Beta Compression= 0.9295 R(k)= 0.6659 R(final)= 0.6659
 Delta k=5.5134e+000 D()/DN=6.2838e-006 Residual K= 0.251
 Max stress = 12.000 r = 0.67 562022 Cycles Subspectra: 0 Pass: 562

C Crack size= 0.75002 Beta Tension= 0.9379 Beta Compression= 0.9379 R(k)= 0.6341 R(final)= 0.6341
 Delta k=5.7588e+000 D()/DN=6.0367e-006 Residual K= 0.255
 Max stress = 12.000 r = 0.67 569781 Cycles Subspectra: 0 Pass: 570

C Crack size= 0.80002 Beta Tension= 0.9502 Beta Compression= 0.9502 R(k)= 0.6105 R(final)= 0.6105
 Delta k=6.0259e+000 D()/DN=6.1825e-006 Residual K= 0.257
 Max stress = 12.000 r = 0.67 576601 Cycles Subspectra: 0 Pass: 577

C Crack size= 0.85003 Beta Tension= 0.9670 Beta Compression= 0.9670 R(k)= 0.6214 R(final)= 0.6214
 Delta k=6.3209e+000 D()/DN=7.5168e-006 Residual K= 0.259
 Max stress = 12.000 r = 0.67 582692 Cycles Subspectra: 0 Pass: 583

C Crack size= 0.90004 Beta Tension= 0.9890 Beta Compression= 0.9890 R(k)= 0.6710 R(final)= 0.6710
 Delta k=6.6520e+000 D()/DN=1.0303e-005 Residual K= 0.260
 Max stress = 12.000 r = 0.67 588102 Cycles Subspectra: 0 Pass: 588

C Crack size= 0.95005 Beta Tension= 1.0173 Beta Compression= 1.0173 R(k)= 0.6440 R(final)= 0.6440
 Delta k=7.0300e+000 D()/DN=1.0596e-005 Residual K= 0.262
 Max stress = 12.000 r = 0.67 592865 Cycles Subspectra: 0 Pass: 593

C Crack size= 1.0001 Beta Tension= 1.0536 Beta Compression= 1.0536 R(k)= 0.6685 R(final)= 0.6685
Delta k=7.4698e+000 D()/DN=1.3312e-005 Residual K= 0.264
Max stress = 12.000 r = 0.67 597032 Cycles Subspectra: 0 Pass: 597

C Crack size= 1.0502 Beta Tension= 1.1002 Beta Compression= 1.1002 R(k)= 0.6038 R(final)= 0.6038
Delta k=7.9935e+000 D()/DN=1.2501e-005 Residual K= 0.267
Max stress = 12.000 r = 0.67 600601 Cycles Subspectra: 0 Pass: 601

C Crack size= 1.1002 Beta Tension= 1.1604 Beta Compression= 1.1604 R(k)= 0.6107 R(final)= 0.6107
Delta k=8.6294e+000 D()/DN=1.5297e-005 Residual K= 0.271
Max stress = 12.000 r = 0.67 603658 Cycles Subspectra: 0 Pass: 604

C Crack size= 1.1502 Beta Tension= 1.2397 Beta Compression= 1.2397 R(k)= 0.6699 R(final)= 0.6699
Delta k=9.4262e+000 D()/DN=2.2837e-005 Residual K= 0.275
Max stress = 12.000 r = 0.67 606203 Cycles Subspectra: 0 Pass: 606

C Crack size= 1.2002 Beta Tension= 1.3477 Beta Compression= 1.3477 R(k)= 0.6696 R(final)= 0.6696
Delta k=1.0468e+001 D()/DN=2.9144e-005 Residual K= 0.279
Max stress = 12.000 r = 0.67 608221 Cycles Subspectra: 0 Pass: 608

C Crack size= 1.2502 Beta Tension= 1.5062 Beta Compression= 1.5062 R(k)= 0.6196 R(final)= 0.6196
Delta k=1.1940e+001 D()/DN=3.4221e-005 Residual K= 0.283
Max stress = 12.000 r = 0.67 609745 Cycles Subspectra: 0 Pass: 610

C Crack size= 1.3002 Beta Tension= 1.7851 Beta Compression= 1.7851 R(k)= 0.6300 R(final)= 0.6300
Delta k=1.4432e+001 D()/DN=5.6208e-005 Residual K= 0.281
Max stress = 12.000 r = 0.67 610805 Cycles Subspectra: 0 Pass: 611

C Crack size= 1.3503 Beta Tension= 2.7122 Beta Compression= 2.7122 R(k)= 0.6680 R(final)= 0.6680
Delta k=2.2344e+001 D()/DN=4.3624e-004 Residual K= 0.279
Max stress = 12.000 r = 0.67 611319 Cycles Subspectra: 0 Pass: 611

*****Fracture based on 'Kmax' Criteria (current maximum stress)

C Crack size= 1.3522 Beta Tension= 2.8209 Beta Compression= 2.8209 R(k)= 0.6680 R(final)= 0.6680
Delta k=2.3257e+001 D()/DN=5.6061e-004 Residual K= 0.279
Max stress = 12.000 r = 0.67 611323 Cycles Subspectra: 0 Pass: 611

Stress State in the 'C' direction (PSC): 2

Fracture has occurred- run time: 0 hour(s) 0 minute(s) 4 second(s)

6.2 Advanced Double Through-the-Thickness Cracks at a Hole

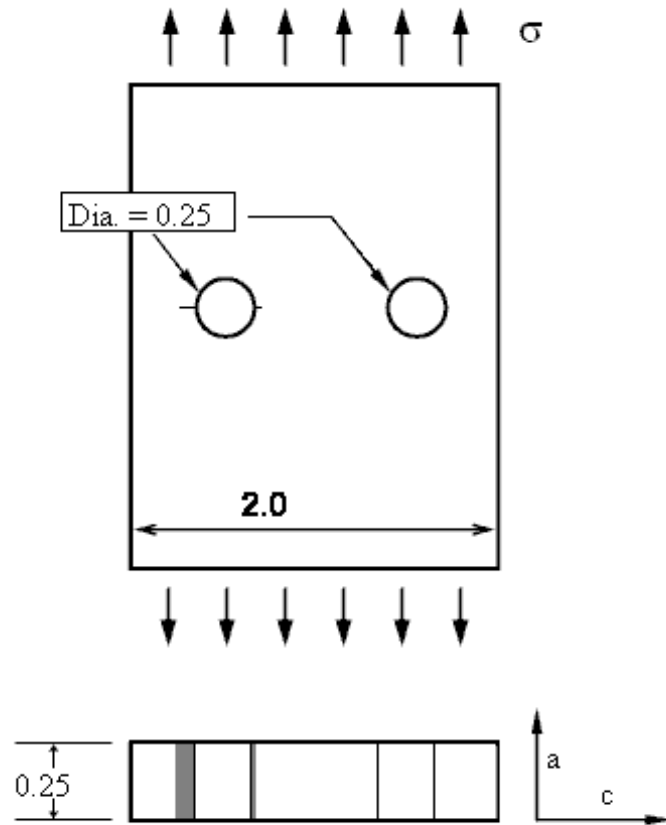


Figure 236: Non-Symmetric Through Crack Geometry

Specimen Geometry:	Unsymmetrical Through Cracks at a Hole
Dimensions:	W = 2.0 in., T = 0.25 in., Dia. = 0.25 in.
Cracked Hole Offset:	B = 0.5 in.
Initial Crack Size:	C11 (left) = 0.05 in., C12 (right) = 0.005 in.
Second Hole Offset:	B = 1.5 in.
Material:	User-Defined Tabular Crack Growth Rate Data
Stress Spectrum:	FALSTAFF
Maximum Applied Stress	25 ksi
Retardation Model:	Hsu Retardation Model
Stress State:	Automatic
Beta Correction:	None
Environment:	N/A

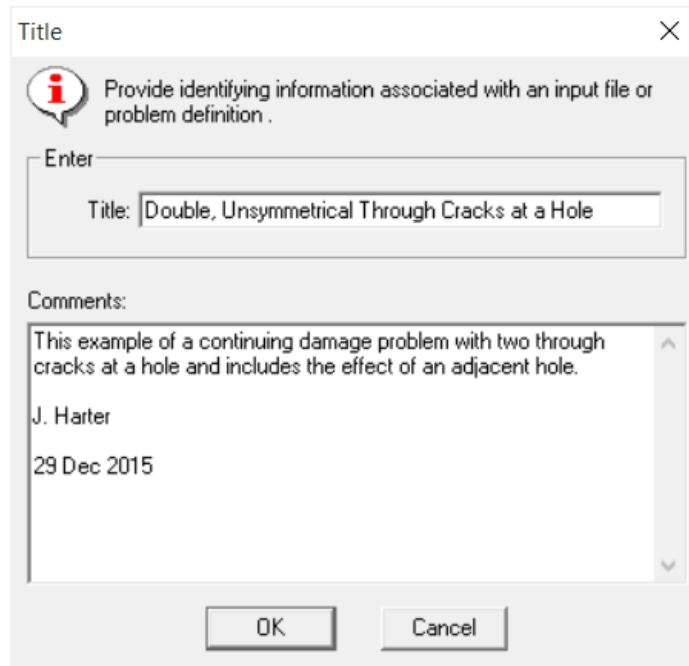
Predict Preferences:

Use the AFGROW defaults except set the print interval to 0.05 inches, and turn off the net section yield failure criterion.

6.2.1 Entering Data

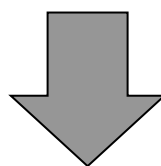
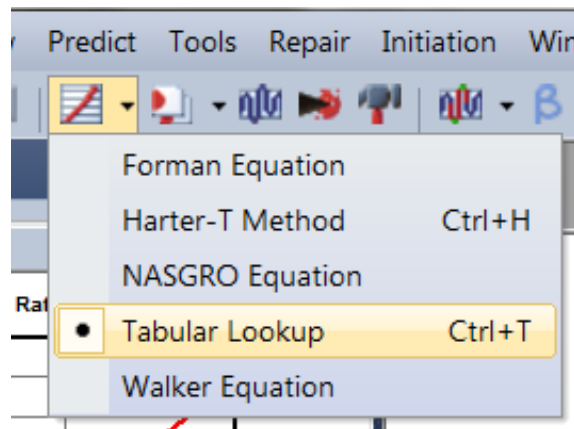
This model uses the AFGROW advanced model interface (see Section 2.2) which allows users to model one or two independent cracks in a plane normal to the applied stresses. The effect of adjacent holes is also included as an option.

6.2.1.1 Input Title



The 'Title' dialog box is shown with a close button (X) in the top right corner. It contains an information icon and the text: 'Provide identifying information associated with an input file or problem definition.' Below this is an 'Enter' section with a text field labeled 'Title:' containing the text 'Double, Unsymmetrical Through Cracks at a Hole'. Underneath is a 'Comments:' section with a text area containing the following text: 'This example of a continuing damage problem with two through cracks at a hole and includes the effect of an adjacent hole.', 'J. Harter', and '29 Dec 2015'. At the bottom are 'OK' and 'Cancel' buttons.

6.2.1.2 Input Material



Tabular LOOKUP Data

Input values of Delta_K for da/dN values and up to 10 different R(stress ratio) values.
Matrix must have at least two R values and two da/dN values.
Input Delta_K for R >= 0, input Kmax for R < 0.0

Number of da/dN Sets: 27 Number of R Sets: 2

		R[1]	R[2]
		0.1	0.6
da/dN[1]	1.00e-009	2.606	1.38
da/dN[2]	3.00e-009	2.636	1.409
da/dN[3]	1.00e-008	2.673	1.503
da/dN[4]	2.00e-008	2.685	1.66
da/dN[5]	1.00e-008	2.770	1.807

Material name: User defined data

Ultimate Strength: 66 Young's Modulus: 10500

Coefficient of Thermal Expansion: 1.25e-005 Poisson's Ratio: 0.33

Upper limit on da/dN, DADNHI: 0.01 Lower limit on da/dN, DADNLO: 1e-009

Plane Stress Fracture Toughness, KC: 62.777 Yield Strength, YLD: 47

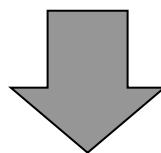
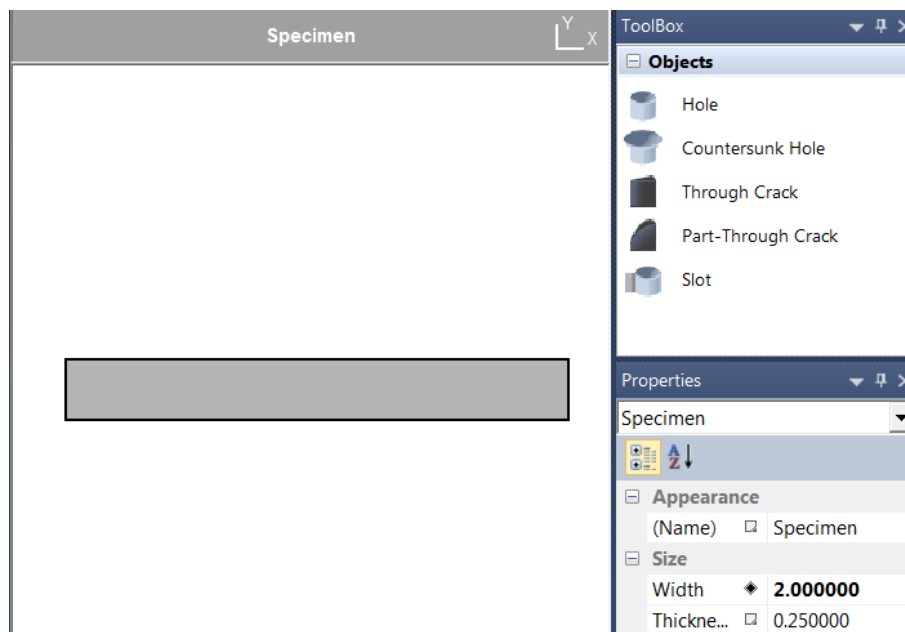
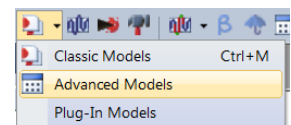
Plane Strain Fracture Toughness, KIC: 35 Lower limit on R shift (Max: 0): -0.3

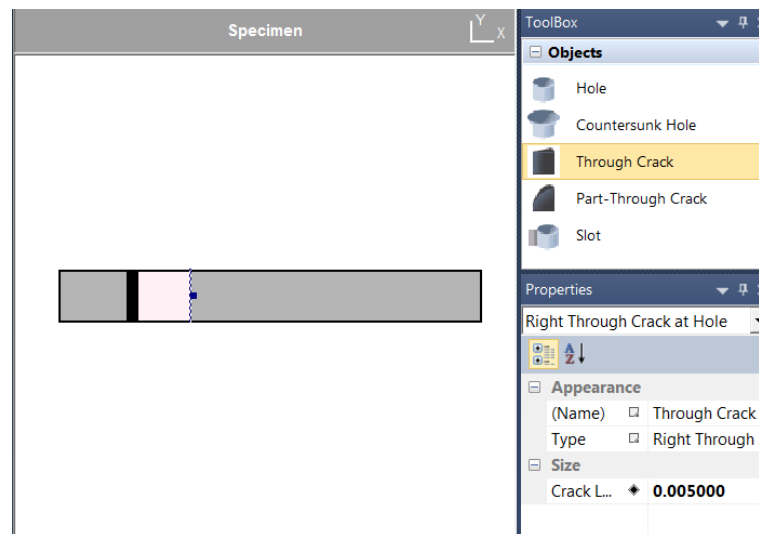
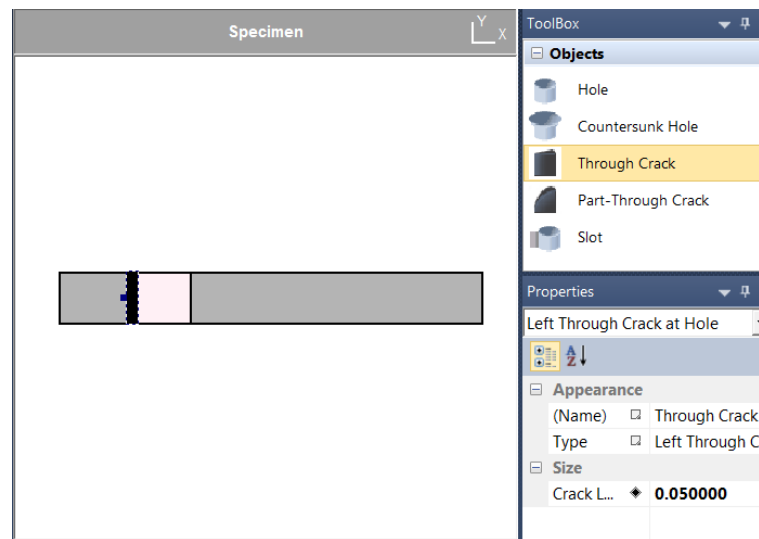
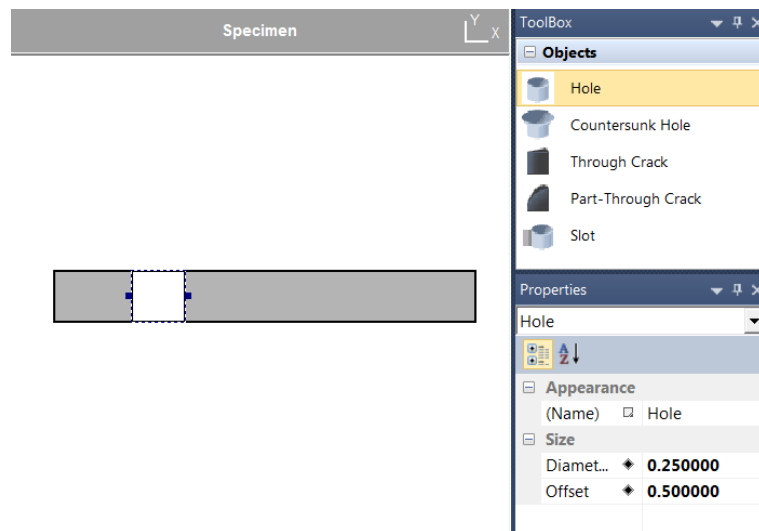
Delta K threshold value @R=0: 2.831 Upper limit on R shift (0, 1): 0.63

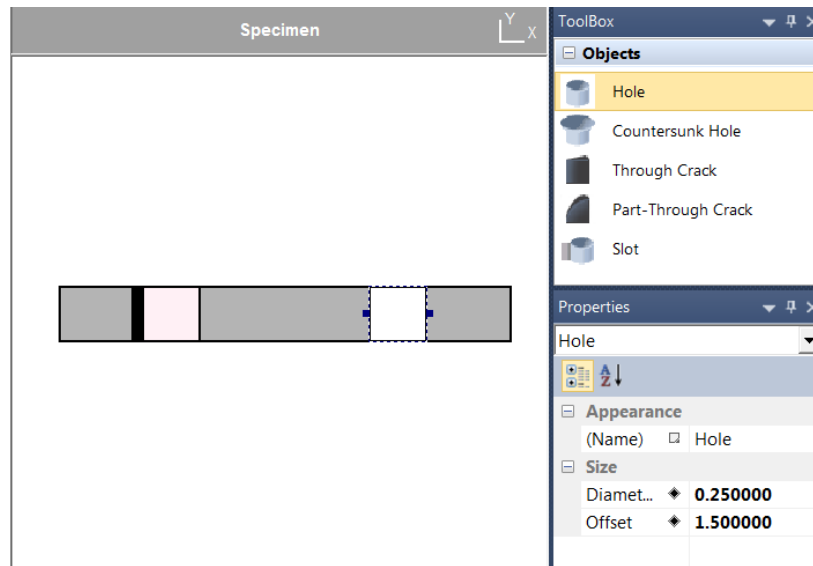
OK Cancel Save Read Apply

(This is the default data provided in the Tabular Look-up dialog)

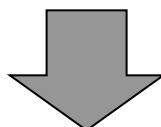
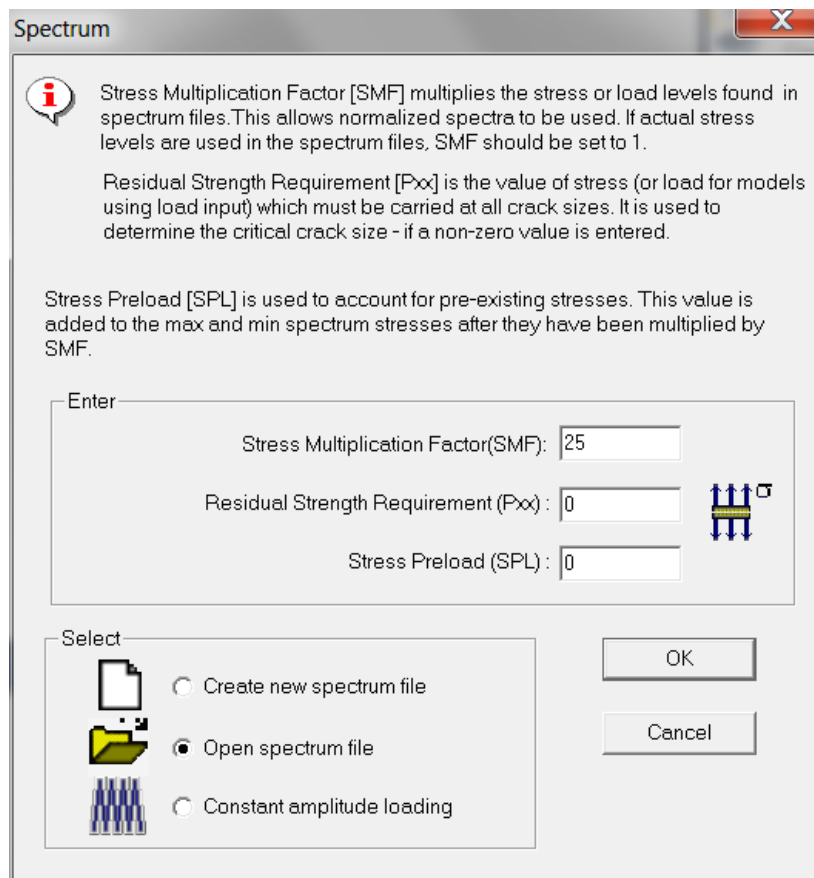
6.2.1.3 Input Model (Advanced Models)



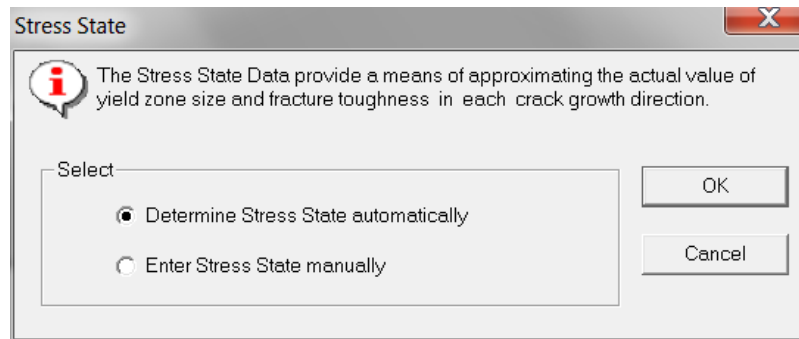




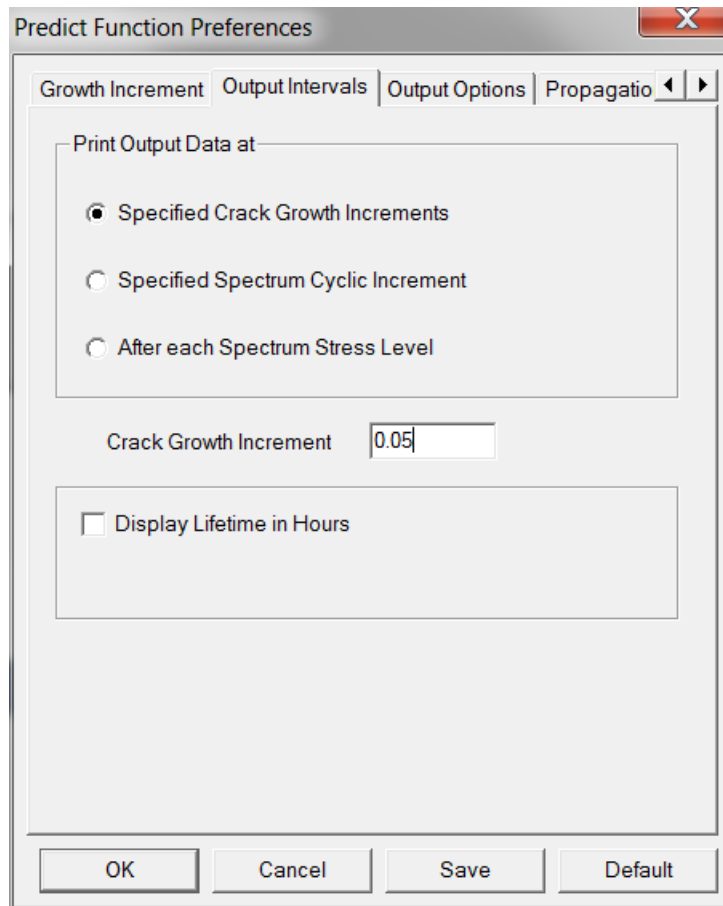
6.2.1.4 Input Spectrum



6.2.1.6 Stress State



6.2.1.7 Predict Preferences



Predict Function Preferences ✕

Output Intervals | Output Options | Propagation Limits | Trc ◀ ▶

Stop Crack Propagation at:

- ☐ Crack Length
- ☐ Cycle Count
- ☒ 'Kmax' Failure Criteria
- ☐ User-Defined 'Kmax'
- ☐ 'Net Section Yield' Failure Criteria
- ☐ Part Through Crack Transition

Number of times the spectrum will be repeated

Spectrum Reps (Max: 9999999):

Minimum crack growth after one pass of the spectrum

Minimum crack growth:

OK Cancel Save Default

6.2.2 AFGROW Output

The results of the AFGROW analysis are given below.

This example of a continuing damage problem with two through
cracks at a hole and includes the effect of an adjacent hole.

J. Harter

29 Dec 2015

#####

AFGROW 5.2.5.19 12/29/2015 14:13

**English Units [Length(in), Stress(Ksi), Temperature(F), Force(Kip)]

Crack Growth Model and Spectrum Information

Title: Double, Unsymmetrical Through Cracks at a Hole

Load: Axial Stress Fraction: 1, Bending Stress Fraction: 0, Bearing Stress Fraction: 0

Thickness : 0.250

Width : 2.000

Hole #1 (Hole)

Diameter = 0.25

Offset = 0.5

Hole #2 (Hole)

Diameter = 0.25

Offset = 1.5

Crack #1 (Through Crack at Hole)

Length = 0.05

Position: Hole Left

Crack #2 (Through Crack at Hole)

Length = 0.005

Position: Hole Right

Young's Modulus =10500 , Poisson's Ratio =0.33 , Coeff. of Thermal Expan. =1.25e-005

Retardation: HSU

Upper limit for the Hsu 'M' (M0): =0.6

Highest R used to calculate Sopen (Rcut) =0.3

Determine Stress State automatically (2 = Plane stress, 6 = Plane strain)

No K-Solution Filters

Tabular Lookup crack growth rate data are being used

For Reff < 0.0, Kmax is used in place of Delta K

Material: User defined data

da/dN values= 27, R values= 2

Lower 'R' value boundary: -0.3

Upper 'R' value boundary: 0.63
Plane strain fracture toughness: 35
Plane stress fracture toughness: 62.777
Delta K threshold value: 2.831
Upper limit on da/dn: 0.01
Lower limit on da/dn: 1e-009
Yield stress: 47
Ultimate strength: 66

Failure is based on the current load in the applied spectrum
Vroman integration at 5% crack length

Spectrum Information : Spectrum title: Falstaff
Spectrum multiplication factor: 25.000
SPL: 0.000
The spectrum will be repeated up to 999999 times
Total Cycles: 17983
Levels: 15674
Subspectra: 200
Max Value: 1
Min Value: -0.2667

No Spectrum Filters

Crack #1

Left Tip C = 0.05 Beta Tension= 2.0383 Beta Compression= 2.0383 R(k)=-5.3523 R(final)=-0.3000
Delta k=3.8978e-001 D()/DN=0.0000e+000
Right Tip C= 0.005 Beta Tension= 3.5945 Beta Compression= 3.5945 R(k)=-5.3523 R(final)=-0.3000
Delta k=2.1737e-001 D()/DN=0.0000e+000
Max stress 0.483, r = -5.35, 0 Cycles, Flight: 1, Pass: 1

Crack #1

Left Tip C = 0.10002 Beta Tension= 1.6442 Beta Compression= 1.6442 R(k)= 1.0000 R(final)=
1.0000 Delta k=0.0000e+000 D()/DN=0.0000e+000
Right Tip C= 0.019009 Beta Tension= 3.1398 Beta Compression= 3.1398 R(k)= 0.5848 R(final)=
0.5848 Delta k=3.4077e+000 D()/DN=4.0237e-007
Max stress 10.698, r = 0.05, 37654 Cycles, Flight: 426, Pass: 3

Crack #1

Left Tip C = 0.15002 Beta Tension= 1.5517 Beta Compression= 1.5517 R(k)= 0.3329 R(final)=
0.3329 Delta k=1.0507e+001 D()/DN=2.1821e-005
Right Tip C= 0.053035 Beta Tension= 2.3804 Beta Compression= 2.3804 R(k)= 0.3288 R(final)=
0.3288 Delta k=9.6415e+000 D()/DN=1.5770e-005
Max stress 14.785, r = 0.31, 61602 Cycles, Flight: 693, Pass: 4

Crack #1

Left Tip C = 0.15282 Beta Tension= 1.5501 Beta Compression= 1.5501 R(k)= 1.0000 R(final)=
1.0000 Delta k=0.0000e+000 D()/DN=0.0000e+000
Right Tip C= 0.055014 Beta Tension= 2.3407 Beta Compression= 2.3407 R(k)= 1.0000 R(final)=
1.0000 Delta k=0.0000e+000 D()/DN=0.0000e+000
Max stress 6.613, r = 0.54, 63242 Cycles, Flight: 707, Pass: 4

Crack #1

Left Tip C = 0.20002 Beta Tension= 1.5714 Beta Compression= 1.5714 R(k)= 0.2927 R(final)= 0.2927 Delta k=1.4826e+001 D()/DN=7.9991e-005
Right Tip C= 0.085064 Beta Tension= 2.0246 Beta Compression= 2.0246 R(k)= 0.2867 R(final)= 0.2867 Delta k=1.2562e+001 D()/DN=4.2101e-005
Max stress 16.828, r = 0.27, 75223 Cycles, Flight: 840, Pass: 5

Crack #1

Left Tip C = 0.24154 Beta Tension= 1.6406 Beta Compression= 1.6406 R(k)= 1.0000 R(final)= 1.0000 Delta k=0.0000e+000 D()/DN=0.0000e+000
Right Tip C= 0.10502 Beta Tension= 1.9259 Beta Compression= 1.9259 R(k)= 1.0000 R(final)= 1.0000 Delta k=0.0000e+000 D()/DN=0.0000e+000
Max stress 0.483, r = -3.23, 83627 Cycles, Flight: 928, Pass: 5

Crack #1

Left Tip C = 0.25003 Beta Tension= 1.6803 Beta Compression= 1.6803 R(k)= 1.0000 R(final)= 1.0000 Delta k=0.0000e+000 D()/DN=0.0000e+000
Right Tip C= 0.10841 Beta Tension= 1.9054 Beta Compression= 1.9054 R(k)= 1.0000 R(final)= 1.0000 Delta k=0.0000e+000 D()/DN=0.0000e+000
Max stress 6.613, r = 0.38, 84297 Cycles, Flight: 932, Pass: 5

++++++Net Section Yield Criteria Failure (current maximum stress)

Crack #1

Left Tip C = 0.2837 Beta Tension= 1.7825 Beta Compression= 1.7825 R(k)= 0.1419 R(final)= 0.1419 Delta k=3.6101e+001 D()/DN=2.9577e-003
Right Tip C= 0.11945 Beta Tension= 1.8804 Beta Compression= 1.8804 R(k)= 0.1419 R(final)= 0.1419 Delta k=2.4712e+001 D()/DN=5.2744e-004
Max stress 25.000, r = 0.14, 86689 Cycles, Flight: 973, Pass: 5

Crack #1

Left Tip C = 0.30109 Beta Tension= 1.9755 Beta Compression= 1.9755 R(k)= 0.8074 R(final)= 0.6300 Delta k=3.2034e+000 D()/DN=3.4218e-007
Right Tip C= 0.12602 Beta Tension= 1.8545 Beta Compression= 1.8545 R(k)= 0.7780 R(final)= 0.6300 Delta k=2.2424e+000 D()/DN=8.9545e-008
Max stress 8.655, r = 0.53, 88523 Cycles, Flight: 991, Pass: 5

++++++Kmax Criteria Failure. Edge 1, Crack 1

Crack #1

Left Tip C = 0.34799 Beta Tension= 2.6385 Beta Compression= 2.6385 R(k)= 0.1101 R(final)= 0.1101 Delta k=5.6362e+001 D()/DN=1.0000e-002
Right Tip C= 0.13871 Beta Tension= 1.9206 Beta Compression= 1.9206 R(k)= 0.1101 R(final)= 0.1101 Delta k=2.5903e+001 D()/DN=6.0886e-004
Max stress 22.958, r = 0.11, 91664 Cycles, Flight: 1026, Pass: 6

*****Transition to slot

Crack #1

C Length = 0.13871 Beta Tension= 2.7102 Beta Compression= 2.7102 R(k)= 0.1101 R(final)= 0.1101 Delta k=3.6551e+001 D()/DN=2.8462e-003
Max stress 22.958, r = 0.11, 91664 Cycles, Flight: 1026, Pass: 6

Crack #1

C Length = 0.15524 Beta Tension= 2.5938 Beta Compression= 2.5938 R(k)= 0.7014 R(final)= 0.6300 Delta k=4.6814e+000 D()/DN=1.3641e-006
Max stress 8.655, r = 0.17, 93898 Cycles, Flight: 1046, Pass: 6

Crack #1

C Length = 0.20584 Beta Tension= 2.3668 Beta Compression= 2.3668 R(k)= 1.0000 R(final)= 1.0000
Delta k=0.0000e+000 D()/DN=0.0000e+000
Max stress 6.613, r = 0.38, 99055 Cycles, Flight: 1105, Pass: 6

Crack #1

C Length = 0.25658 Beta Tension= 2.1772 Beta Compression= 2.1772 R(k)= 0.7787 R(final)= 0.6300
Delta k=6.8354e+000 D()/DN=5.7466e-006
Max stress 15.805, r = 0.74, 102881 Cycles, Flight: 1139, Pass: 6

Crack #1

C Length = 0.30705 Beta Tension= 2.0753 Beta Compression= 2.0753 R(k)= 1.0000 R(final)= 1.0000
Delta k=0.0000e+000 D()/DN=0.0000e+000
Max stress 6.613, r = 0.38, 107862 Cycles, Flight: 1200, Pass: 6

Crack #1

C Length = 0.35708 Beta Tension= 1.9937 Beta Compression= 1.9937 R(k)= 0.4855 R(final)= 0.4855
Delta k=1.3843e+001 D()/DN=7.5709e-005
Max stress 12.743, r = 0.20, 112281 Cycles, Flight: 1252, Pass: 7

Crack #1

C Length = 0.41088 Beta Tension= 1.9503 Beta Compression= 1.9503 R(k)= 0.5333 R(final)= 0.5333
Delta k=1.5289e+001 D()/DN=1.2509e-004
Max stress 14.785, r = 0.10, 115697 Cycles, Flight: 1295, Pass: 7

Crack #1

C Length = 0.46341 Beta Tension= 1.9166 Beta Compression= 1.9166 R(k)= 0.8897 R(final)= 0.6300
Delta k=2.7295e+000 D()/DN=1.8906e-007
Max stress 10.698, r = 0.71, 119185 Cycles, Flight: 1325, Pass: 7

Crack #1

C Length = 0.51587 Beta Tension= 1.8996 Beta Compression= 1.8996 R(k)= 0.8734 R(final)= 0.6300
Delta k=2.3380e+000 D()/DN=1.0562e-007
Max stress 7.635, r = 0.60, 121699 Cycles, Flight: 1358, Pass: 7

Crack #1

C Length = 0.56741 Beta Tension= 1.9128 Beta Compression= 1.9128 R(k)= 0.0934 R(final)= 0.0934
Delta k=3.8958e+001 D()/DN=3.7581e-003
Max stress 16.828, r = 0.09, 123130 Cycles, Flight: 1379, Pass: 7

Crack #1

C Length = 0.61764 Beta Tension= 1.9923 Beta Compression= 1.9923 R(k)= 0.4198 R(final)= 0.4198
Delta k=2.3806e+001 D()/DN=7.8737e-004
Max stress 14.785, r = 0.31, 124206 Cycles, Flight: 1389, Pass: 7

Crack #1

C Length = 0.6732 Beta Tension= 2.1058 Beta Compression= 2.1058 R(k)= 0.7521 R(final)= 0.6300
Delta k=8.8990e+000 D()/DN=1.5486e-005
Max stress 11.720, r = 0.56, 124432 Cycles, Flight: 1391, Pass: 7

+++++Kmax Criteria Failure. Edge 1, Crack 1

Crack #1

C Length = 0.69974 Beta Tension= 2.3645 Beta Compression= 2.3645 R(k)= 0.1339 R(final)= 0.1339

Delta k=5.7293e+001 D()/DN=1.0000e-002

Max stress 18.870, r = 0.13, 124488 Cycles, Flight: 1391, Pass: 7

*****Crack[0] Dim[0] transitioned to a hole [1]

Stress State in the 'C' direction (PSC): 2

Fracture has occurred- run time: 0 hour(s) 0 minute(s) 0 second(s)

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